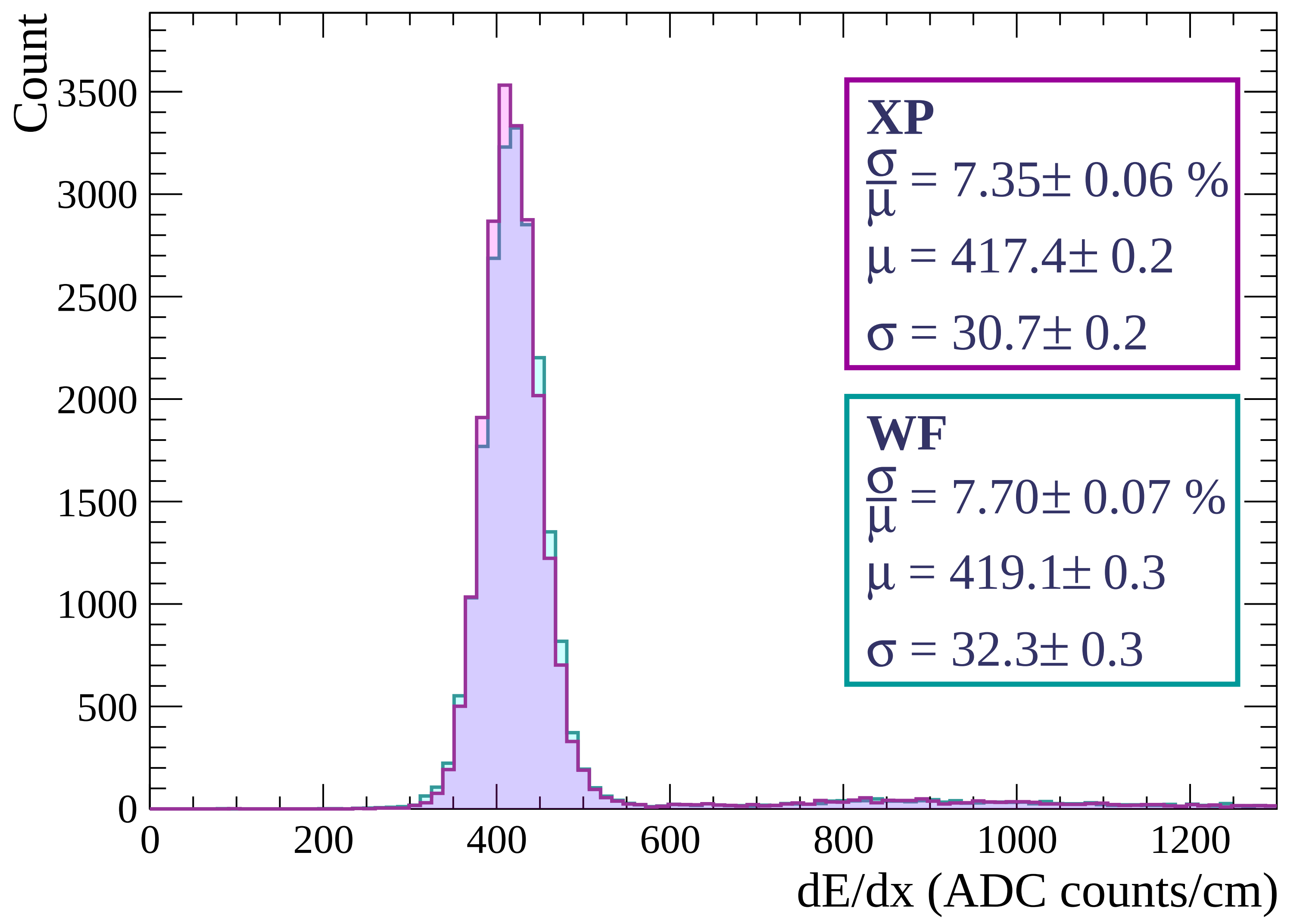
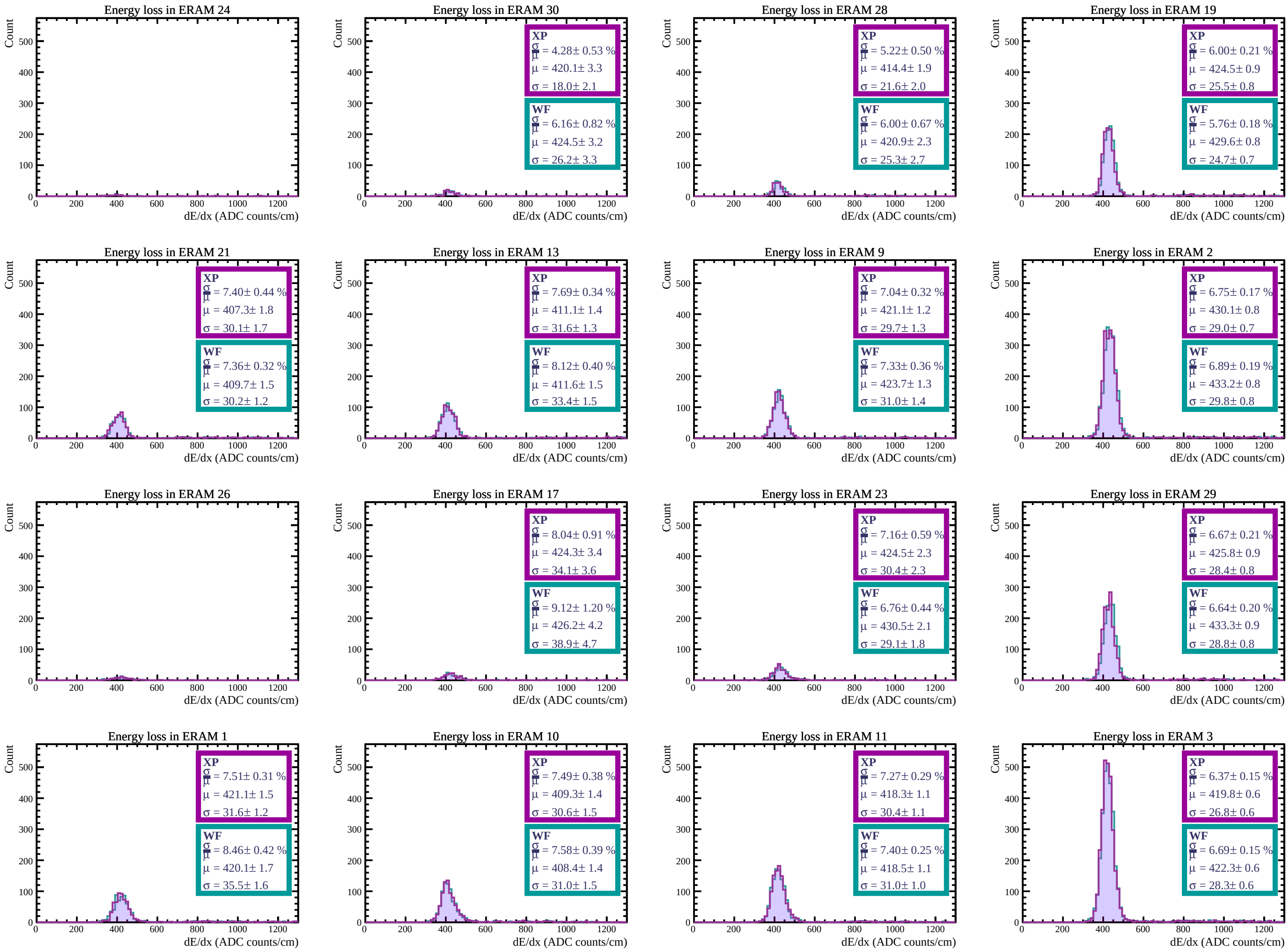
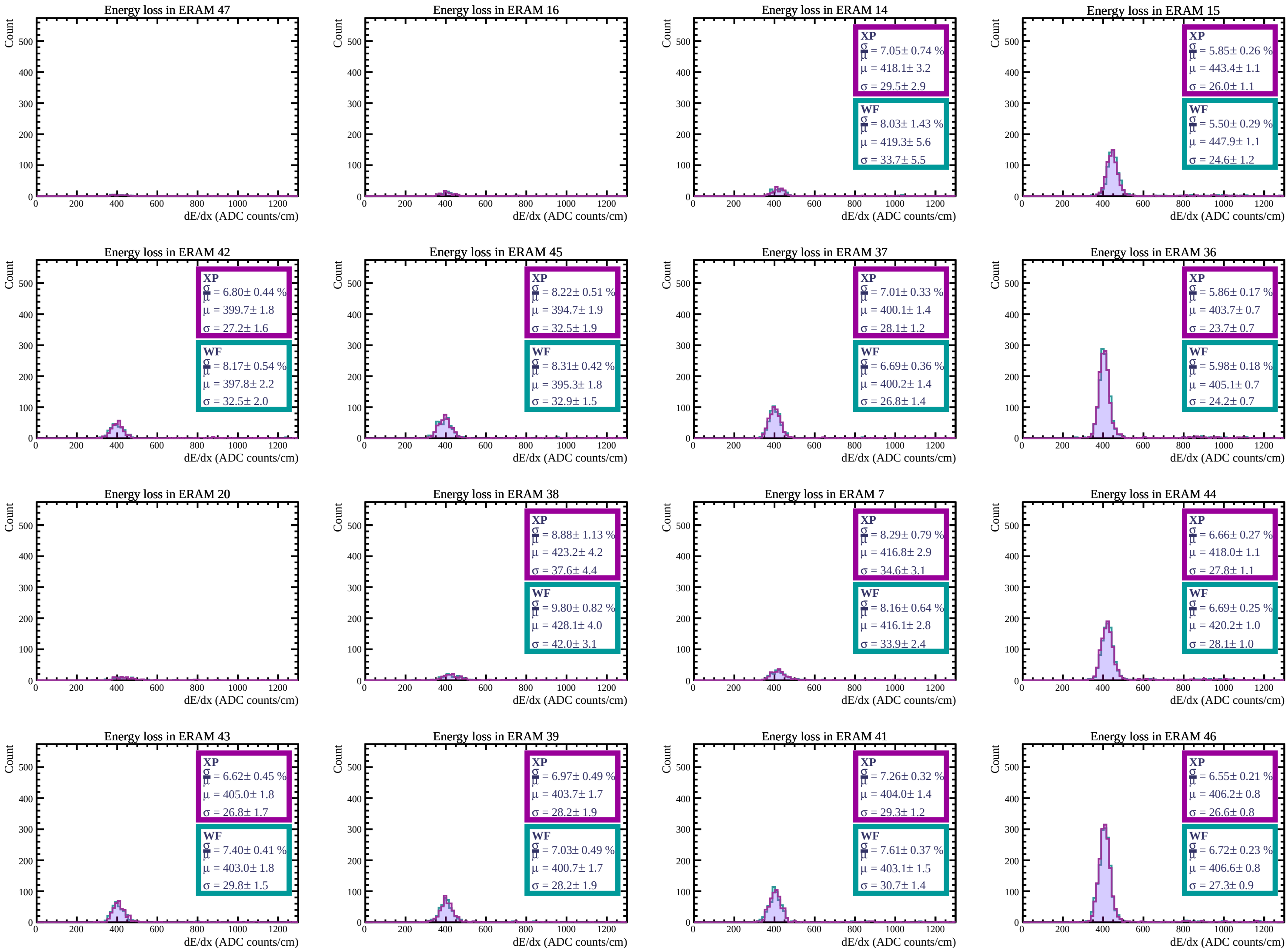
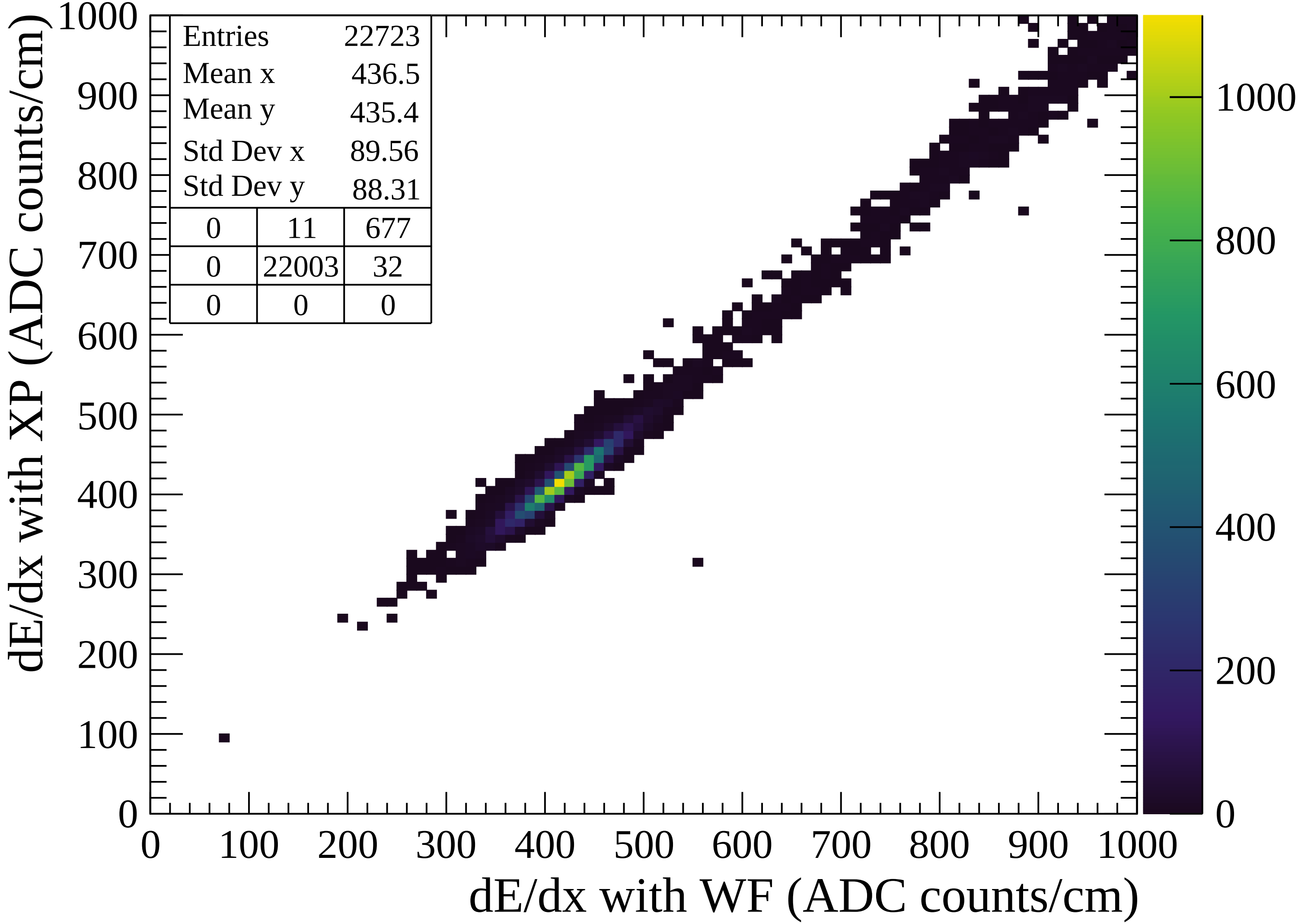
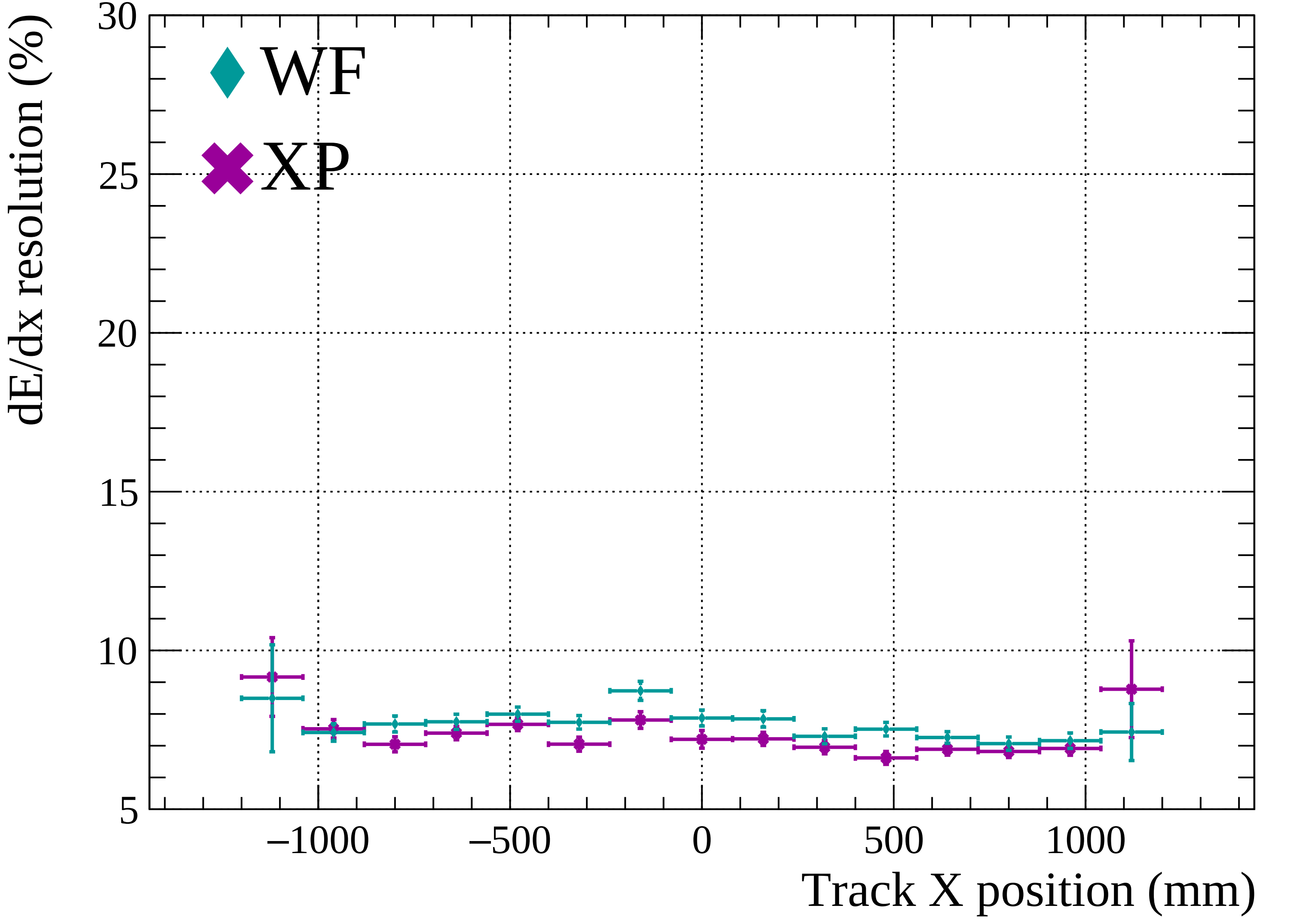
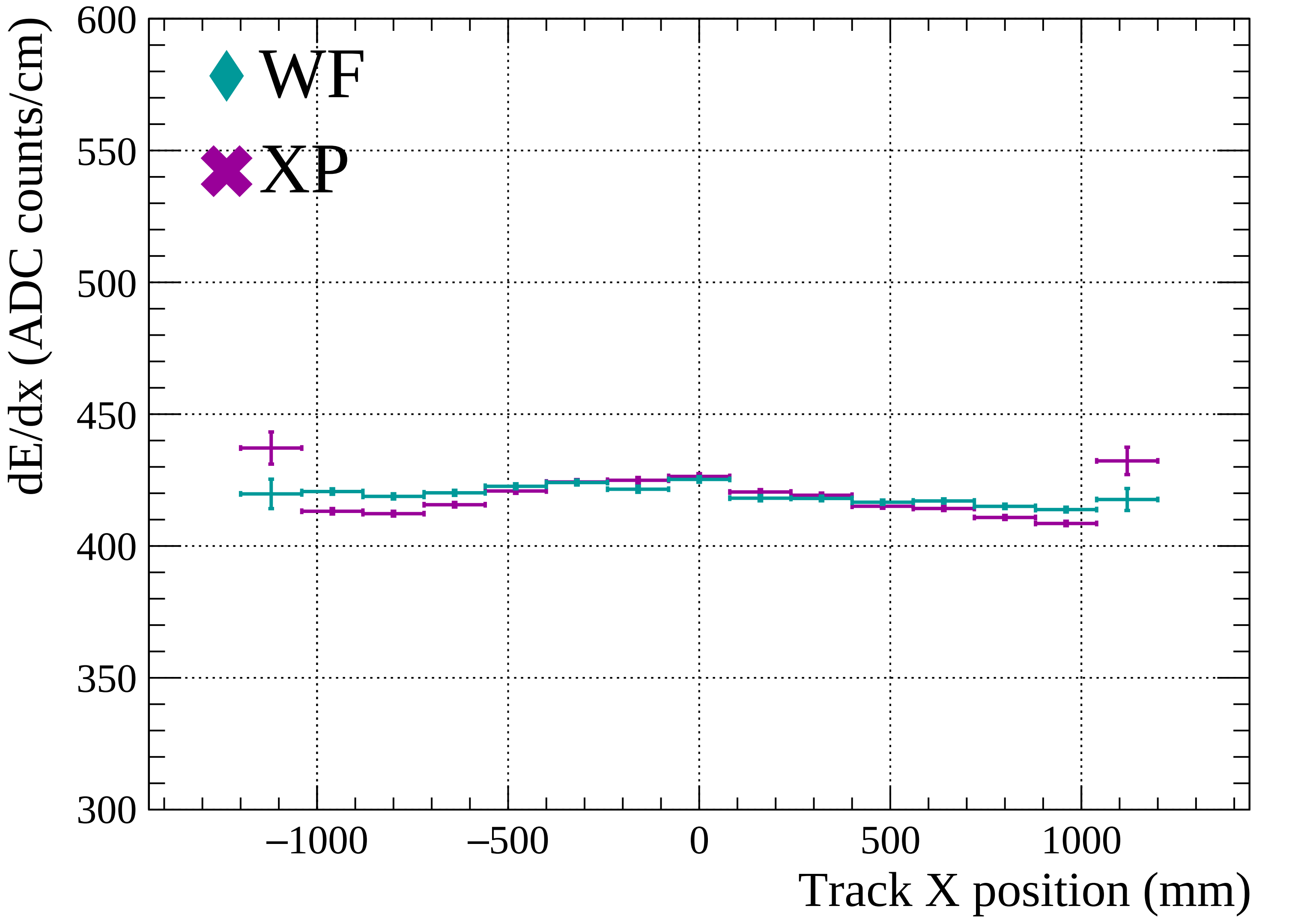

Count
3500
3000
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.35± 0.06 %
μ = 417.4± 0.2
σ = 30.7± 0.2
WF
$\frac{\sigma}{\mu}$ = 7.70± 0.07 %
μ = 419.1± 0.3
σ = 32.3± 0.3


Energy loss in ERAM 24
Energy loss in ERAM 30
XP
σ/μ = 4.28± 0.53 %
μ = 420.1± 3.3
σ = 18.0± 2.1
WF
σ/μ = 6.16± 0.82 %
μ = 424.5± 3.2
σ = 26.2± 3.3
Energy loss in ERAM 28
XP
σ/μ = 5.22± 0.50 %
μ = 414.4± 1.9
σ = 21.6± 2.0
WF
σ/μ = 6.00± 0.67 %
μ = 420.9± 2.3
σ = 25.3± 2.7
Energy loss in ERAM 19
XP
σ/μ = 6.00± 0.21 %
μ = 424.5± 0.9
σ = 25.5± 0.8
WF
σ/μ = 5.76± 0.18 %
μ = 429.6± 0.8
σ = 24.7± 0.7
Energy loss in ERAM 21
XP
σ/μ = 7.40± 0.44 %
μ = 407.3± 1.8
σ = 30.1± 1.7
WF
σ/μ = 7.36± 0.32 %
μ = 409.7± 1.5
σ = 30.2± 1.2
Energy loss in ERAM 13
XP
σ/μ = 7.69± 0.34 %
μ = 411.1± 1.4
σ = 31.6± 1.3
WF
σ/μ = 8.12± 0.40 %
μ = 411.6± 1.5
σ = 33.4± 1.5
Energy loss in ERAM 9
XP
σ/μ = 7.04± 0.32 %
μ = 421.1± 1.2
σ = 29.7± 1.3
WF
σ/μ = 7.33± 0.36 %
μ = 423.7± 1.3
σ = 31.0± 1.4
Energy loss in ERAM 2
XP
σ/μ = 6.75± 0.17 %
μ = 430.1± 0.8
σ = 29.0± 0.7
WF
σ/μ = 6.89± 0.19 %
μ = 433.2± 0.8
σ = 29.8± 0.8
Energy loss in ERAM 26
Energy loss in ERAM 17
XP
σ/μ = 8.04± 0.91 %
μ = 424.3± 3.4
σ = 34.1± 3.6
WF
σ/μ = 9.12± 1.20 %
μ = 426.2± 4.2
σ = 38.9± 4.7
Energy loss in ERAM 23
XP
σ/μ = 7.16± 0.59 %
μ = 424.5± 2.3
σ = 30.4± 2.3
WF
σ/μ = 6.76± 0.44 %
μ = 430.5± 2.1
σ = 29.1± 1.8
Energy loss in ERAM 29
XP
σ/μ = 6.67± 0.21 %
μ = 425.8± 0.9
σ = 28.4± 0.8
WF
σ/μ = 6.64± 0.20 %
μ = 433.3± 0.9
σ = 28.8± 0.8
Energy loss in ERAM 1
XP
σ/μ = 7.51± 0.31 %
μ = 421.1± 1.5
σ = 31.6± 1.2
WF
σ/μ = 8.46± 0.42 %
μ = 420.1± 1.7
σ = 35.5± 1.6
Energy loss in ERAM 10
XP
σ/μ = 7.49± 0.38 %
μ = 409.3± 1.4
σ = 30.6± 1.5
WF
σ/μ = 7.58± 0.39 %
μ = 408.4± 1.4
σ = 31.0± 1.5
Energy loss in ERAM 11
XP
σ/μ = 7.27± 0.29 %
μ = 418.3± 1.1
σ = 30.4± 1.1
WF
σ/μ = 7.40± 0.25 %
μ = 418.5± 1.1
σ = 31.0± 1.0
Energy loss in ERAM 3
XP
σ/μ = 6.37± 0.15 %
μ = 419.8± 0.6
σ = 26.8± 0.6
WF
σ/μ = 6.69± 0.15 %
μ = 422.3± 0.6
σ = 28.3± 0.6
Count
dE/dx (ADC counts/cm)

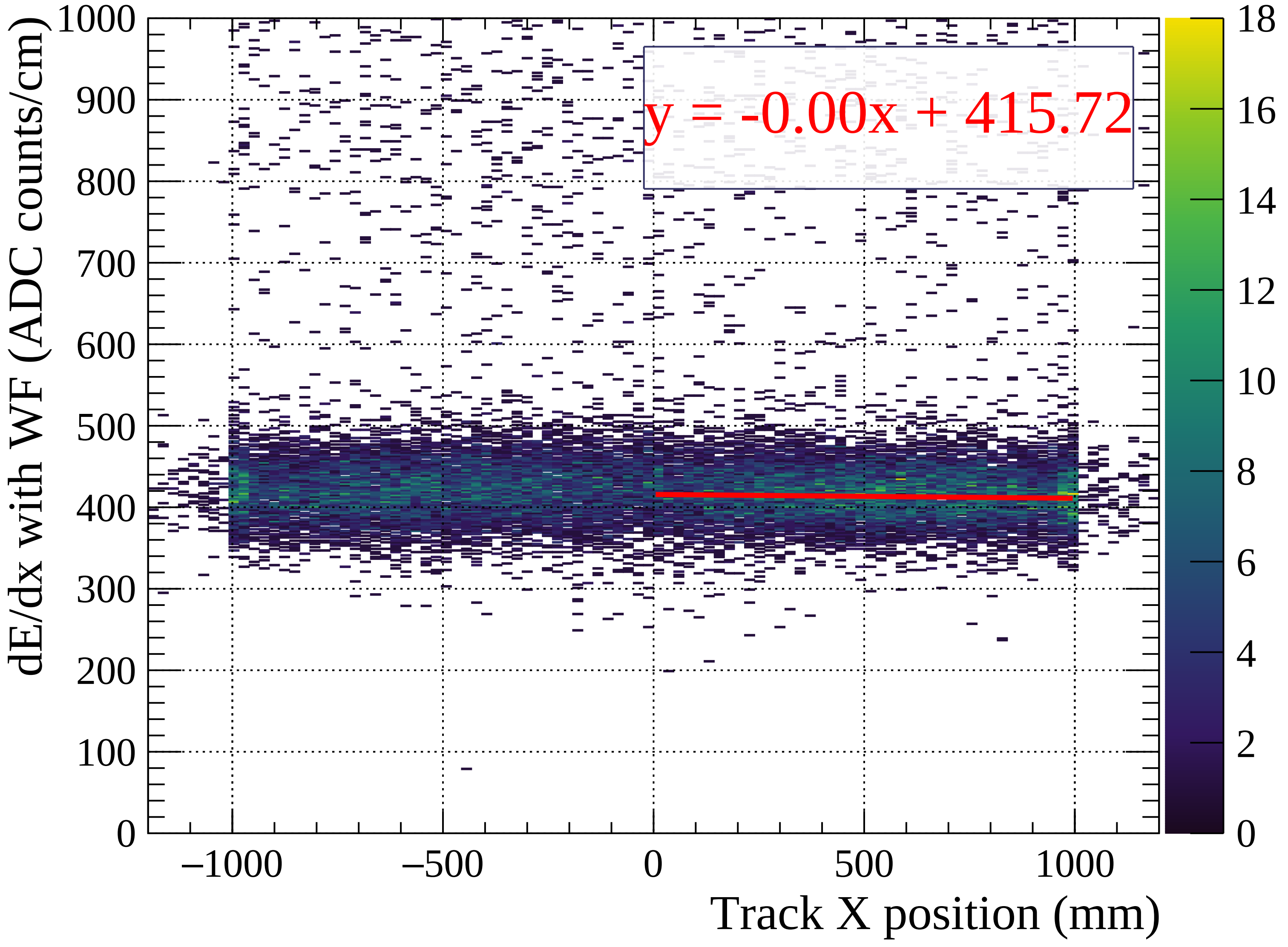

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
XP
σ/μ = 7.05± 0.74 %
μ = 418.1± 3.2
σ = 29.5± 2.9
WF
σ/μ = 8.03± 1.43 %
μ = 419.3± 5.6
σ = 33.7± 5.5
Energy loss in ERAM 15
XP
σ/μ = 5.85± 0.26 %
μ = 443.4± 1.1
σ = 26.0± 1.1
WF
σ/μ = 5.50± 0.29 %
μ = 447.9± 1.1
σ = 24.6± 1.2
Energy loss in ERAM 42
XP
σ/μ = 6.80± 0.44 %
μ = 399.7± 1.8
σ = 27.2± 1.6
WF
σ/μ = 8.17± 0.54 %
μ = 397.8± 2.2
σ = 32.5± 2.0
Energy loss in ERAM 45
XP
σ/μ = 8.22± 0.51 %
μ = 394.7± 1.9
σ = 32.5± 1.9
WF
σ/μ = 8.31± 0.42 %
μ = 395.3± 1.8
σ = 32.9± 1.5
Energy loss in ERAM 37
XP
σ/μ = 7.01± 0.33 %
μ = 400.1± 1.4
σ = 28.1± 1.2
WF
σ/μ = 6.69± 0.36 %
μ = 400.2± 1.4
σ = 26.8± 1.4
Energy loss in ERAM 36
XP
σ/μ = 5.86± 0.17 %
μ = 403.7± 0.7
σ = 23.7± 0.7
WF
σ/μ = 5.98± 0.18 %
μ = 405.1± 0.7
σ = 24.2± 0.7
Energy loss in ERAM 20
Energy loss in ERAM 38
XP
σ/μ = 8.88± 1.13 %
μ = 423.2± 4.2
σ = 37.6± 4.4
WF
σ/μ = 9.80± 0.82 %
μ = 428.1± 4.0
σ = 42.0± 3.1
Energy loss in ERAM 7
XP
σ/μ = 8.29± 0.79 %
μ = 416.8± 2.9
σ = 34.6± 3.1
WF
σ/μ = 8.16± 0.64 %
μ = 416.1± 2.8
σ = 33.9± 2.4
Energy loss in ERAM 44
XP
σ/μ = 6.66± 0.27 %
μ = 418.0± 1.1
σ = 27.8± 1.1
WF
σ/μ = 6.69± 0.25 %
μ = 420.2± 1.0
σ = 28.1± 1.0
Energy loss in ERAM 43
XP
σ/μ = 6.62± 0.45 %
μ = 405.0± 1.8
σ = 26.8± 1.7
WF
σ/μ = 7.40± 0.41 %
μ = 403.0± 1.8
σ = 29.8± 1.5
Energy loss in ERAM 39
XP
σ/μ = 6.97± 0.49 %
μ = 403.7± 1.7
σ = 28.2± 1.9
WF
σ/μ = 7.03± 0.49 %
μ = 400.7± 1.7
σ = 28.2± 1.9
Energy loss in ERAM 41
XP
σ/μ = 7.26± 0.32 %
μ = 404.0± 1.4
σ = 29.3± 1.2
WF
σ/μ = 7.61± 0.37 %
μ = 403.1± 1.5
σ = 30.7± 1.4
Energy loss in ERAM 46
XP
σ/μ = 6.55± 0.21 %
μ = 406.2± 0.8
σ = 26.6± 0.8
WF
σ/μ = 6.72± 0.23 %
μ = 406.6± 0.8
σ = 27.3± 0.9
Count
dE/dx (ADC counts/cm)


Entries 22723
Mean x 436.5
Mean y 435.4
Std Dev x 89.56
Std Dev y 88.31
0 11 677
0 22003 32
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
200
400
600
800
1000

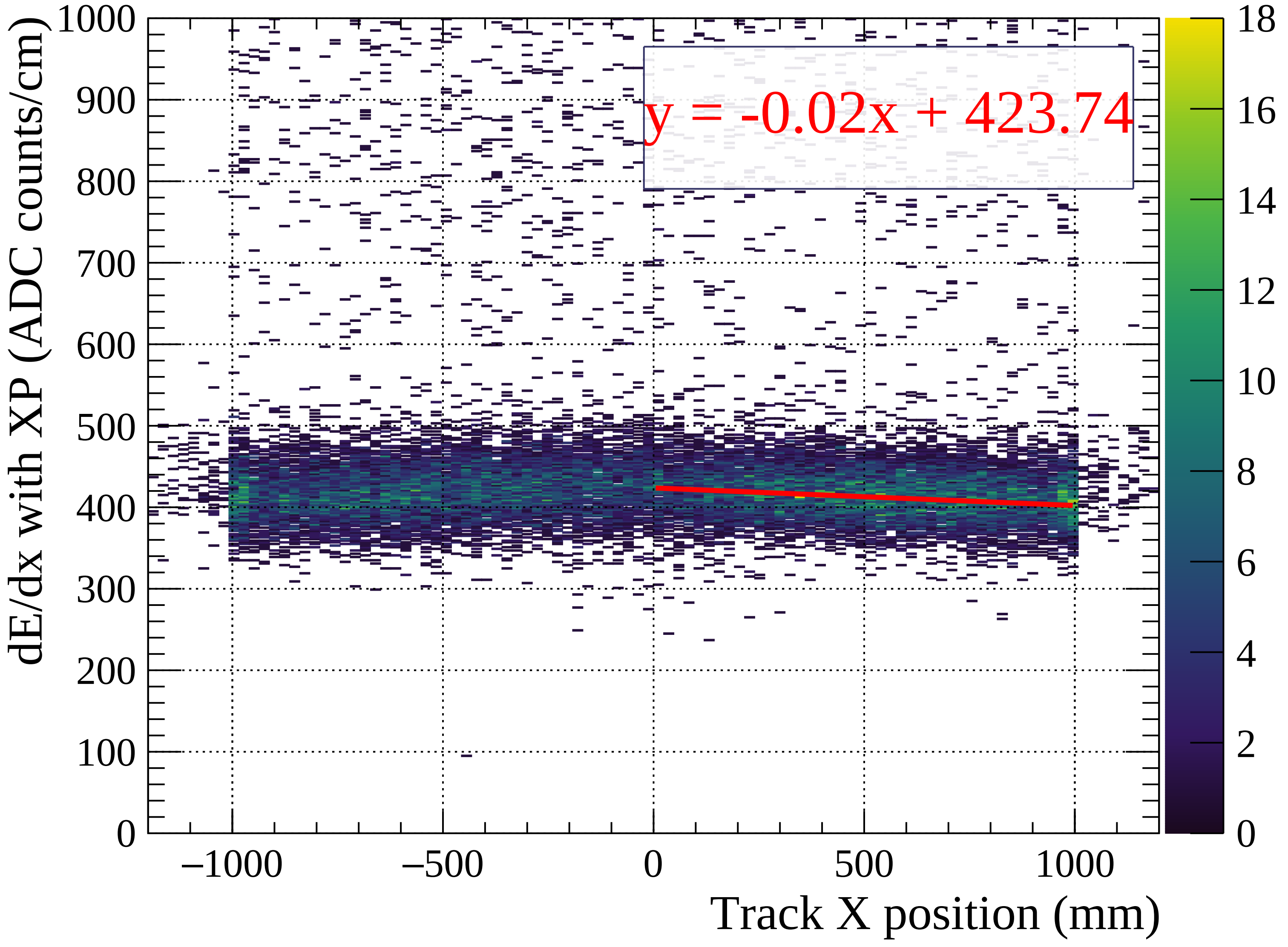

WF
XP
30
25
20
15
10
5
dE/dx resolution (%)
−1000
−500
0
500
1000
Track X position (mm)



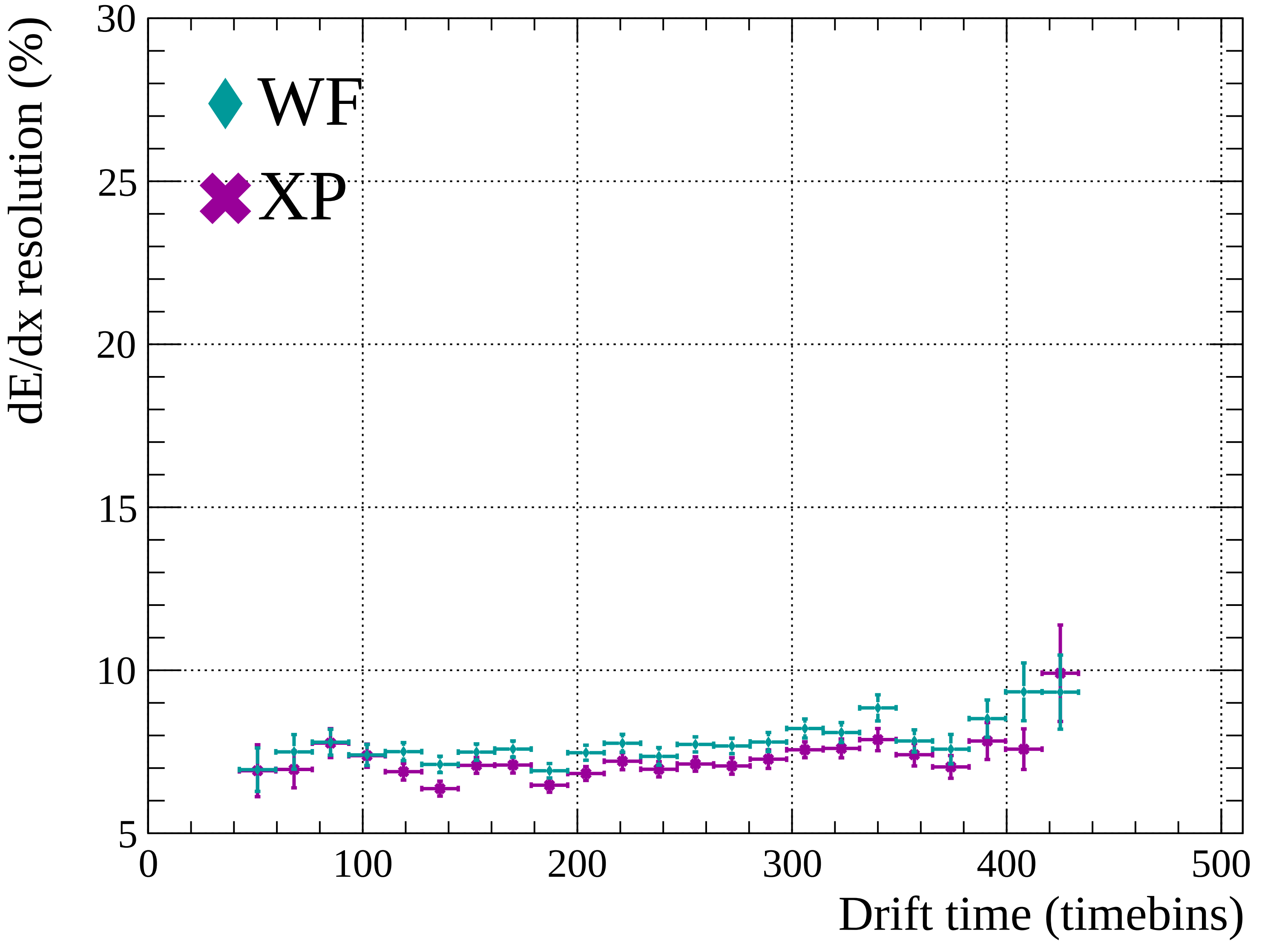
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)

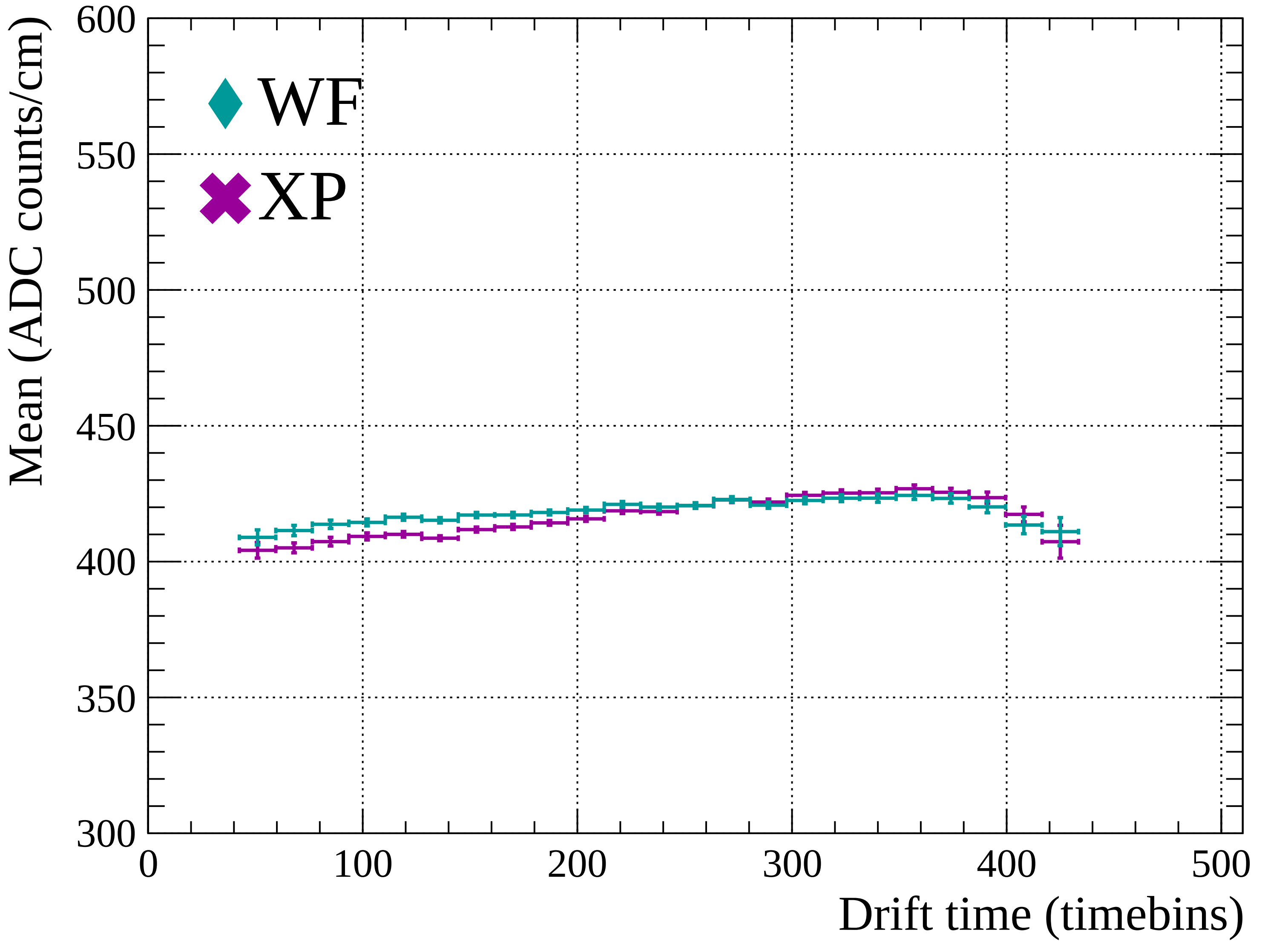


y = -0.00x + 415.72
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
2
4
6
8
10
12
14
16
18

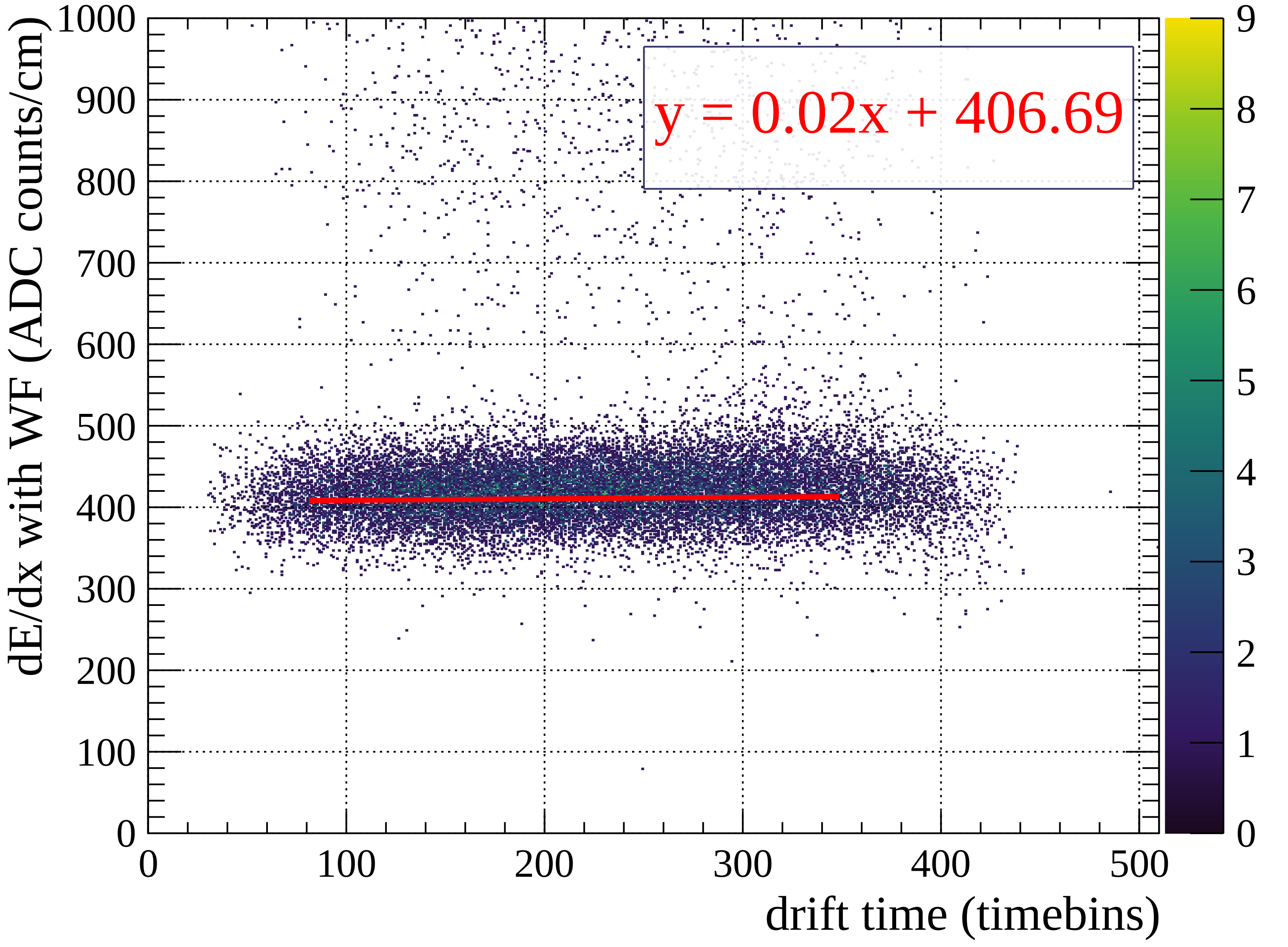


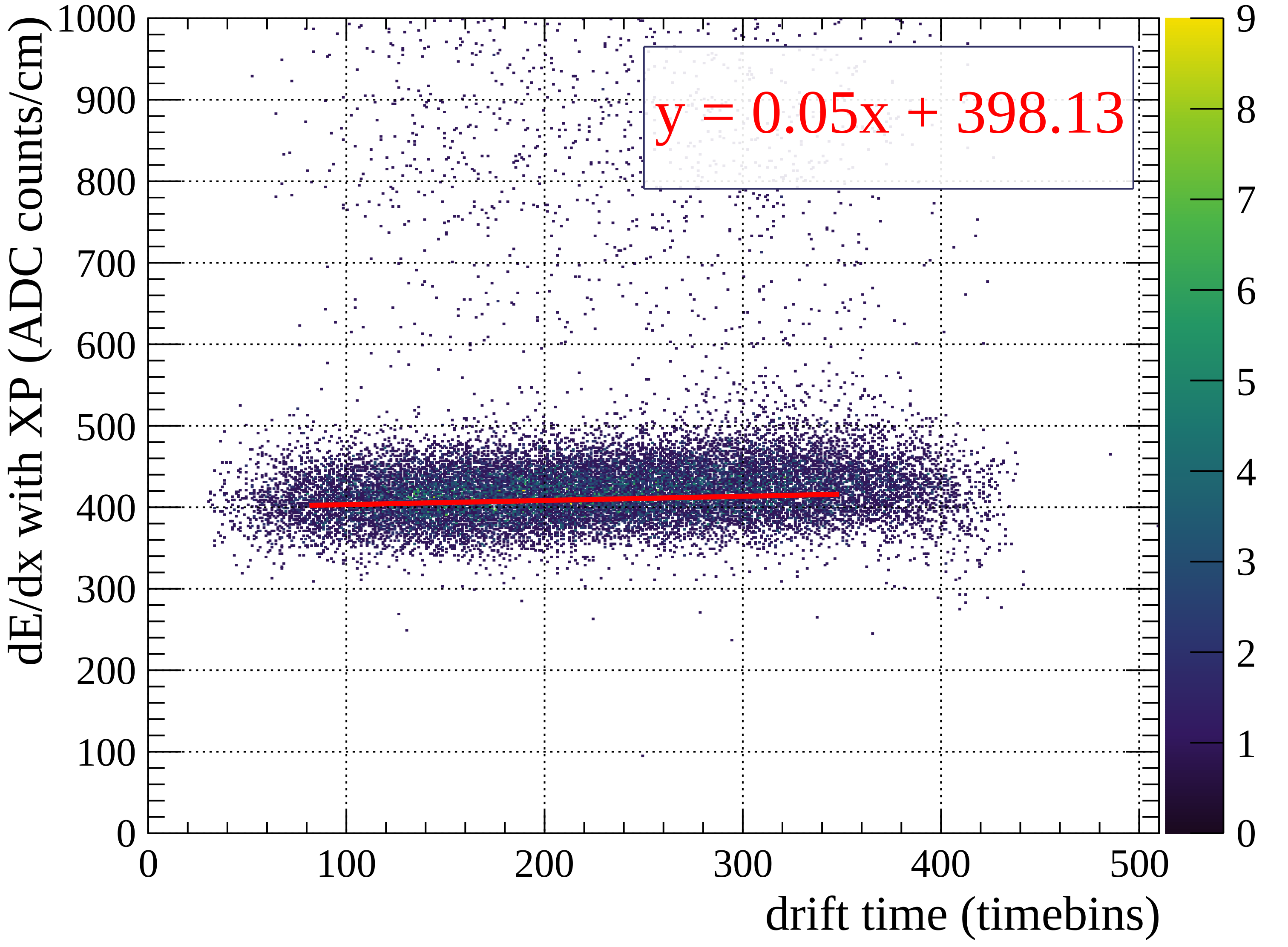
y = -0.02x + 423.74
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
2
4
6
8
10
12
14
16
18

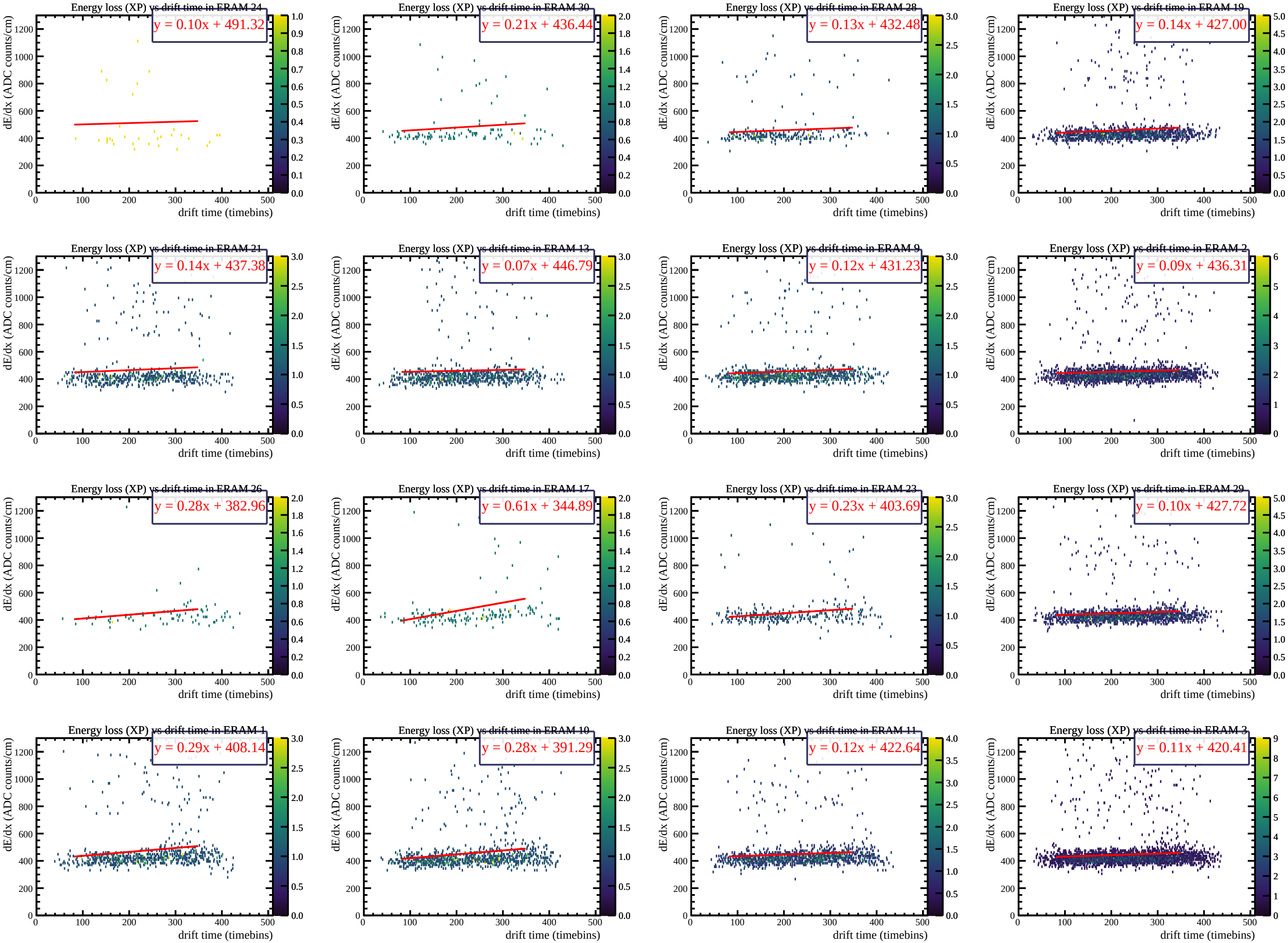

dE/dx resolution (%)
Drift time (timebins)
WF
XP
5
10
15
20
25
30
0
100
200
300
400
500

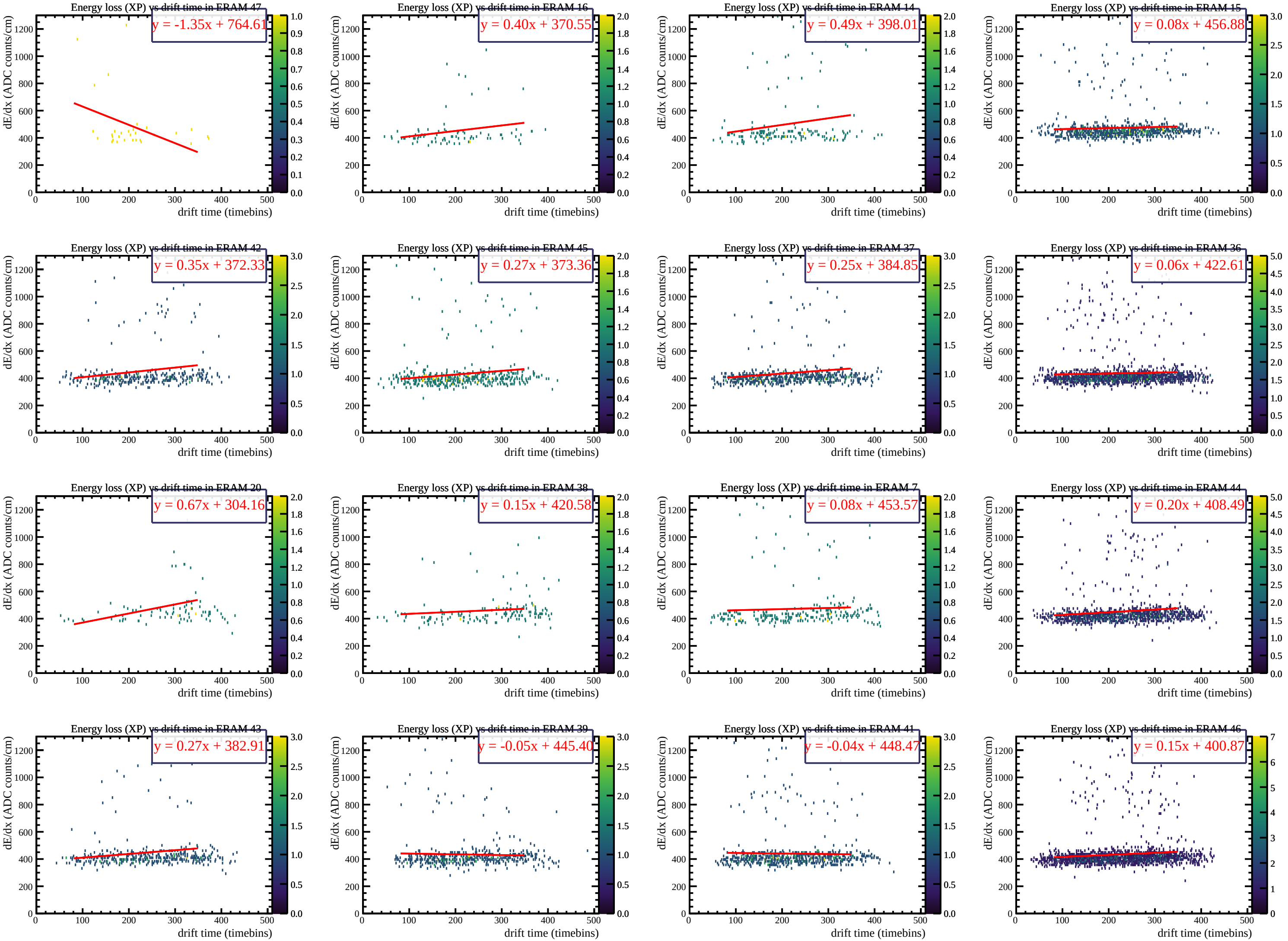


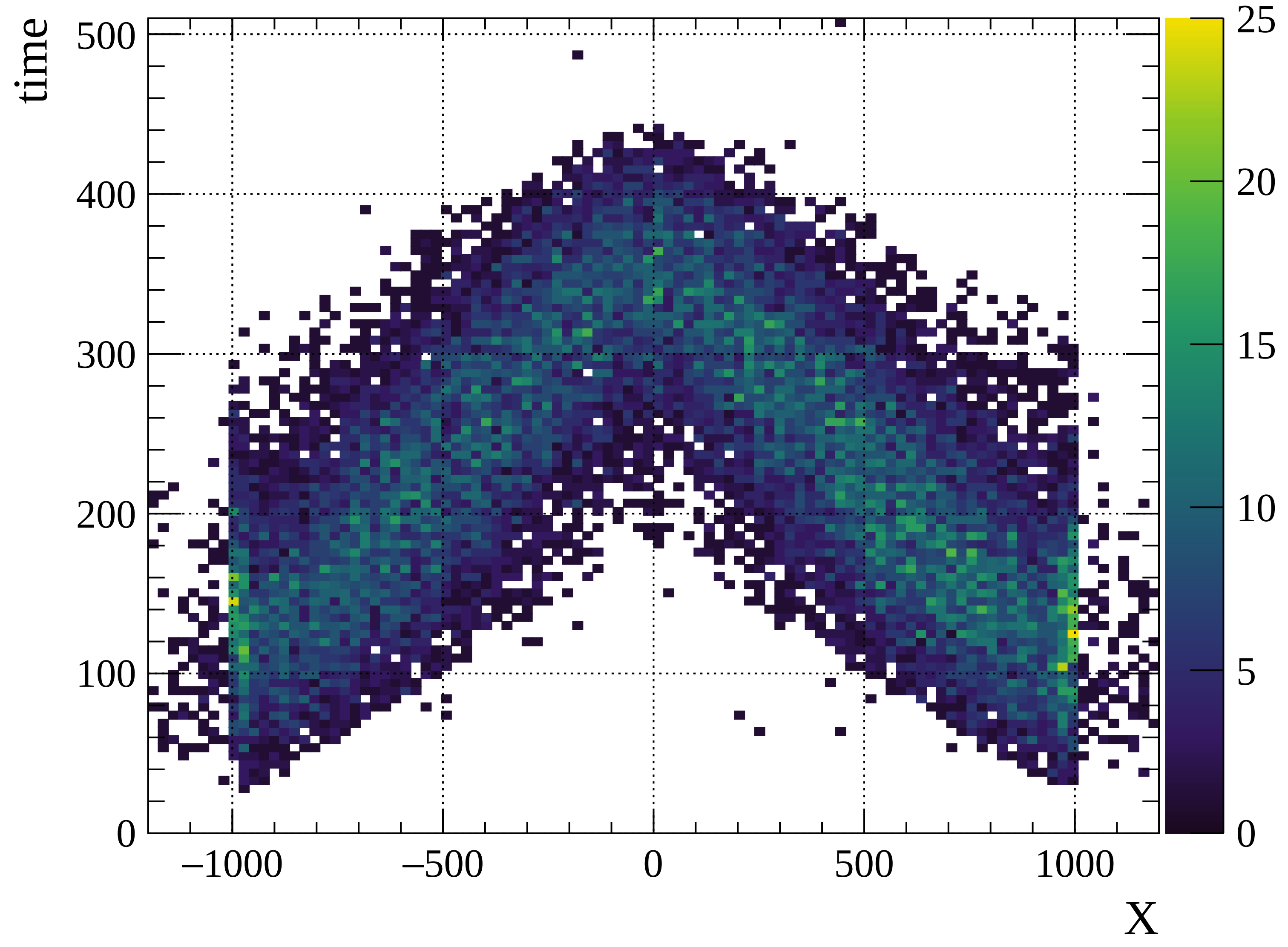
Mean (ADC counts/cm)
Drift time (timebins)
WF
XP
600
550
500
450
400
350
300
0
100
200
300
400
500


y = 0.02x + 406.69
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6
7
8
9

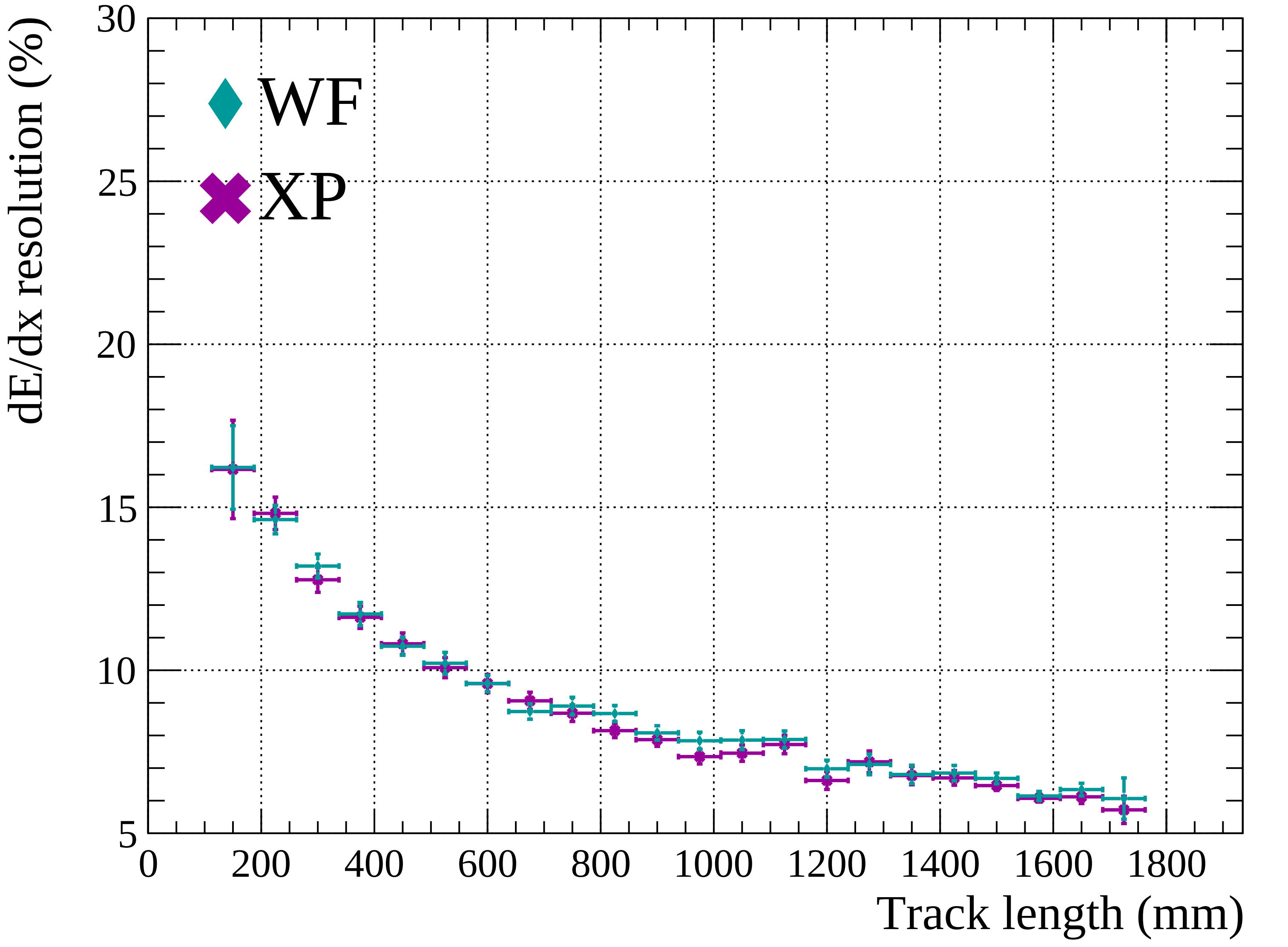

y = 0.05x + 398.13
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6
7
8
9

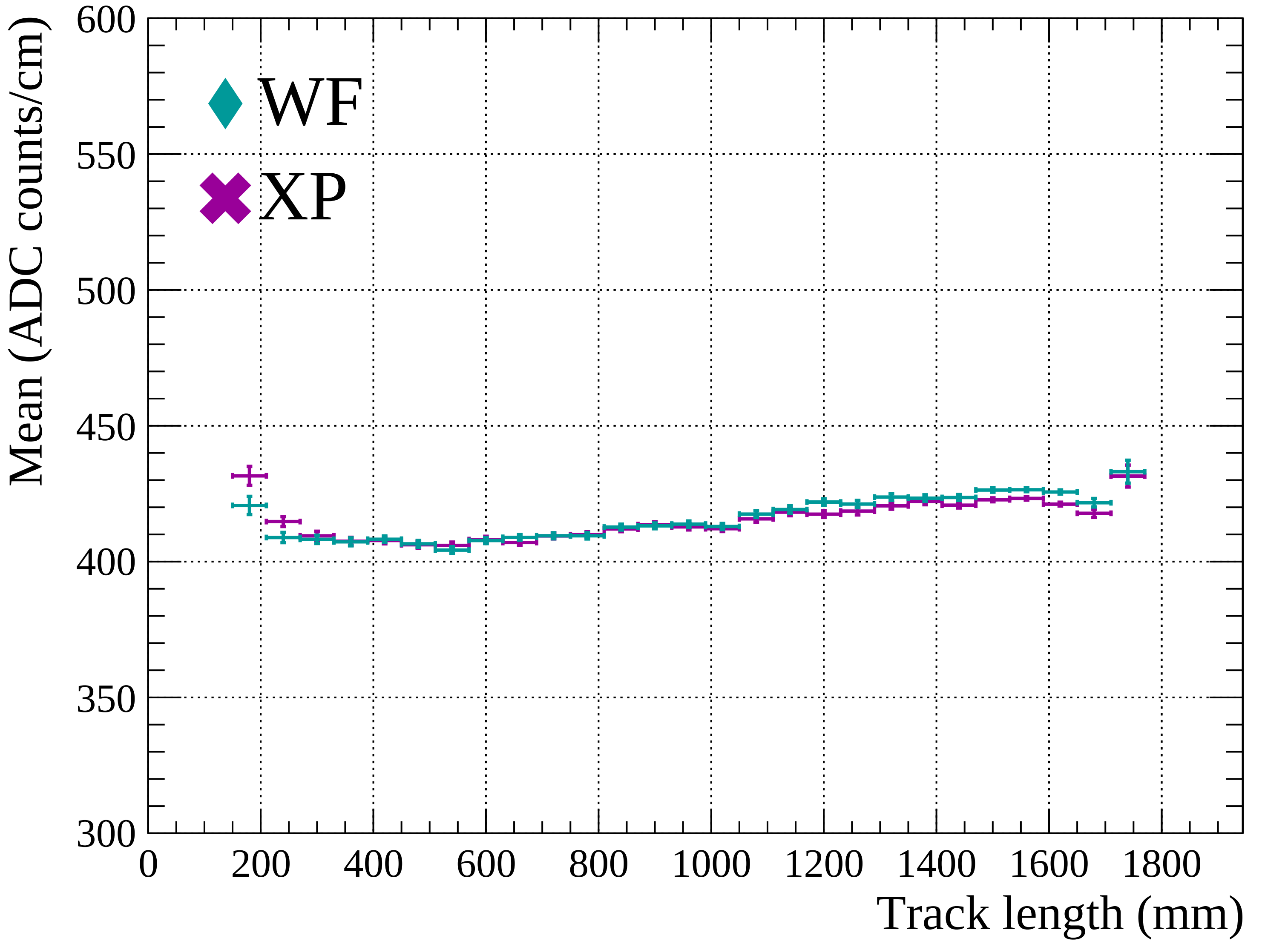

Energy loss (XP) vs drift time in ERAM 24
y = 0.10x + 491.32
Energy loss (XP) vs drift time in ERAM 30
y = 0.21x + 436.44
Energy loss (XP) vs drift time in ERAM 28
y = 0.13x + 432.48
Energy loss (XP) vs drift time in ERAM 19
y = 0.14x + 427.00
Energy loss (XP) vs drift time in ERAM 21
y = 0.14x + 437.38
Energy loss (XP) vs drift time in ERAM 13
y = 0.07x + 446.79
Energy loss (XP) vs drift time in ERAM 9
y = 0.12x + 431.23
Energy loss (XP) vs drift time in ERAM 2
y = 0.09x + 436.31
Energy loss (XP) vs drift time in ERAM 26
y = 0.28x + 382.96
Energy loss (XP) vs drift time in ERAM 17
y = 0.61x + 344.89
Energy loss (XP) vs drift time in ERAM 23
y = 0.23x + 403.69
Energy loss (XP) vs drift time in ERAM 29
y = 0.10x + 427.72
Energy loss (XP) vs drift time in ERAM 1
y = 0.29x + 408.14
Energy loss (XP) vs drift time in ERAM 10
y = 0.28x + 391.29
Energy loss (XP) vs drift time in ERAM 11
y = 0.12x + 422.64
Energy loss (XP) vs drift time in ERAM 3
y = 0.11x + 420.41
dE/dx (ADC counts/cm)
drift time (timebins)

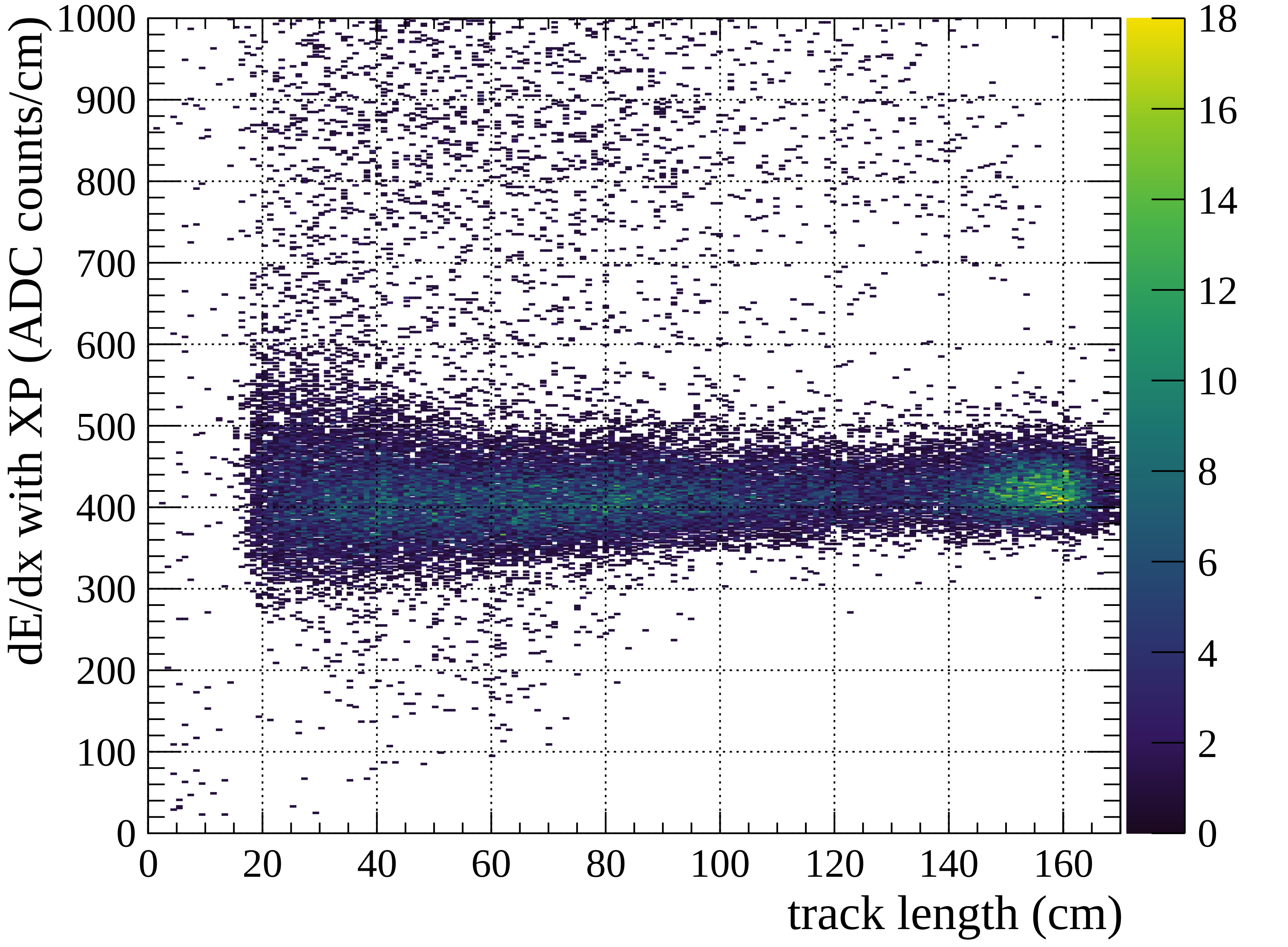

Energy loss (XP) vs drift time in ERAM 47
y = -1.35x + 764.61
Energy loss (XP) vs drift time in ERAM 16
y = 0.40x + 370.55
Energy loss (XP) vs drift time in ERAM 14
y = 0.49x + 398.01
Energy loss (XP) vs drift time in ERAM 15
y = 0.08x + 456.88
Energy loss (XP) vs drift time in ERAM 42
y = 0.35x + 372.33
Energy loss (XP) vs drift time in ERAM 45
y = 0.27x + 373.36
Energy loss (XP) vs drift time in ERAM 37
y = 0.25x + 384.85
Energy loss (XP) vs drift time in ERAM 36
y = 0.06x + 422.61
Energy loss (XP) vs drift time in ERAM 20
y = 0.67x + 304.16
Energy loss (XP) vs drift time in ERAM 38
y = 0.15x + 420.58
Energy loss (XP) vs drift time in ERAM 7
y = 0.08x + 453.57
Energy loss (XP) vs drift time in ERAM 44
y = 0.20x + 408.49
Energy loss (XP) vs drift time in ERAM 43
y = 0.27x + 382.91
Energy loss (XP) vs drift time in ERAM 39
y = -0.05x + 445.40
Energy loss (XP) vs drift time in ERAM 41
y = -0.04x + 448.47
Energy loss (XP) vs drift time in ERAM 46
y = 0.15x + 400.87
dE/dx (ADC counts/cm)
drift time (timebins)

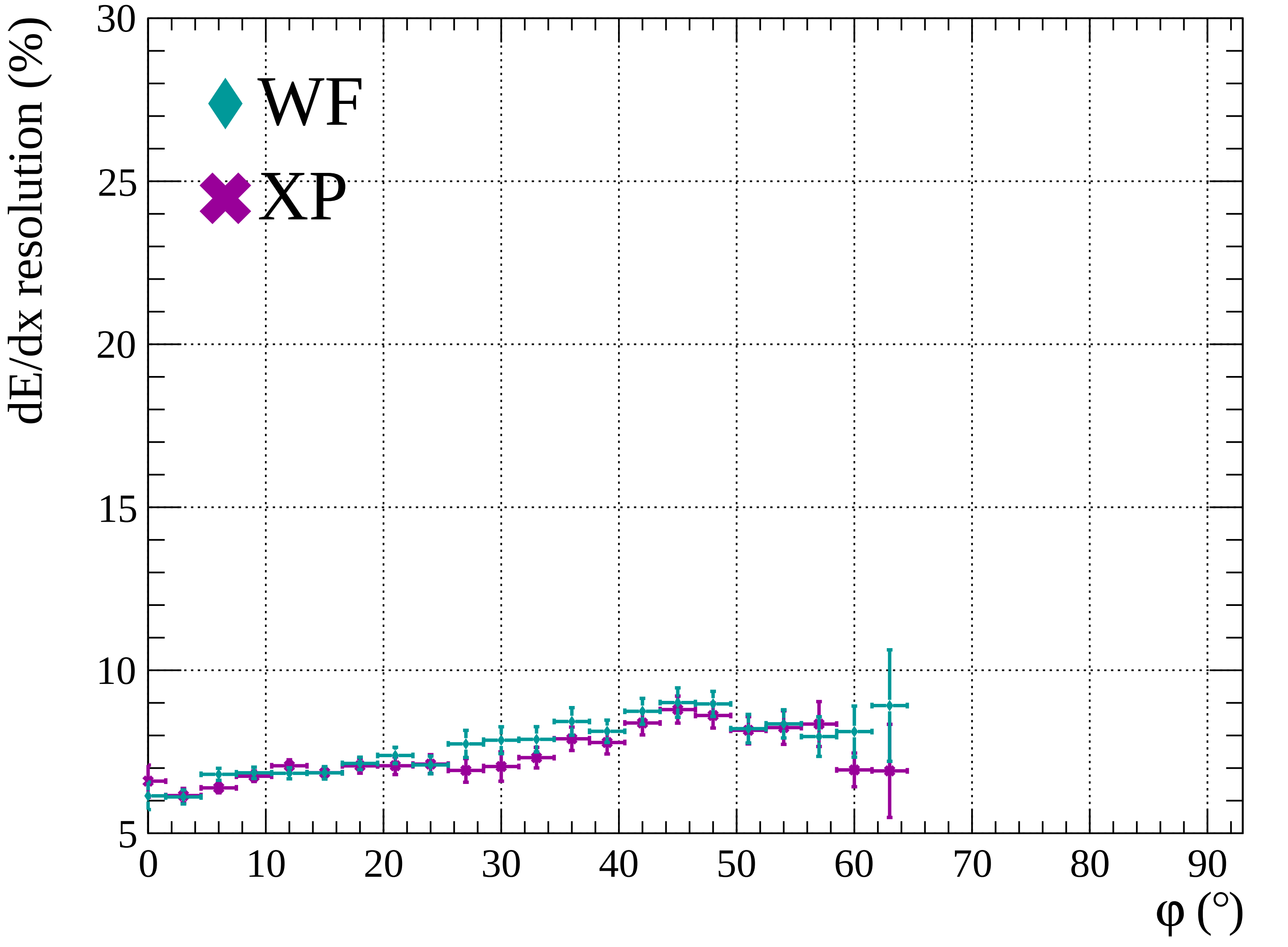


time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
25
20
15
10
5
0

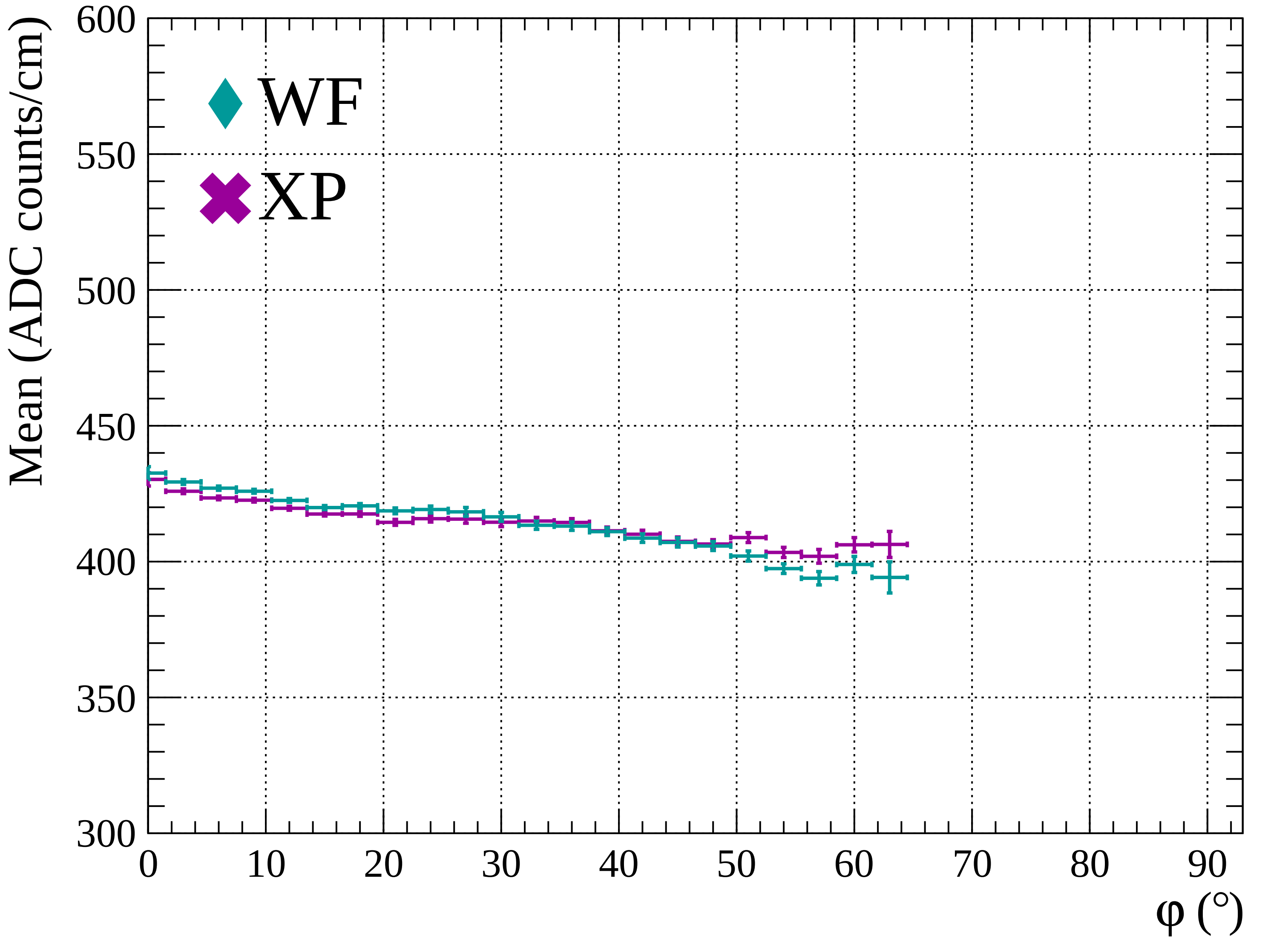

dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)

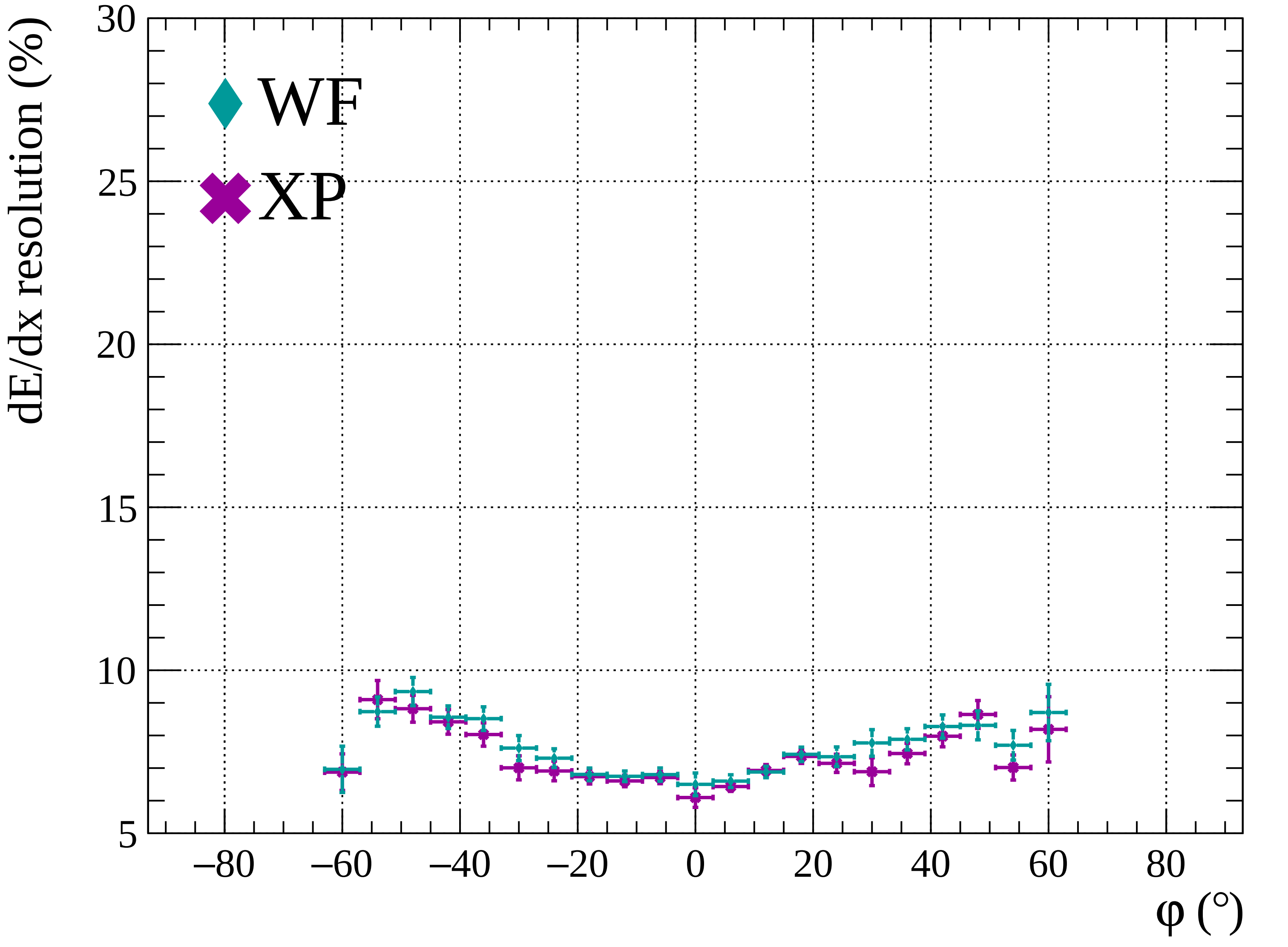

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
2
4
6
8
10
12
14
16
18



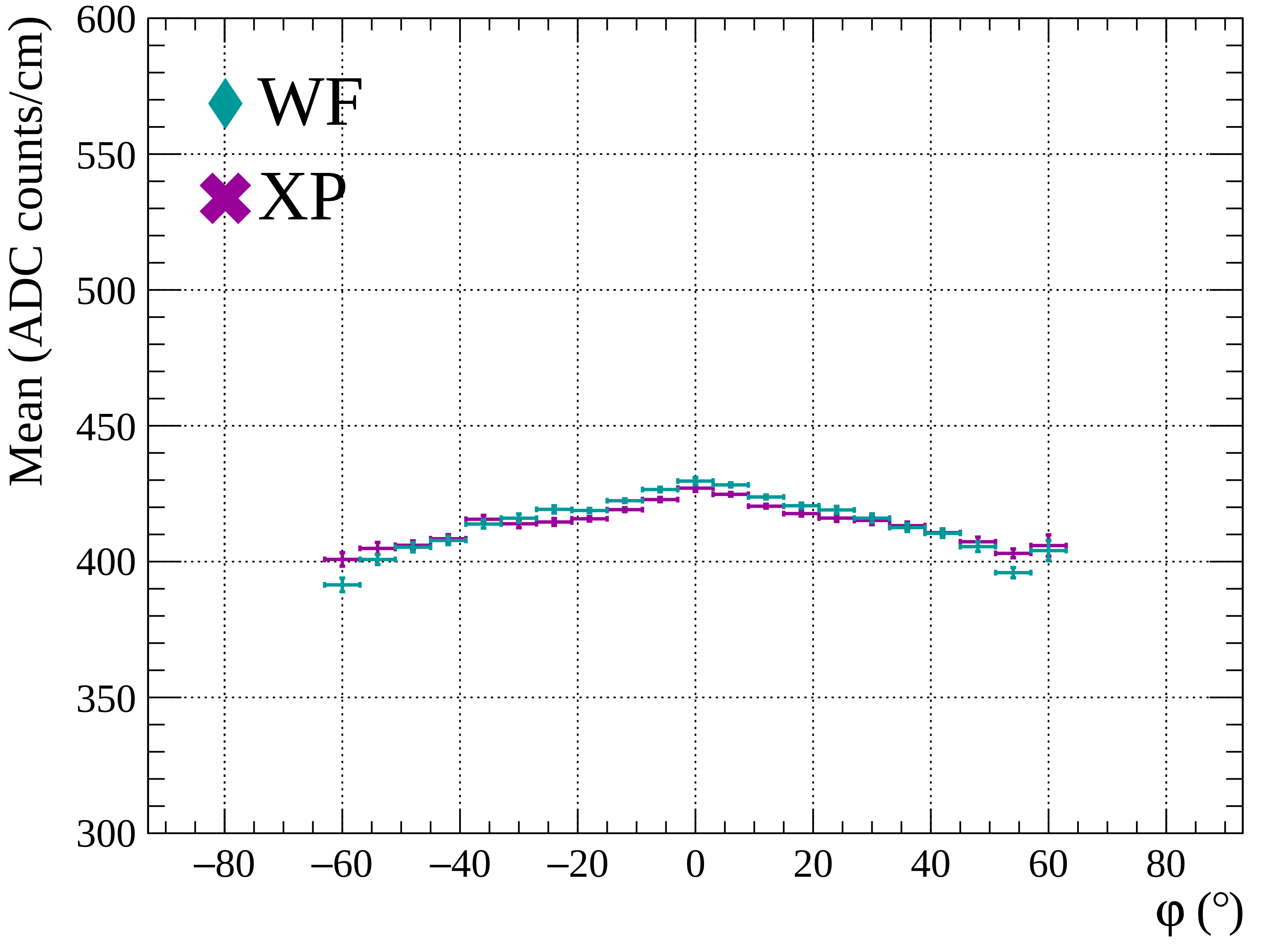
dE/dx resolution (%)
φ (°)
WF
XP



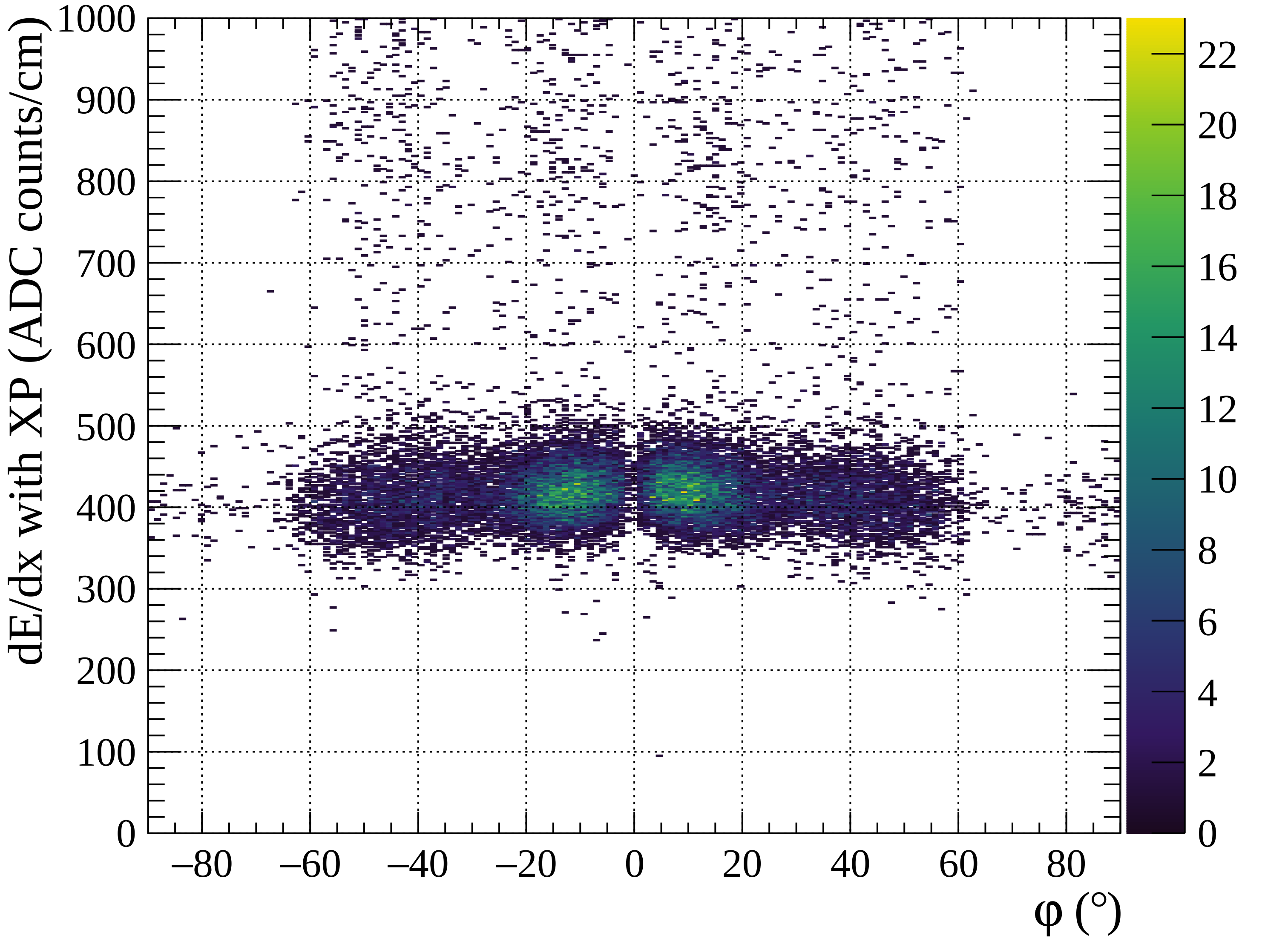
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
0
10
20
30
40
50
60
70
80
90
φ (°)

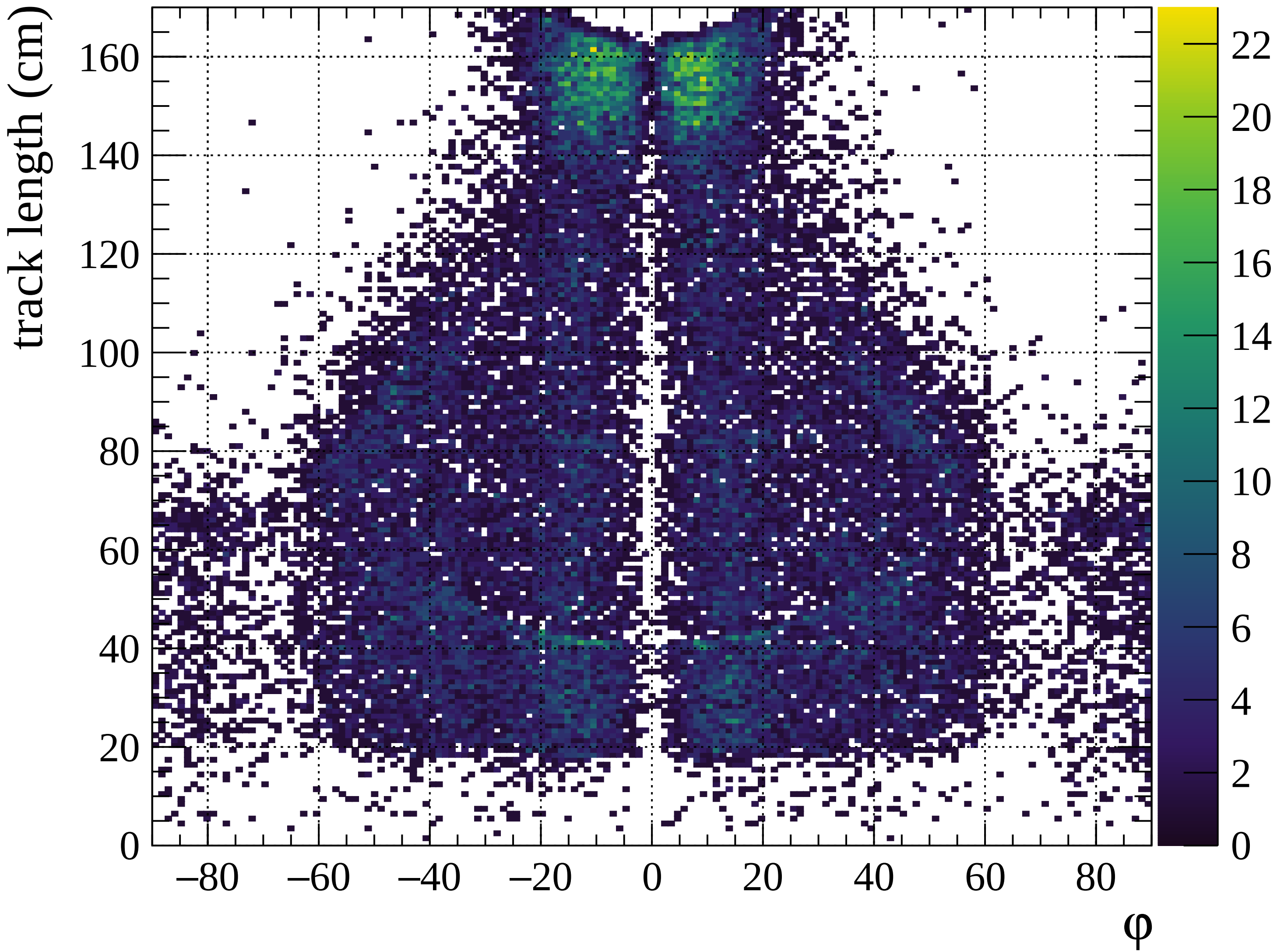


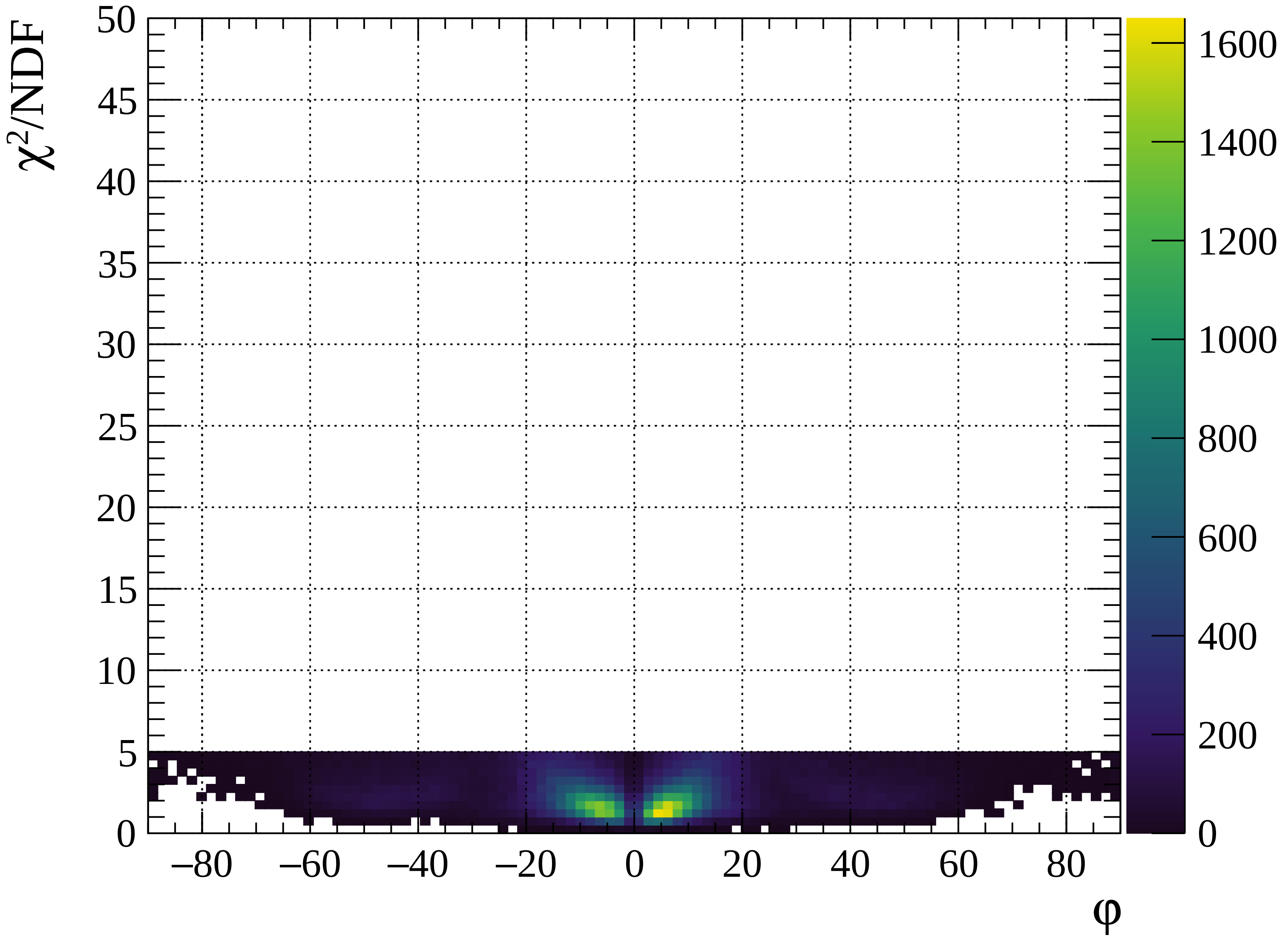
dE/dx resolution (%)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80
φ (°)
WF
XP



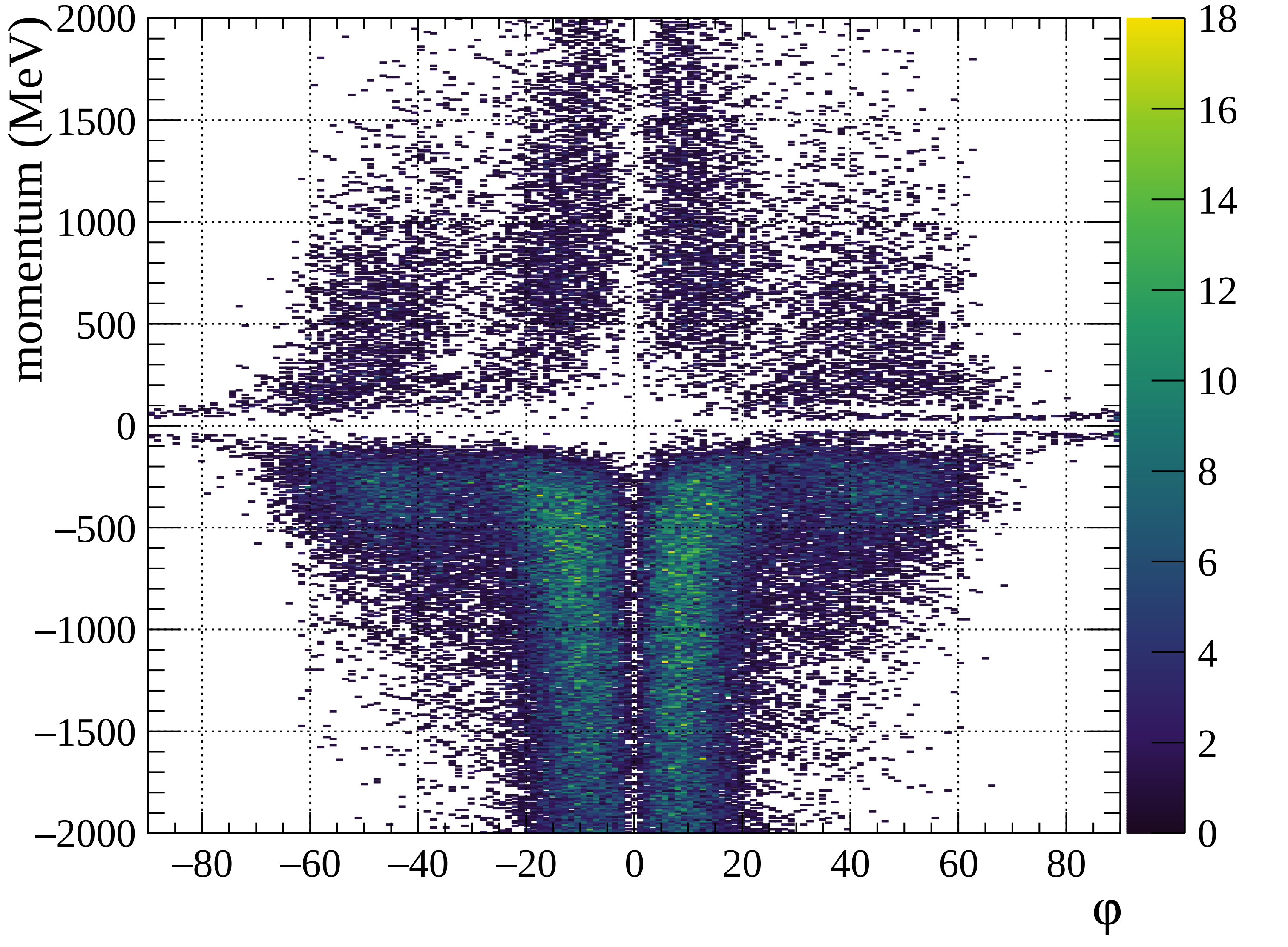
WF
XP
Mean (ADC counts/cm)
φ (°)
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80

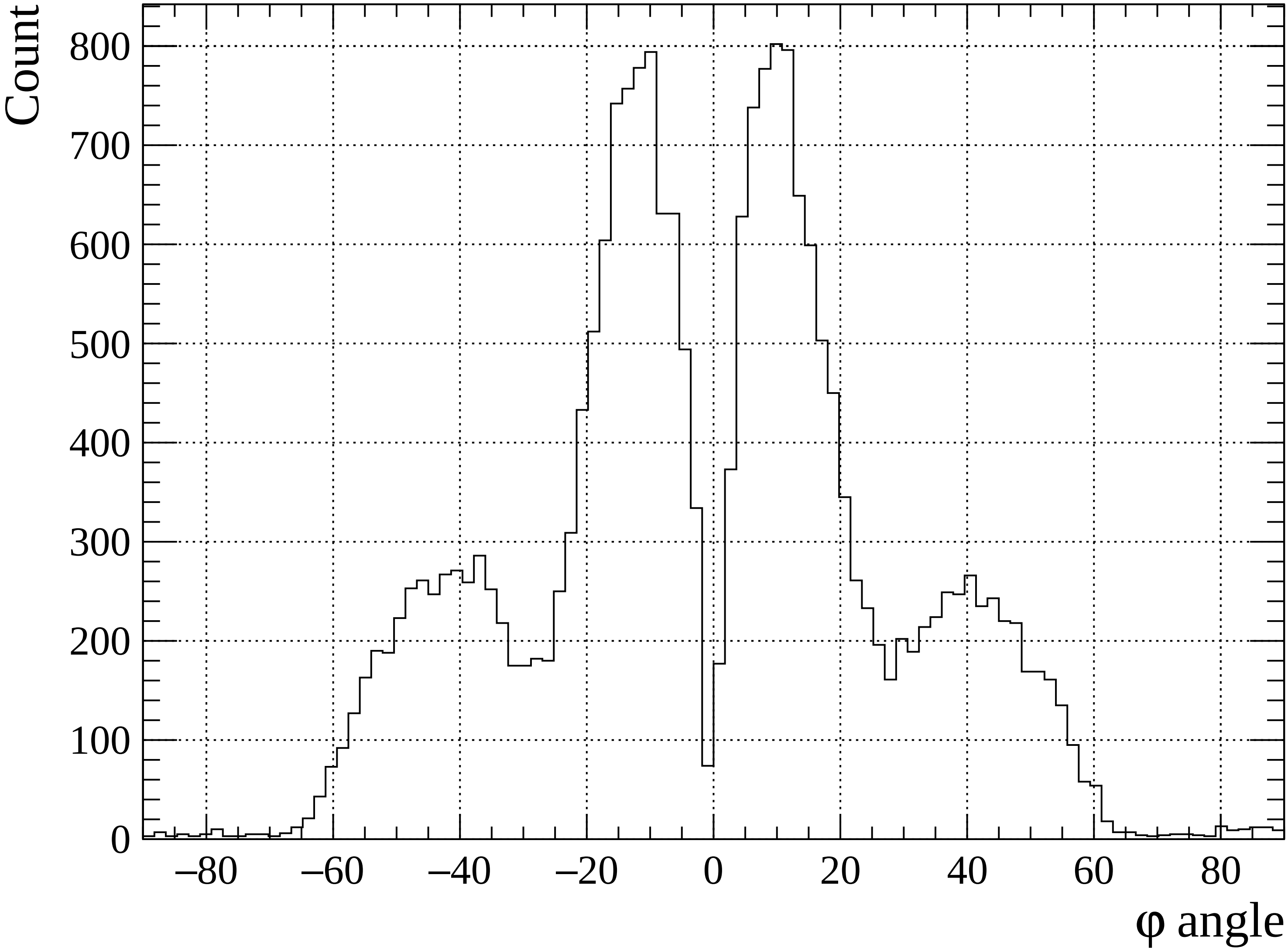

dE/dx with XP (ADC counts/cm)
φ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
2
4
6
8
10
12
14
16
18
20
22

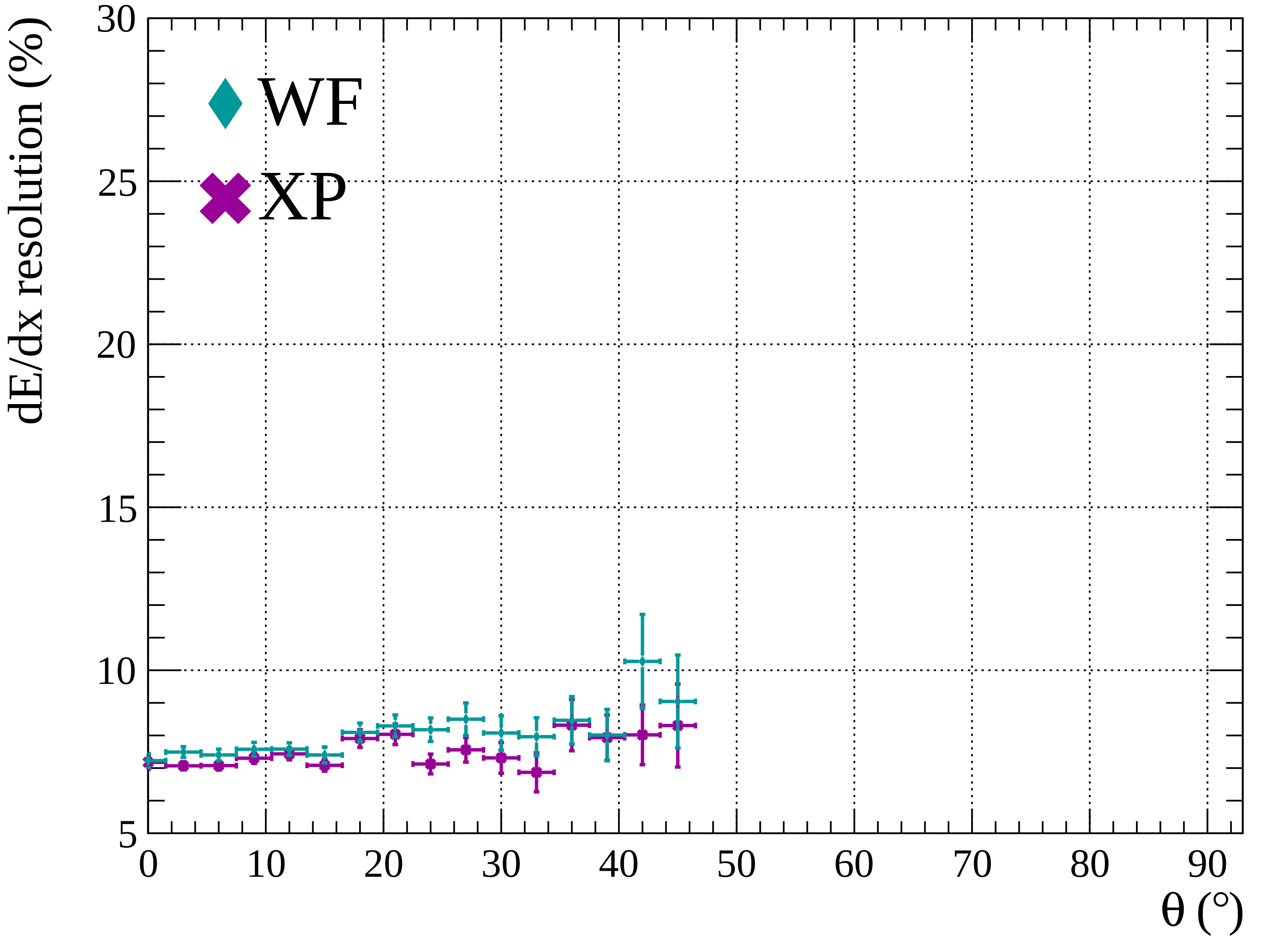

track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
22
20
18
16
14
12
10
8
6
4
2
0



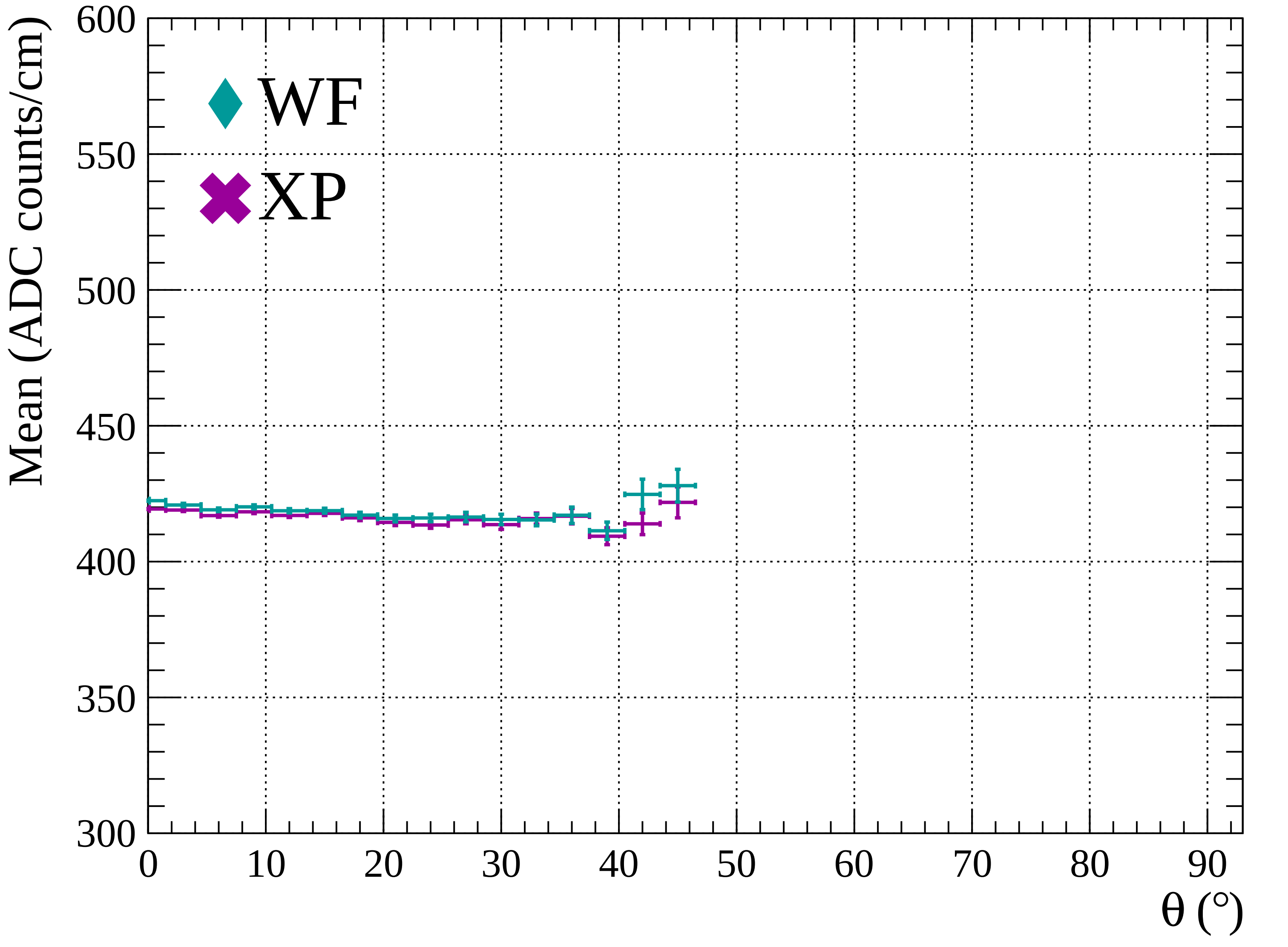
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
1600
1400
1200
1000
800
600
400
200
0


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
18
16
14
12
10
8
6
4
2
0

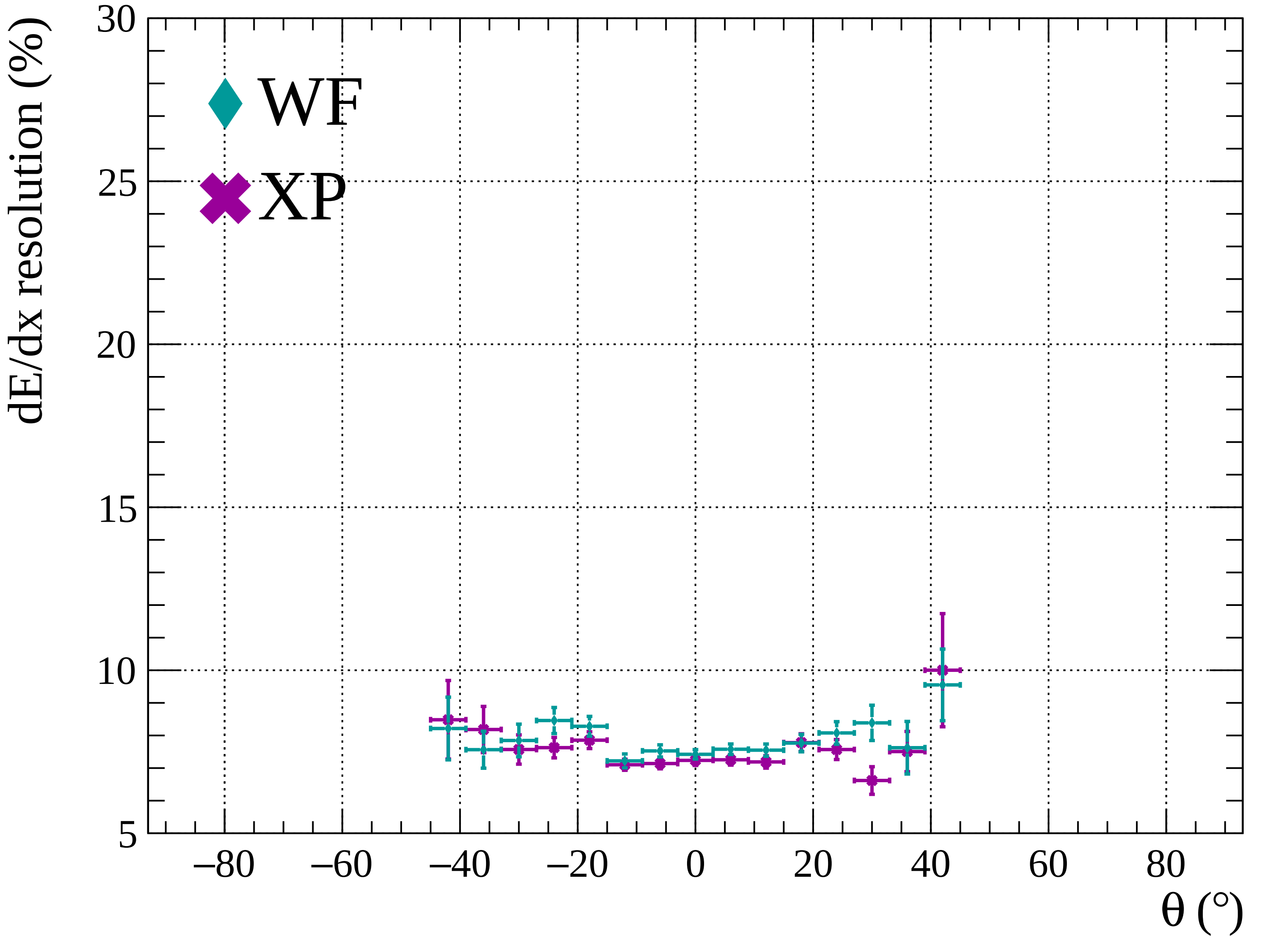

Count
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ angle



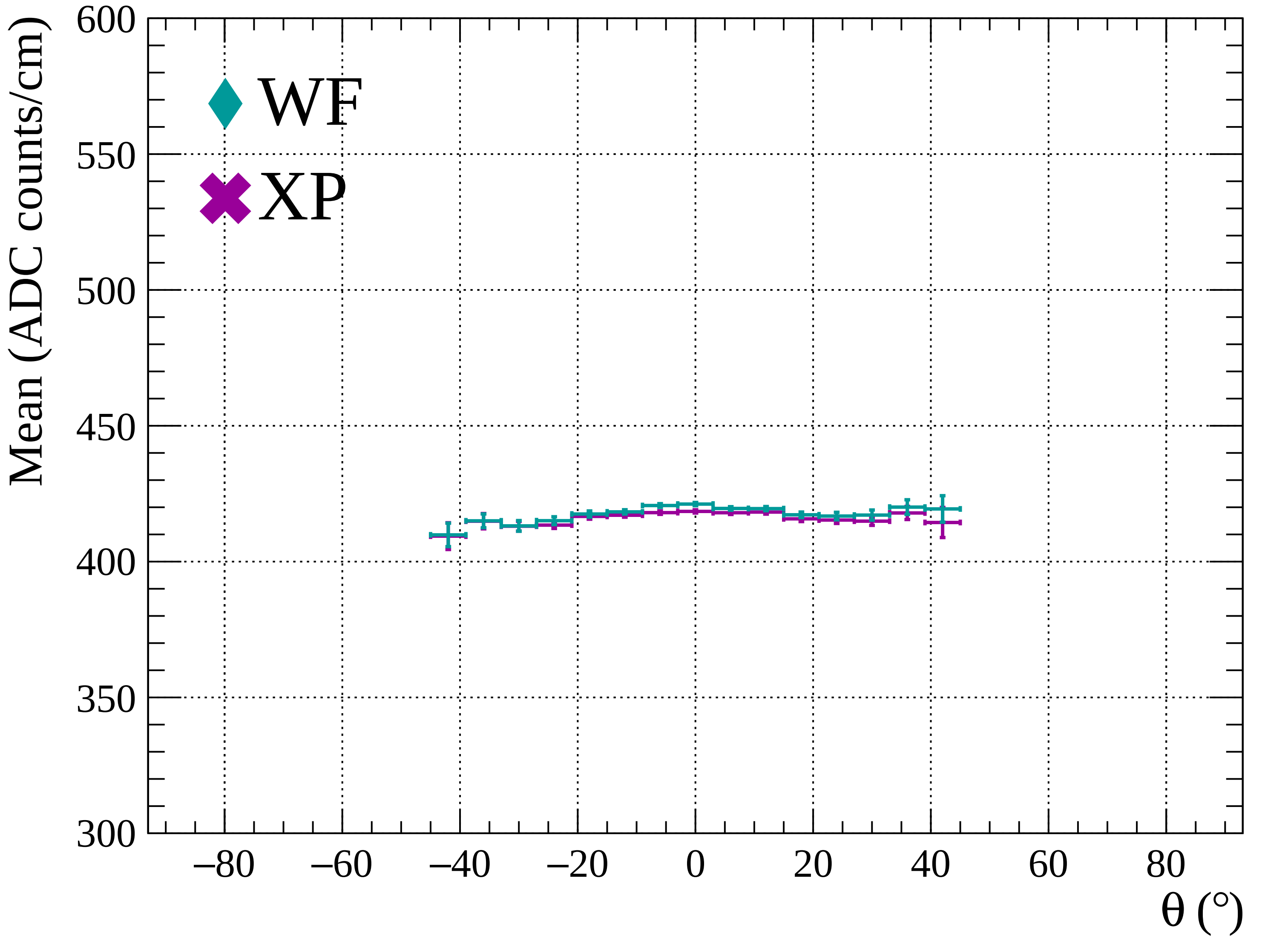
WF
XP
dE/dx resolution (%)
θ (°)
5
10
15
20
25
30
0
10
20
30
40
50
60
70
80
90



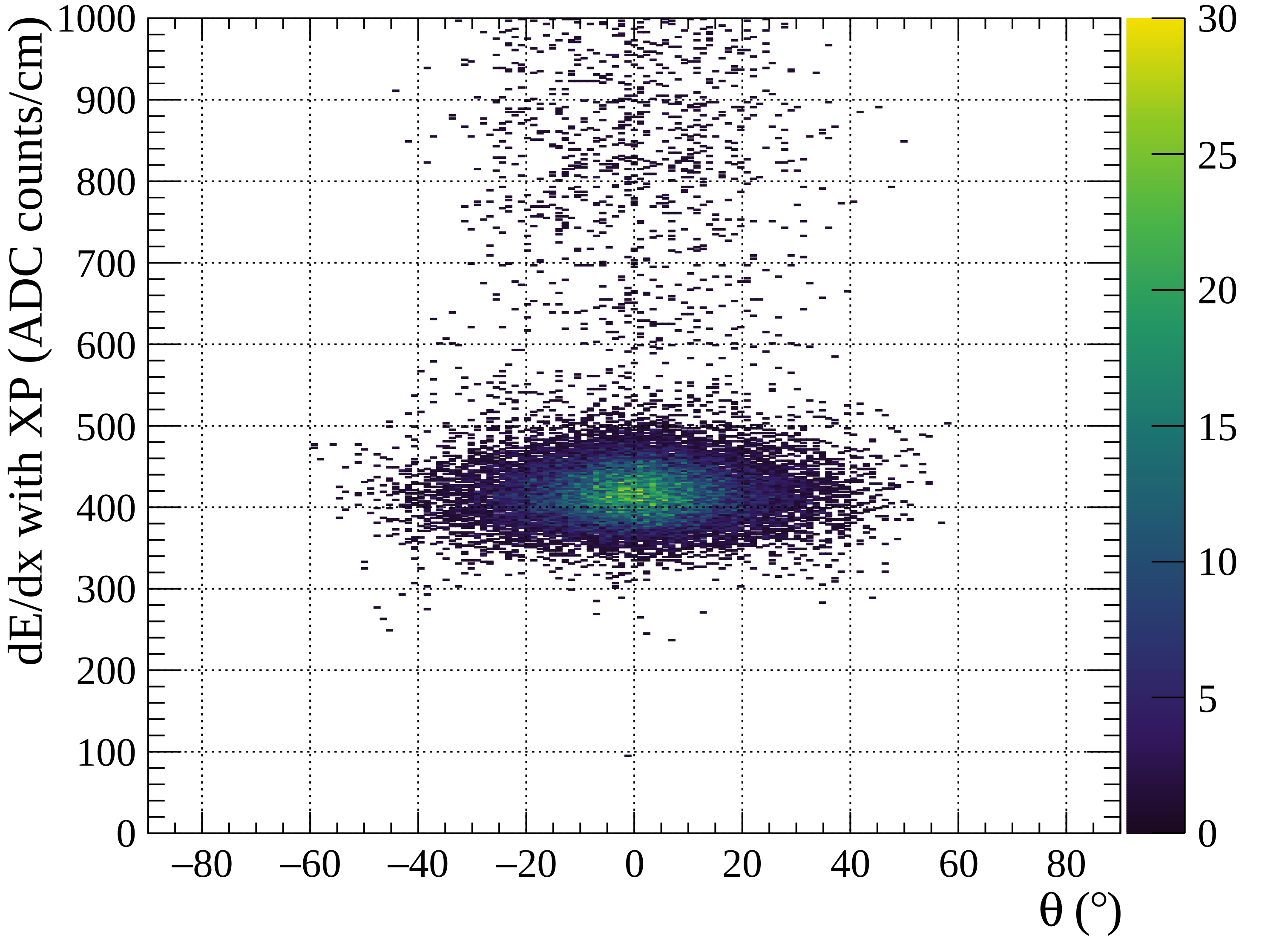
WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90



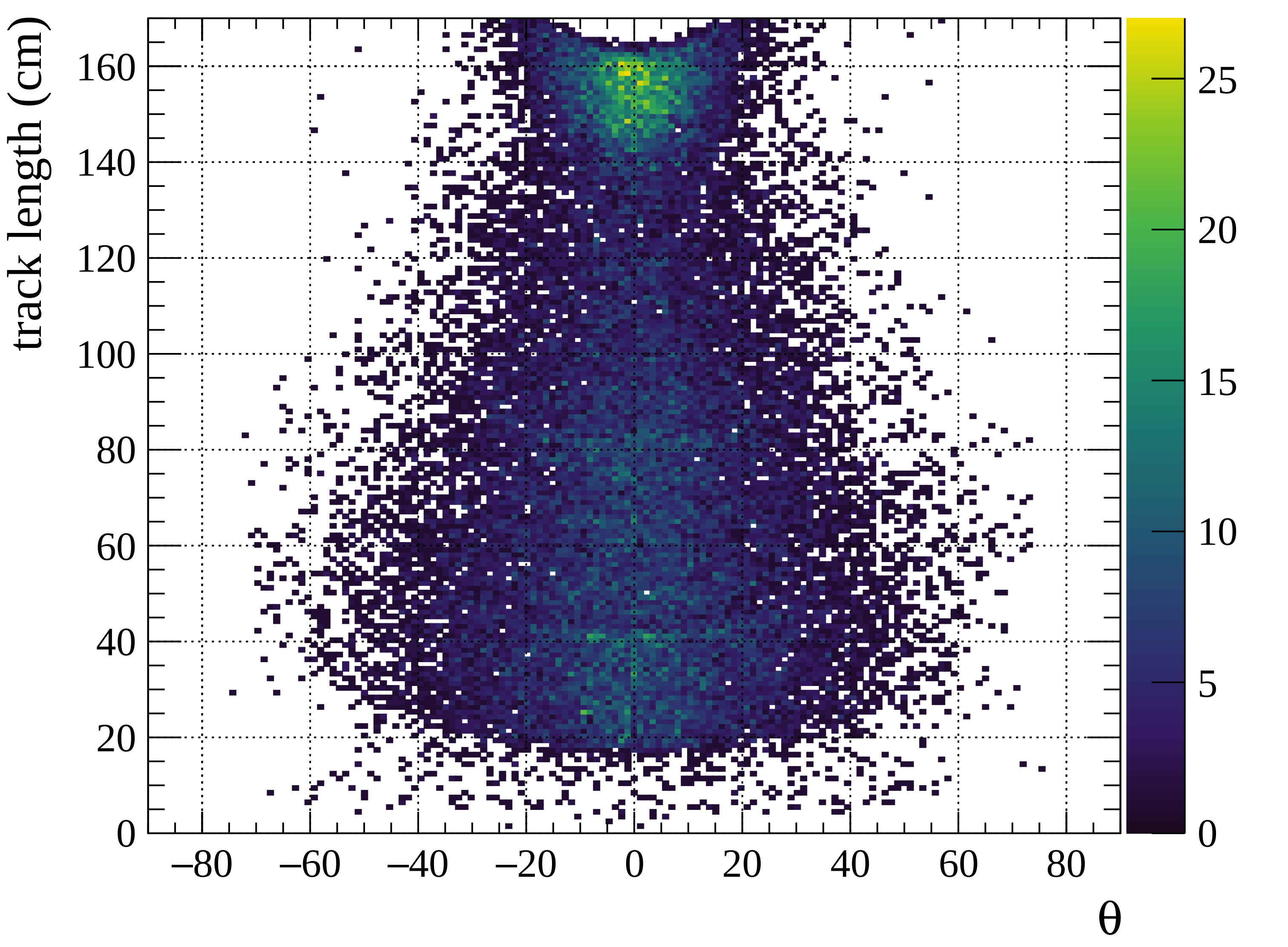
dE/dx resolution (%)
θ (°)
WF
XP
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



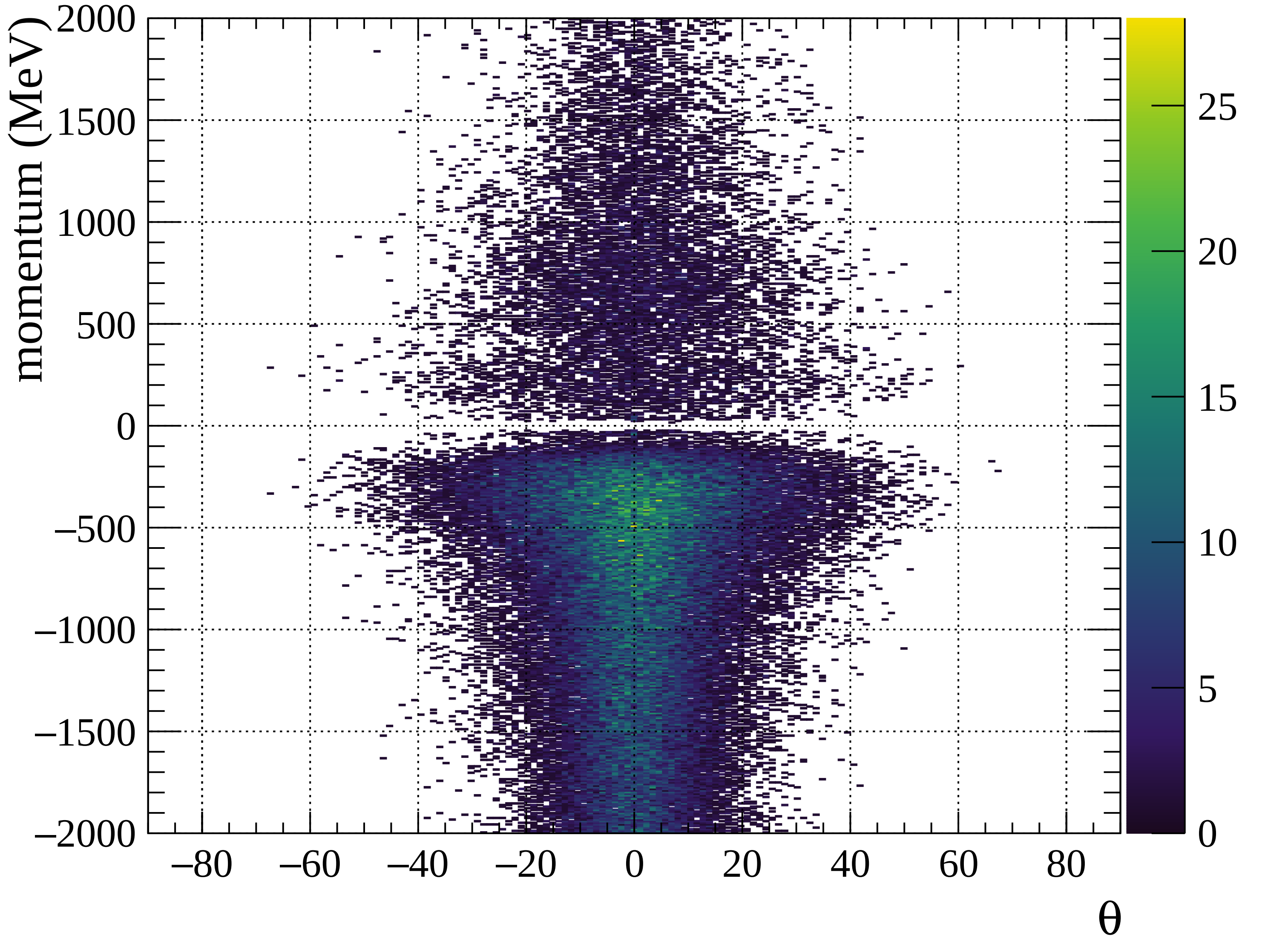
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



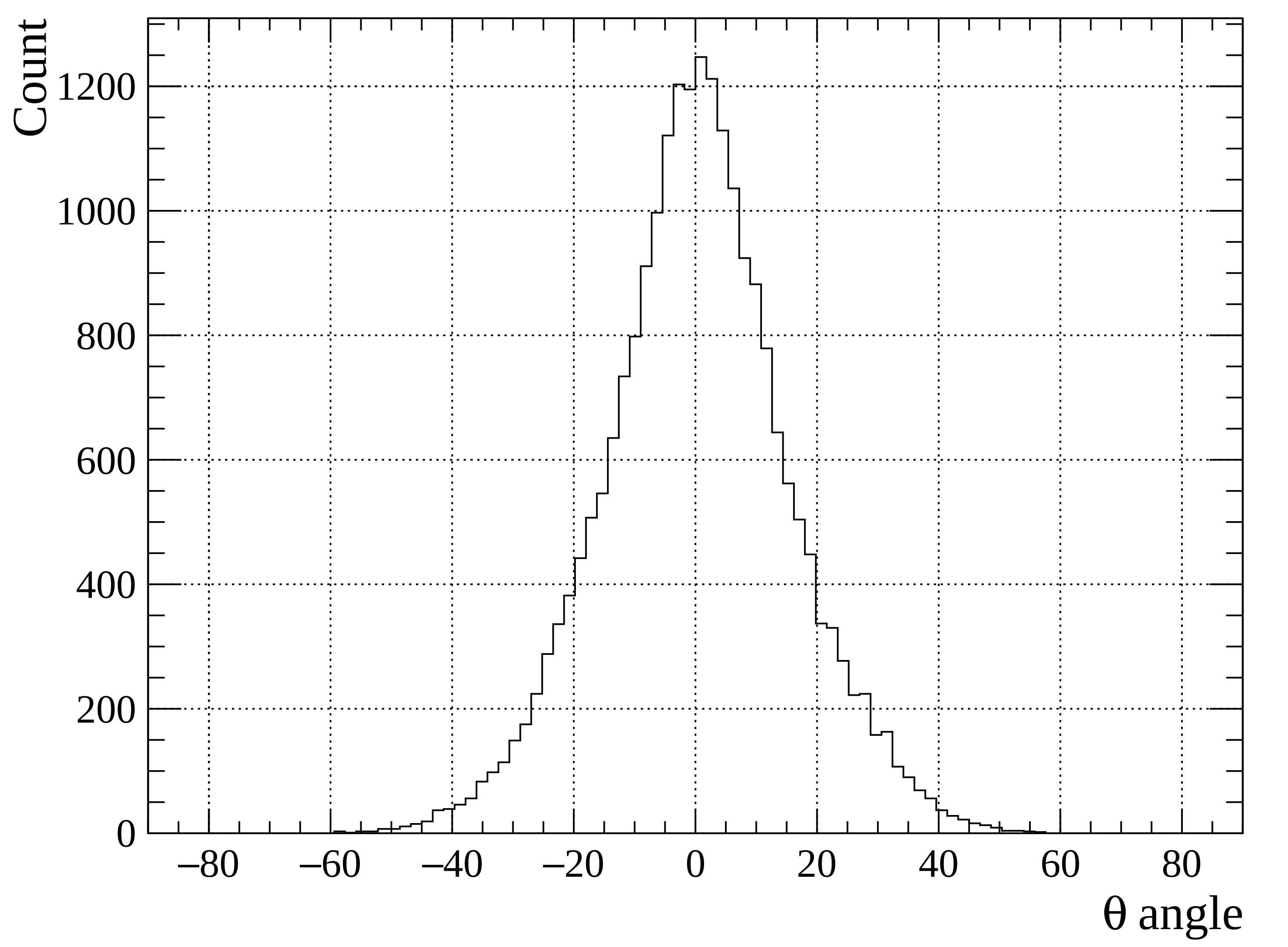
dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
5
10
15
20
25
30



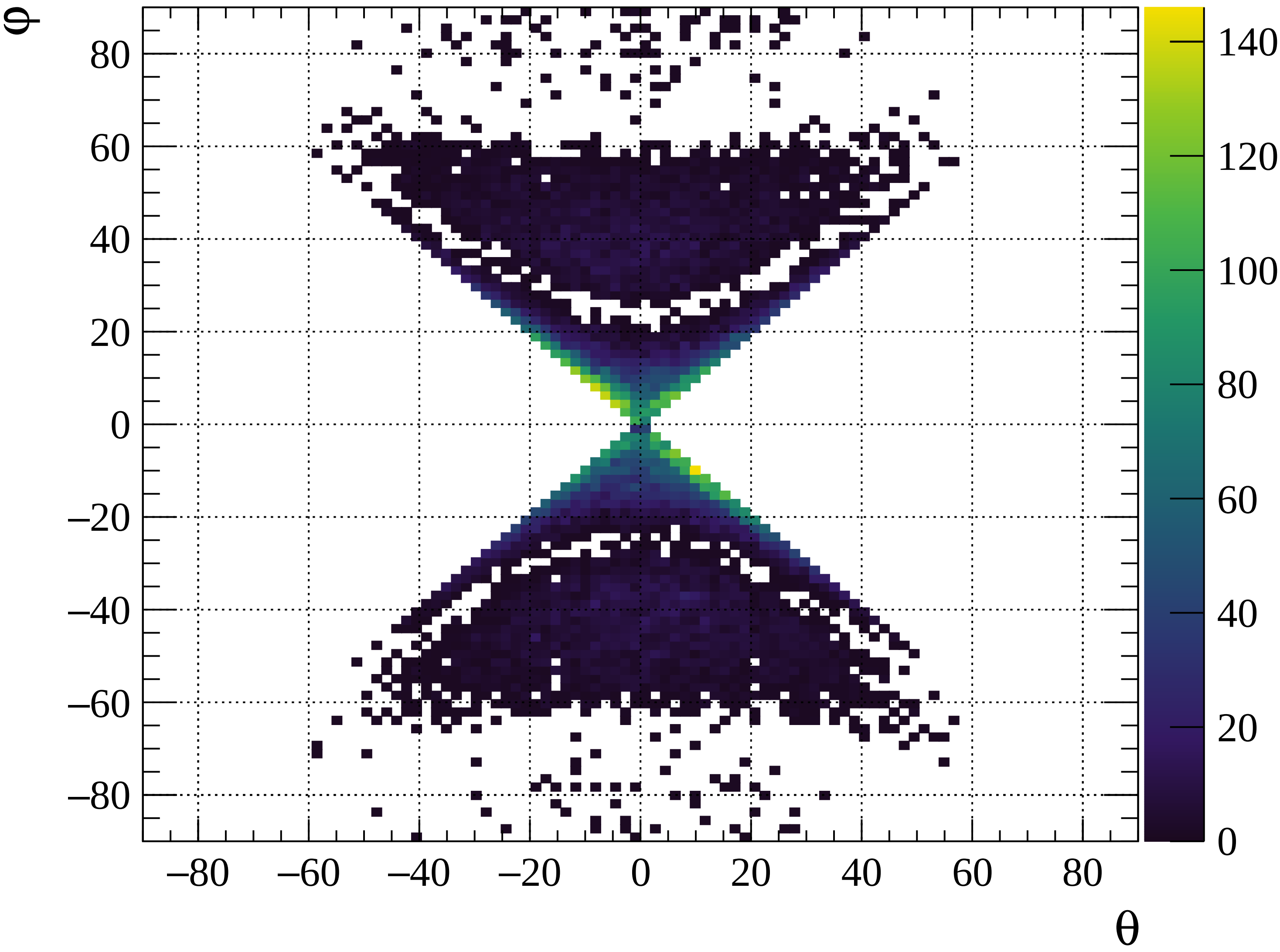
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
25
20
15
10
5
0

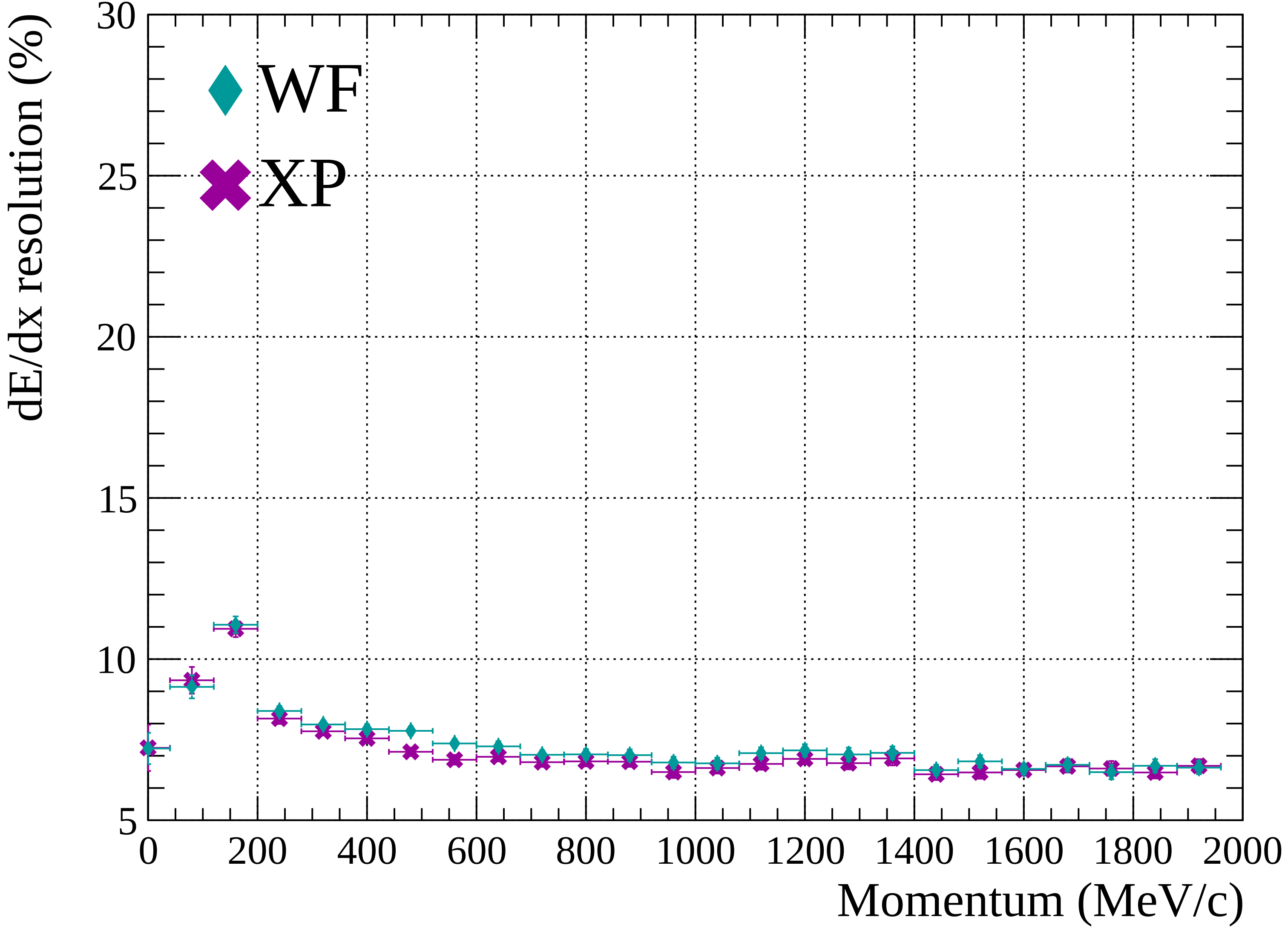


momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
25
20
15
10
5
0

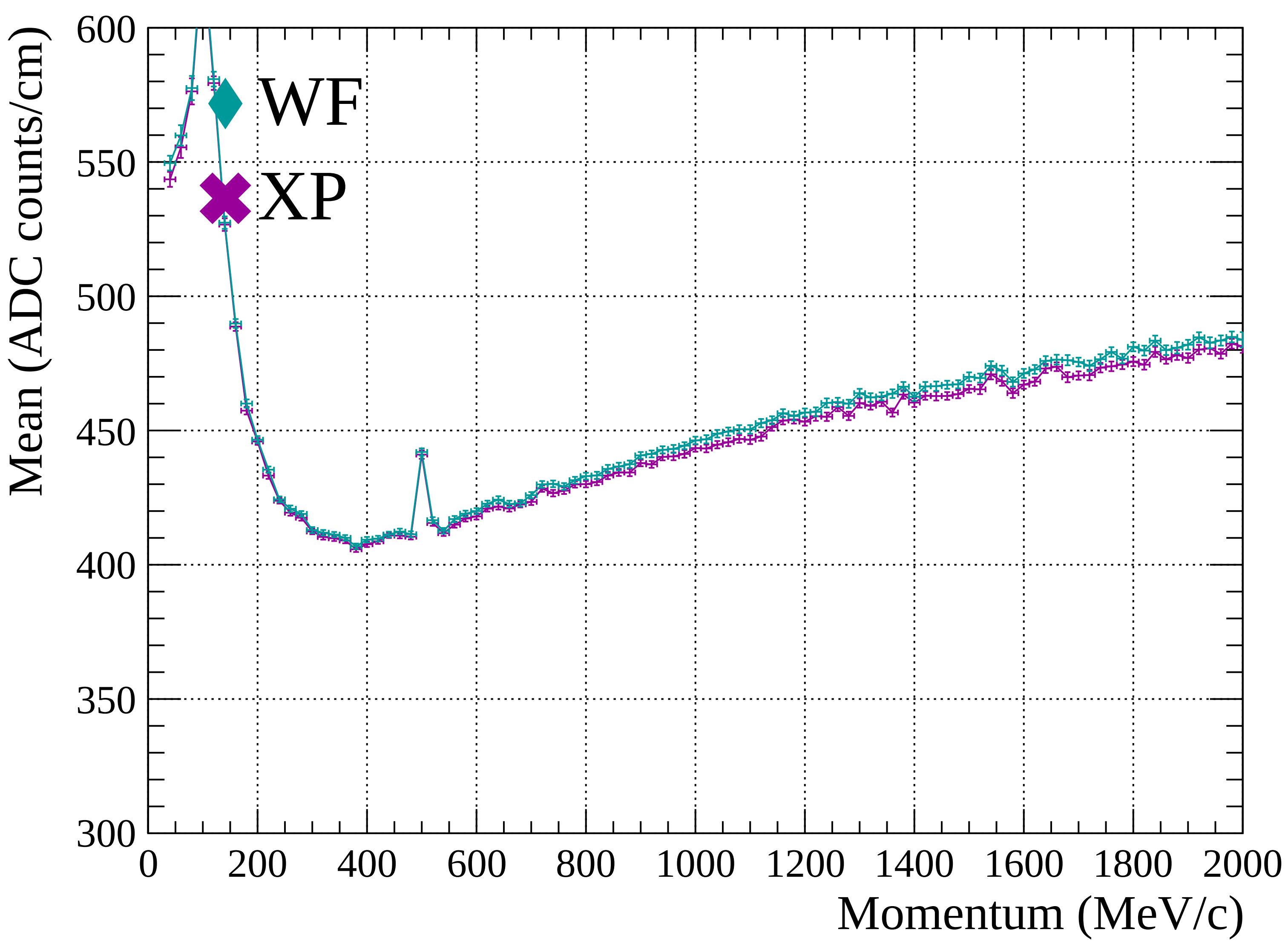


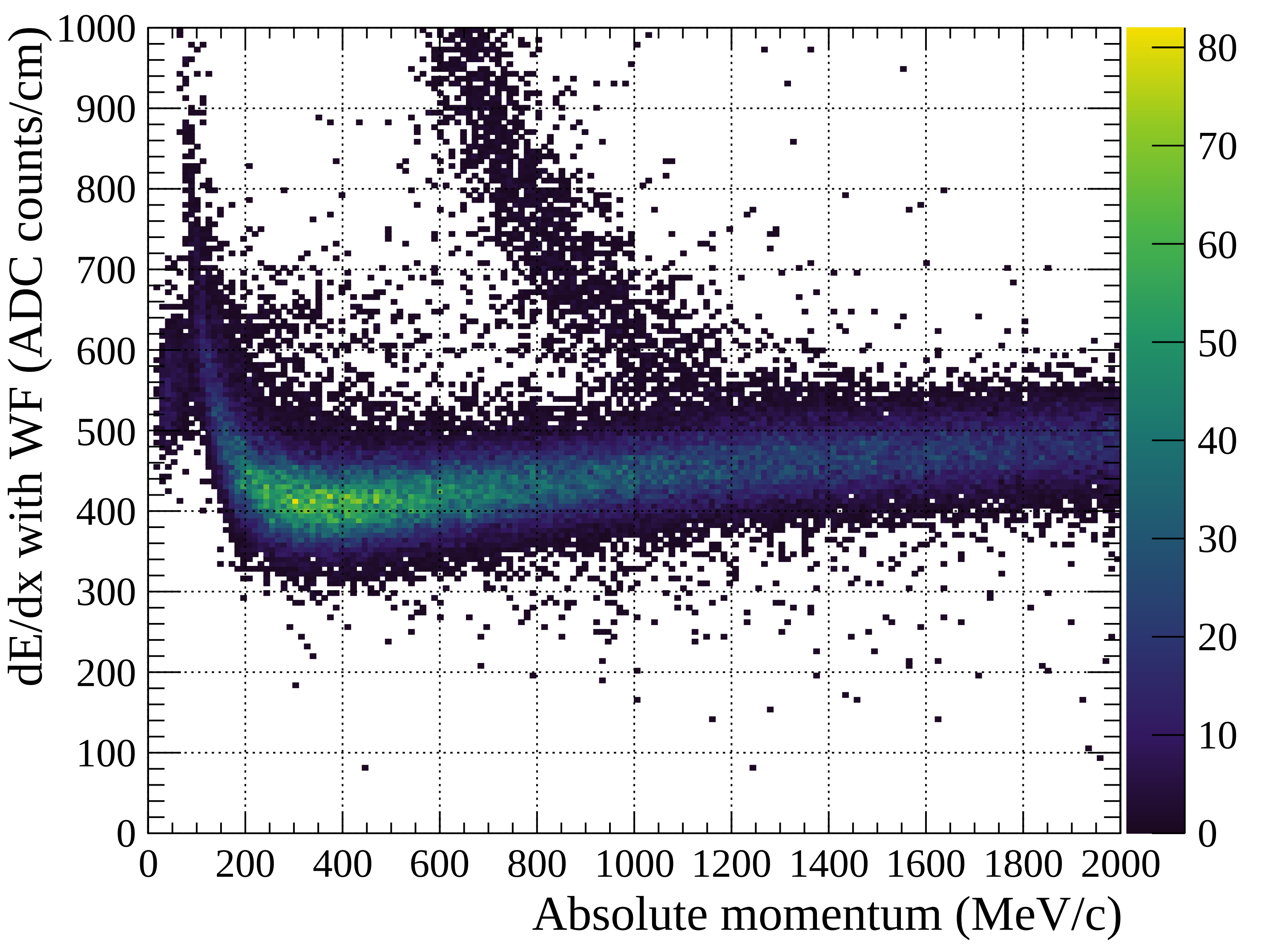
Count
1200
1000
800
600
400
200
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

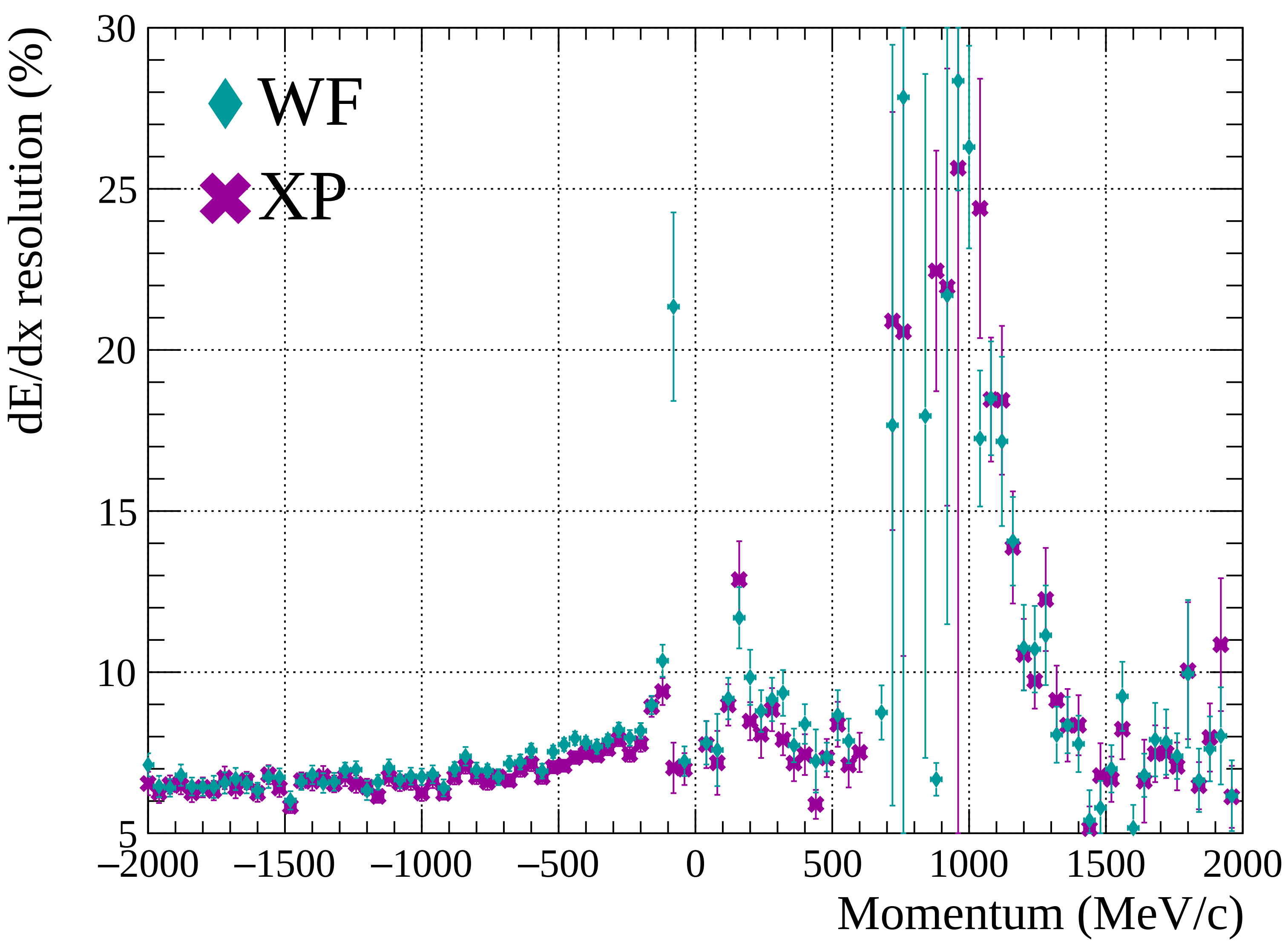

φ
θ
80
60
40
20
0
−20
−40
−60
−80
140
120
100
80
60
40
20
0



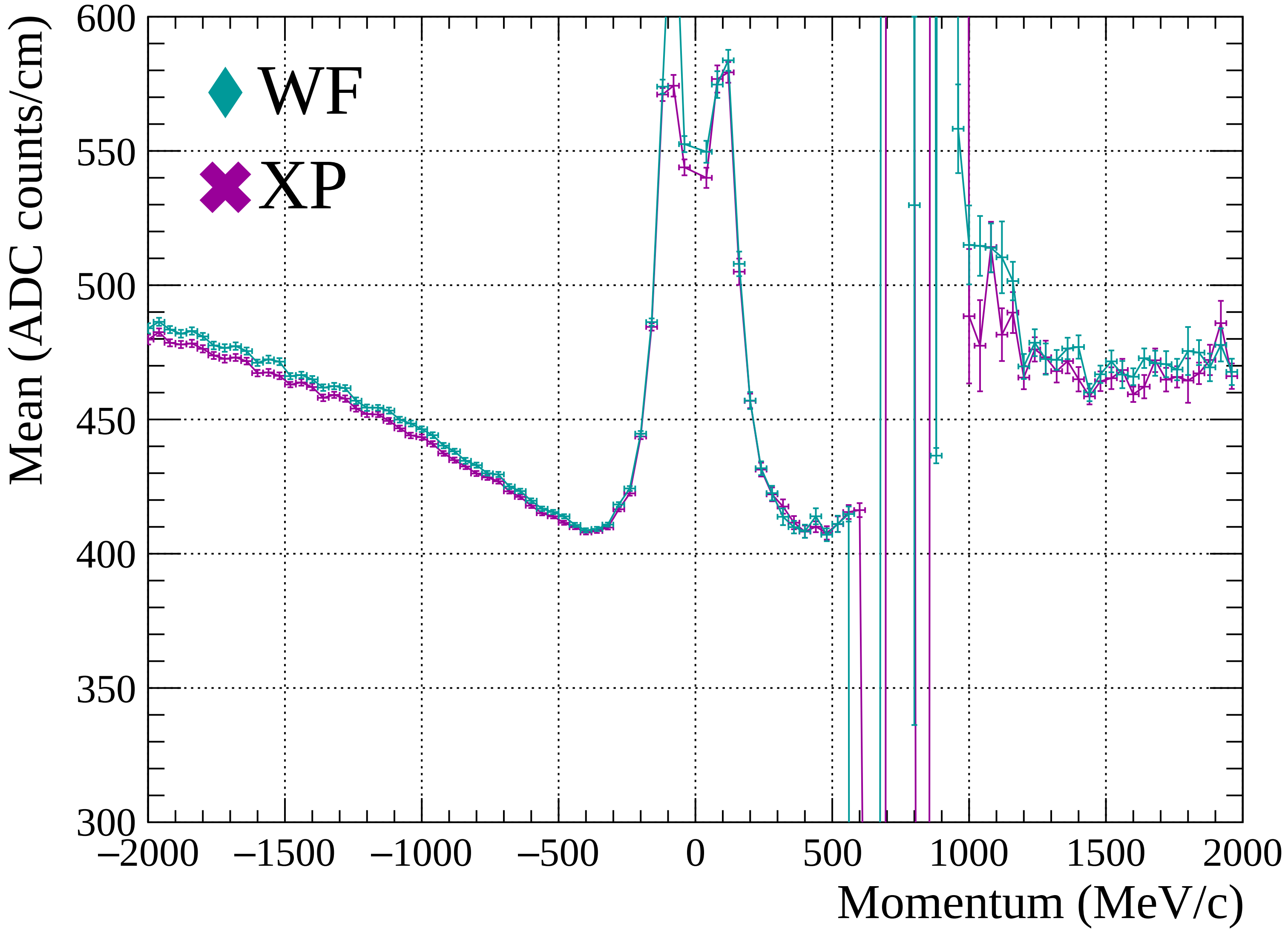
dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000

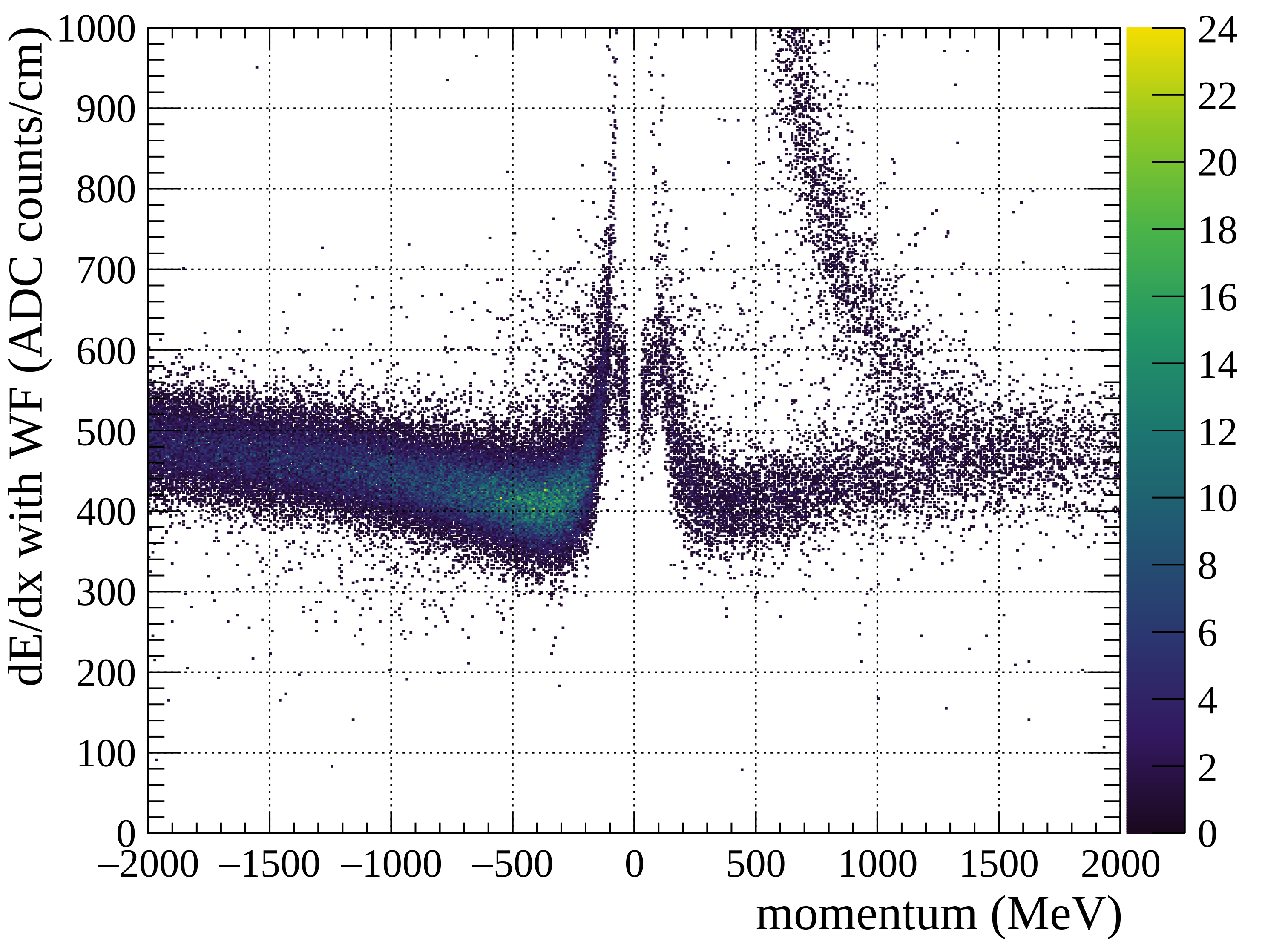

WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

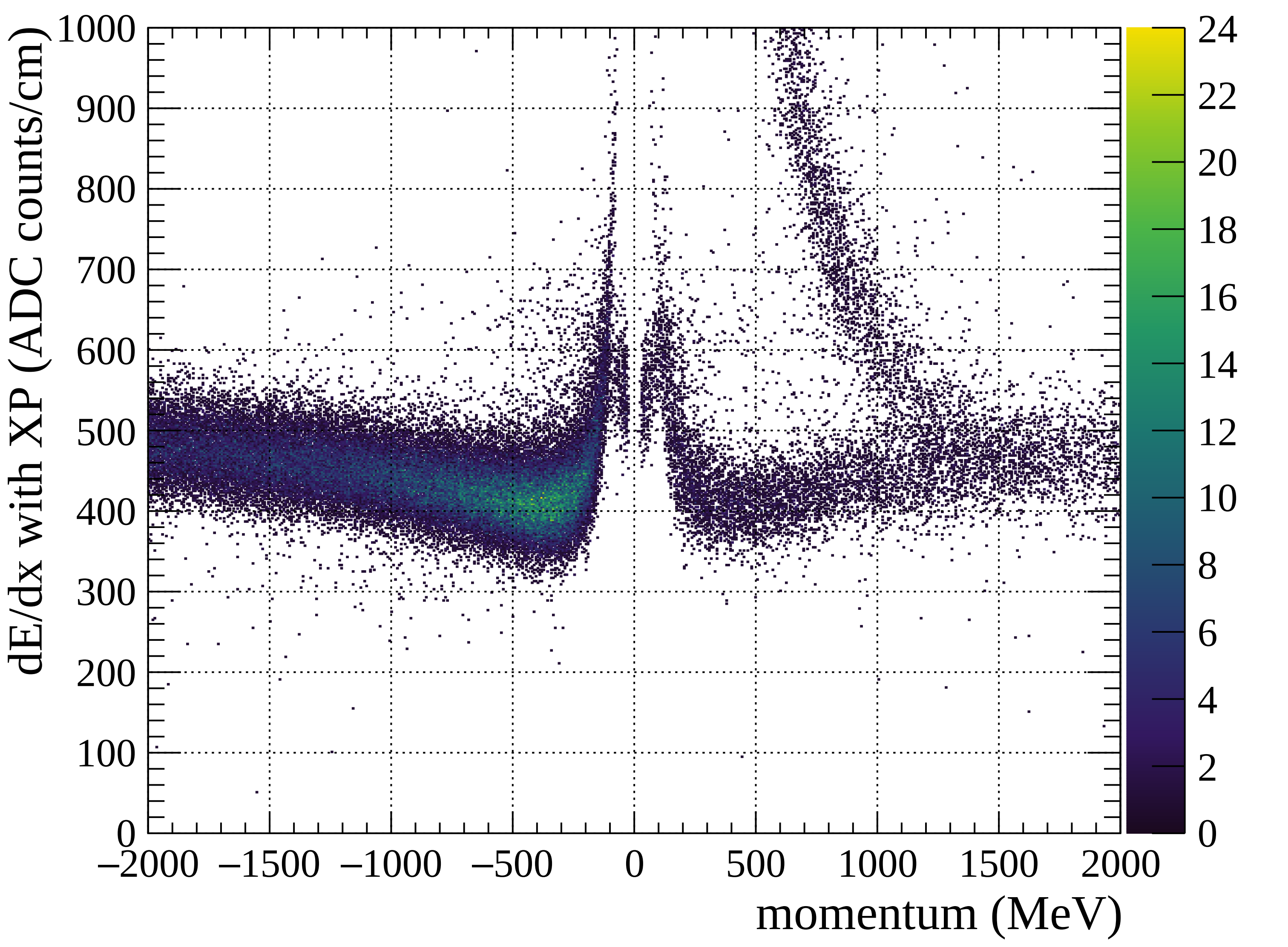

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Absolute momentum (MeV/c)
80
70
60
50
40
30
20
10
0



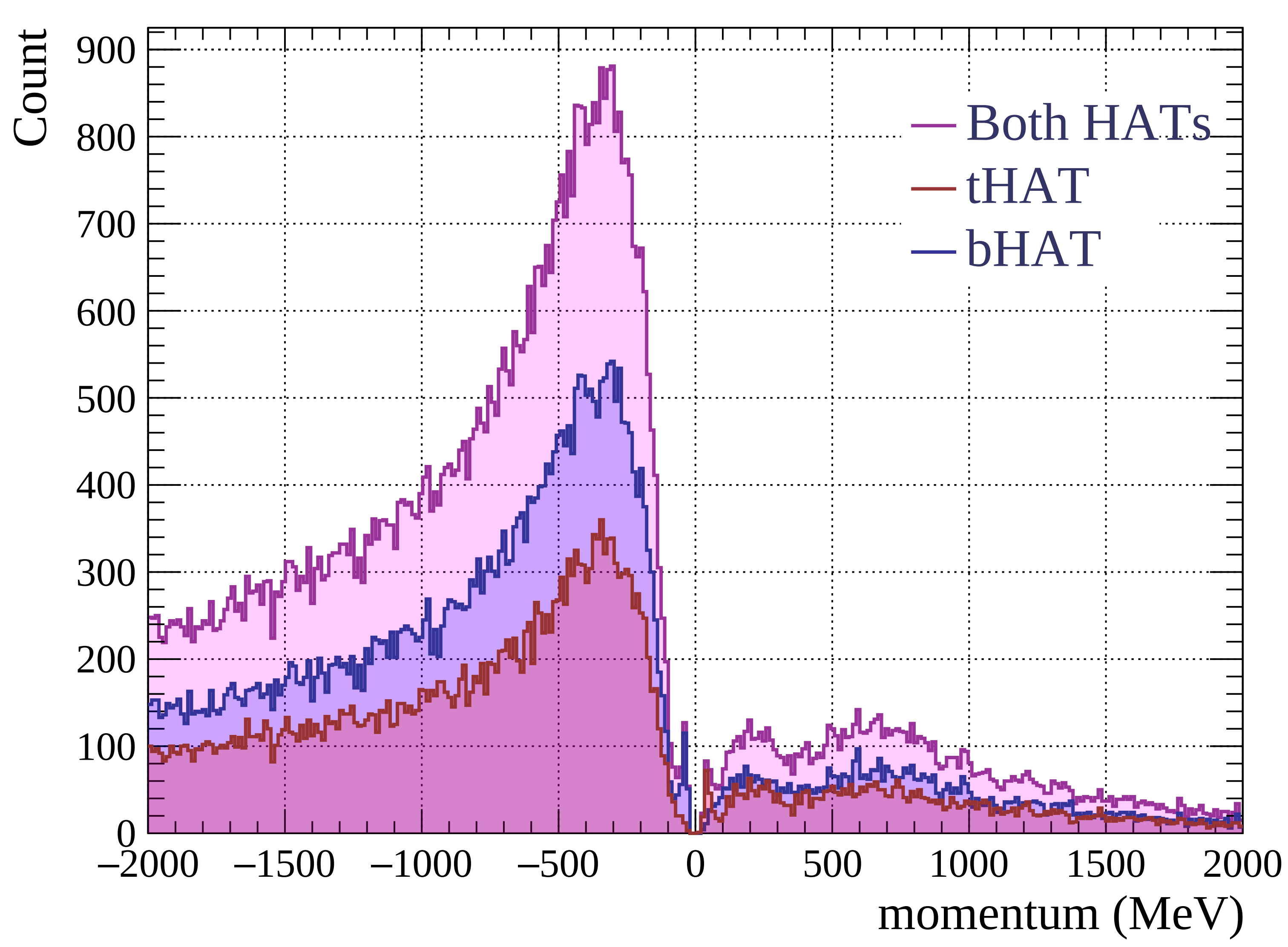
WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000


WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)

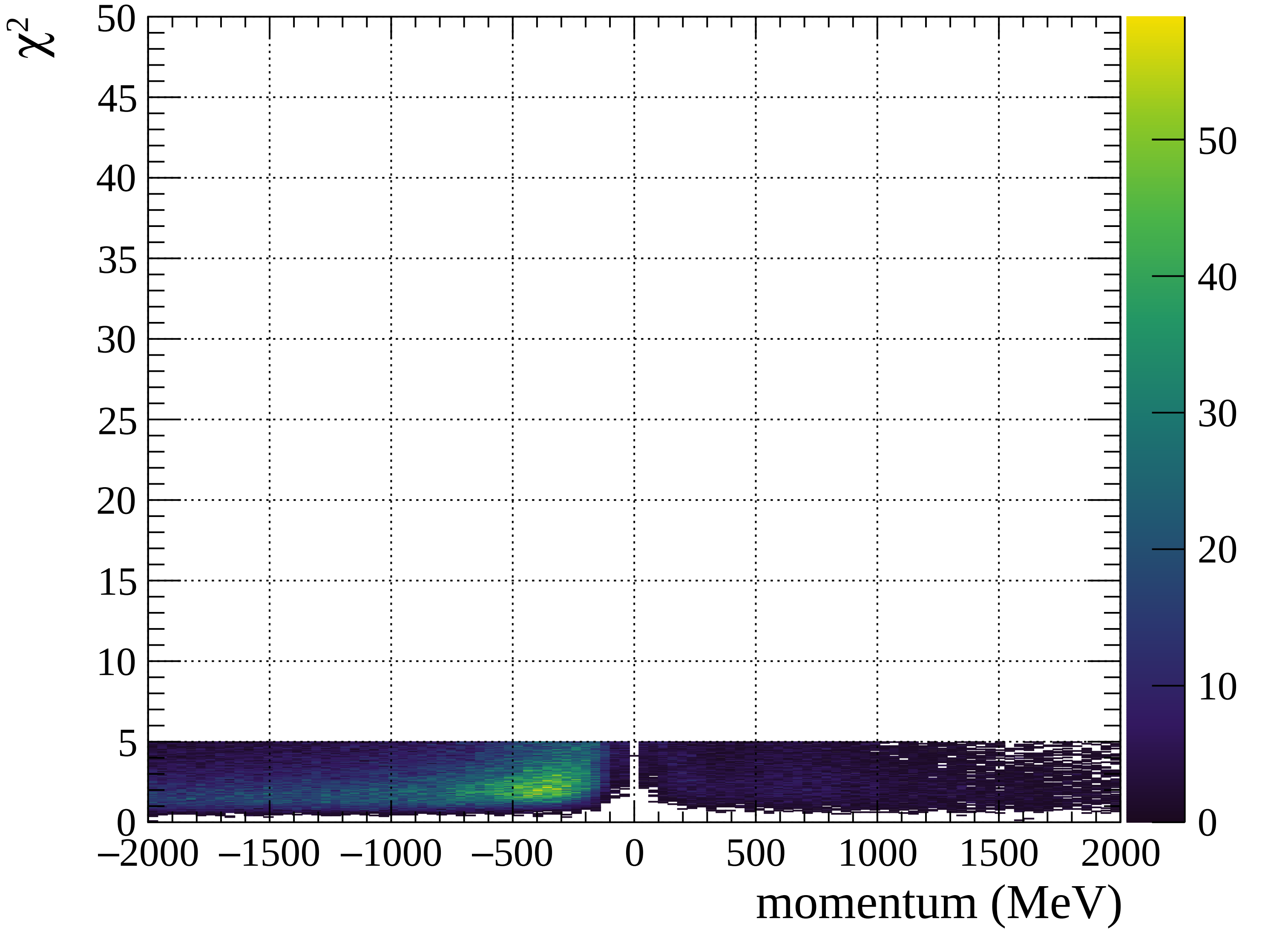

dE/dx with WF (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
24
22
20
18
16
14
12
10
8
6
4
2
0



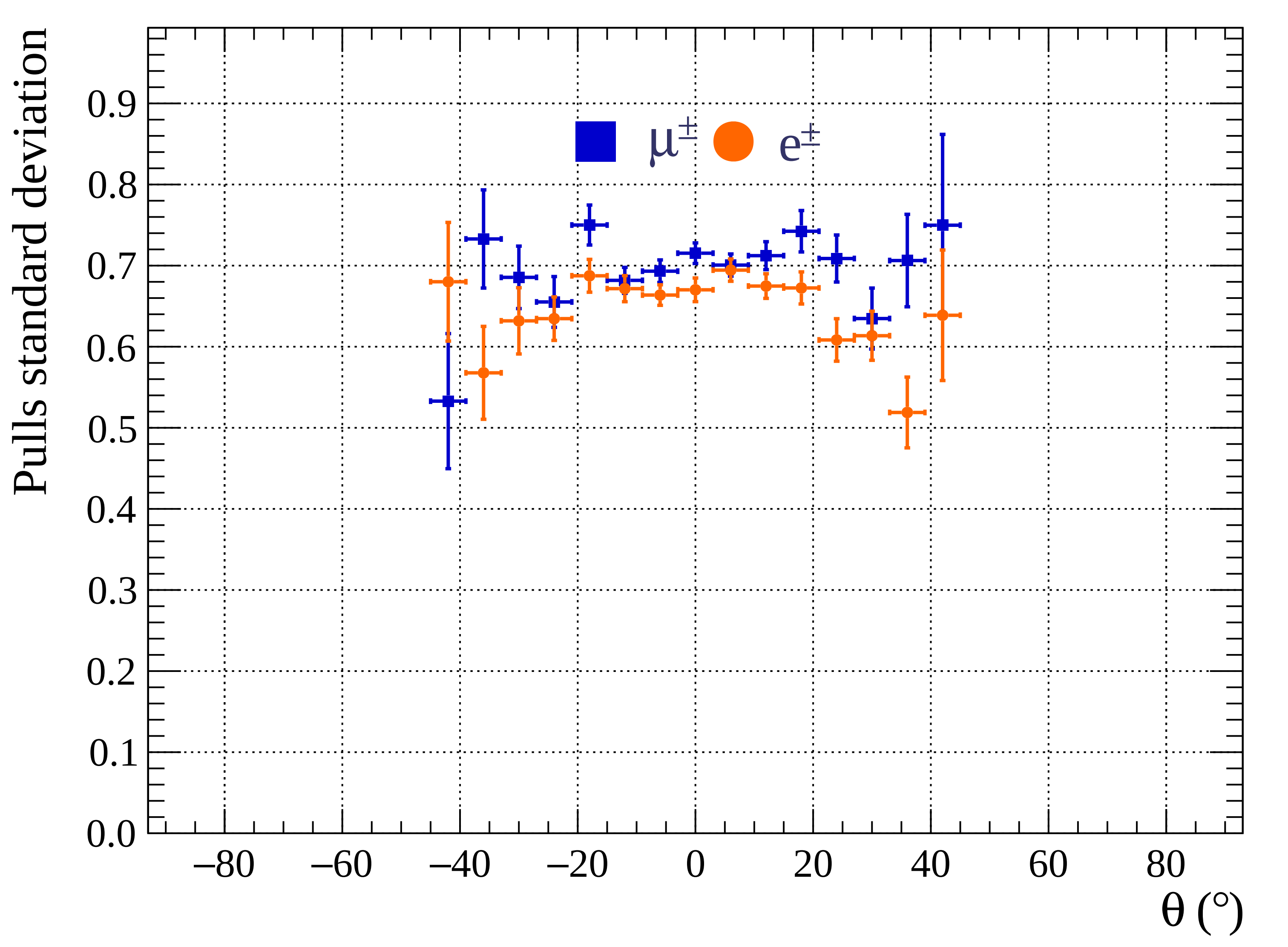
dE/dx with XP (ADC counts/cm)
momentum (MeV)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
24
22
20
18
16
14
12
10
8
6
4
2
0

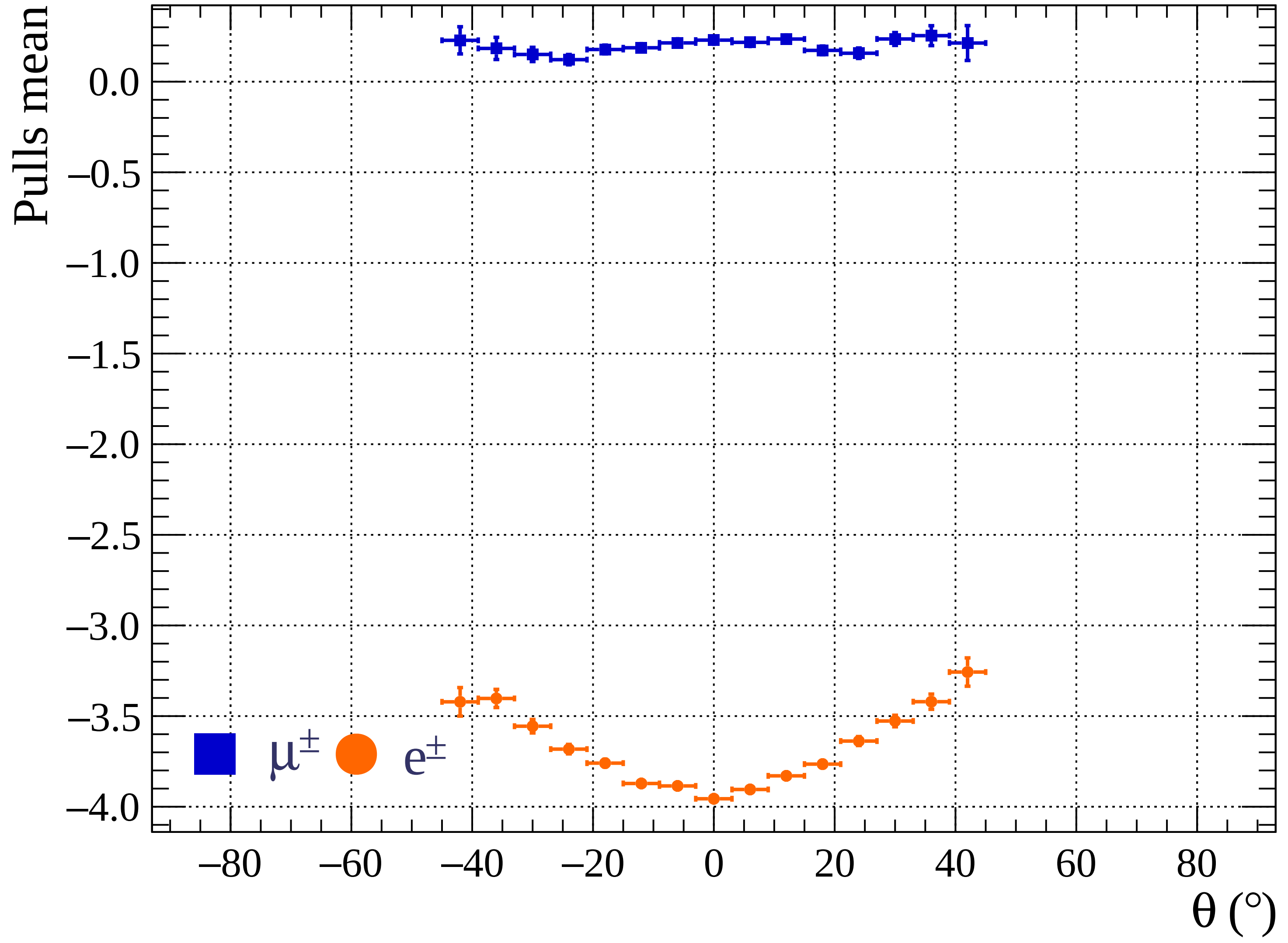


Count
900
800
700
600
500
400
300
200
100
0
–2000
–1500
–1000
–500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT

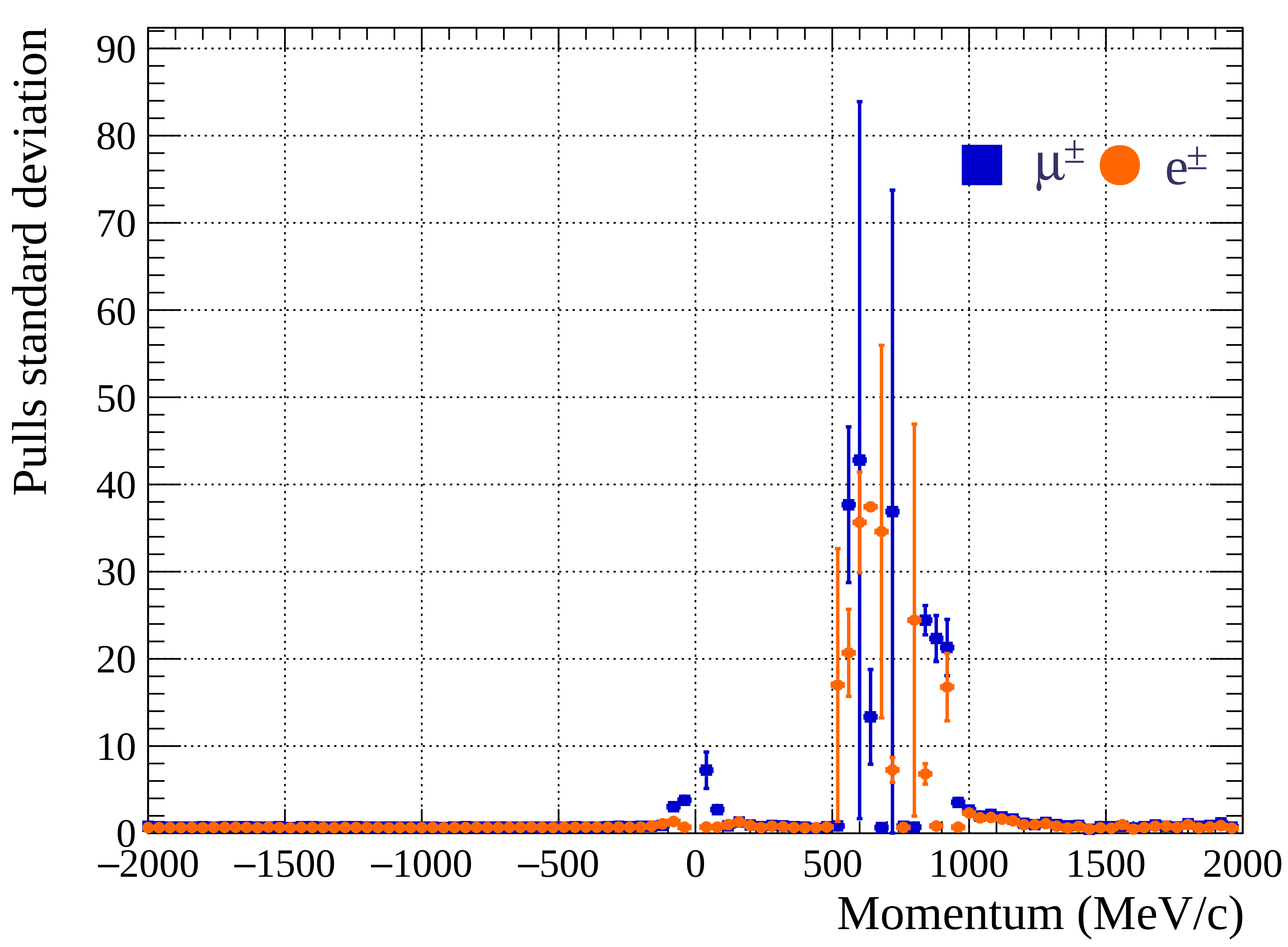


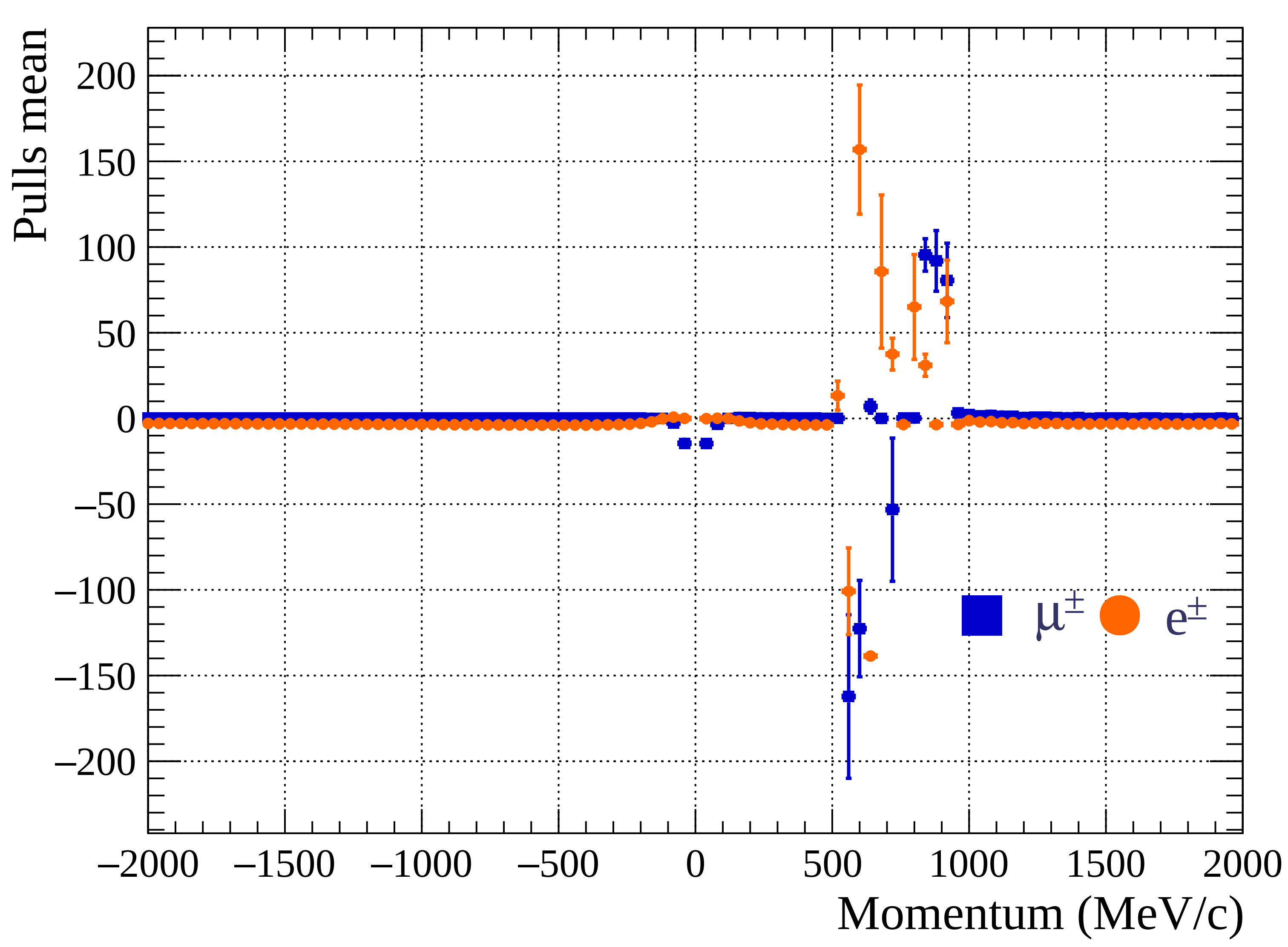
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
50
40
30
20
10
0


Pulls standard deviation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−80
−60
−40
−20
0
20
40
60
80
θ (°)
$\mu^{\pm}$
$e^{\pm}$

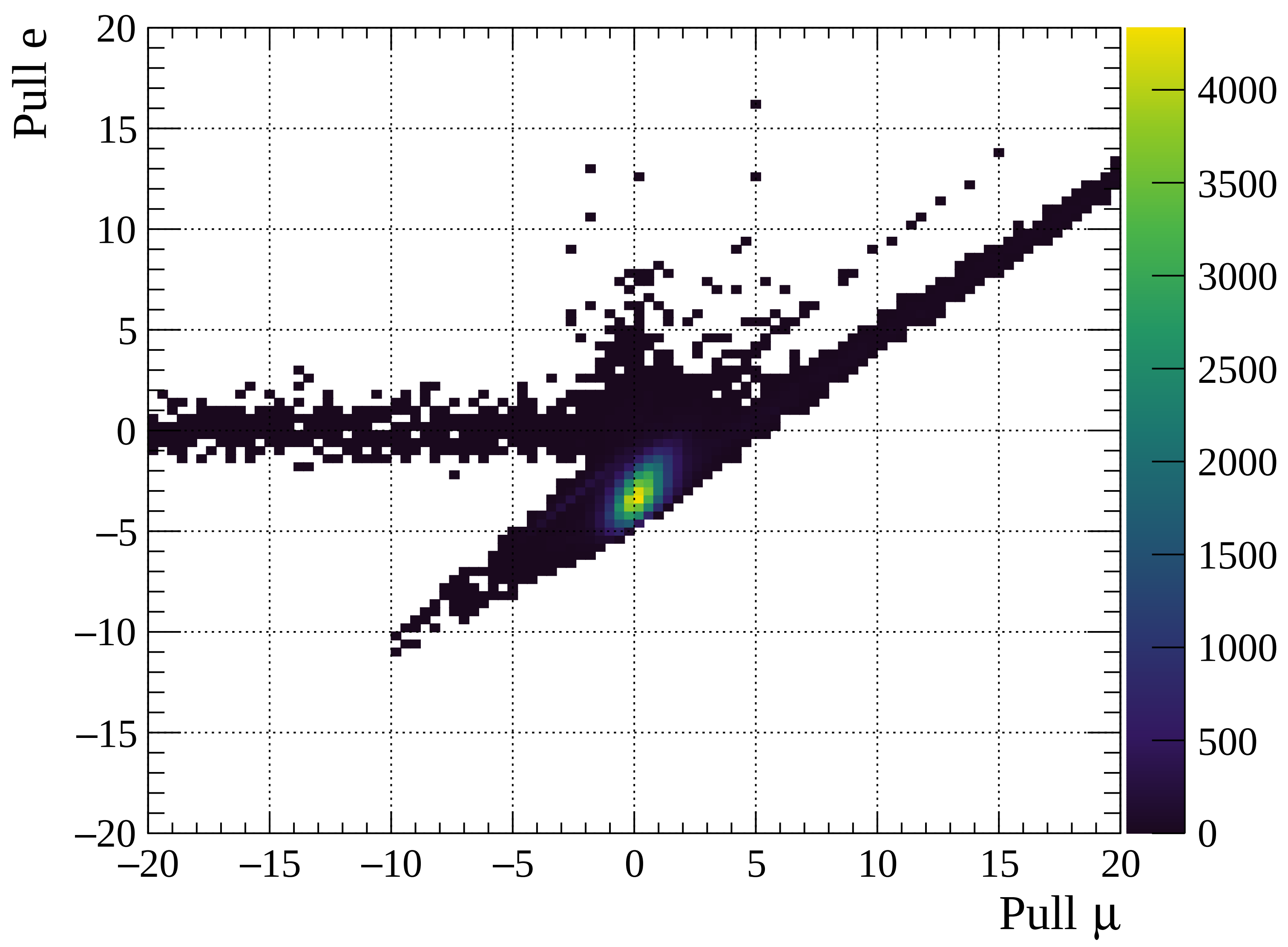


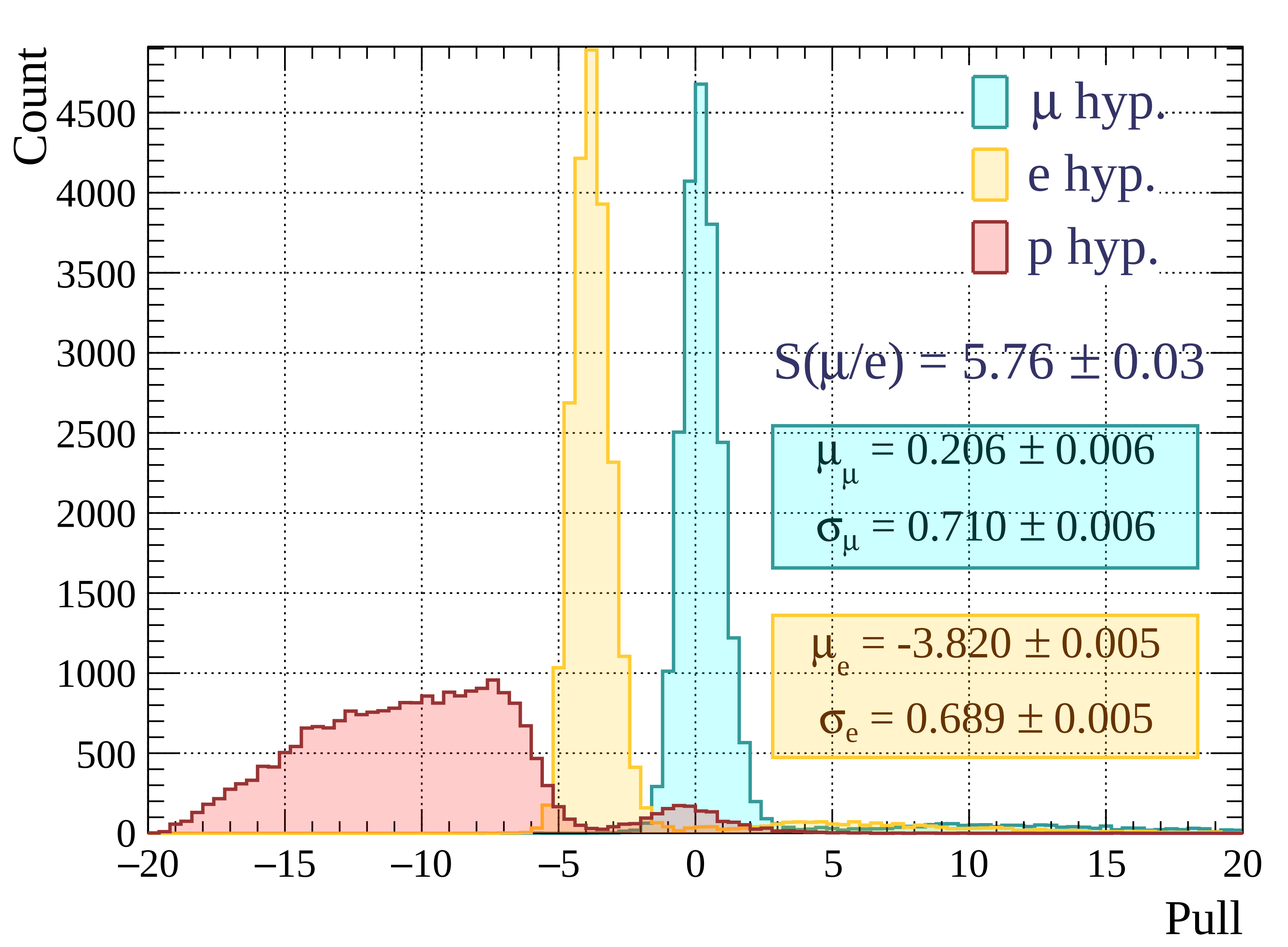
Pulls mean
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±


Pulls standard deviation
Momentum (MeV/c)
μ±
e±
0
10
20
30
40
50
60
70
80
90
−2000
−1500
−1000
−500
0
500
1000
1500
2000

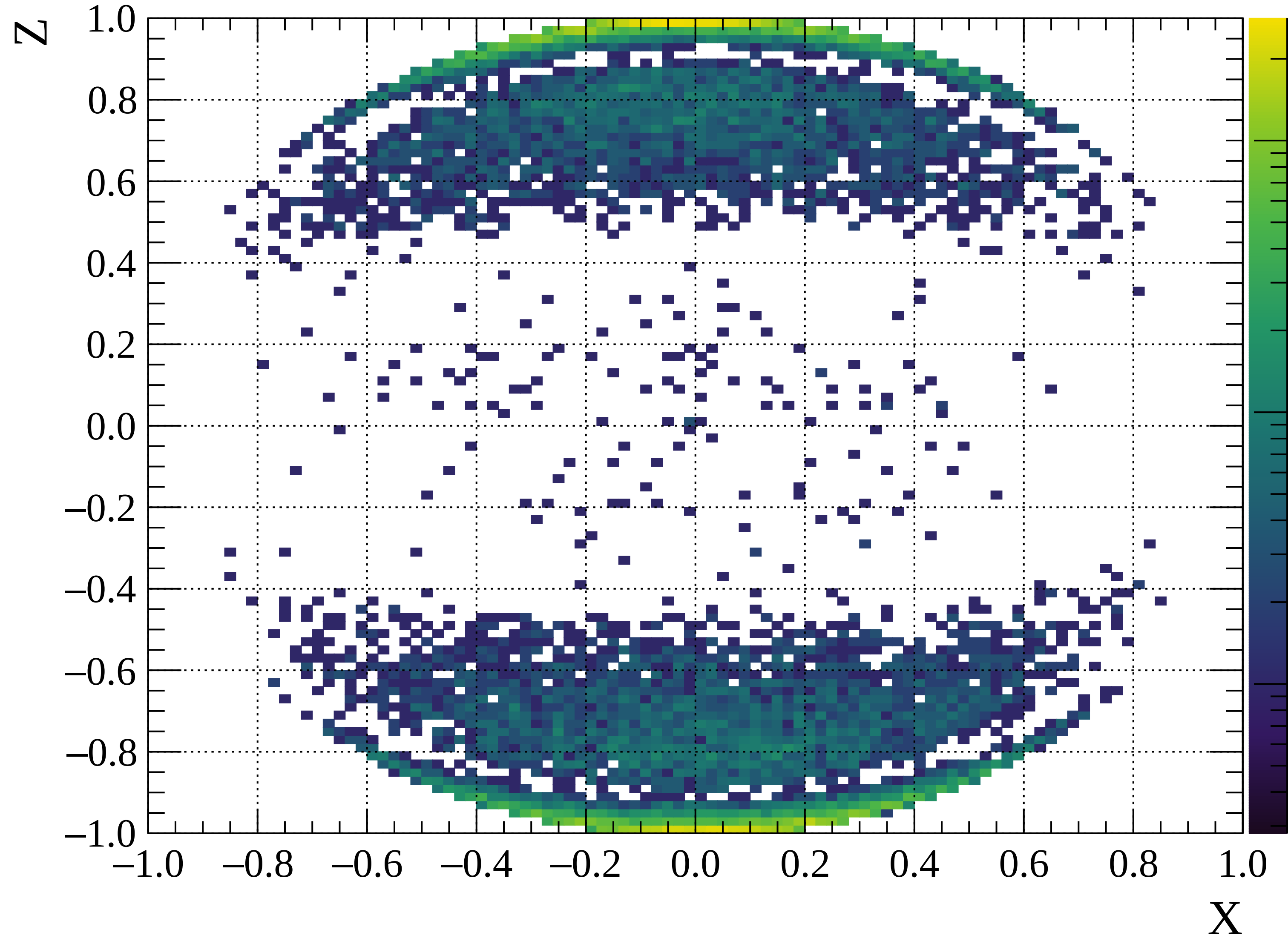


Pulls mean
200
150
100
50
0
−50
−100
−150
−200
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±

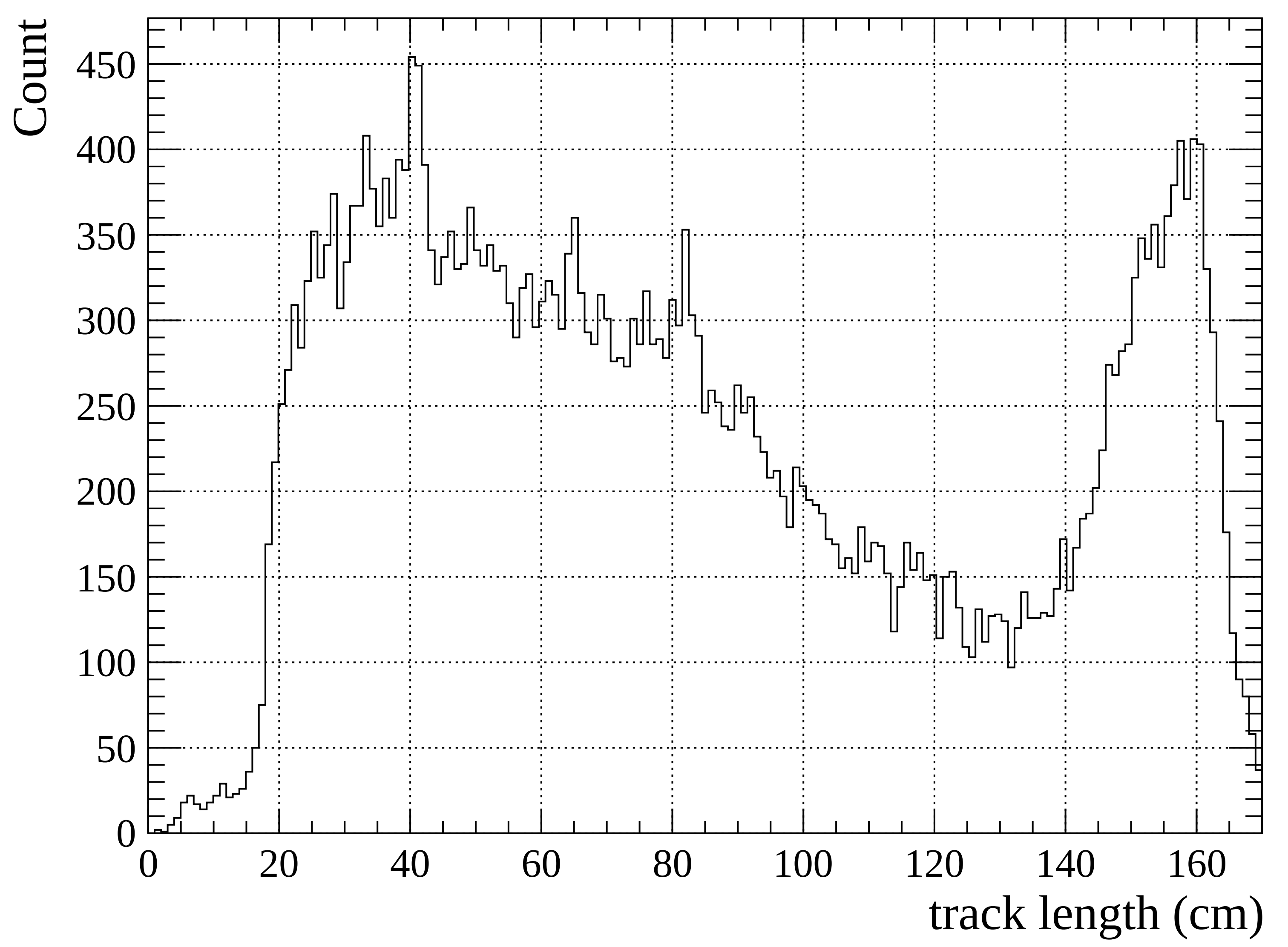

Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
4000
3500
3000
2500
2000
1500
1000
500
0

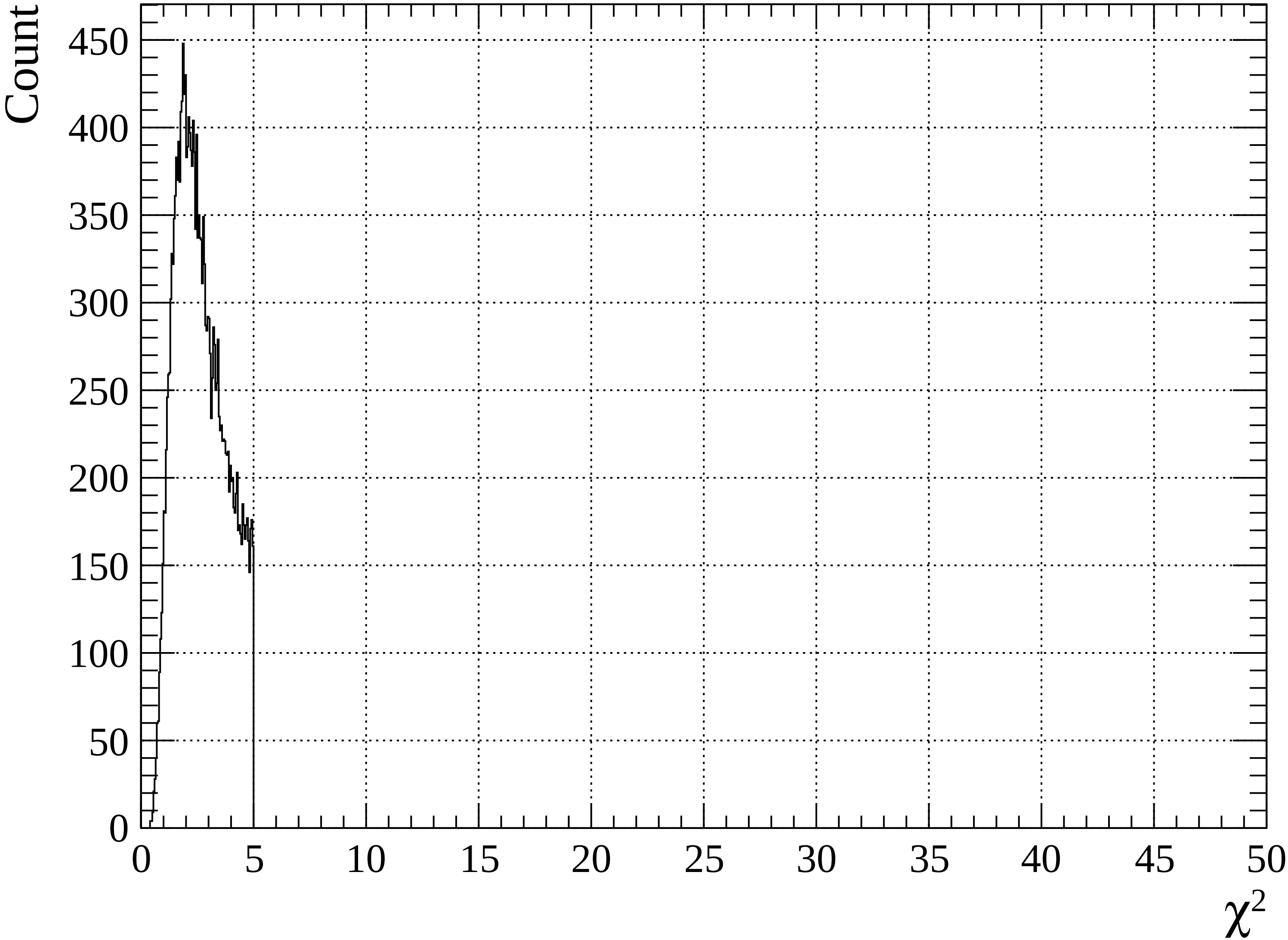


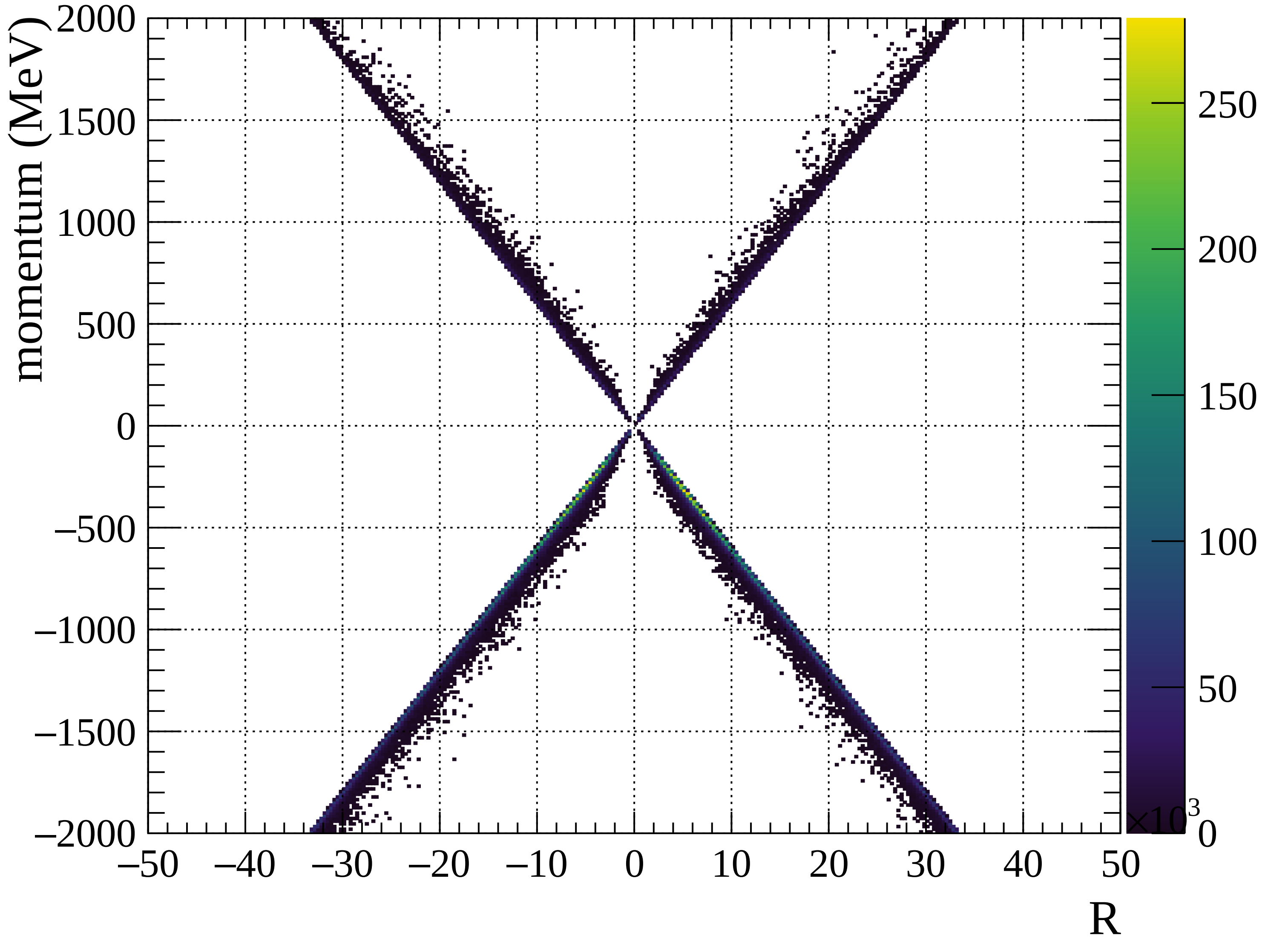
Count
4500
4000
3500
3000
2500
2000
1500
1000
500
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 5.76 ± 0.03
μ_μ = 0.206 ± 0.006
σ_μ = 0.710 ± 0.006
μ_e = -3.820 ± 0.005
σ_e = 0.689 ± 0.005



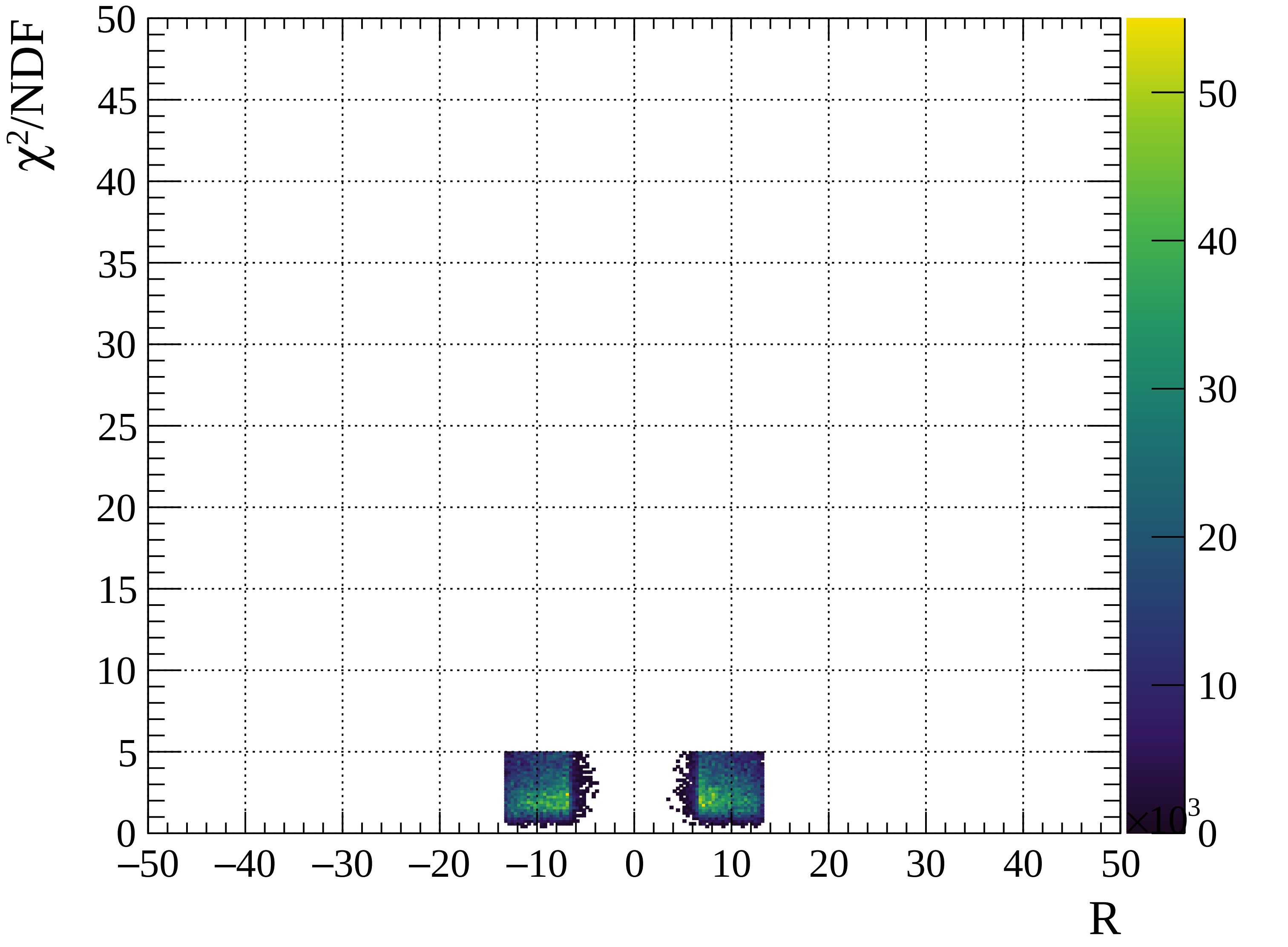
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X


Count
0
50
100
150
200
250
300
350
400
450
0
20
40
60
80
100
120
140
160
track length (cm)

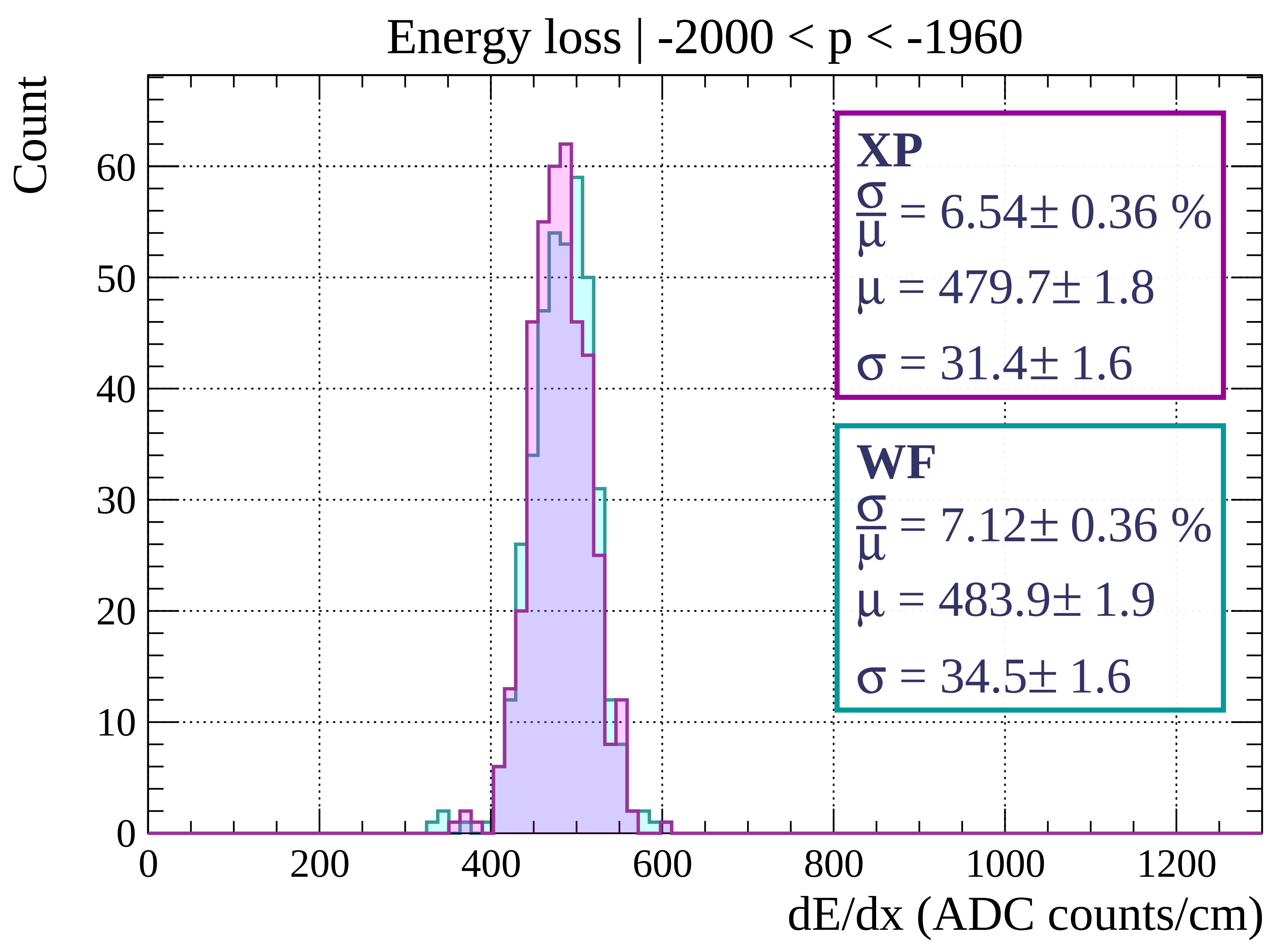

Count
450
400
350
300
250
200
150
100
50
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



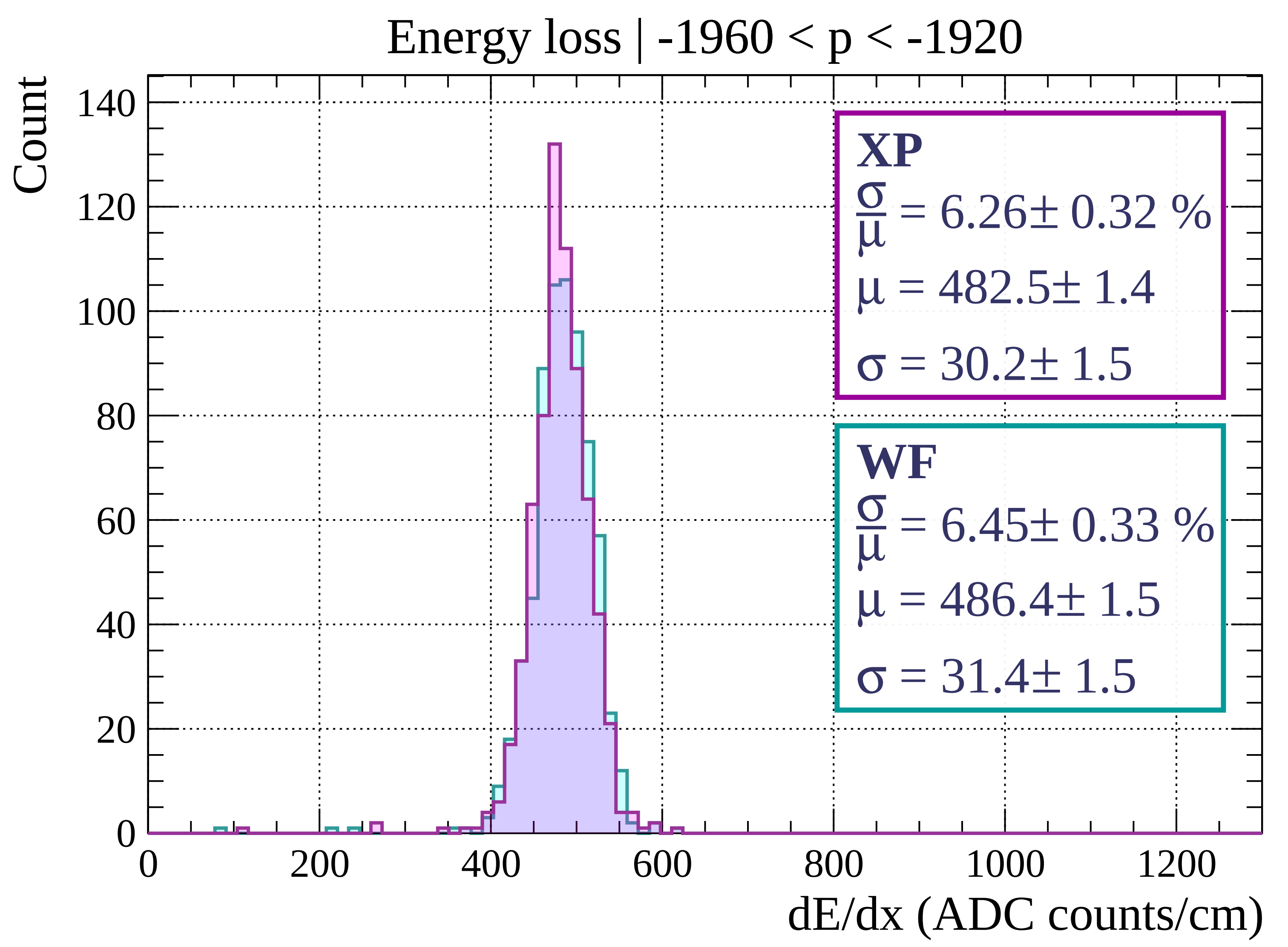
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
250
200
150
100
50
0
×10^3

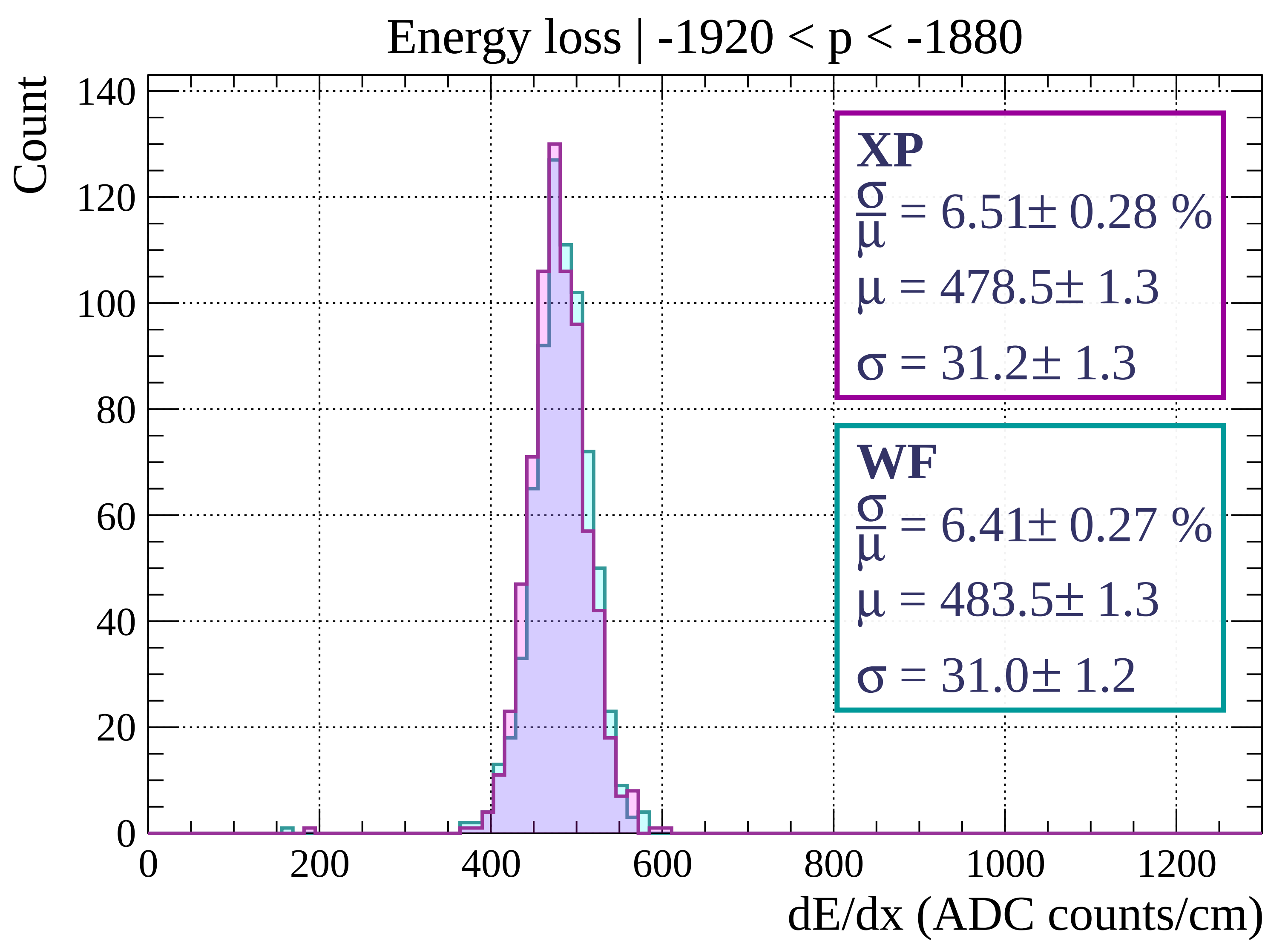


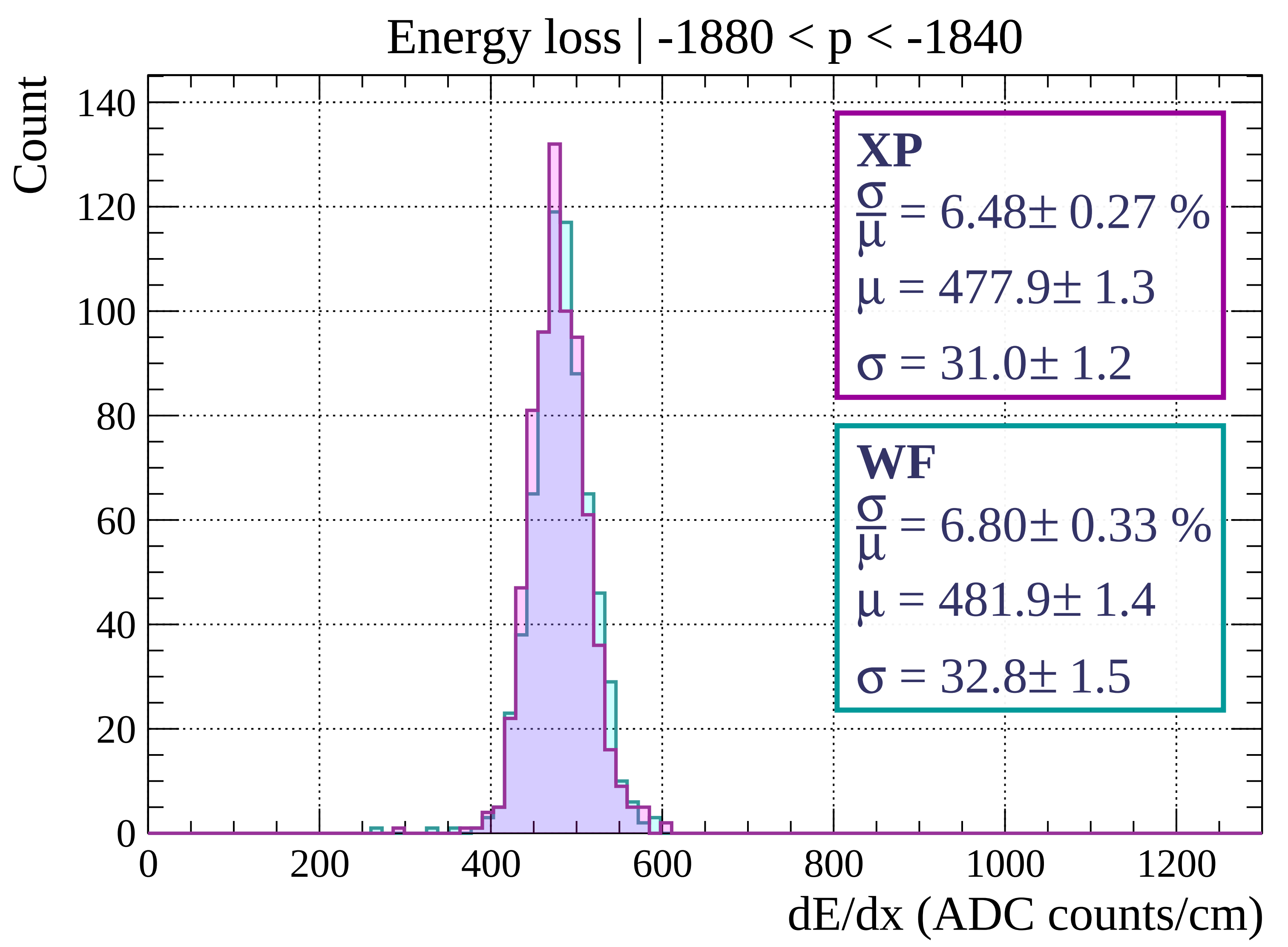
χ^2/NDF
R
$\times10^3$



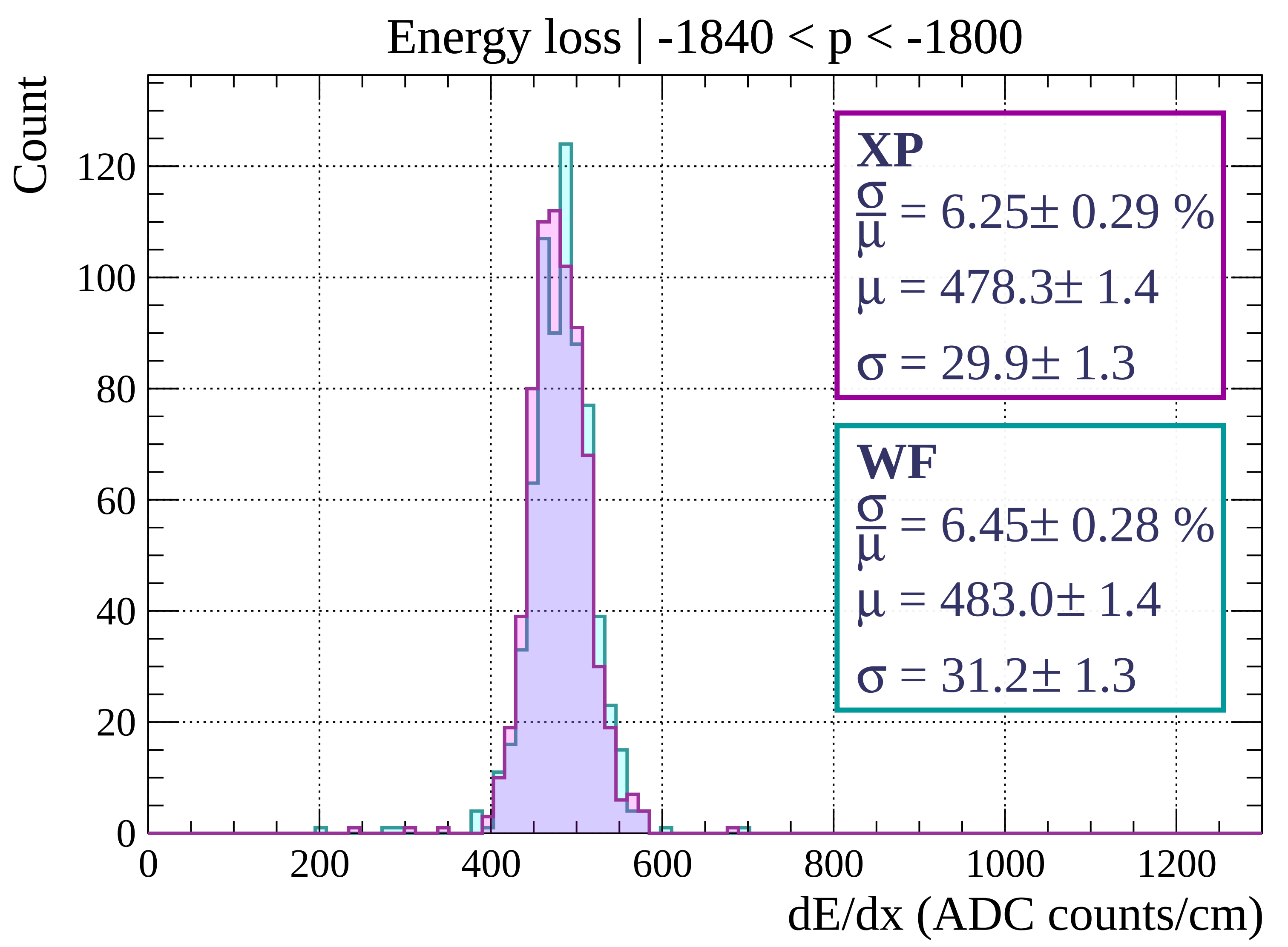
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.54± 0.36 %
μ = 479.7± 1.8
σ = 31.4± 1.6
WF
σ/μ = 7.12± 0.36 %
μ = 483.9± 1.9
σ = 34.5± 1.6
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200

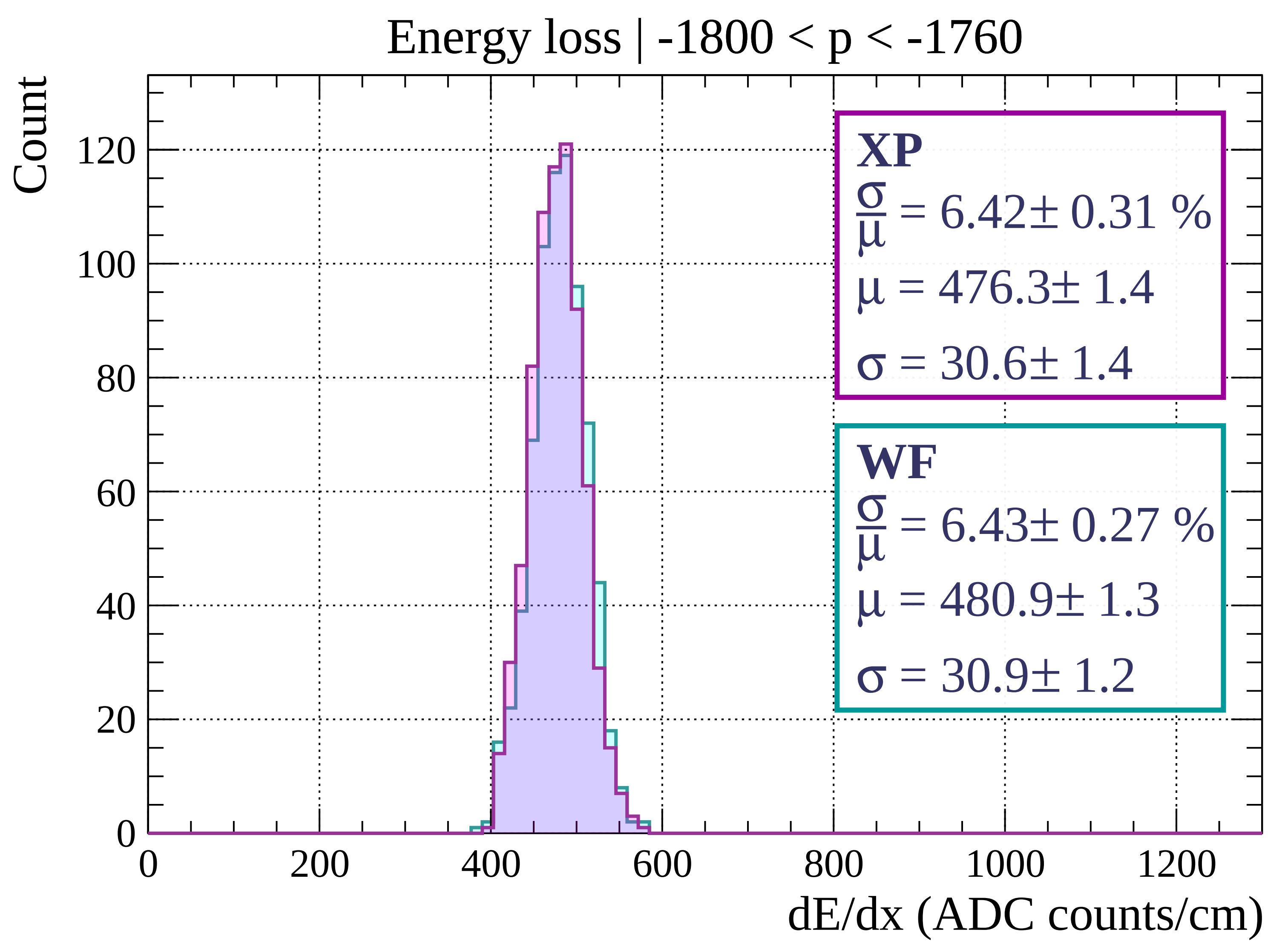

Energy loss | -1960 < p < -1920
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.26± 0.32 %
μ = 482.5± 1.4
σ = 30.2± 1.5
WF
σ/μ = 6.45± 0.33 %
μ = 486.4± 1.5
σ = 31.4± 1.5


Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 6.51± 0.28 %
μ = 478.5± 1.3
σ = 31.2± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.41± 0.27 %
μ = 483.5± 1.3
σ = 31.0± 1.2

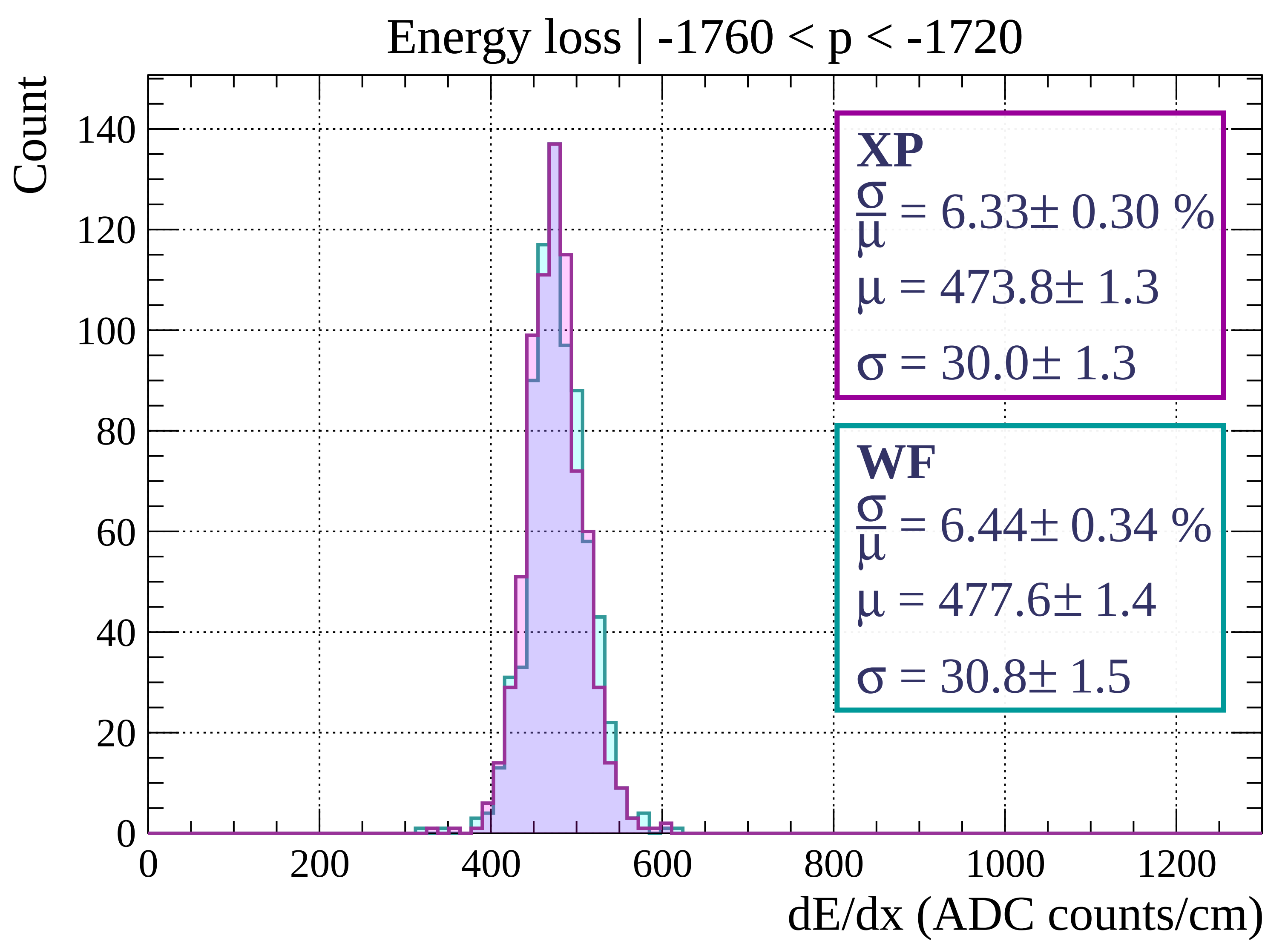


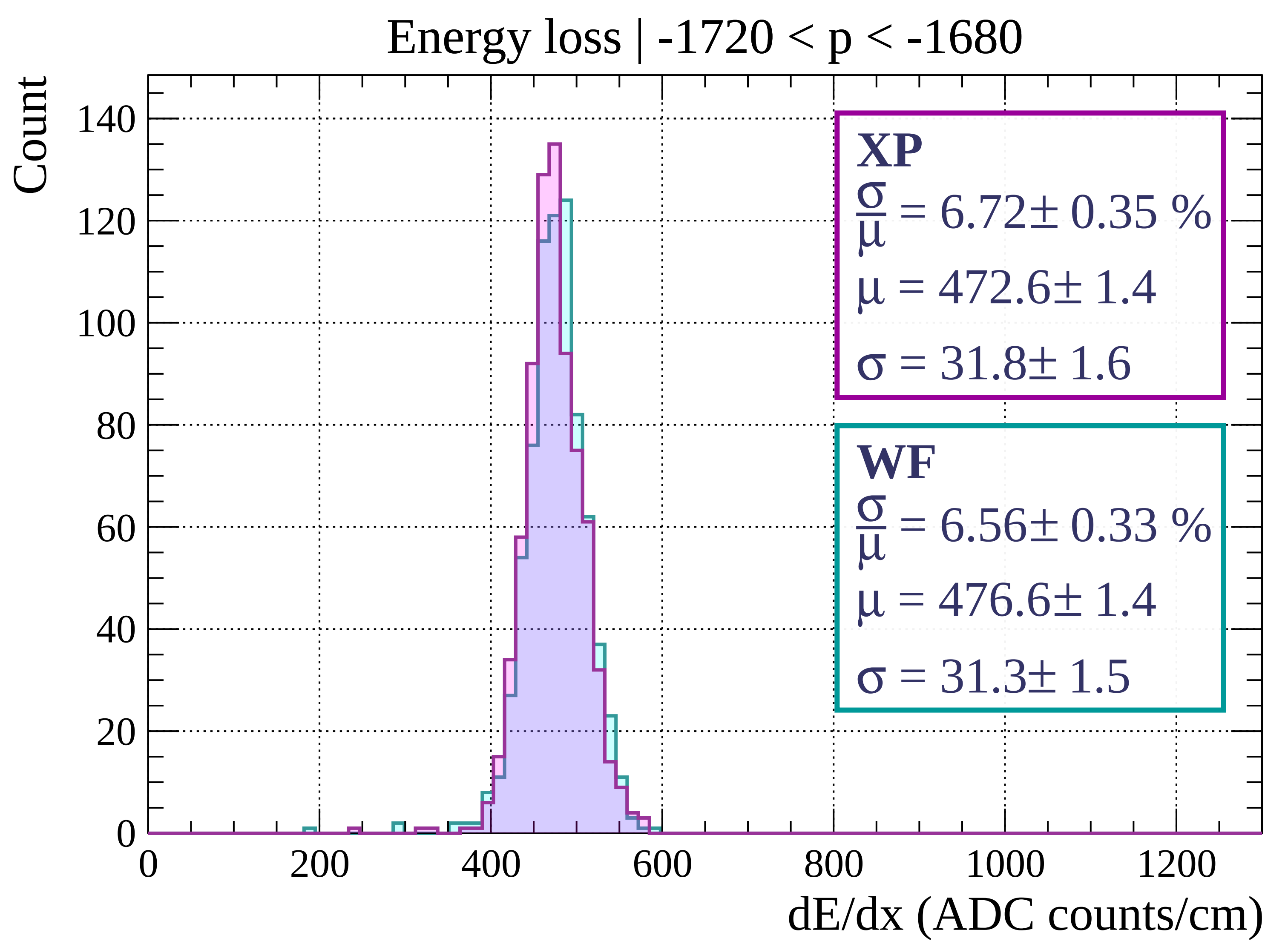
Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.48± 0.27 %
μ = 477.9± 1.3
σ = 31.0± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.80± 0.33 %
μ = 481.9± 1.4
σ = 32.8± 1.5

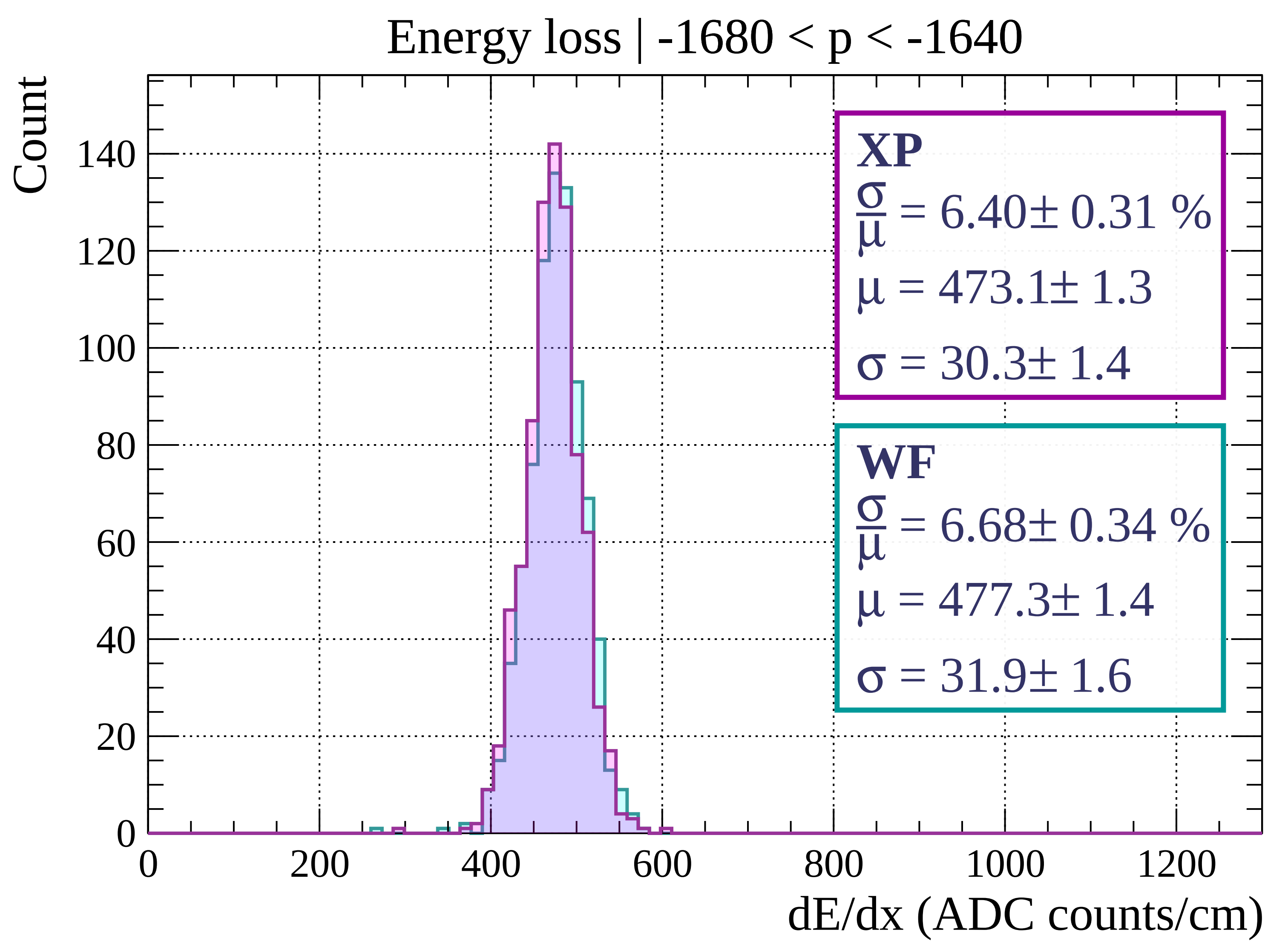

Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 6.25± 0.29 %
μ = 478.3± 1.4
σ = 29.9± 1.3
WF
σ/μ = 6.45± 0.28 %
μ = 483.0± 1.4
σ = 31.2± 1.3

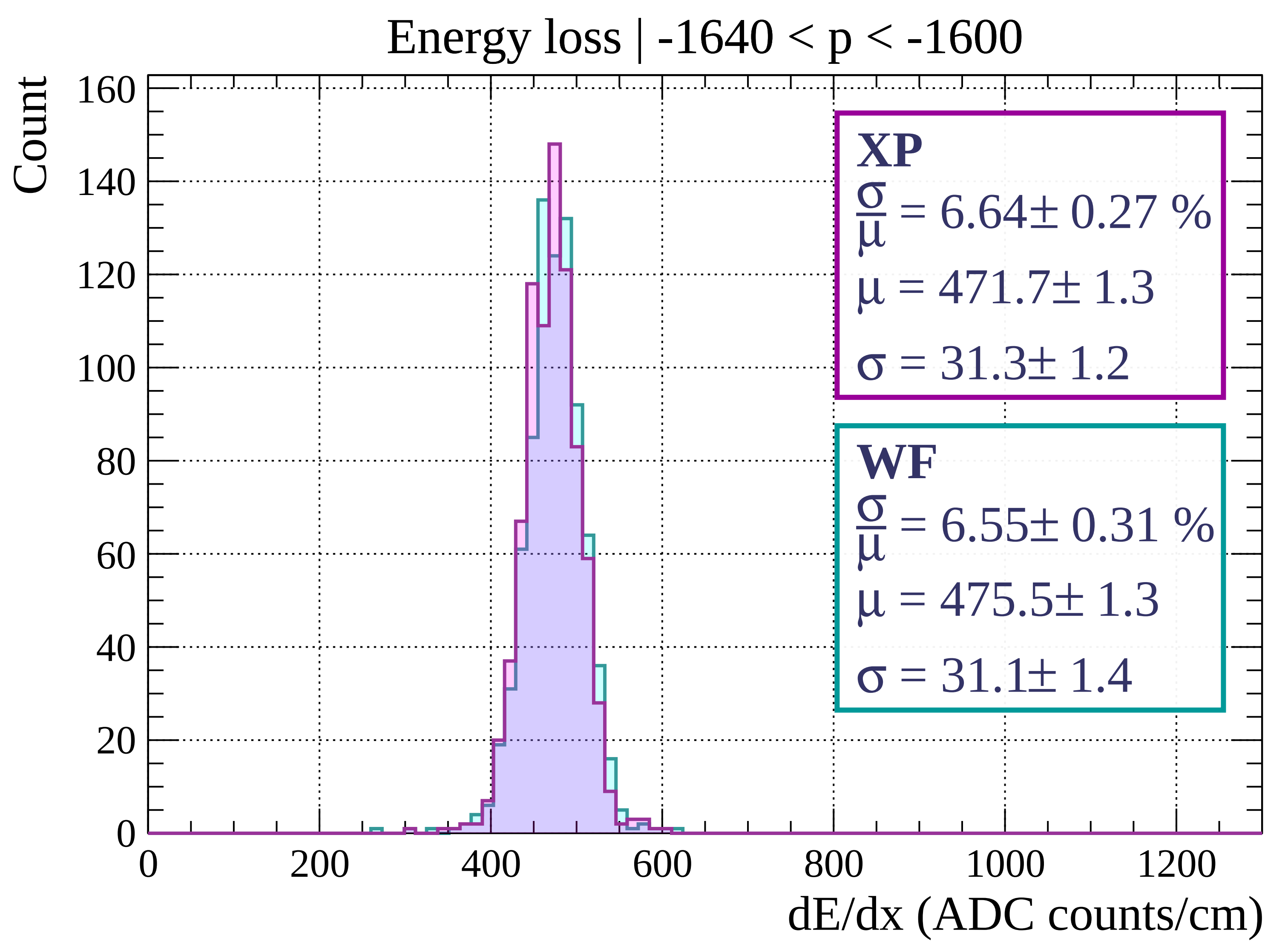


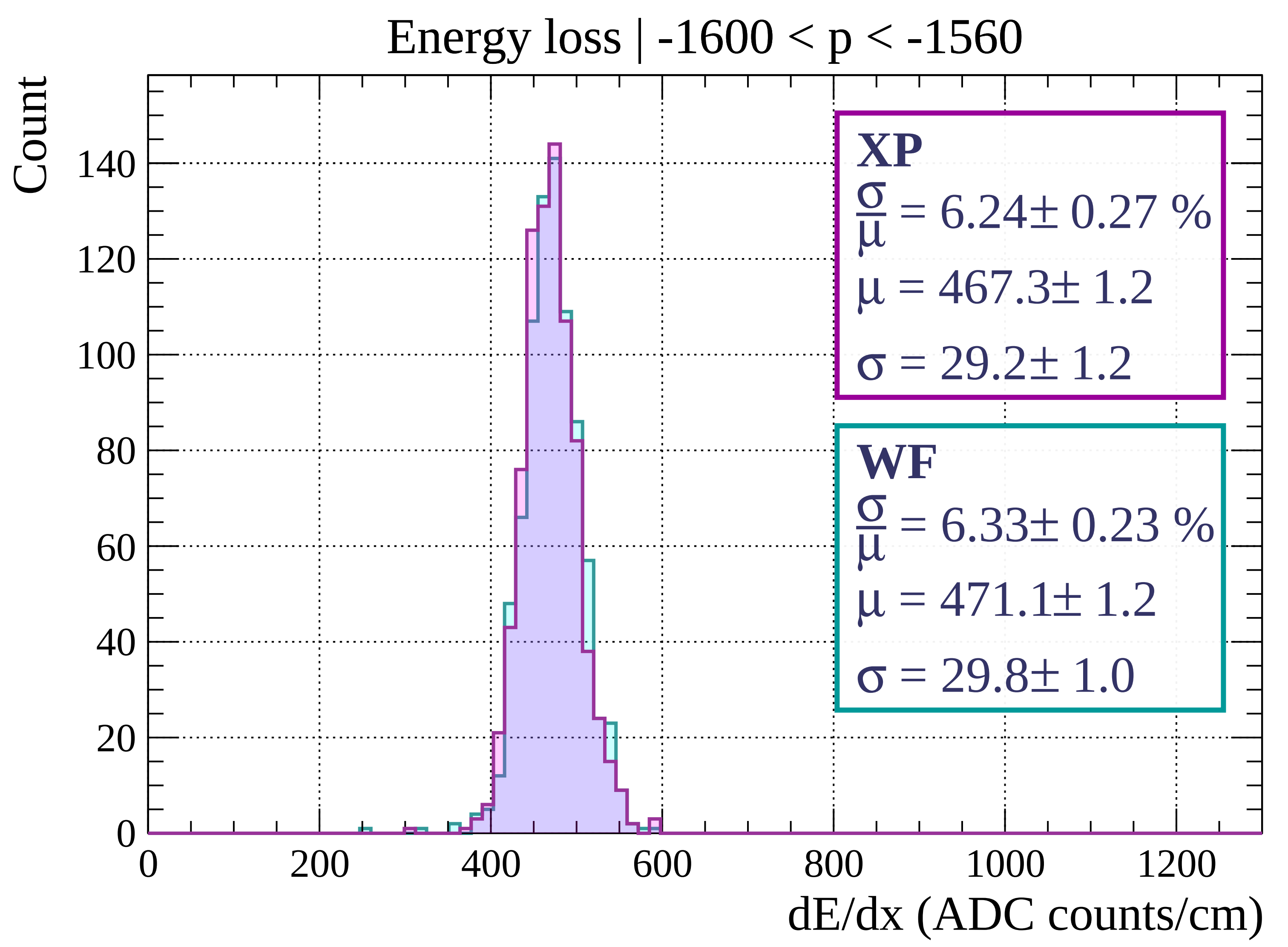
Energy loss | -1800 < p < -1760
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.42± 0.31 %
μ = 476.3± 1.4
σ = 30.6± 1.4
WF
$\frac{\sigma}{\mu}$ = 6.43± 0.27 %
μ = 480.9± 1.3
σ = 30.9± 1.2


Energy loss | -1760 < p < -1720
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.33± 0.30 %
μ = 473.8± 1.3
σ = 30.0± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.44± 0.34 %
μ = 477.6± 1.4
σ = 30.8± 1.5

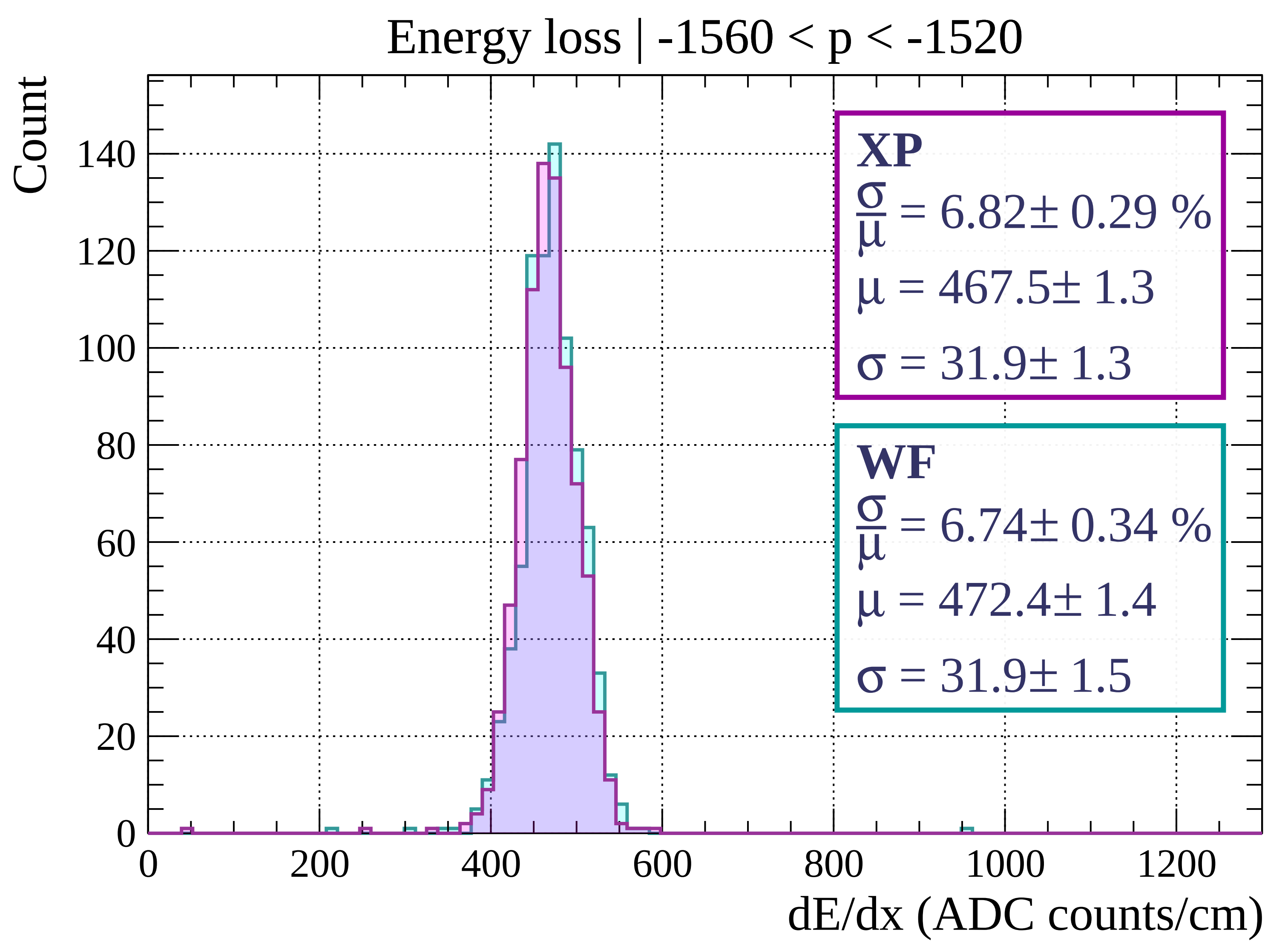

Energy loss | -1720 < p < -1680
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.72± 0.35 %
μ = 472.6± 1.4
σ = 31.8± 1.6
WF
$\frac{\sigma}{\mu}$ = 6.56± 0.33 %
μ = 476.6± 1.4
σ = 31.3± 1.5


Energy loss | -1680 < p < -1640
Count
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.40± 0.31 %
μ = 473.1± 1.3
σ = 30.3± 1.4
WF
σ/μ = 6.68± 0.34 %
μ = 477.3± 1.4
σ = 31.9± 1.6

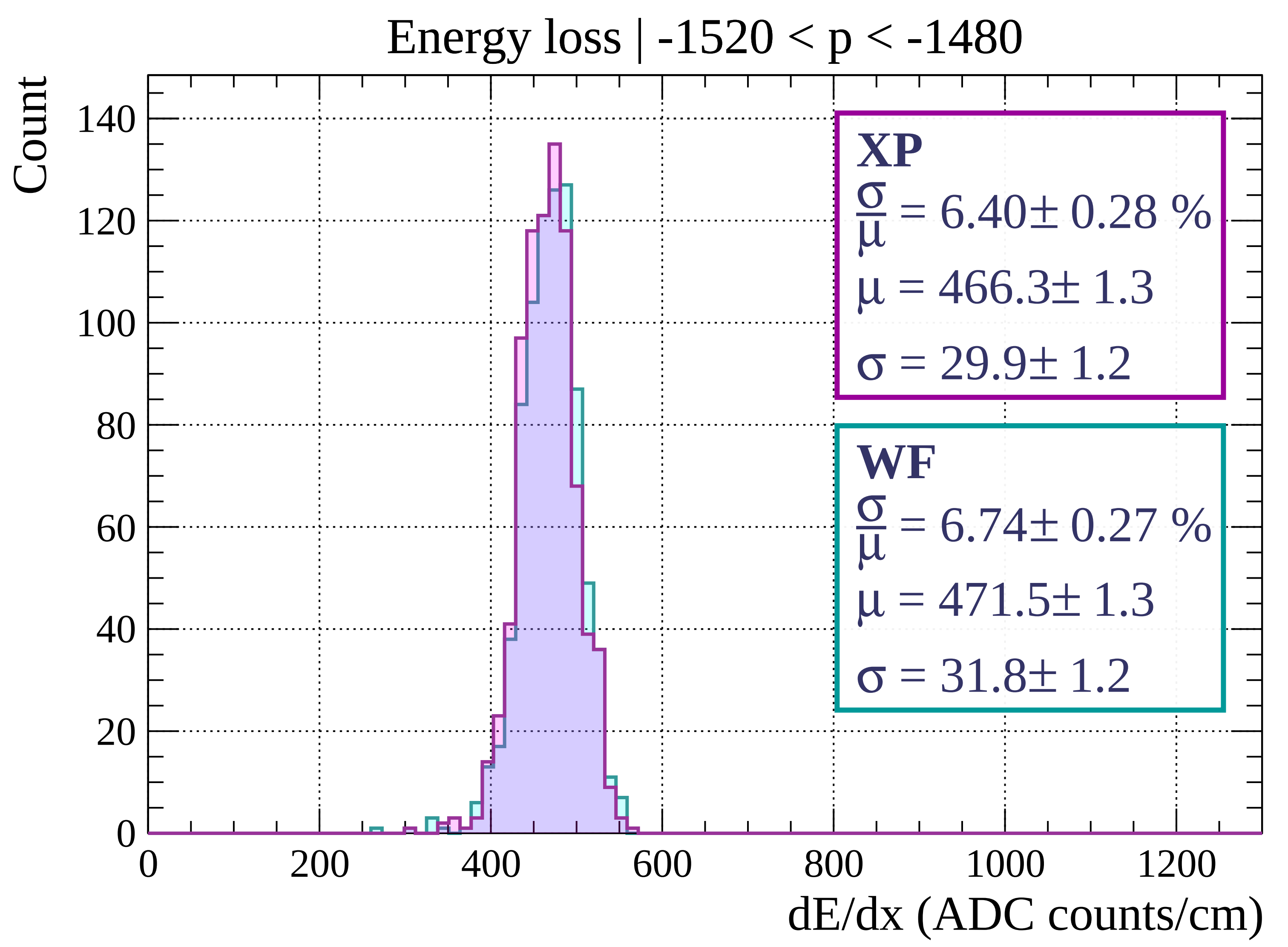

Energy loss | -1640 < p < -1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 6.64± 0.27 %
μ = 471.7± 1.3
σ = 31.3± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.55± 0.31 %
μ = 475.5± 1.3
σ = 31.1± 1.4

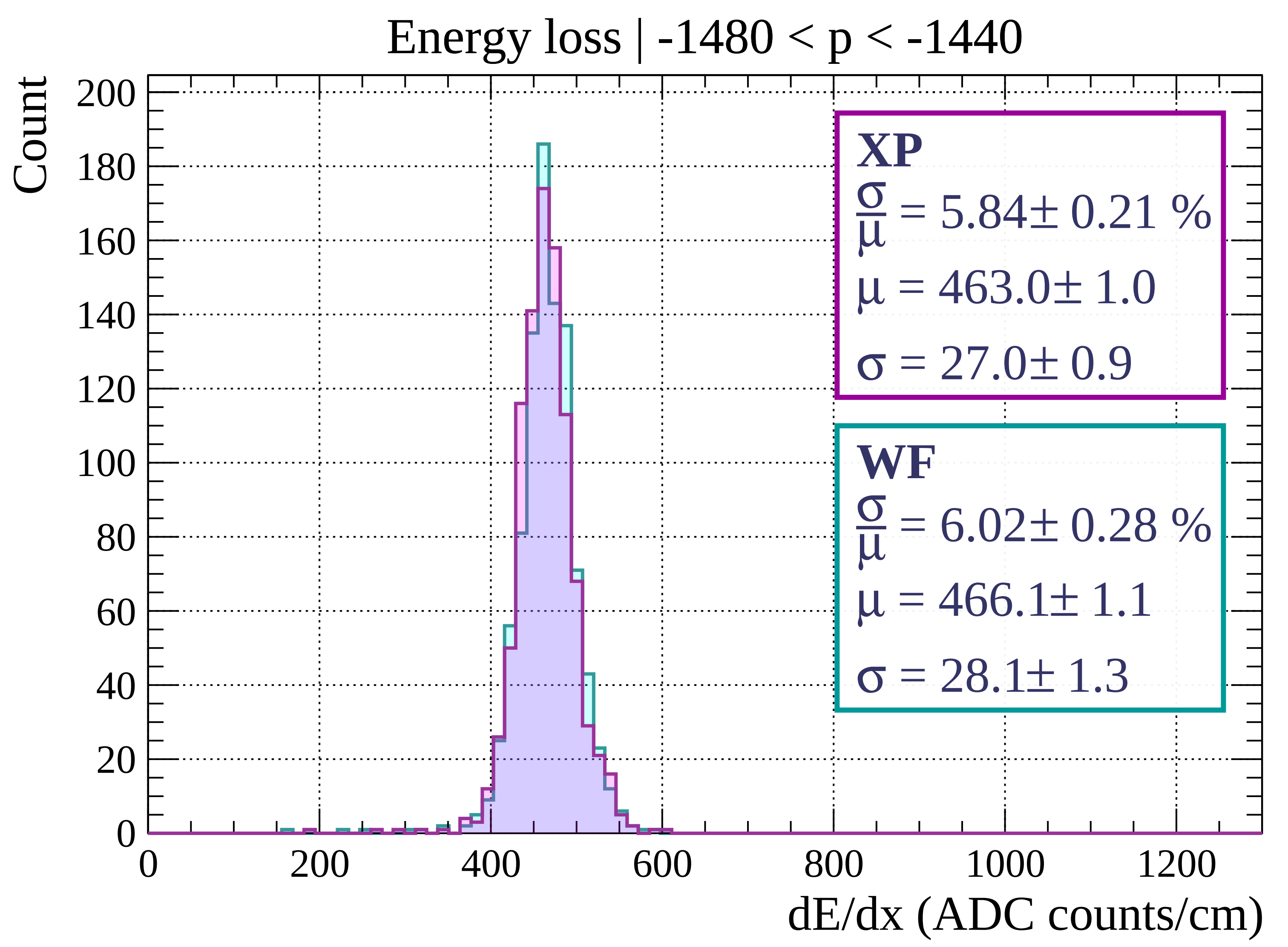


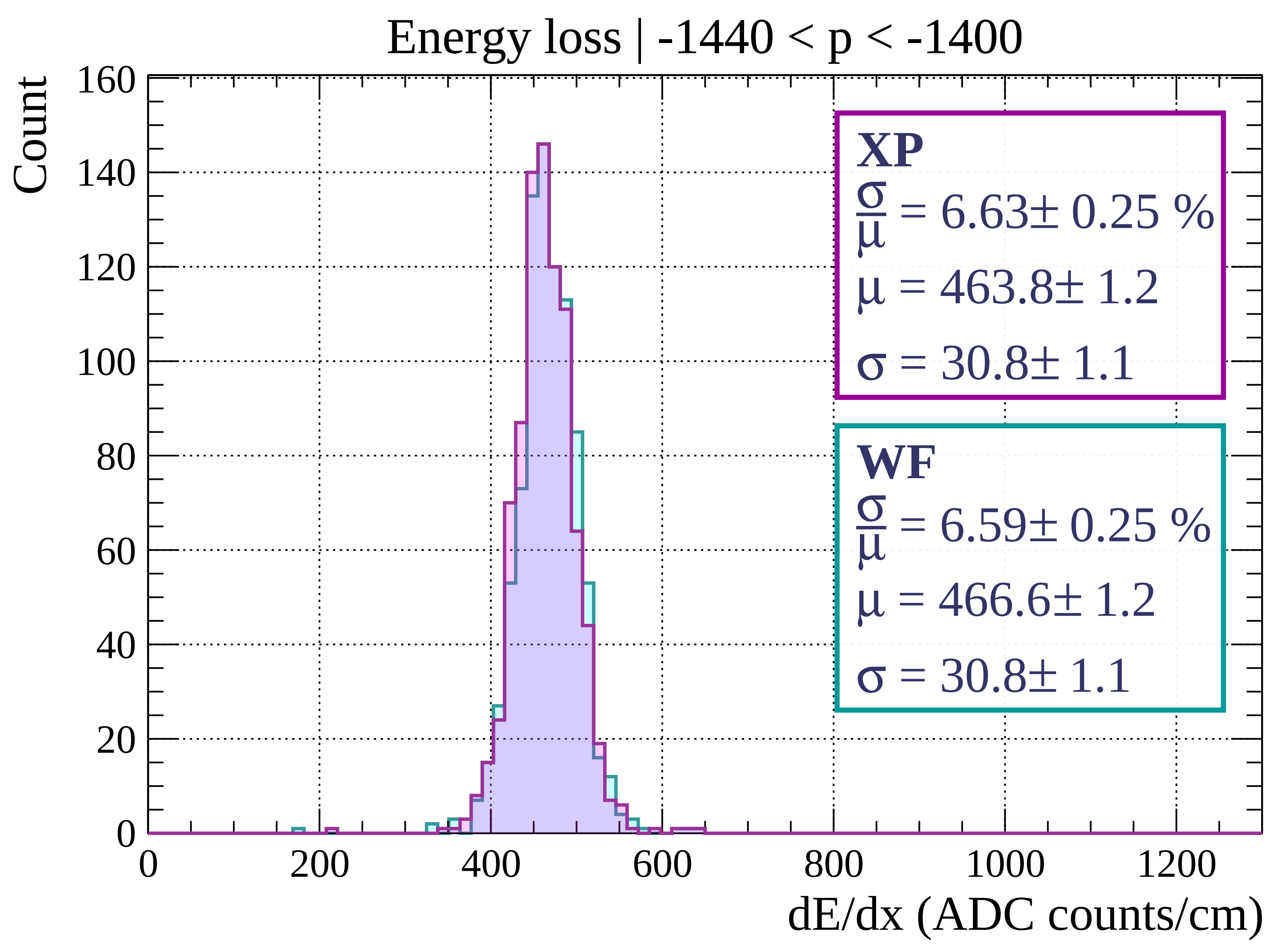
Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 6.24± 0.27 %
μ = 467.3± 1.2
σ = 29.2± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.33± 0.23 %
μ = 471.1± 1.2
σ = 29.8± 1.0



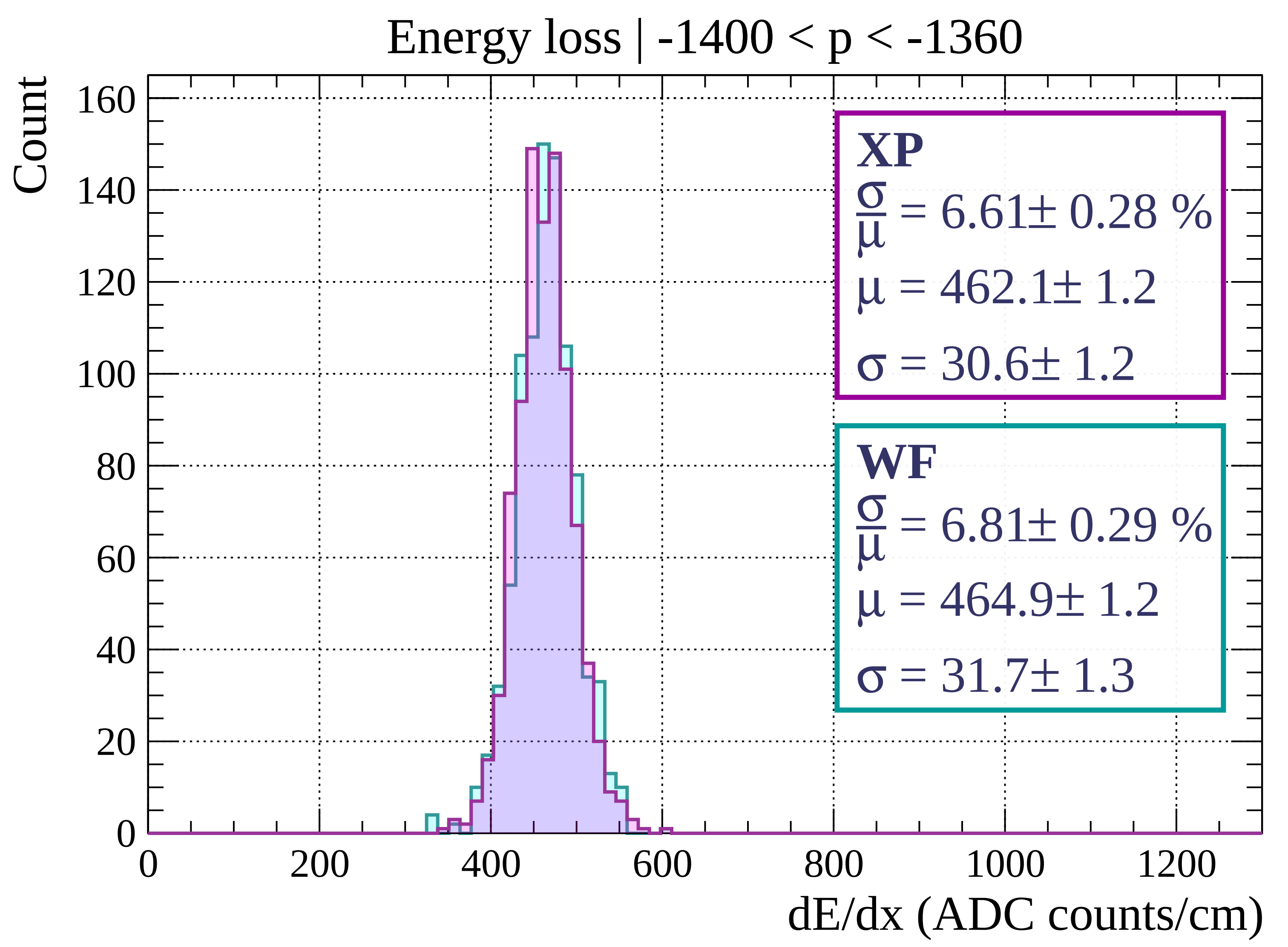
Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
$\frac{\sigma}{\mu}$ = 6.82± 0.29 %
μ = 467.5± 1.3
σ = 31.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.74± 0.34 %
μ = 472.4± 1.4
σ = 31.9± 1.5


Energy loss | -1520 < p < -1480
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.40± 0.28 %
μ = 466.3± 1.3
σ = 29.9± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.74± 0.27 %
μ = 471.5± 1.3
σ = 31.8± 1.2

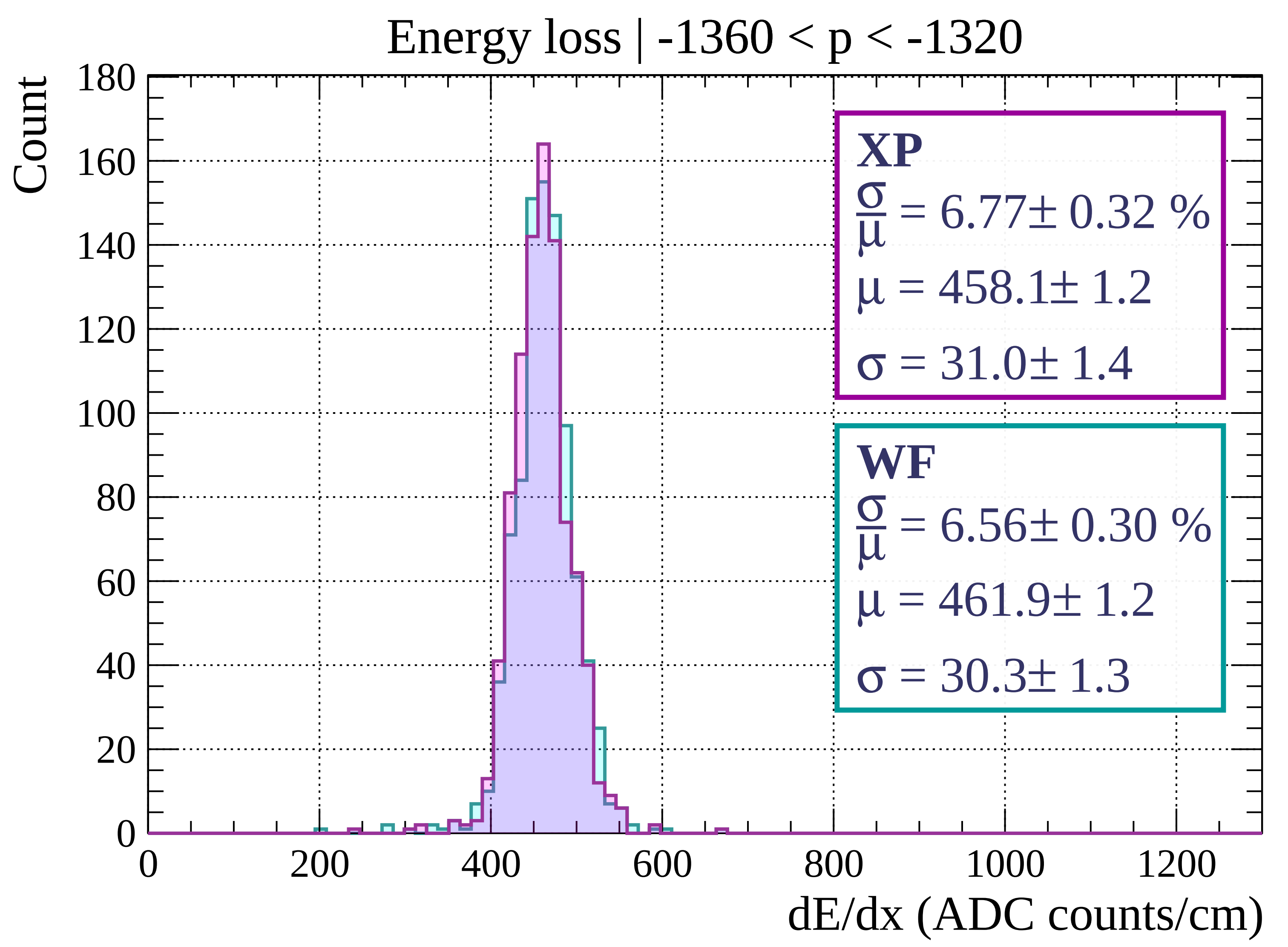

Energy loss | -1480 < p < -1440
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.84± 0.21 %
μ = 463.0± 1.0
σ = 27.0± 0.9
WF
σ/μ = 6.02± 0.28 %
μ = 466.1± 1.1
σ = 28.1± 1.3



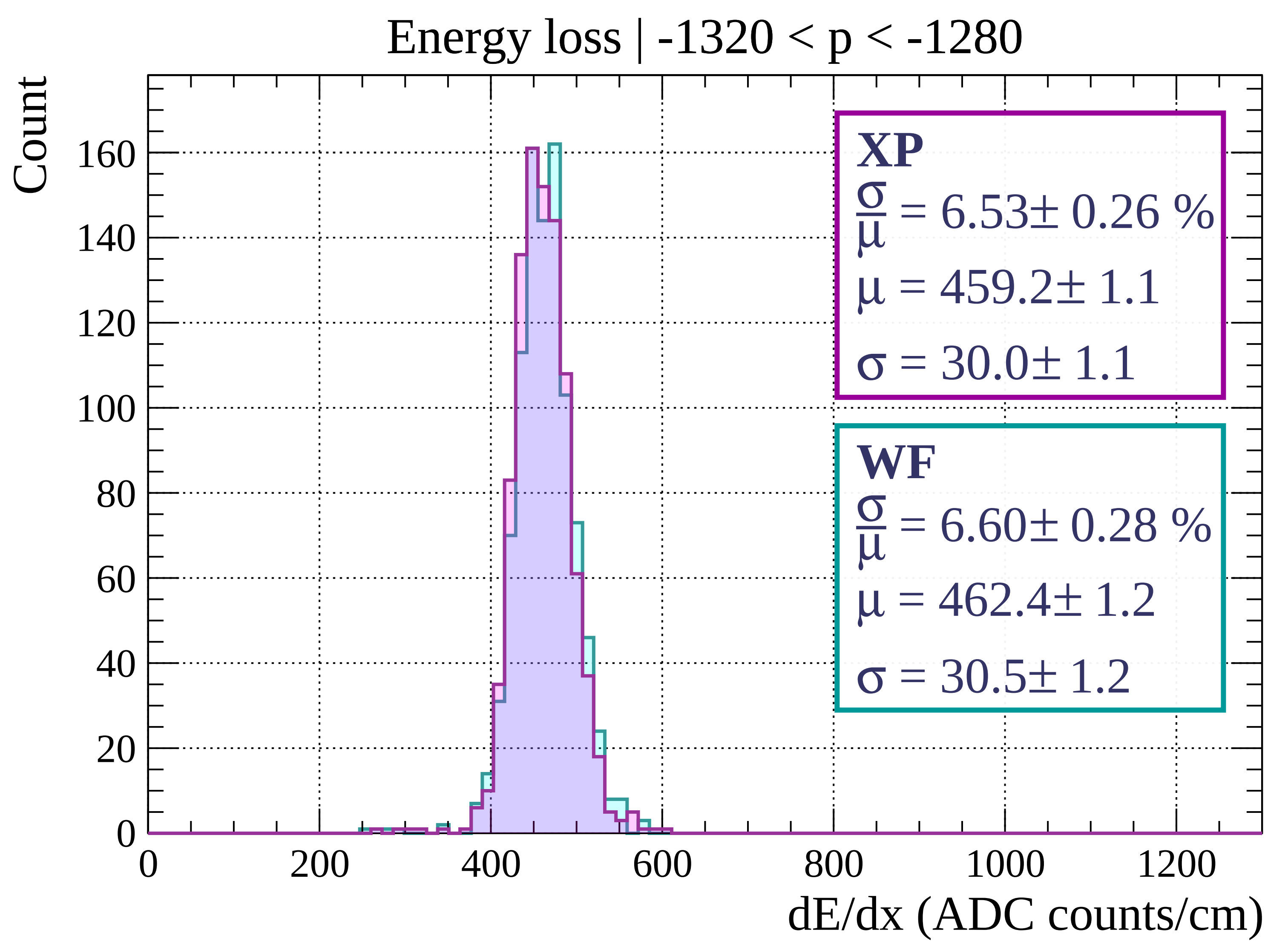
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.63± 0.25 %
μ = 463.8± 1.2
σ = 30.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.59± 0.25 %
μ = 466.6± 1.2
σ = 30.8± 1.1
0
20
40
60
80
100
120
140
160
200
400
600
800
1000
1200



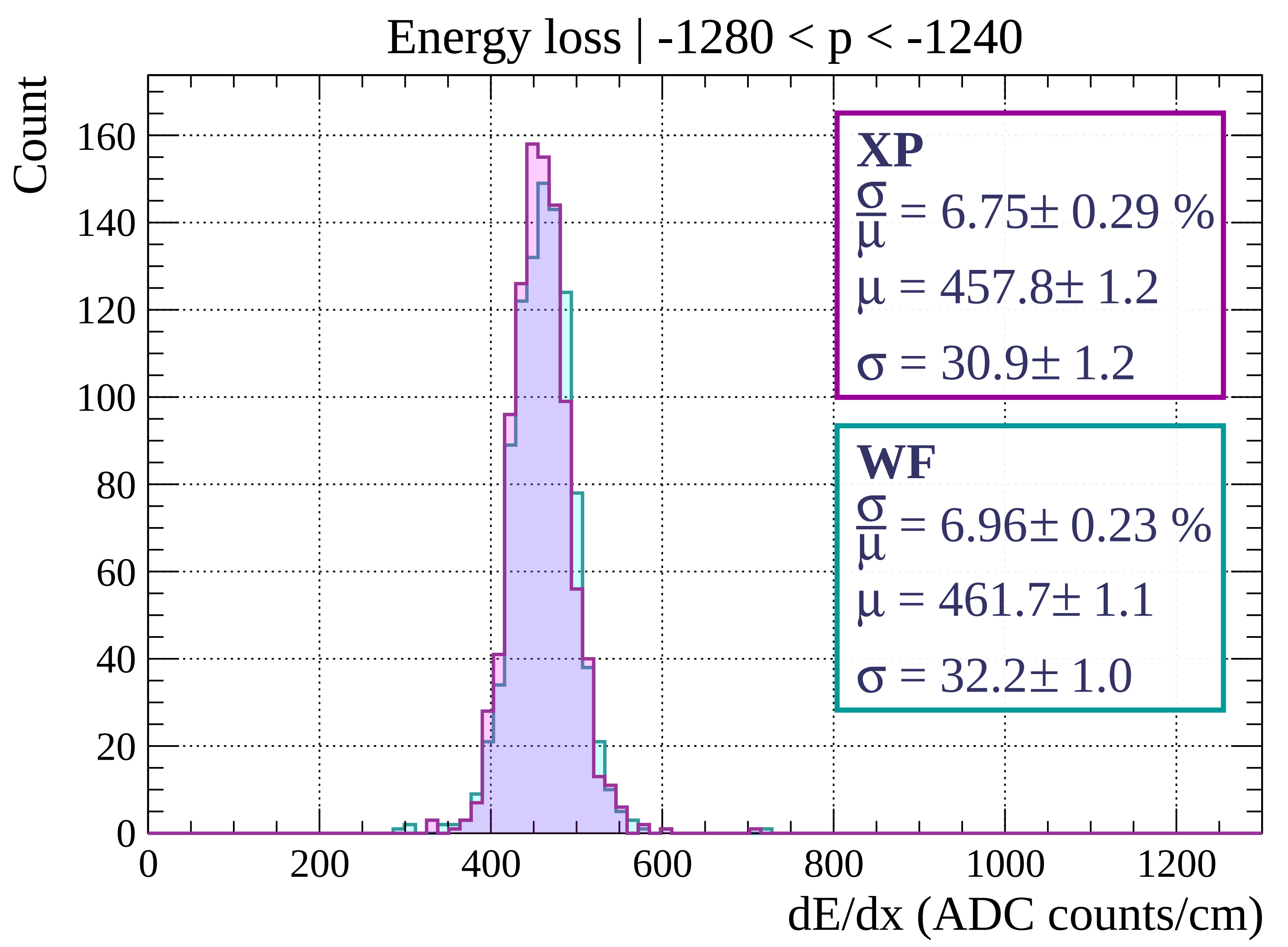
Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
XP
$\frac{\sigma}{\mu}$ = 6.61± 0.28 %
μ = 462.1± 1.2
σ = 30.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 6.81± 0.29 %
μ = 464.9± 1.2
σ = 31.7± 1.3



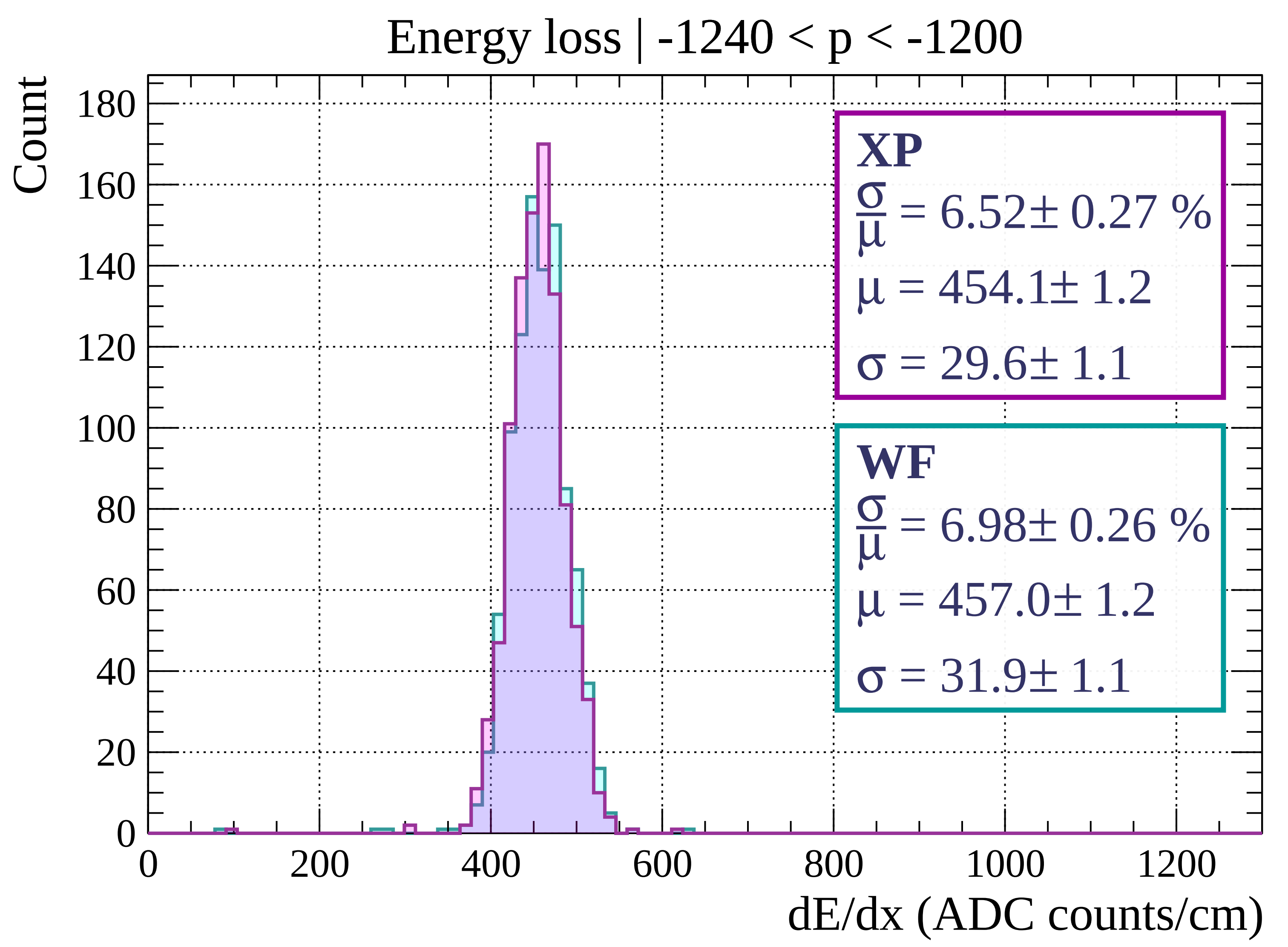
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
σ/μ = 6.77± 0.32 %
μ = 458.1± 1.2
σ = 31.0± 1.4
WF
σ/μ = 6.56± 0.30 %
μ = 461.9± 1.2
σ = 30.3± 1.3



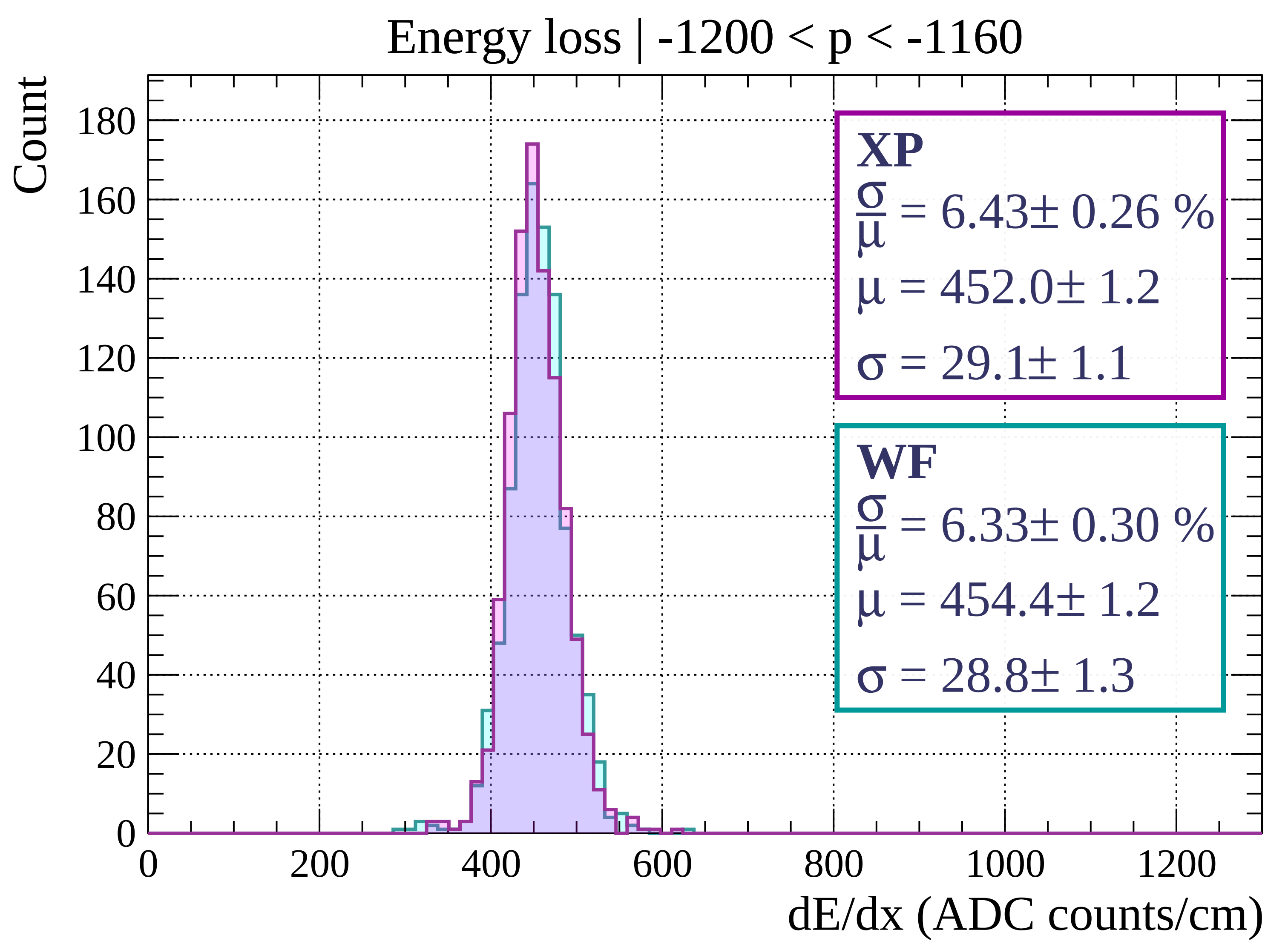
Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.53± 0.26 %
μ = 459.2± 1.1
σ = 30.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.60± 0.28 %
μ = 462.4± 1.2
σ = 30.5± 1.2

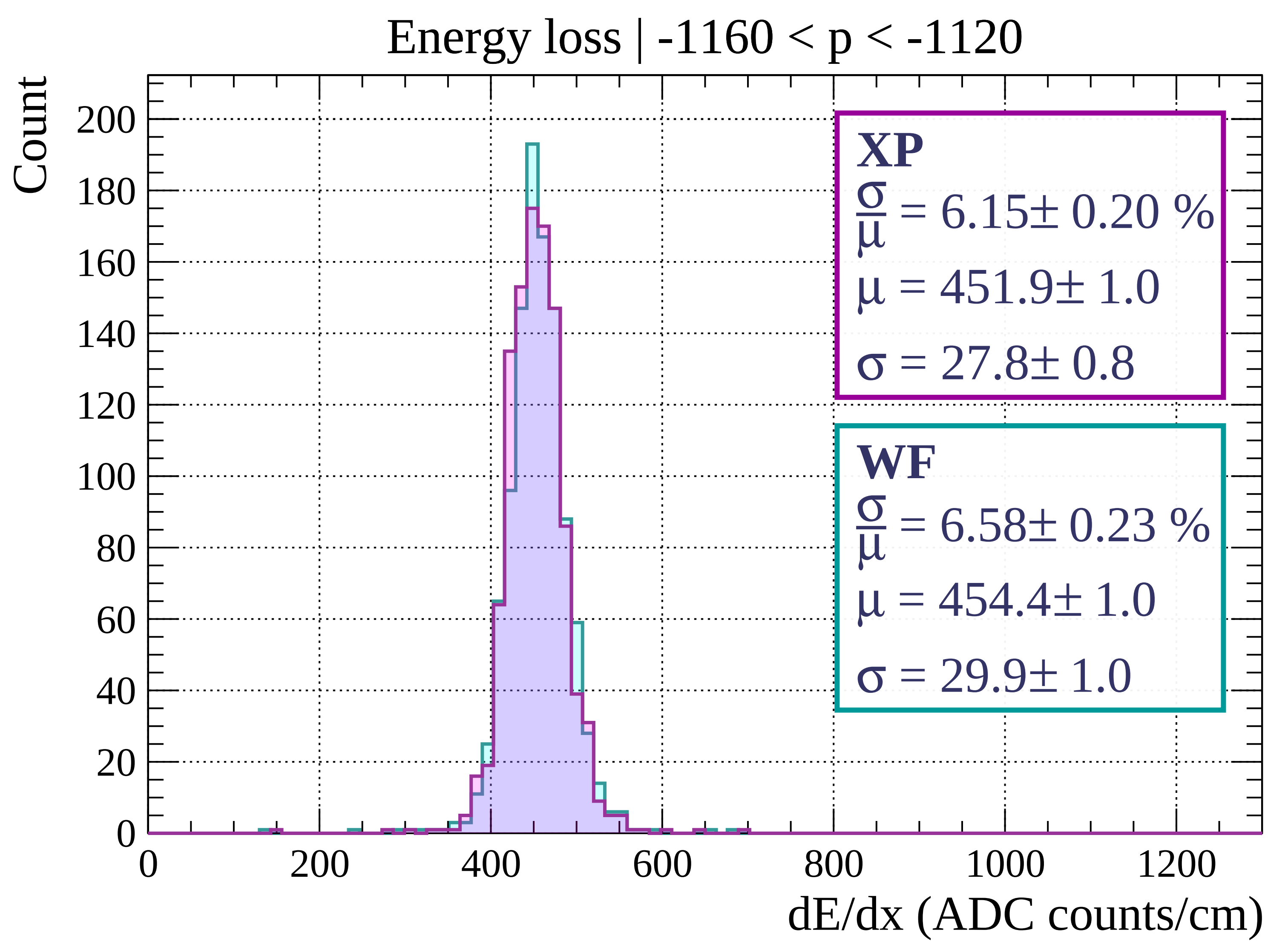


Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 6.75± 0.29 %
μ = 457.8± 1.2
σ = 30.9± 1.2
WF
σ/μ = 6.96± 0.23 %
μ = 461.7± 1.1
σ = 32.2± 1.0

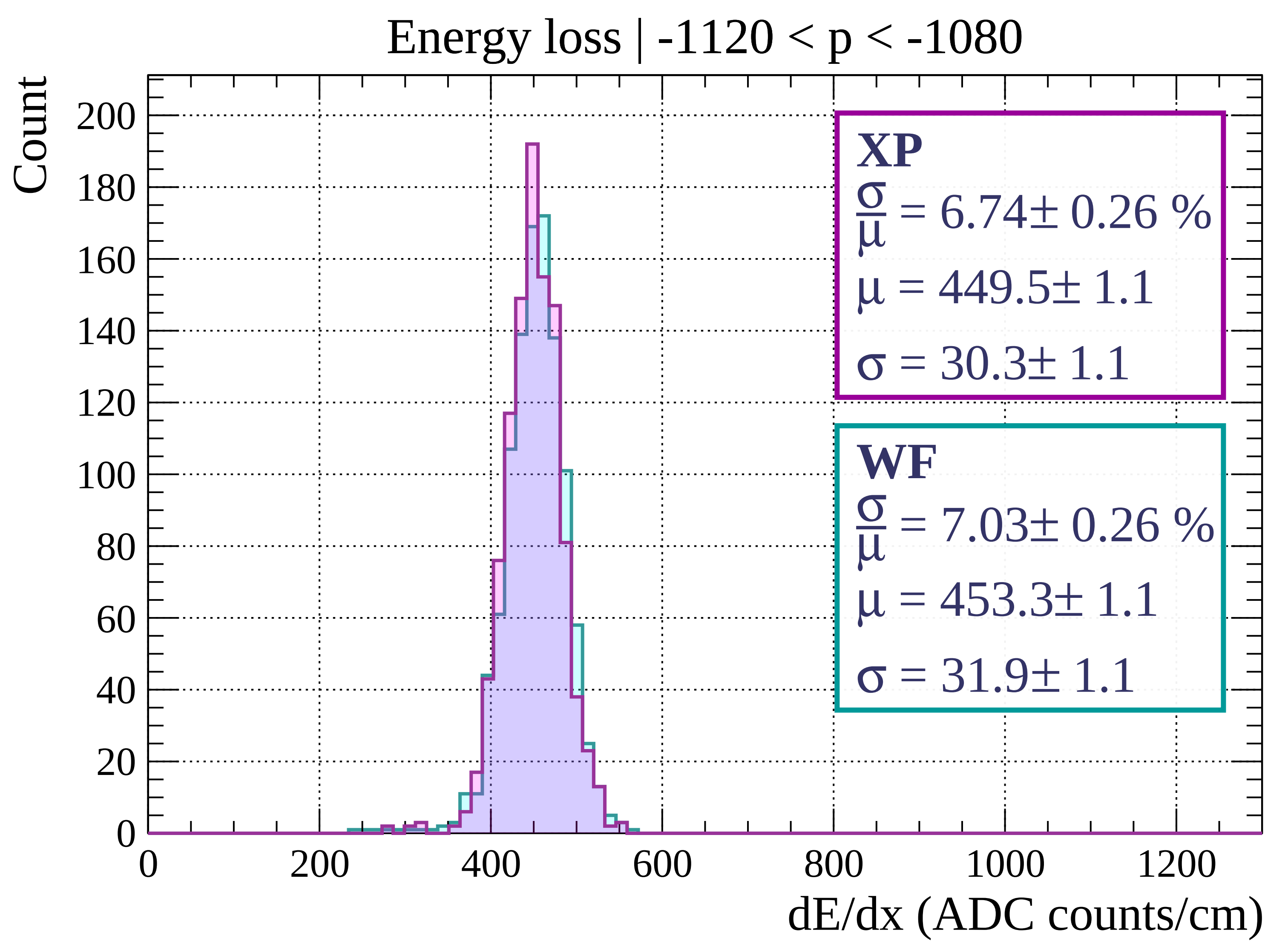


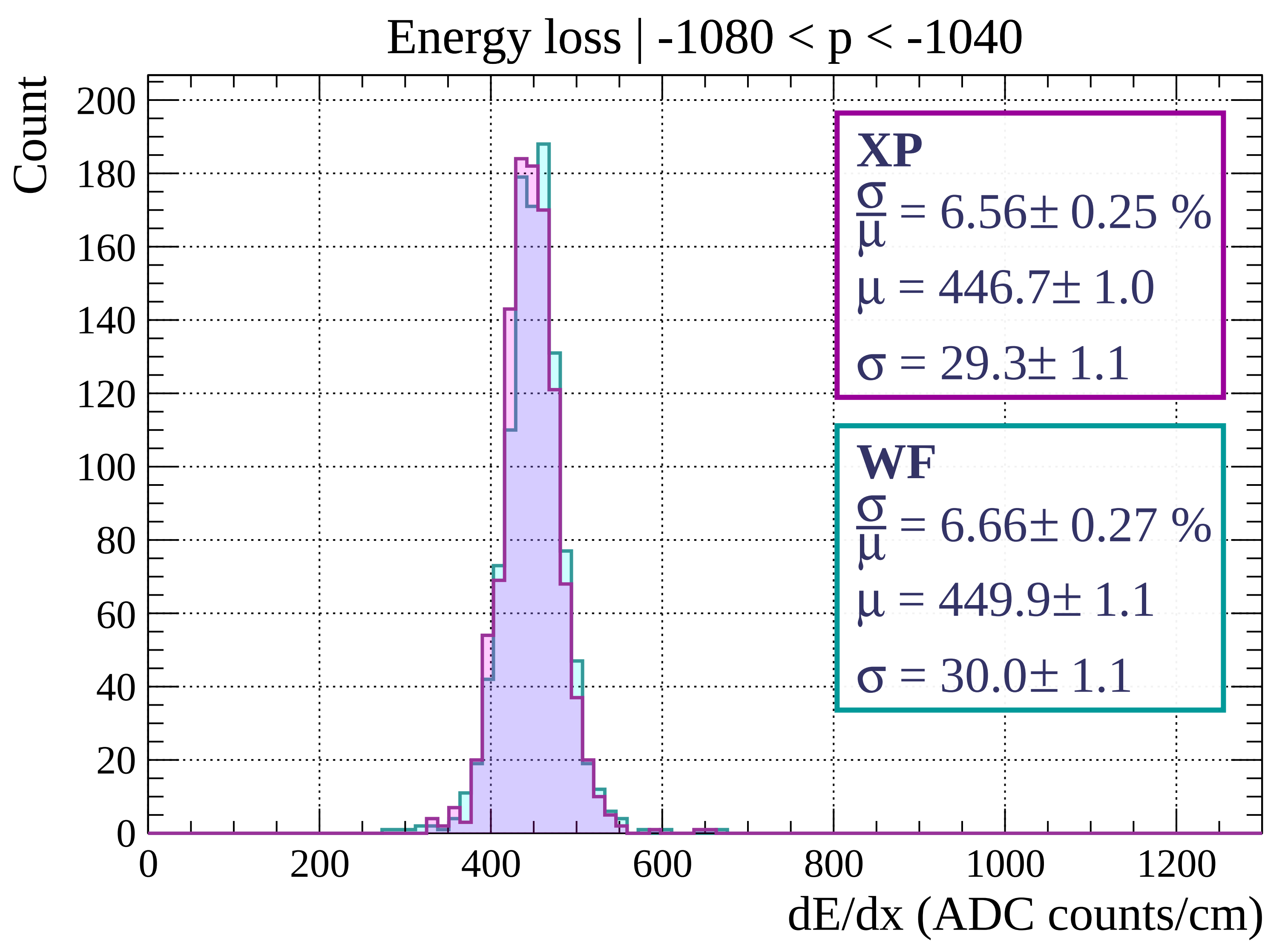
Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 6.52± 0.27 %
μ = 454.1± 1.2
σ = 29.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.98± 0.26 %
μ = 457.0± 1.2
σ = 31.9± 1.1


Energy loss | -1200 < p < -1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 6.43± 0.26 %
μ = 452.0± 1.2
σ = 29.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.33± 0.30 %
μ = 454.4± 1.2
σ = 28.8± 1.3

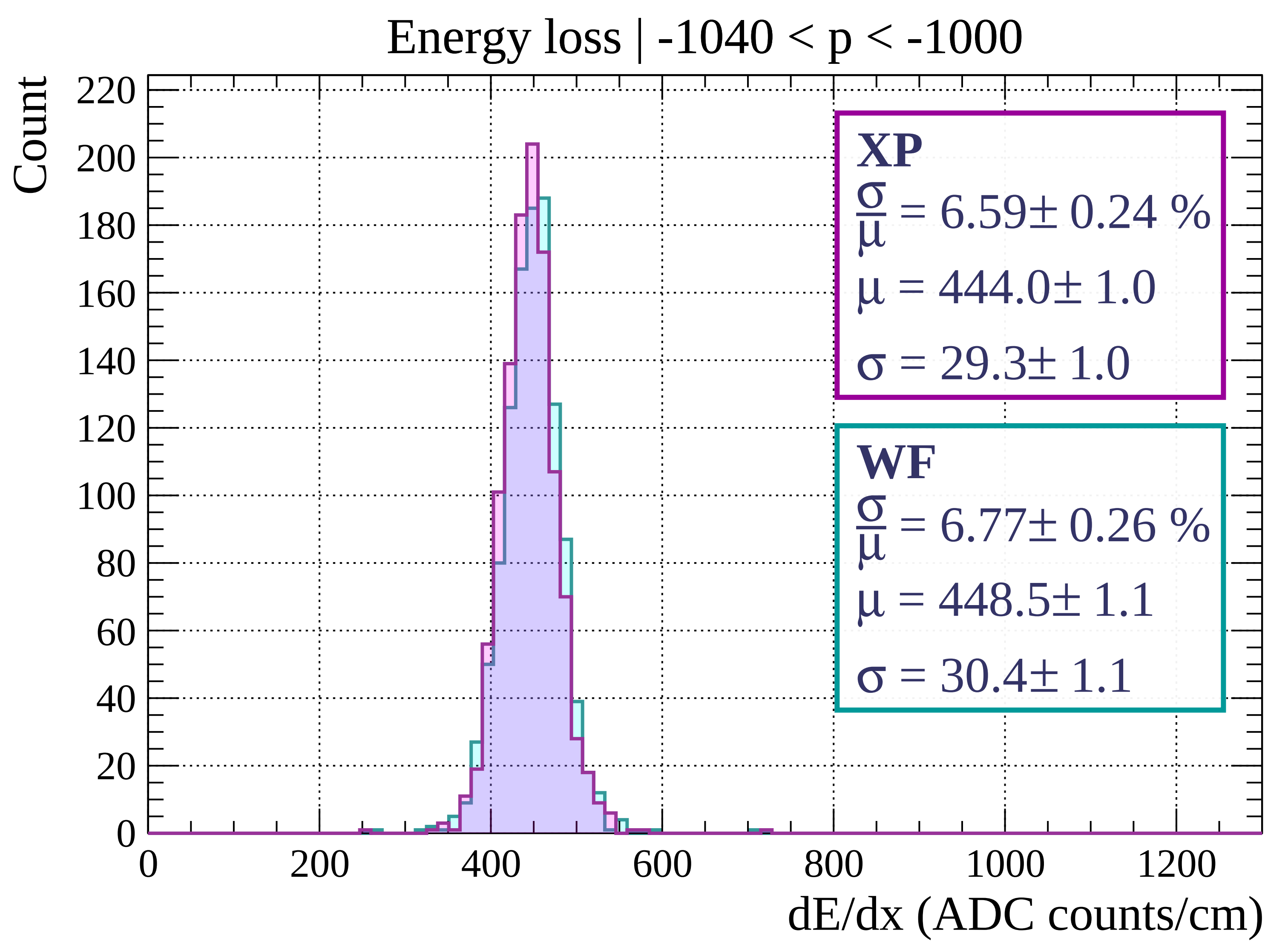


Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.15± 0.20 %
μ = 451.9± 1.0
σ = 27.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.58± 0.23 %
μ = 454.4± 1.0
σ = 29.9± 1.0

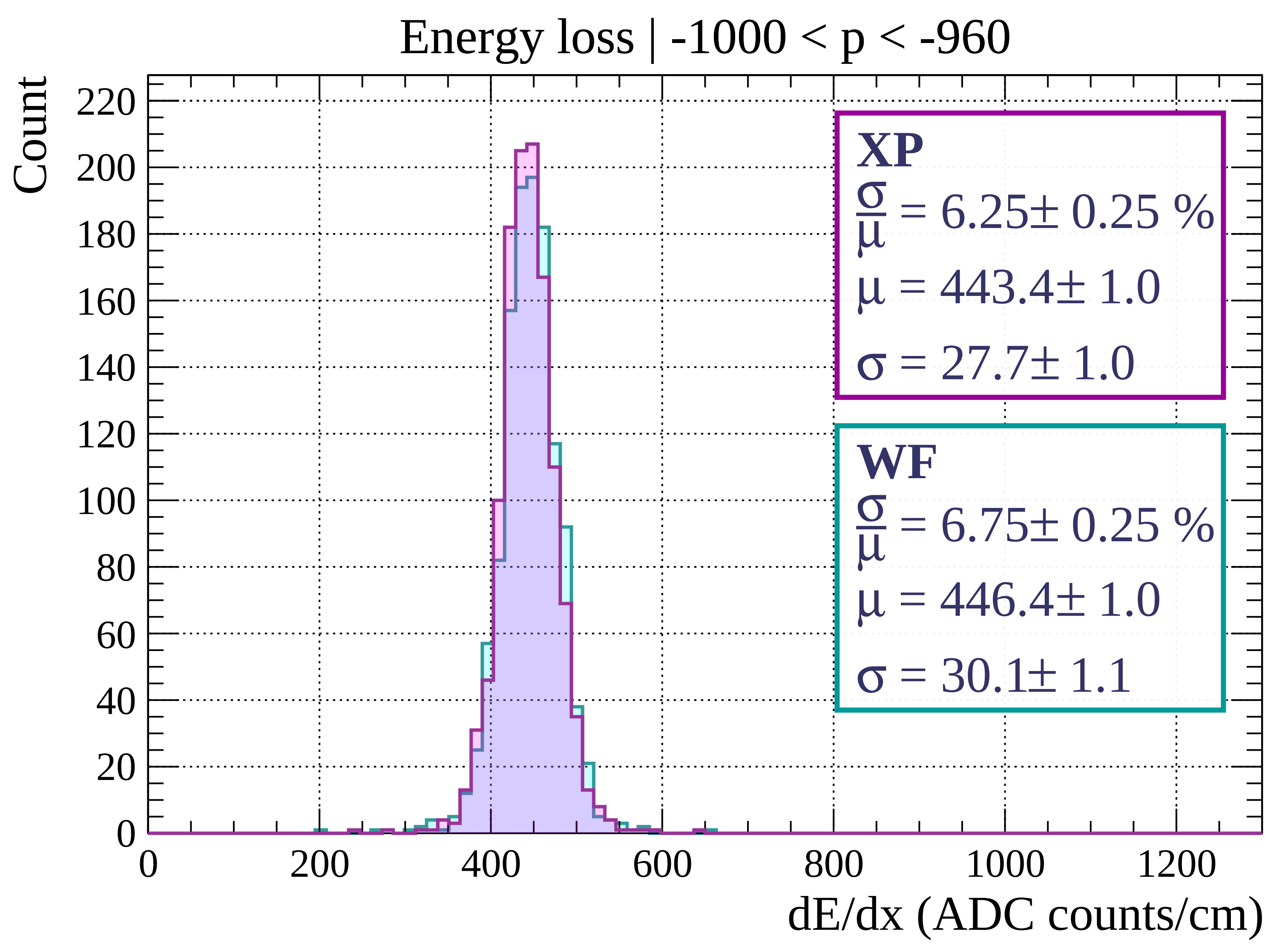

Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
σ/μ = 6.74± 0.26 %
μ = 449.5± 1.1
σ = 30.3± 1.1
WF
σ/μ = 7.03± 0.26 %
μ = 453.3± 1.1
σ = 31.9± 1.1

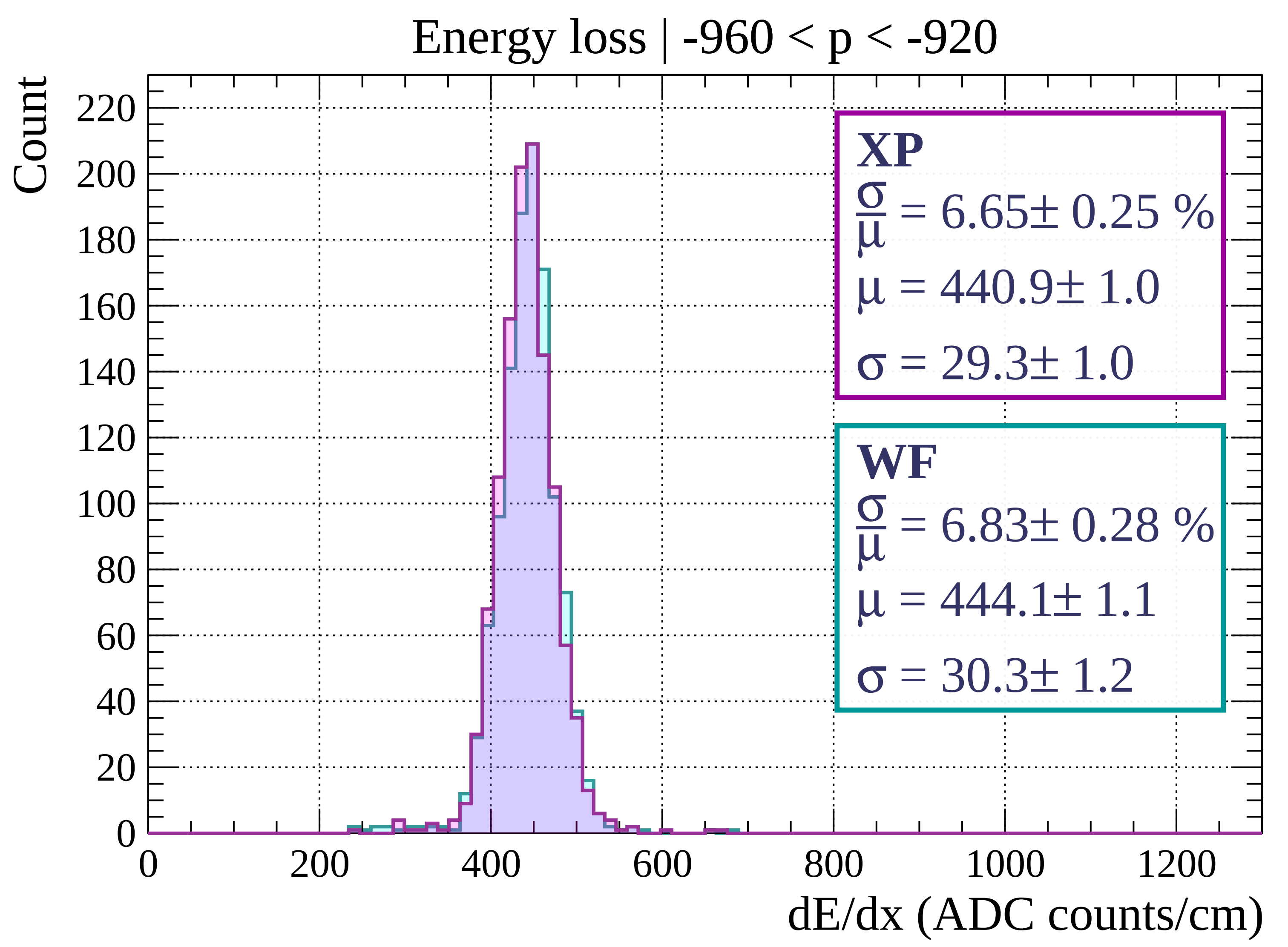


Energy loss | -1080 < p < -1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 6.56± 0.25 %
μ = 446.7± 1.0
σ = 29.3± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.66± 0.27 %
μ = 449.9± 1.1
σ = 30.0± 1.1

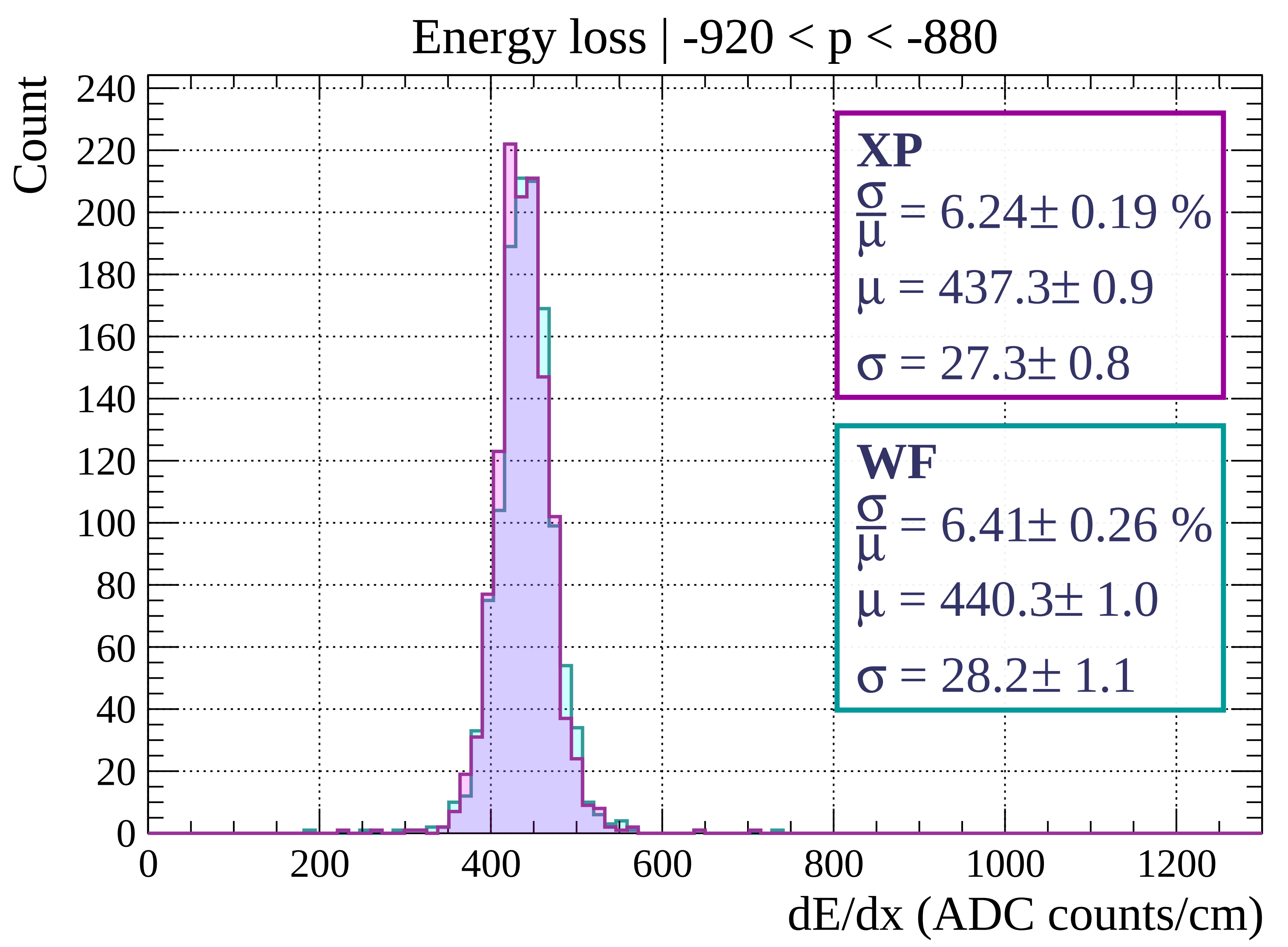


Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu}$ = 6.59± 0.24 %
μ = 444.0± 1.0
σ = 29.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.77± 0.26 %
μ = 448.5± 1.1
σ = 30.4± 1.1

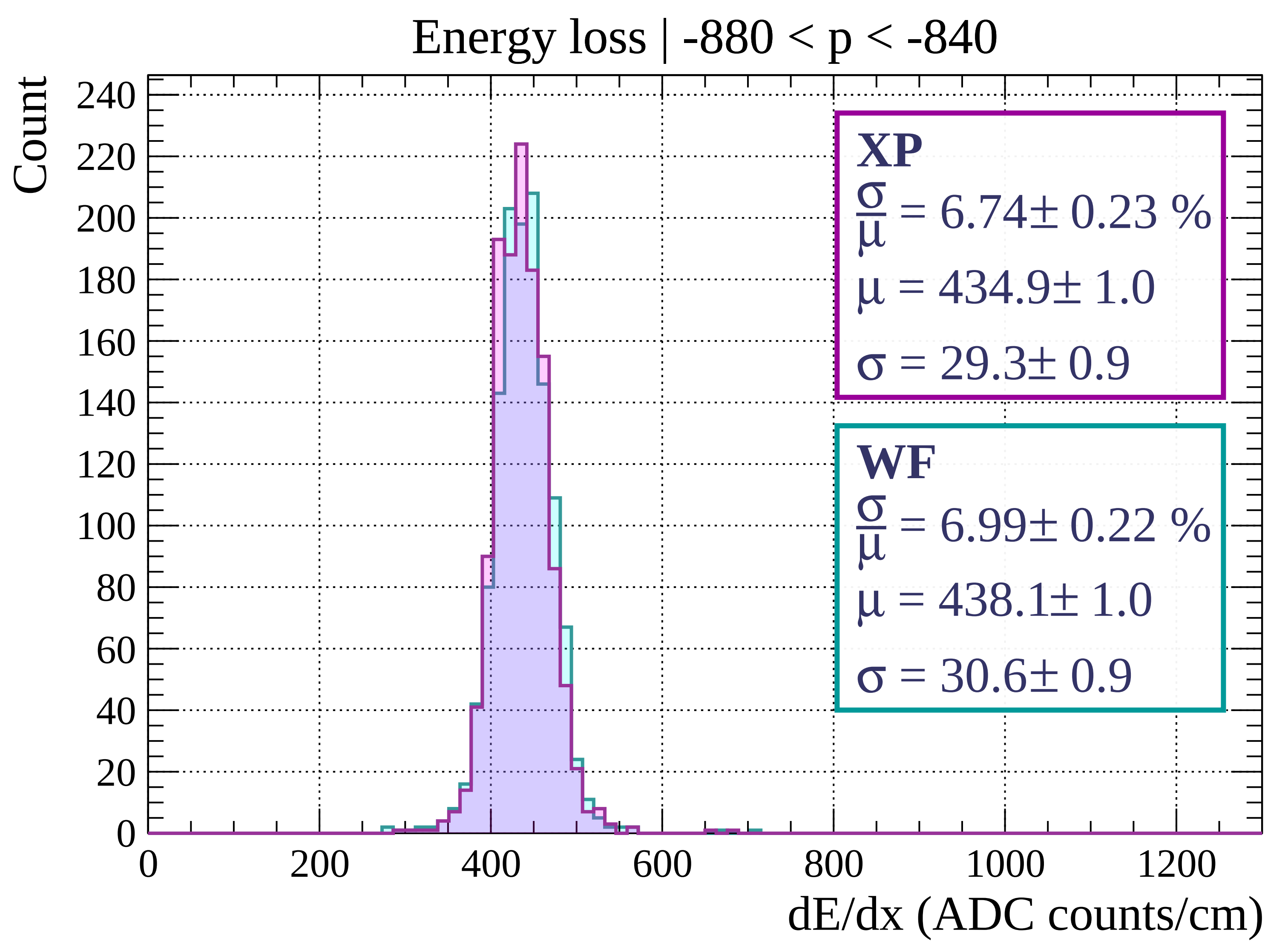

Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.25± 0.25 %
μ = 443.4± 1.0
σ = 27.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.75± 0.25 %
μ = 446.4± 1.0
σ = 30.1± 1.1

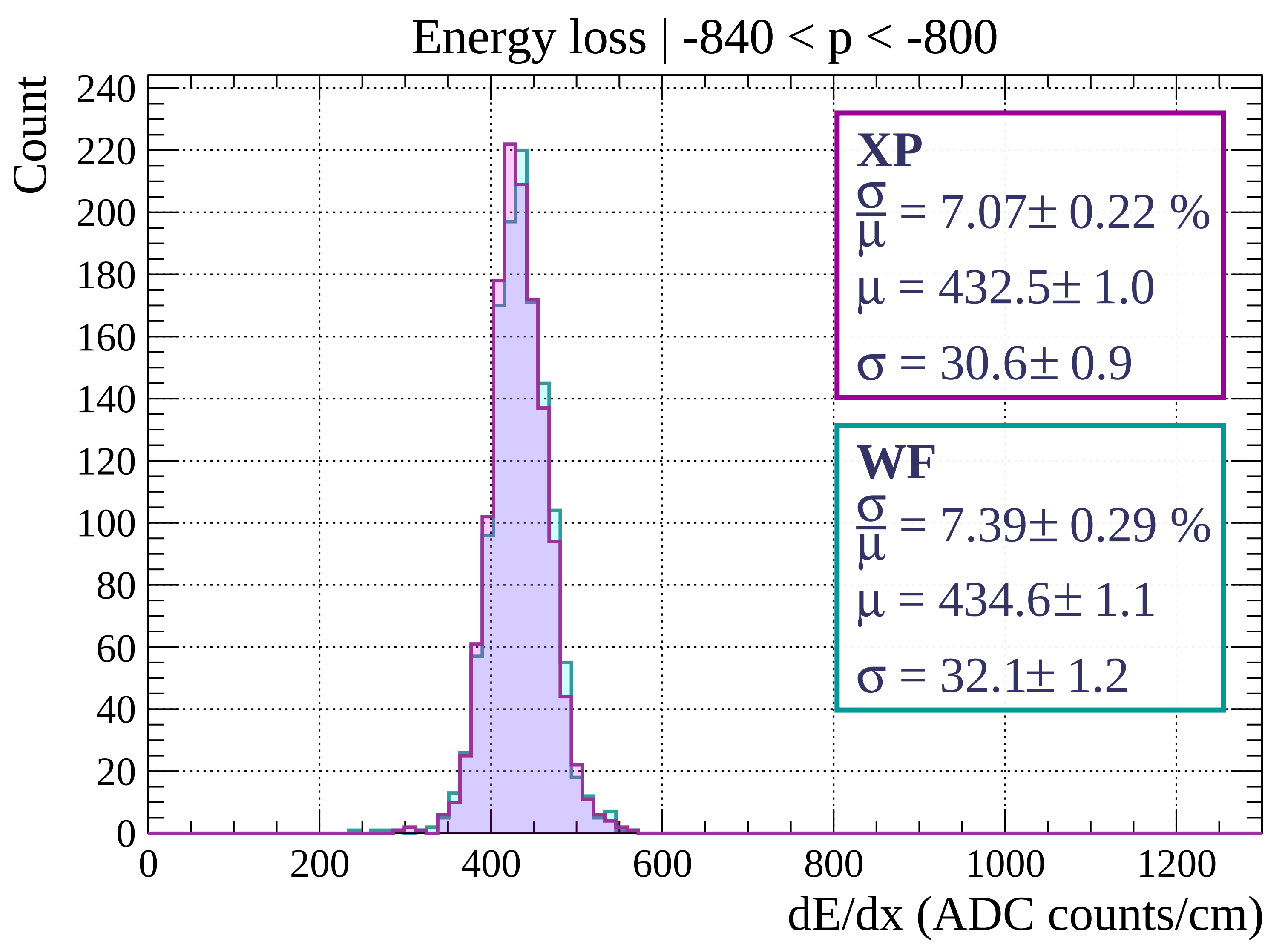


Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu}$ = 6.65± 0.25 %
μ = 440.9± 1.0
σ = 29.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.83± 0.28 %
μ = 444.1± 1.1
σ = 30.3± 1.2

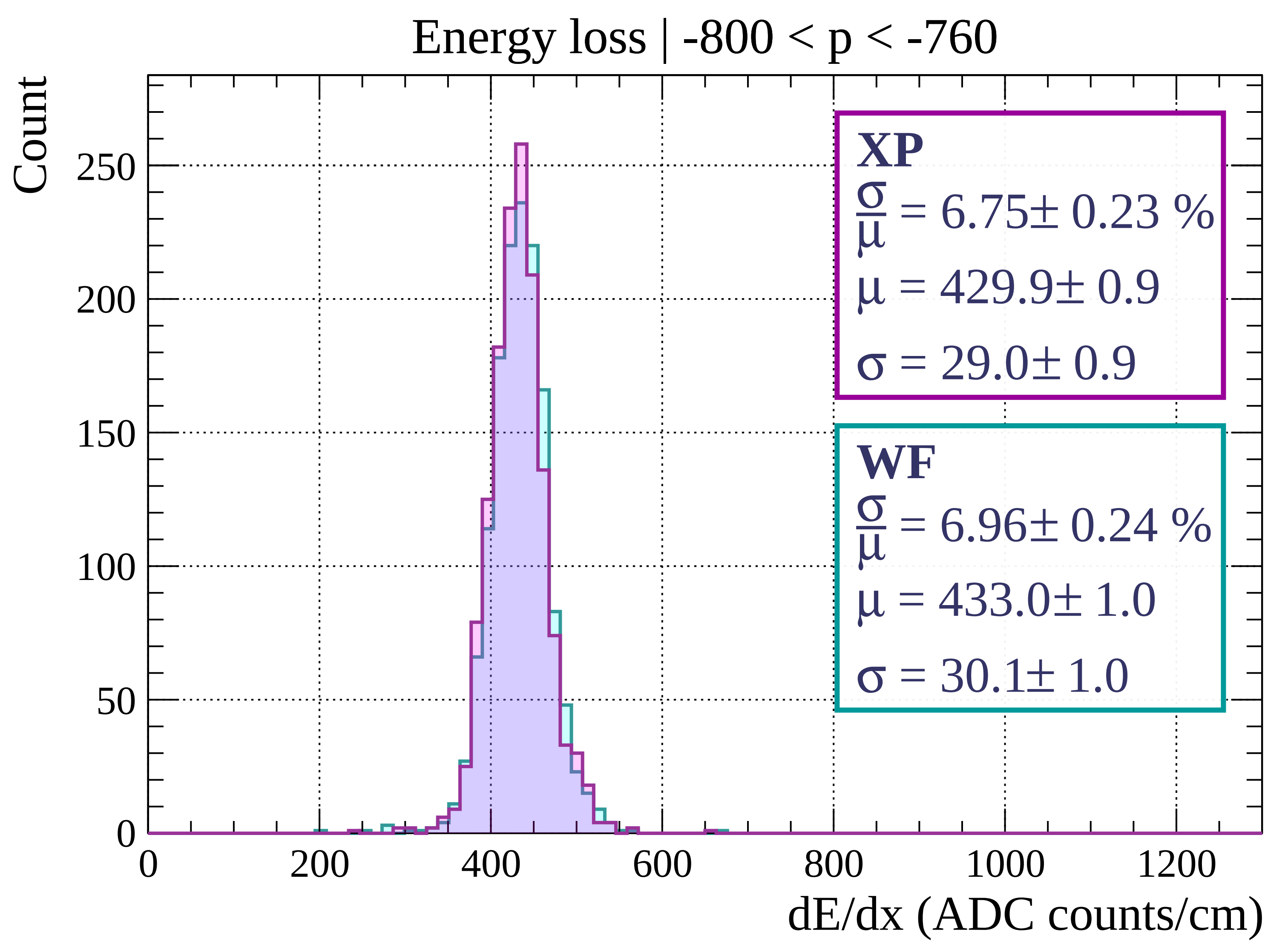


Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.24± 0.19 %
μ = 437.3± 0.9
σ = 27.3± 0.8
WF
σ/μ = 6.41± 0.26 %
μ = 440.3± 1.0
σ = 28.2± 1.1

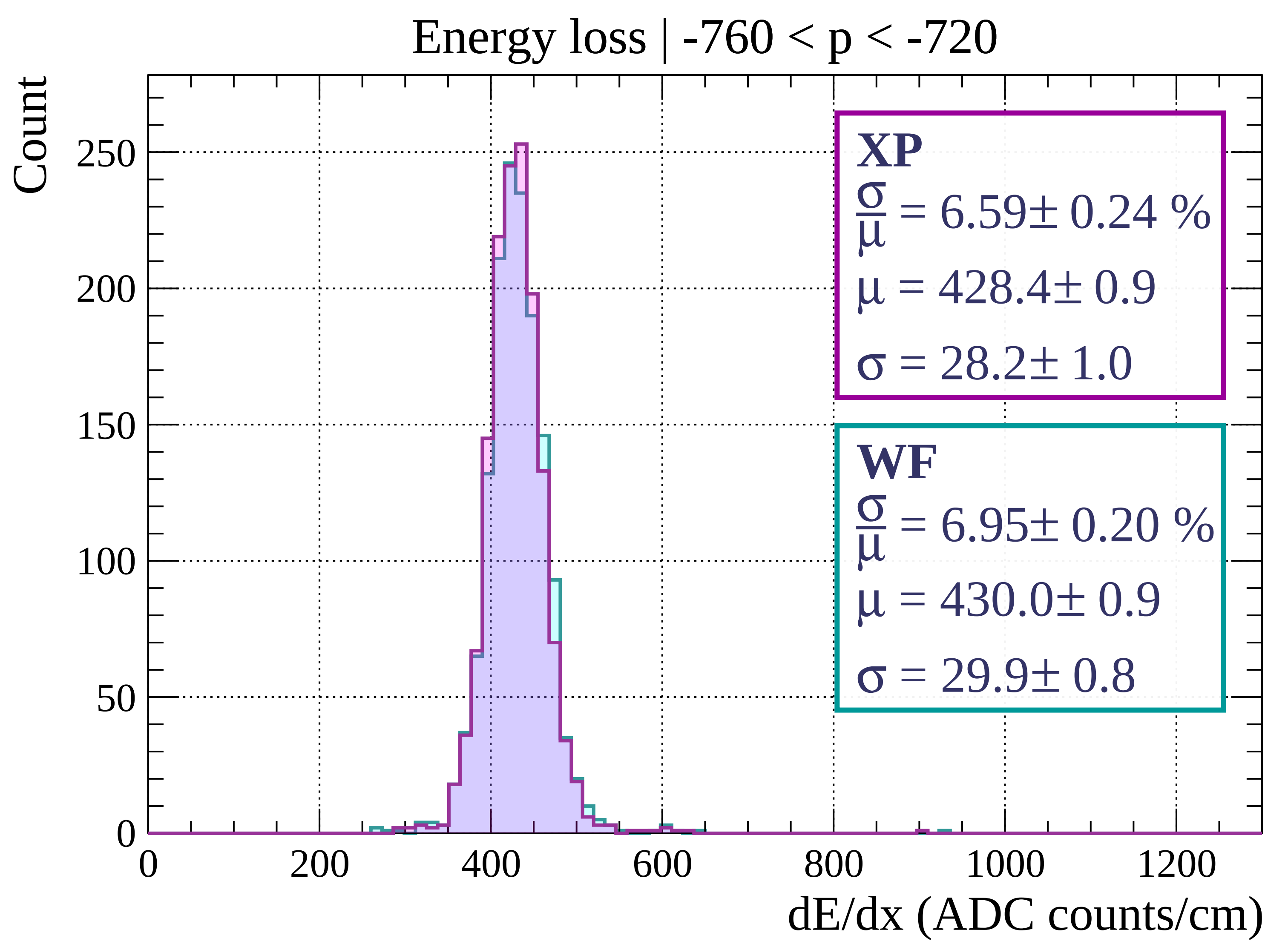

Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.74± 0.23 %
μ = 434.9± 1.0
σ = 29.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.99± 0.22 %
μ = 438.1± 1.0
σ = 30.6± 0.9



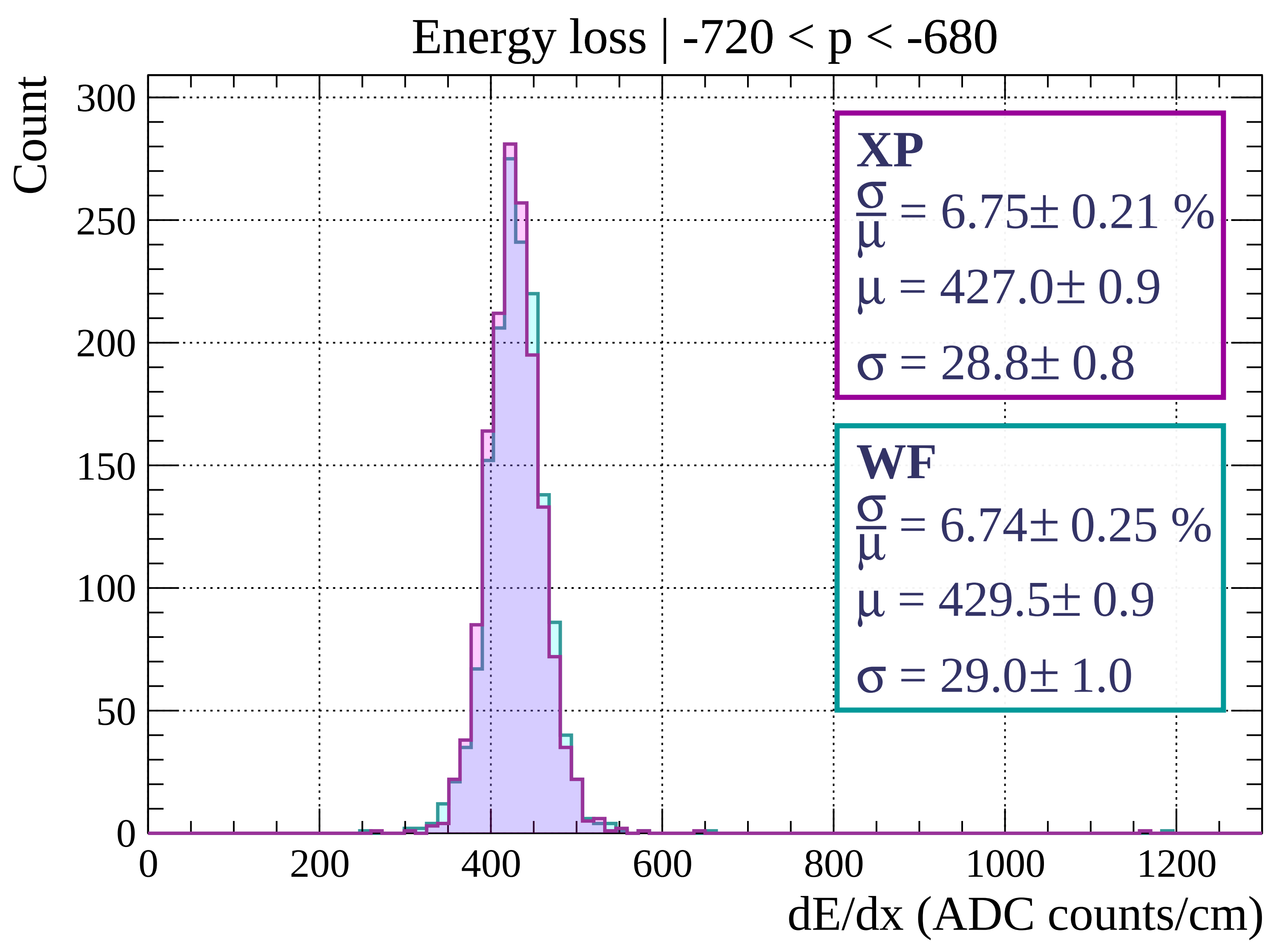
Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.07± 0.22 %
μ = 432.5± 1.0
σ = 30.6± 0.9
WF
σ/μ = 7.39± 0.29 %
μ = 434.6± 1.1
σ = 32.1± 1.2



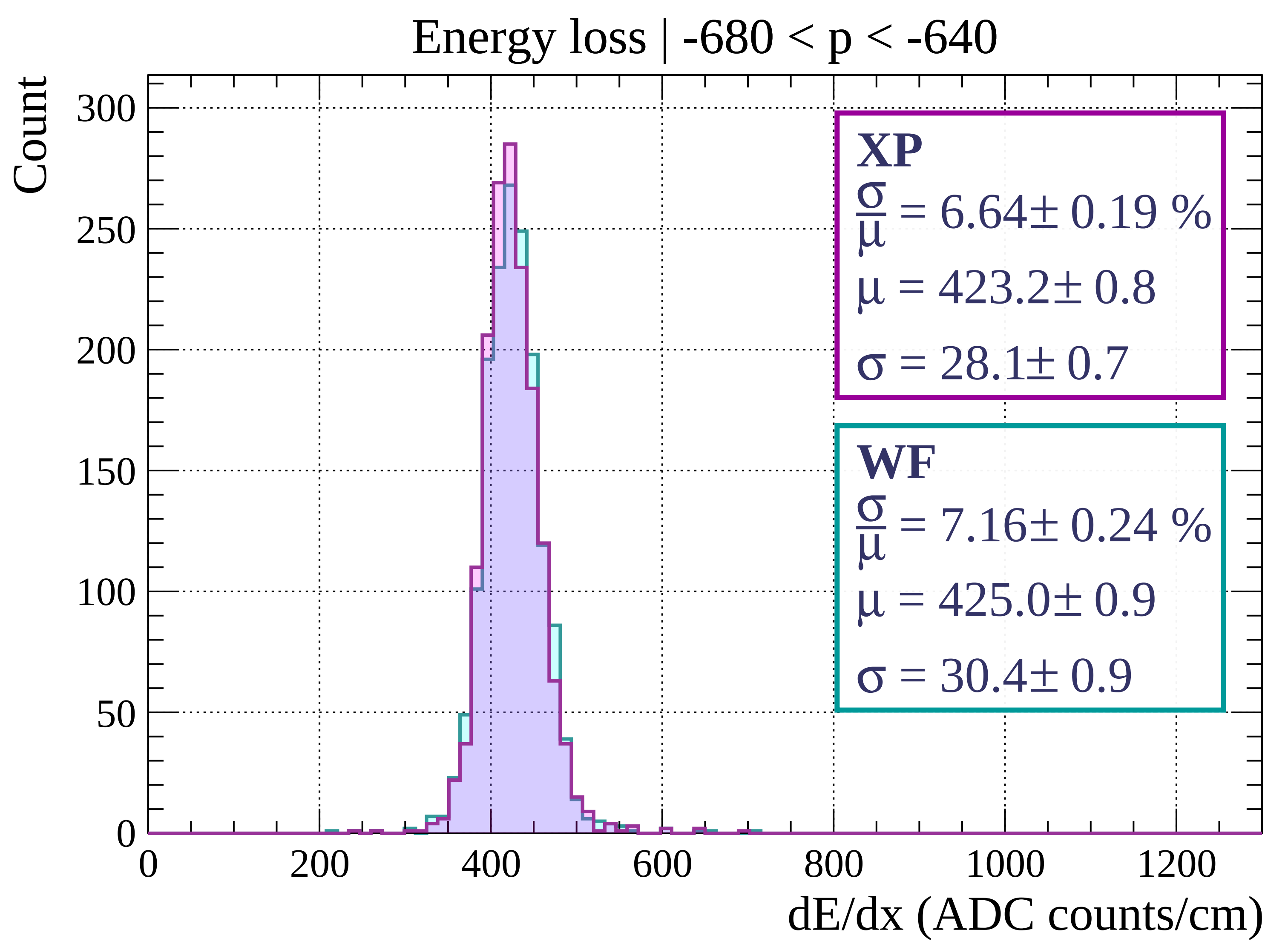
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.75± 0.23 %
μ = 429.9± 0.9
σ = 29.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.96± 0.24 %
μ = 433.0± 1.0
σ = 30.1± 1.0

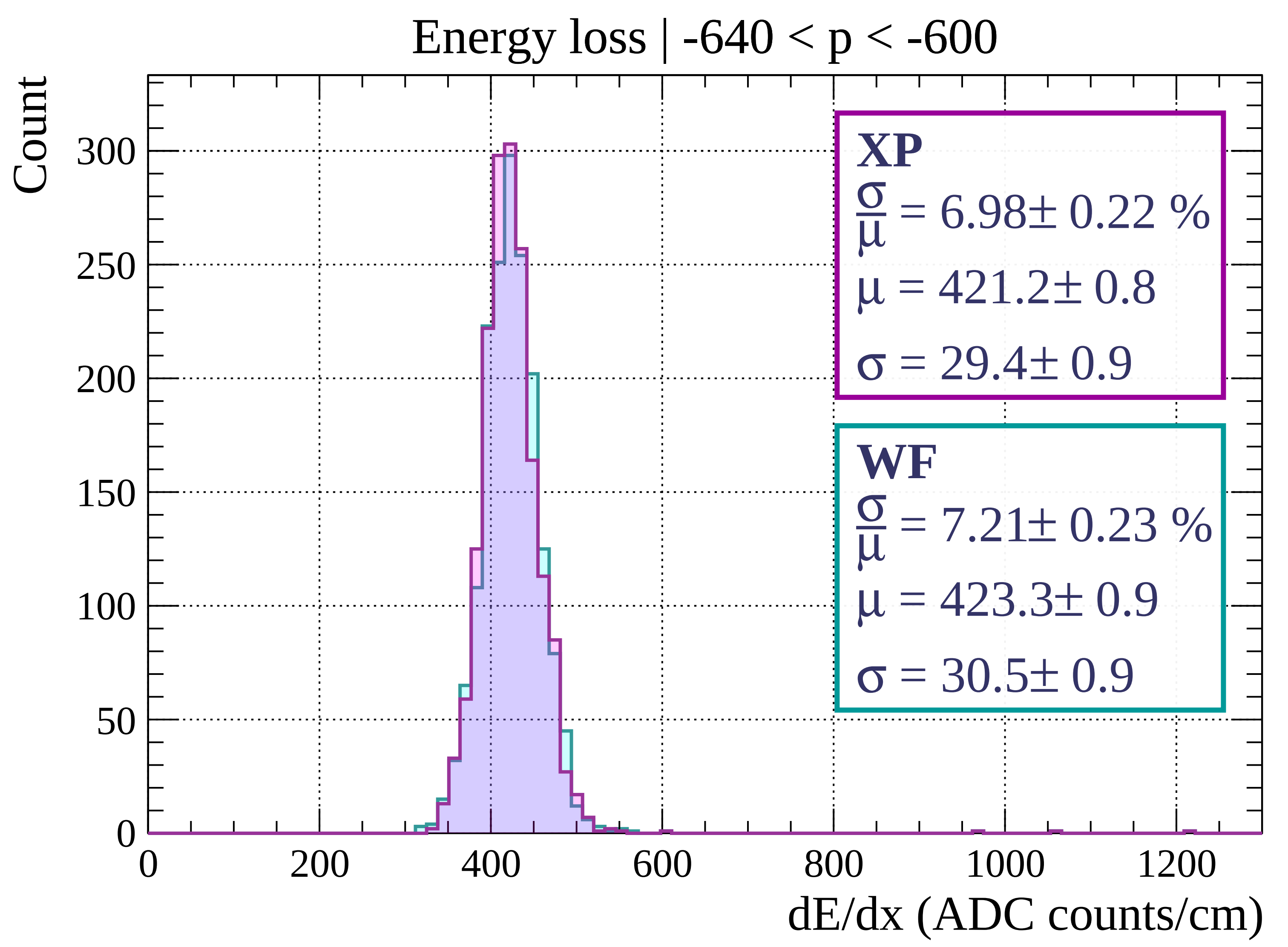


Energy loss | -760 < p < -720
Count
XP
$\frac{\sigma}{\mu}$ = 6.59± 0.24 %
μ = 428.4± 0.9
σ = 28.2± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.95± 0.20 %
μ = 430.0± 0.9
σ = 29.9± 0.8
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

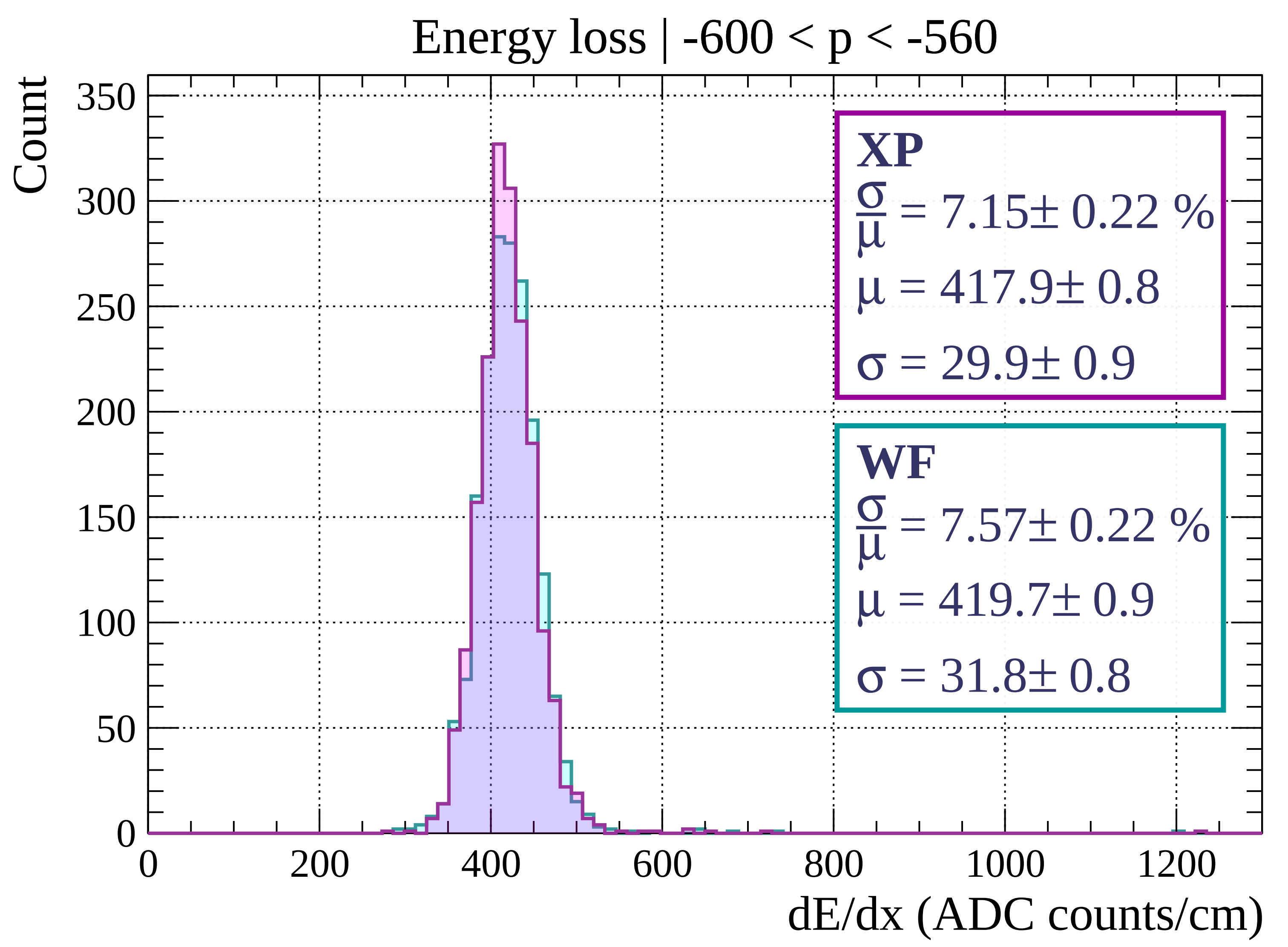


Energy loss | -720 < p < -680
Count
XP
$\frac{\sigma}{\mu}$ = 6.75± 0.21 %
μ = 427.0± 0.9
σ = 28.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.74± 0.25 %
μ = 429.5± 0.9
σ = 29.0± 1.0
dE/dx (ADC counts/cm)

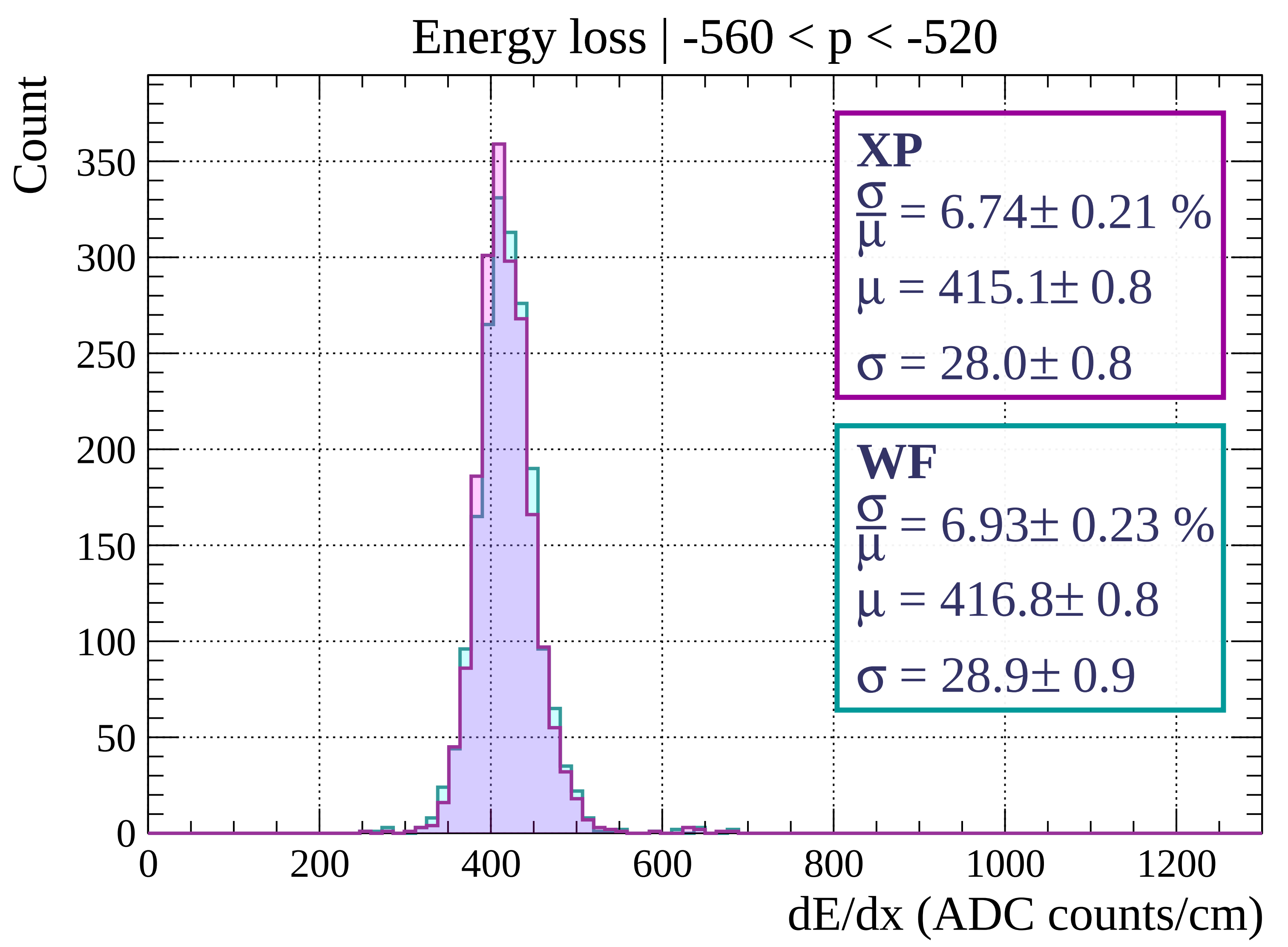

Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.64± 0.19 %
μ = 423.2± 0.8
σ = 28.1± 0.7
WF
σ/μ = 7.16± 0.24 %
μ = 425.0± 0.9
σ = 30.4± 0.9

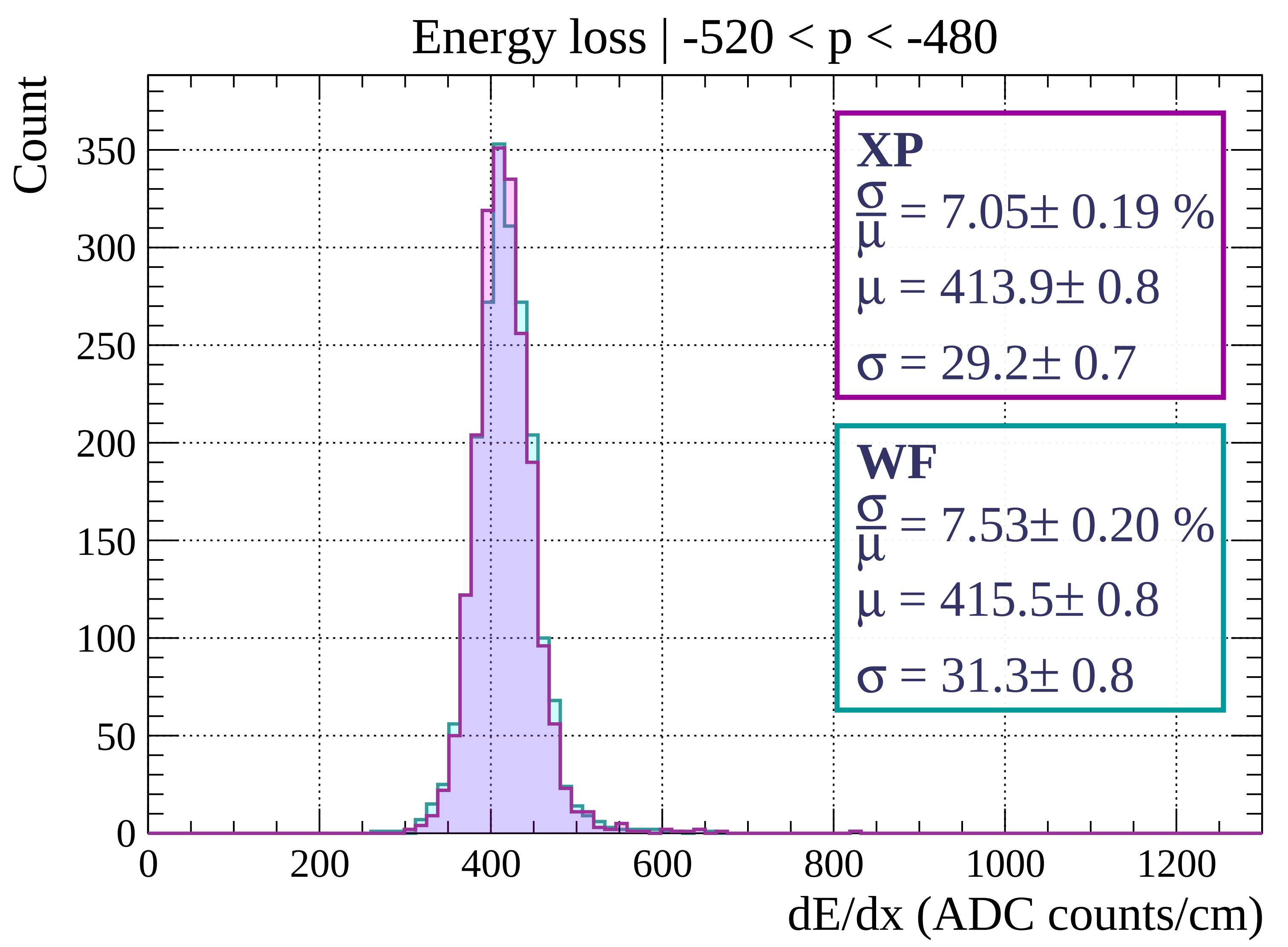


Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 6.98± 0.22 %
μ = 421.2± 0.8
σ = 29.4± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.21± 0.23 %
μ = 423.3± 0.9
σ = 30.5± 0.9

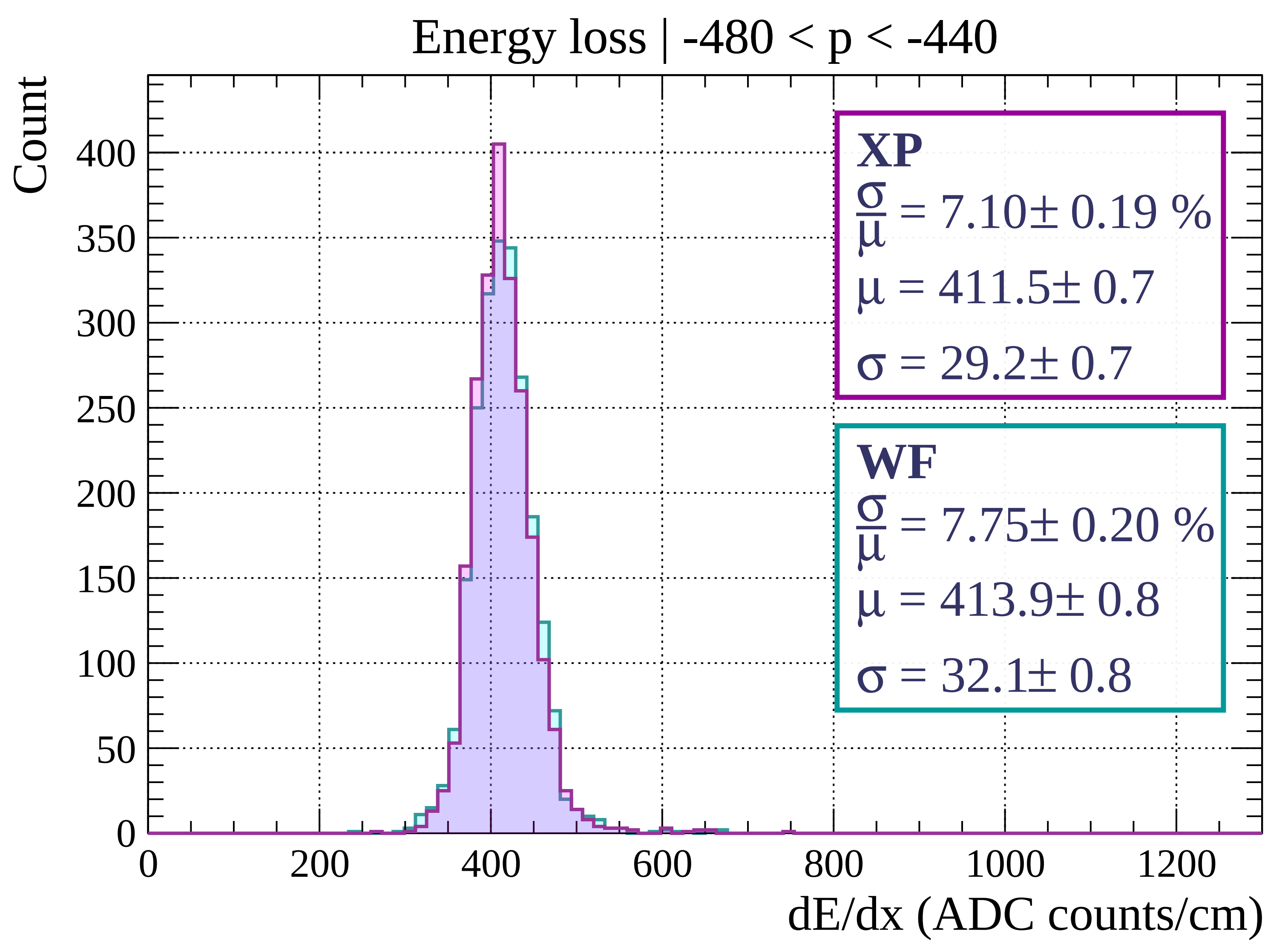


Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 7.15± 0.22 %
μ = 417.9± 0.8
σ = 29.9± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.57± 0.22 %
μ = 419.7± 0.9
σ = 31.8± 0.8

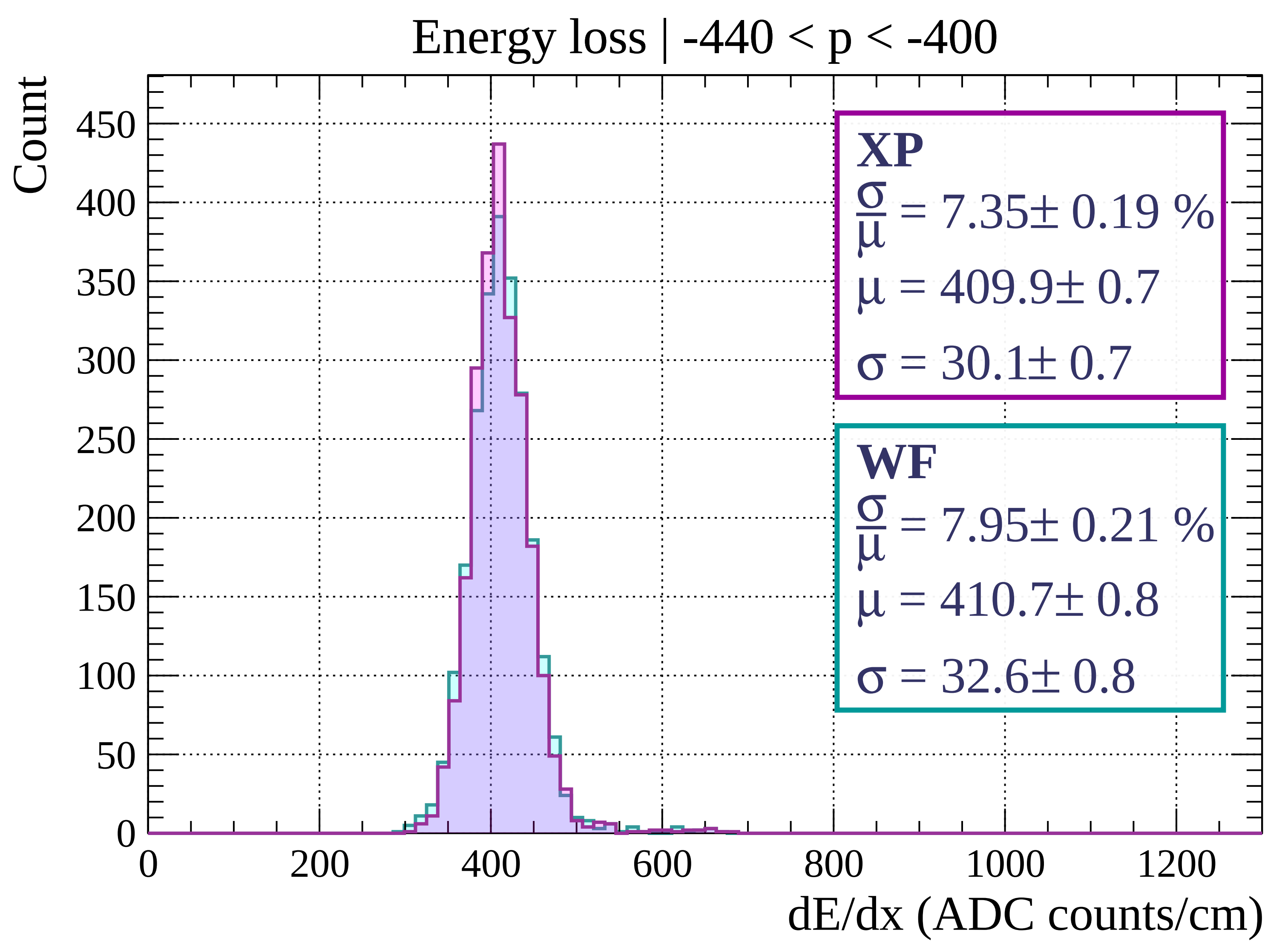

Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 6.74± 0.21 %
μ = 415.1± 0.8
σ = 28.0± 0.8
WF
σ/μ = 6.93± 0.23 %
μ = 416.8± 0.8
σ = 28.9± 0.9



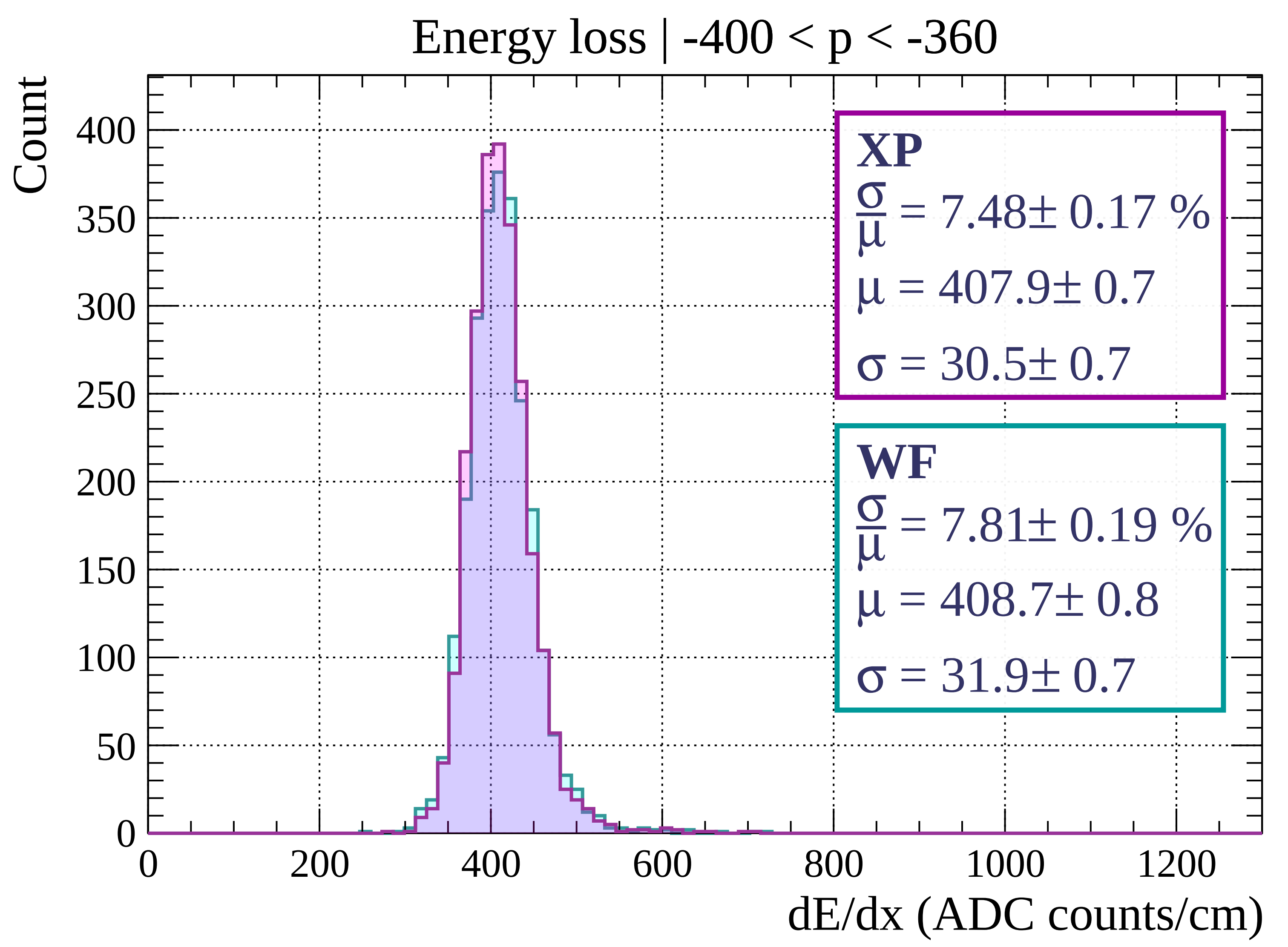
Energy loss | -520 < p < -480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.05± 0.19 %
μ = 413.9± 0.8
σ = 29.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.53± 0.20 %
μ = 415.5± 0.8
σ = 31.3± 0.8



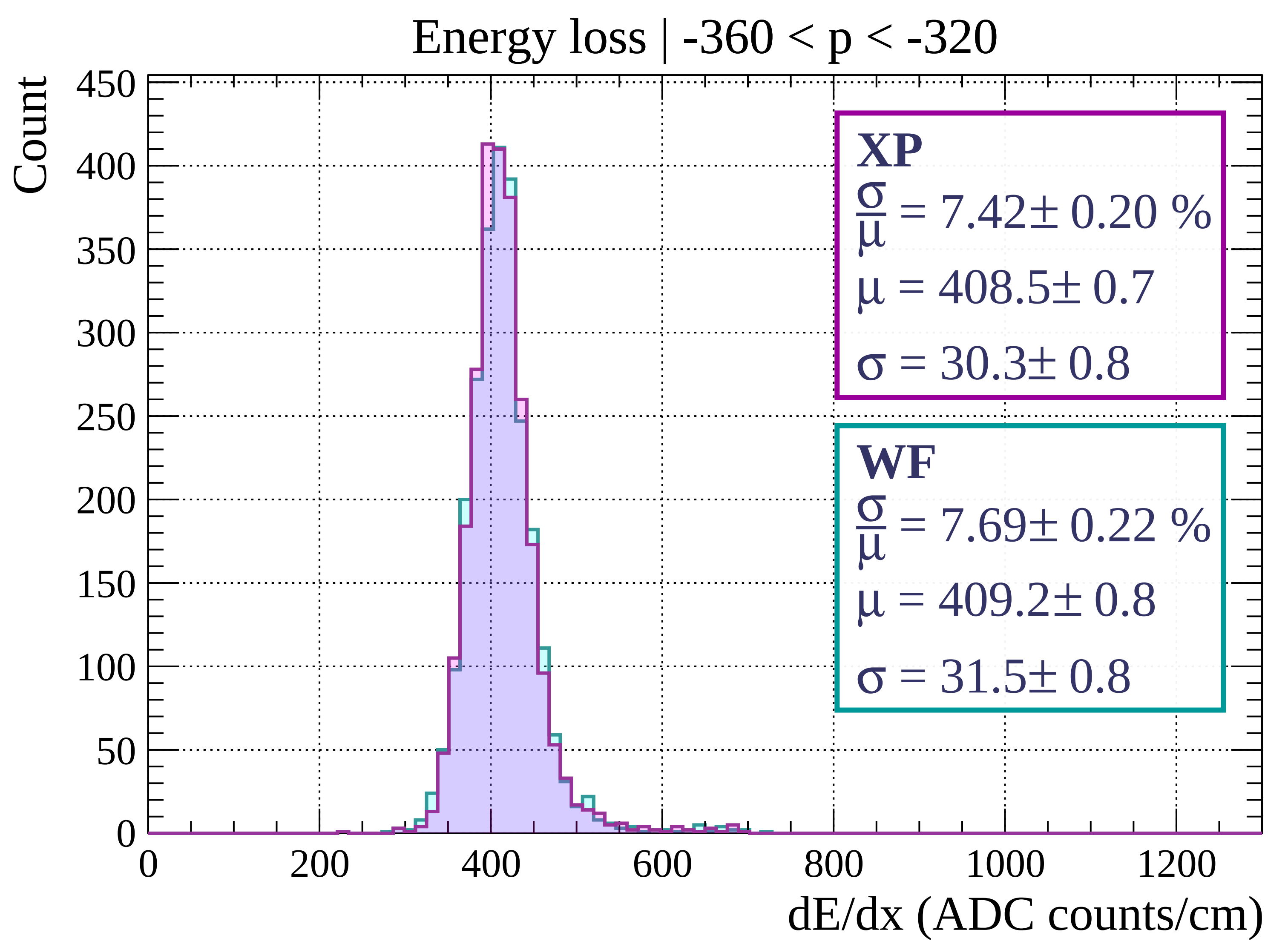
Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.10± 0.19 %
μ = 411.5± 0.7
σ = 29.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.75± 0.20 %
μ = 413.9± 0.8
σ = 32.1± 0.8



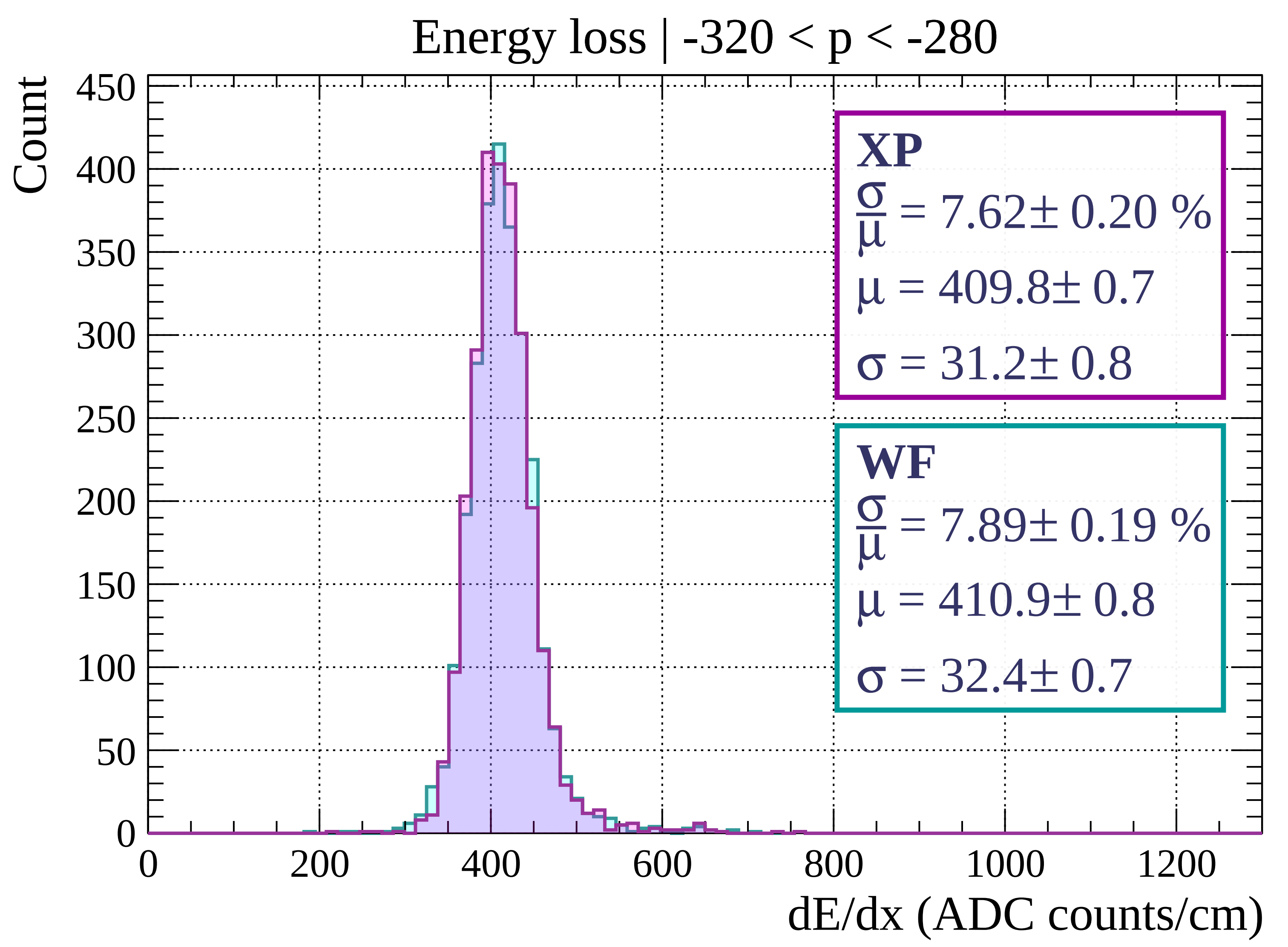
Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.35± 0.19 %
μ = 409.9± 0.7
σ = 30.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.95± 0.21 %
μ = 410.7± 0.8
σ = 32.6± 0.8



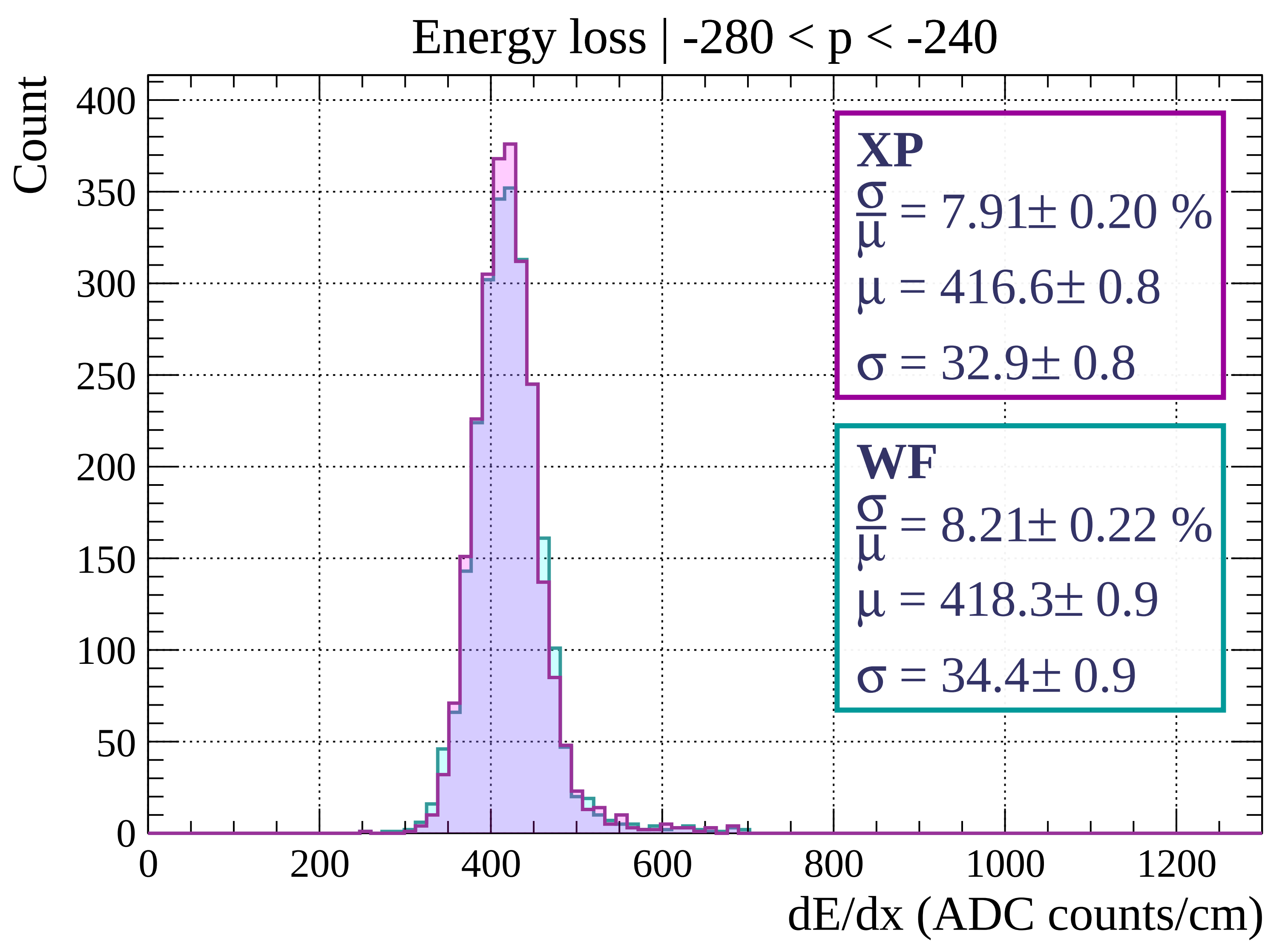
Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 7.48 \pm 0.17$ %
$\mu = 407.9 \pm 0.7$
$\sigma = 30.5 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 7.81 \pm 0.19$ %
$\mu = 408.7 \pm 0.8$
$\sigma = 31.9 \pm 0.7$

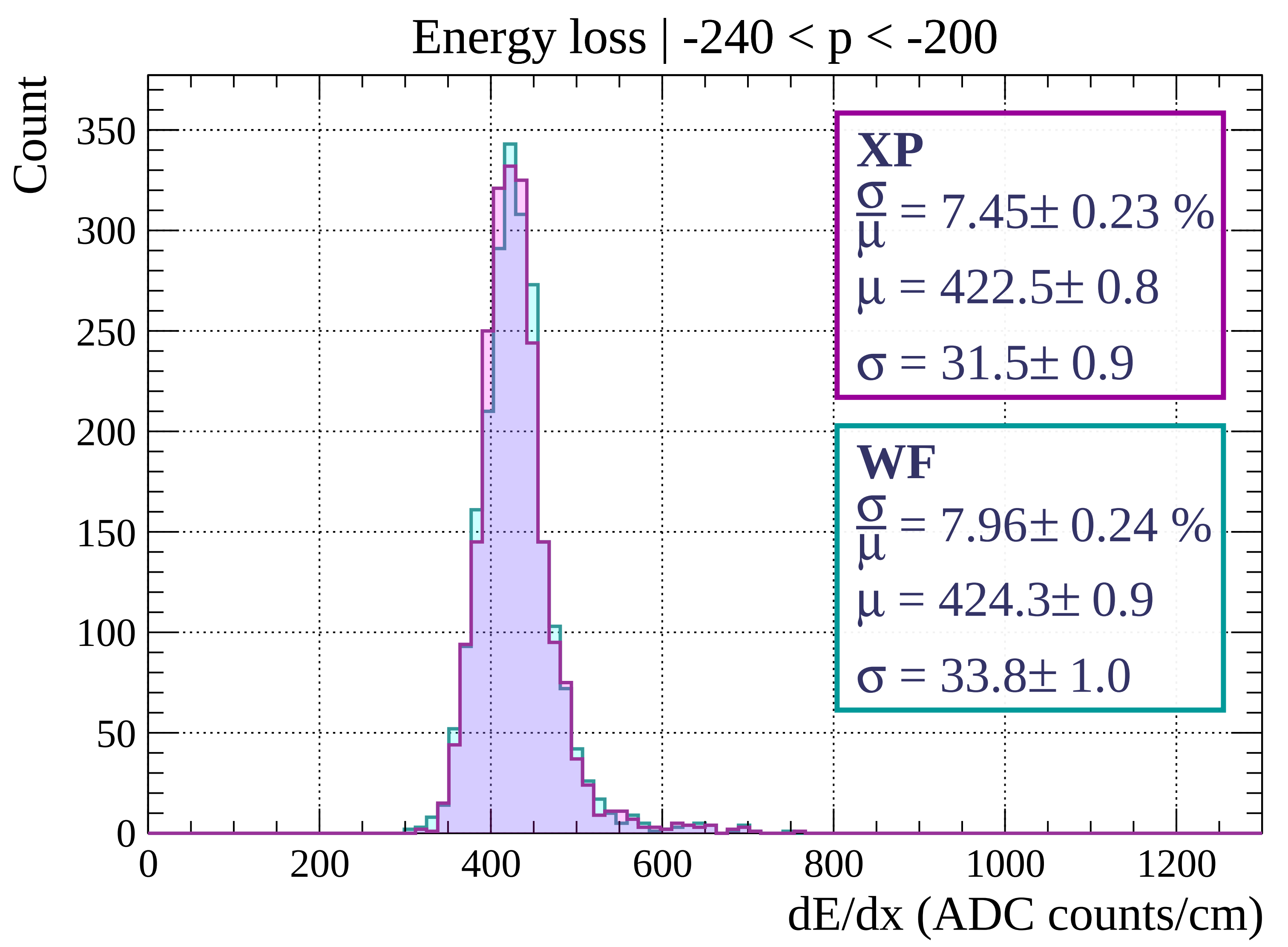


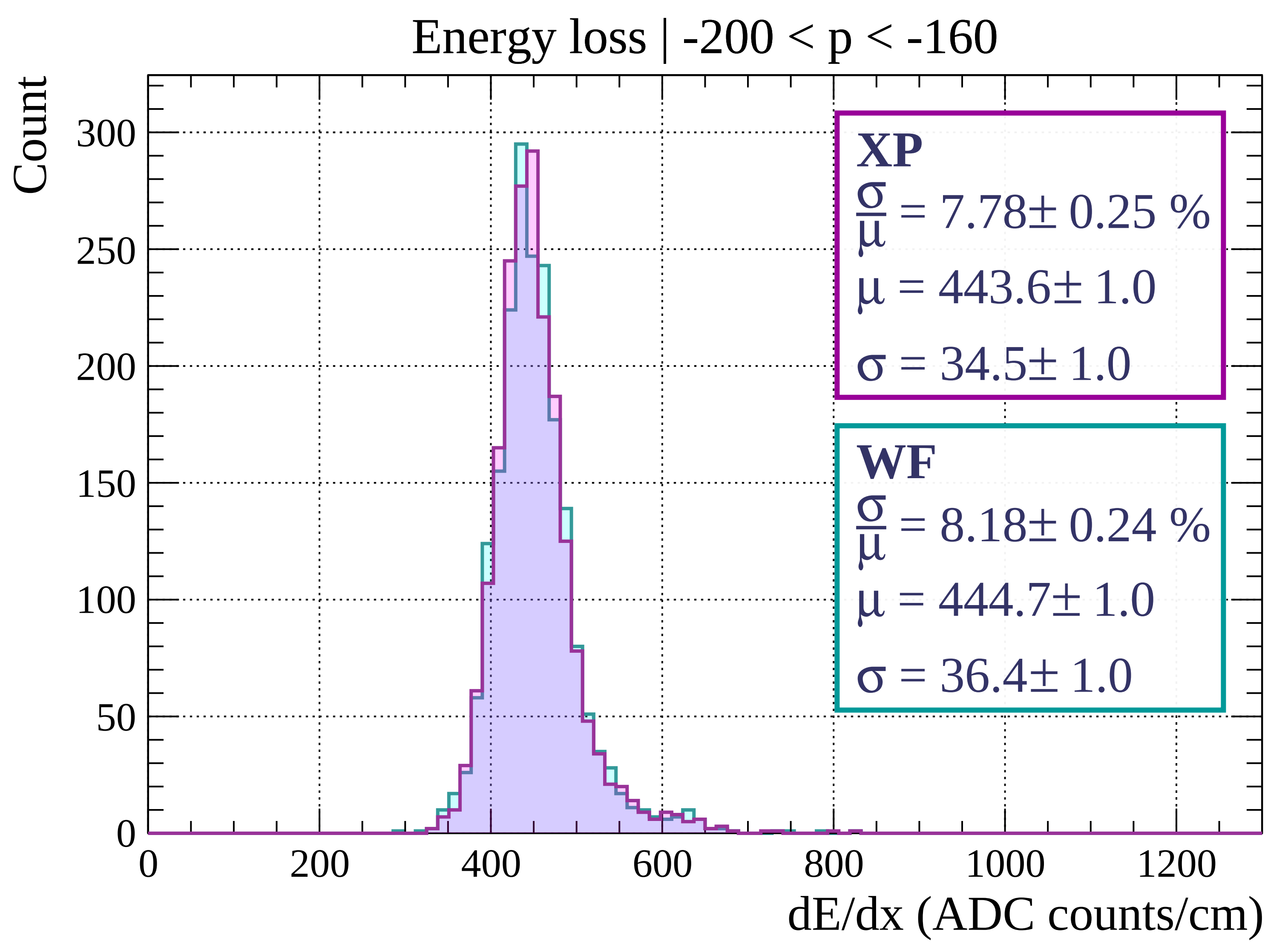
Energy loss | -360 < p < -320
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.42± 0.20 %
μ = 408.5± 0.7
σ = 30.3± 0.8
WF
σ/μ = 7.69± 0.22 %
μ = 409.2± 0.8
σ = 31.5± 0.8



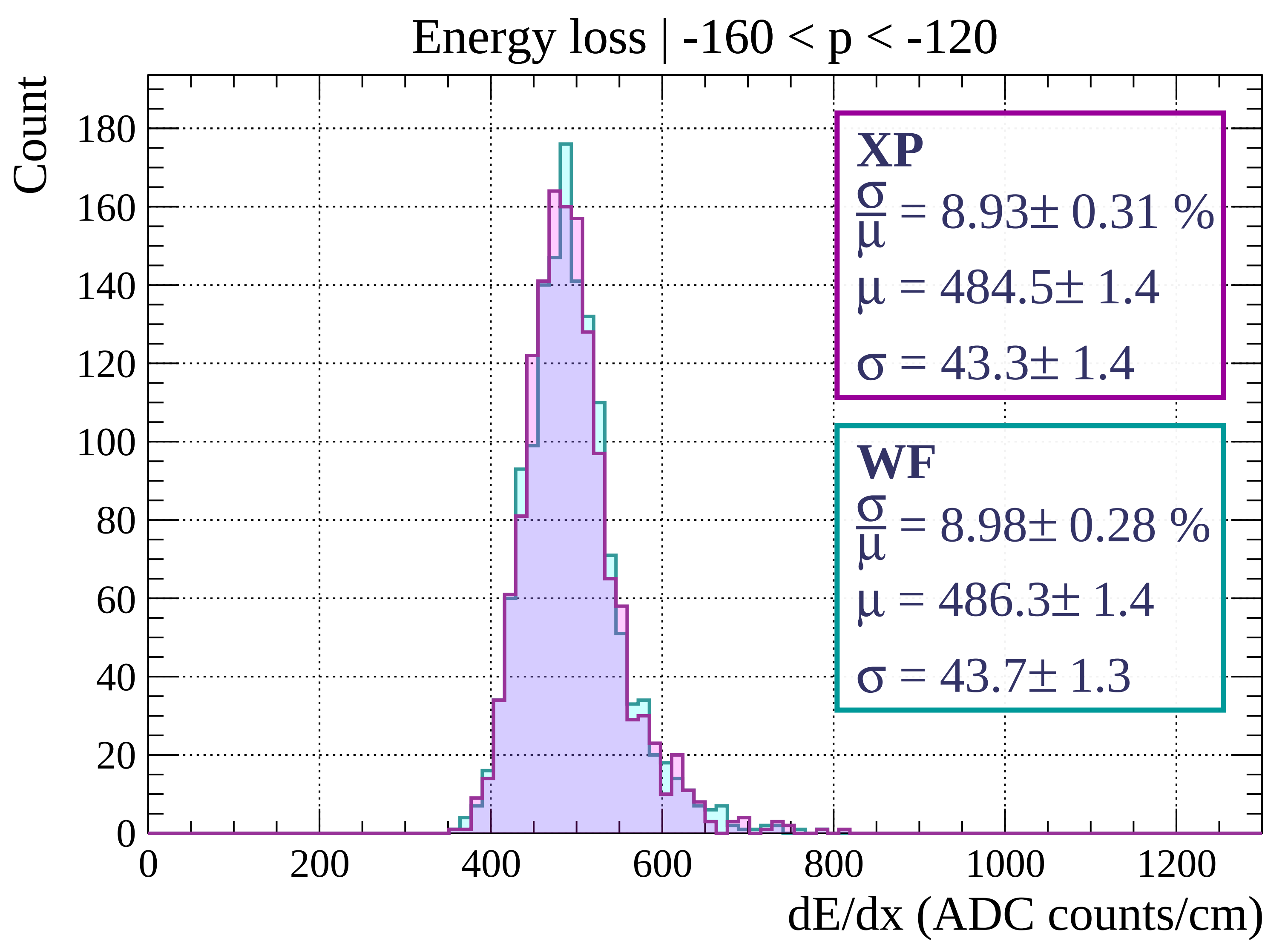
Energy loss | -320 < p < -280
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.62± 0.20 %
μ = 409.8± 0.7
σ = 31.2± 0.8
WF
σ/μ = 7.89± 0.19 %
μ = 410.9± 0.8
σ = 32.4± 0.7


Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.91± 0.20 %
μ = 416.6± 0.8
σ = 32.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 8.21± 0.22 %
μ = 418.3± 0.9
σ = 34.4± 0.9
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

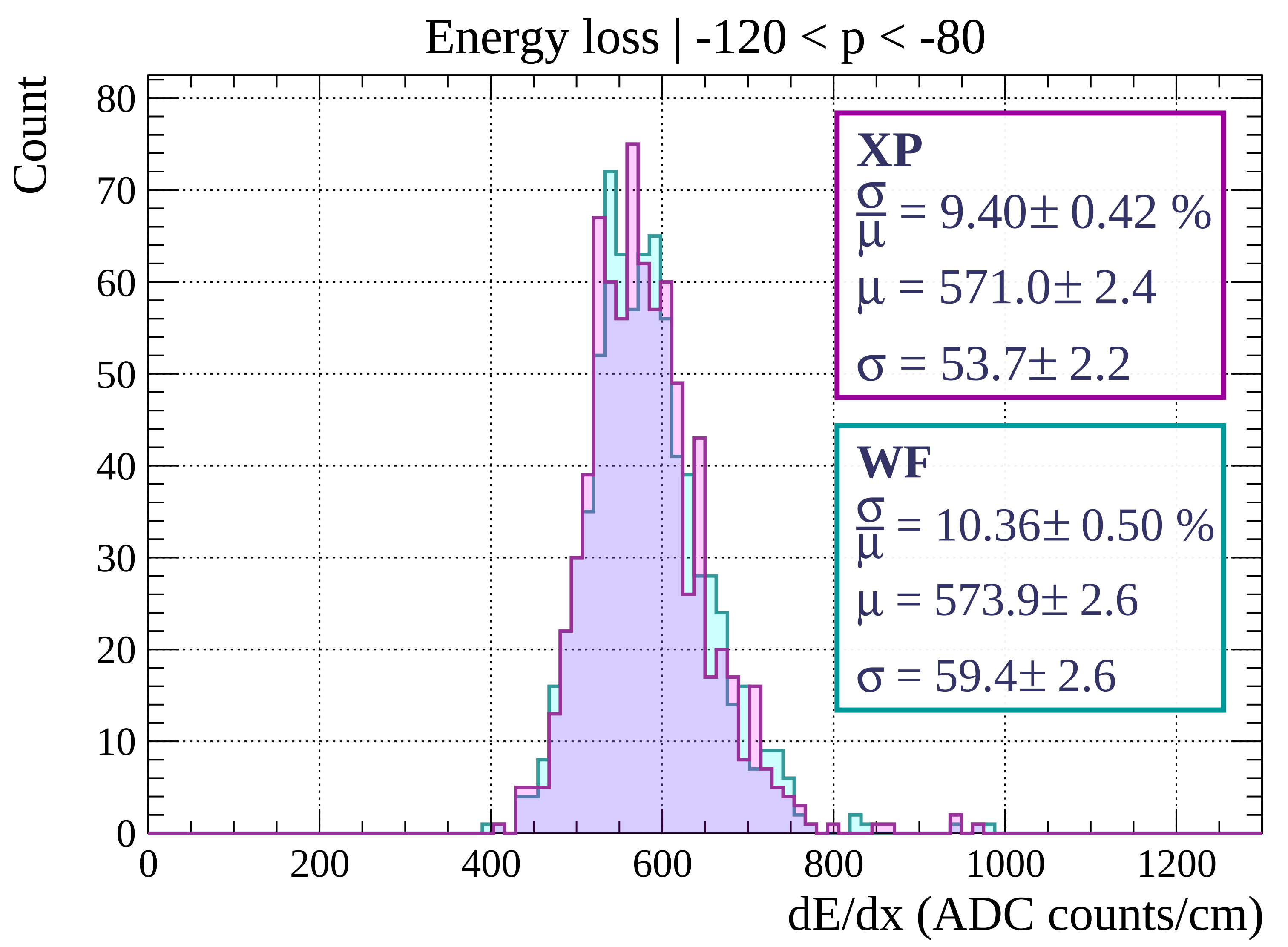

Energy loss | -240 < p < -200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.45± 0.23 %
μ = 422.5± 0.8
σ = 31.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.96± 0.24 %
μ = 424.3± 0.9
σ = 33.8± 1.0

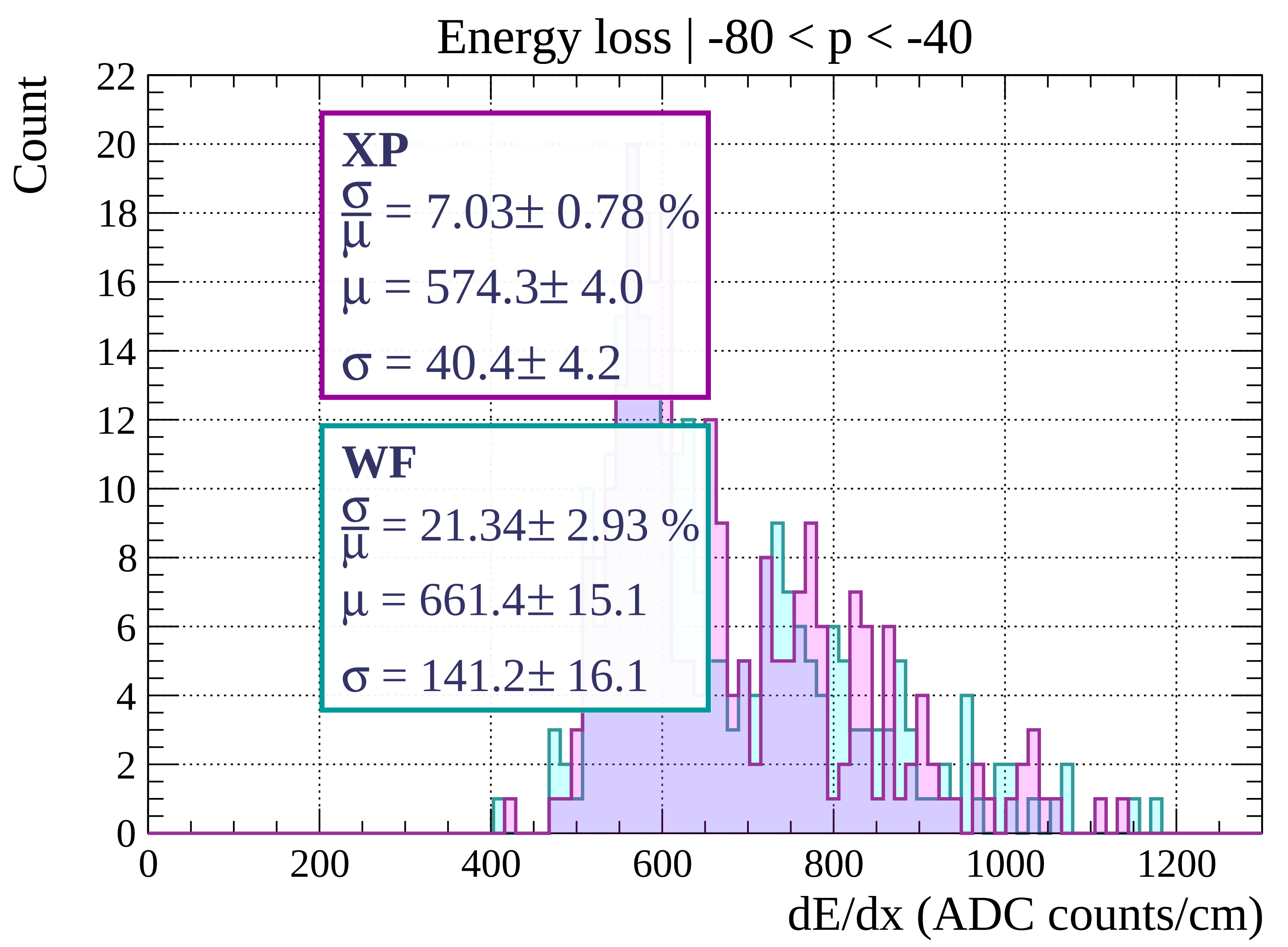


Energy loss | -200 < p < -160
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.78± 0.25 %
μ = 443.6± 1.0
σ = 34.5± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.18± 0.24 %
μ = 444.7± 1.0
σ = 36.4± 1.0

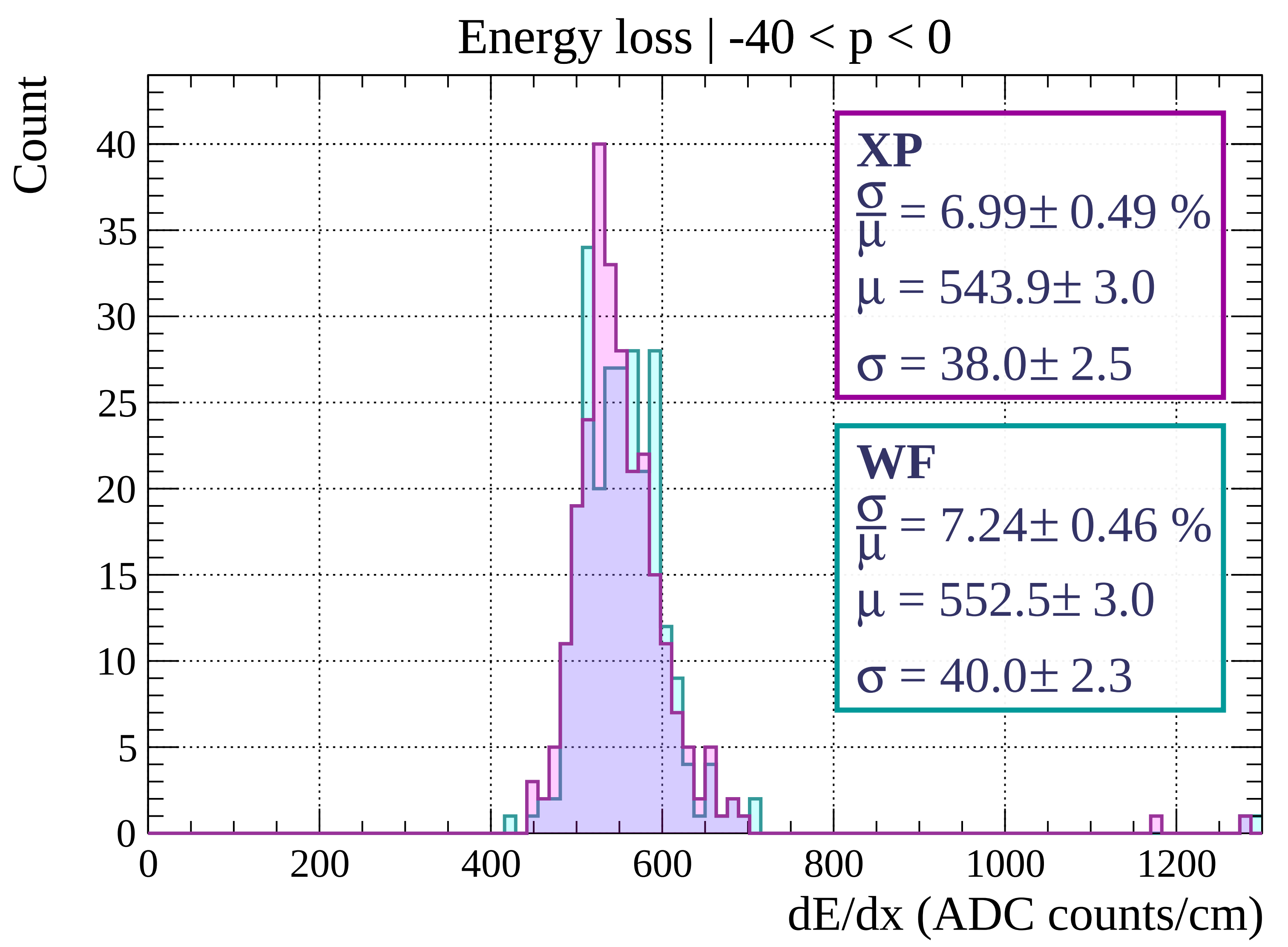


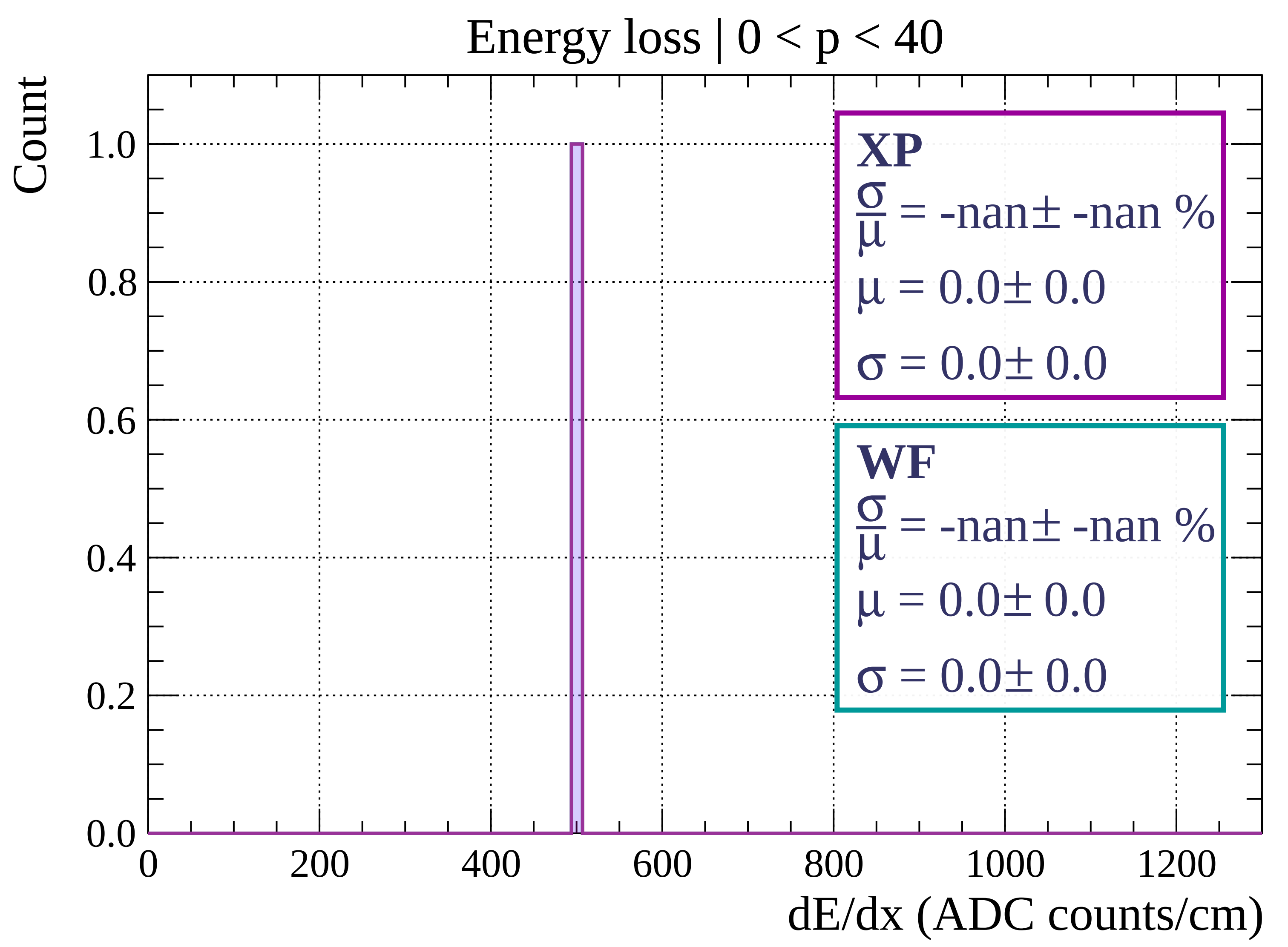
Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.93± 0.31 %
μ = 484.5± 1.4
σ = 43.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 8.98± 0.28 %
μ = 486.3± 1.4
σ = 43.7± 1.3


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
$\frac{\sigma}{\mu}$ = 9.40± 0.42 %
μ = 571.0± 2.4
σ = 53.7± 2.2
WF
$\frac{\sigma}{\mu}$ = 10.36± 0.50 %
μ = 573.9± 2.6
σ = 59.4± 2.6

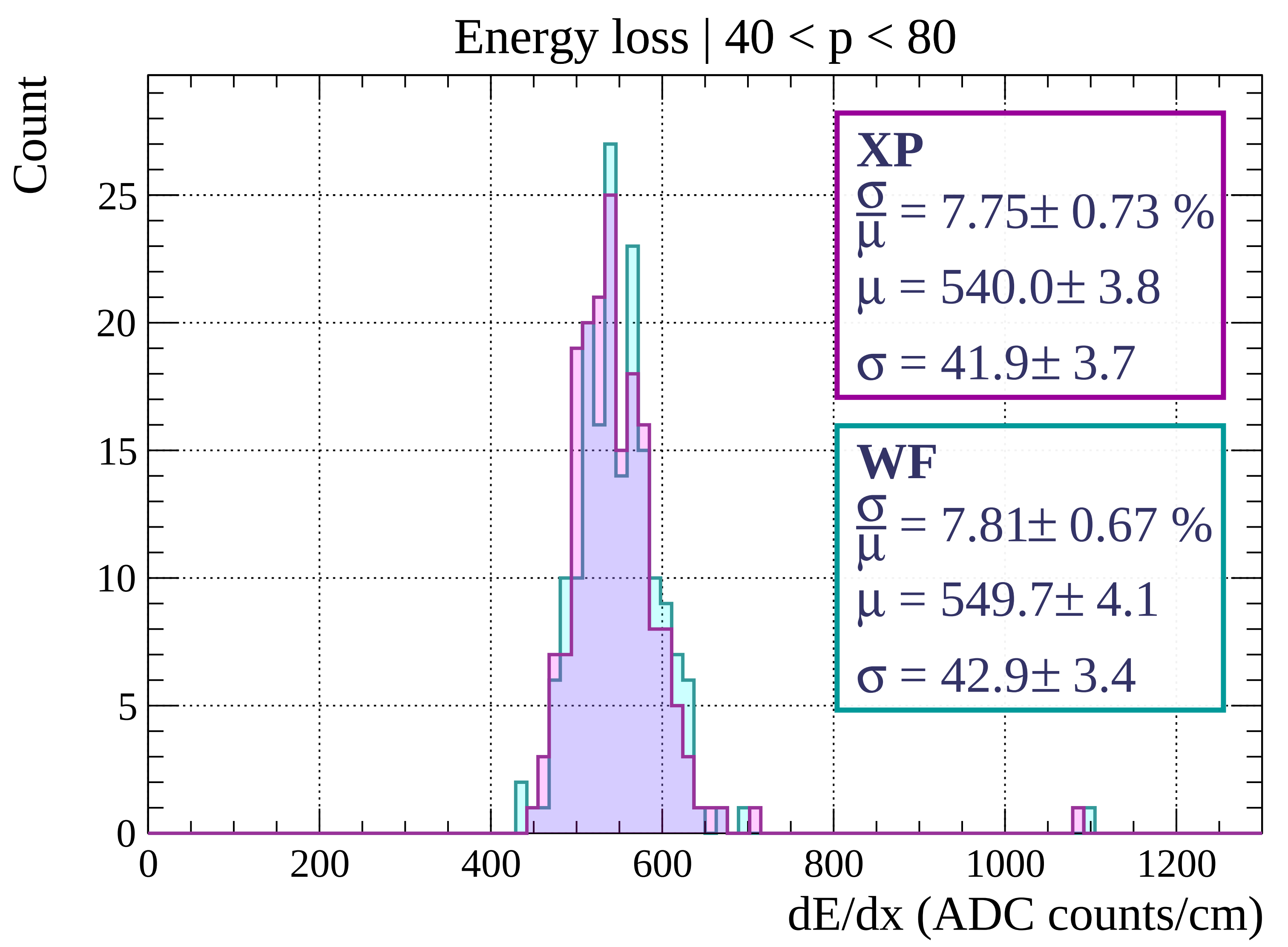


Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.03± 0.78 %
μ = 574.3± 4.0
σ = 40.4± 4.2
WF
$\frac{\sigma}{\mu}$ = 21.34± 2.93 %
μ = 661.4± 15.1
σ = 141.2± 16.1
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22

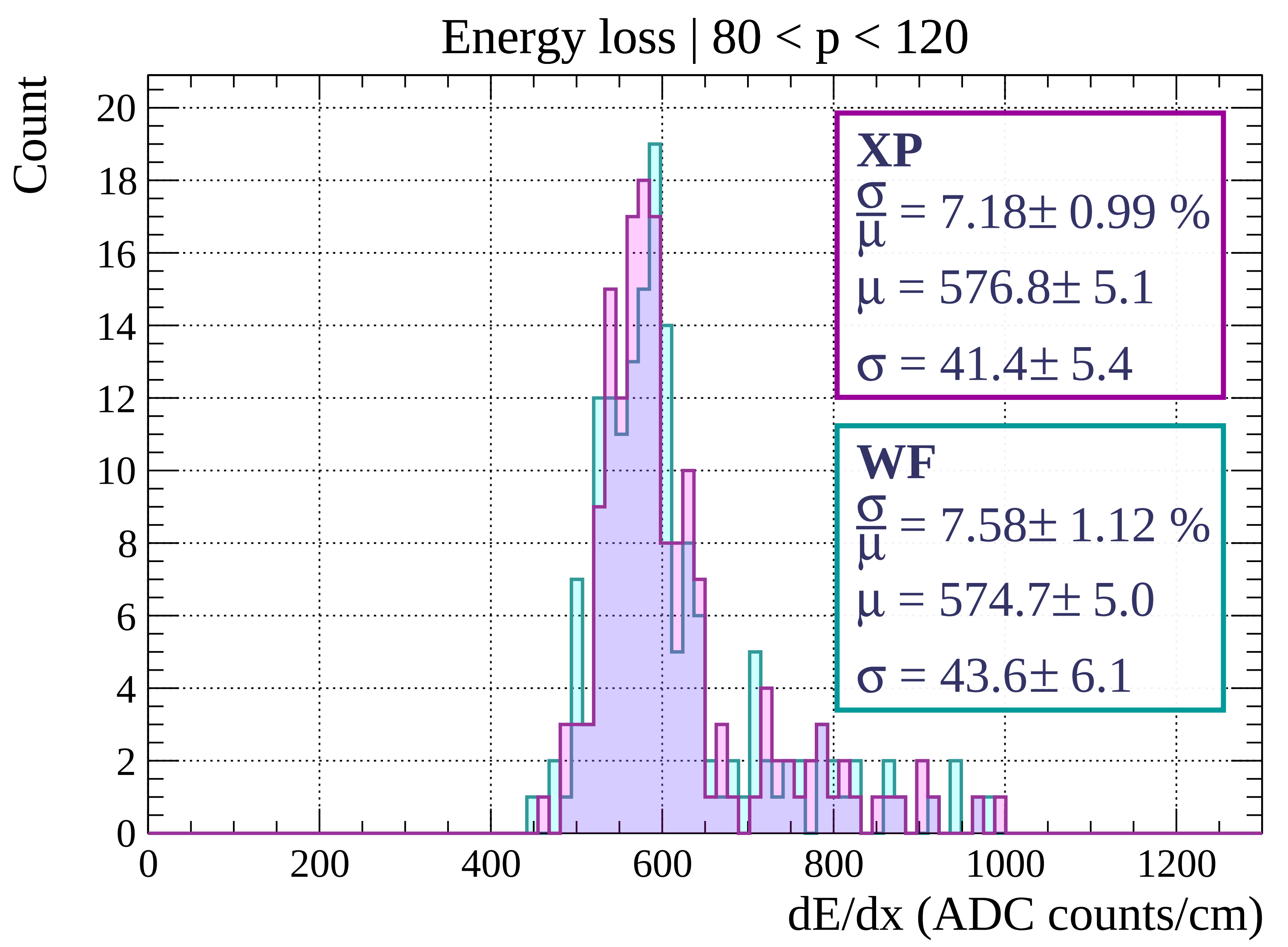

Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.99± 0.49 %
μ = 543.9± 3.0
σ = 38.0± 2.5
WF
$\frac{\sigma}{\mu}$ = 7.24± 0.46 %
μ = 552.5± 3.0
σ = 40.0± 2.3

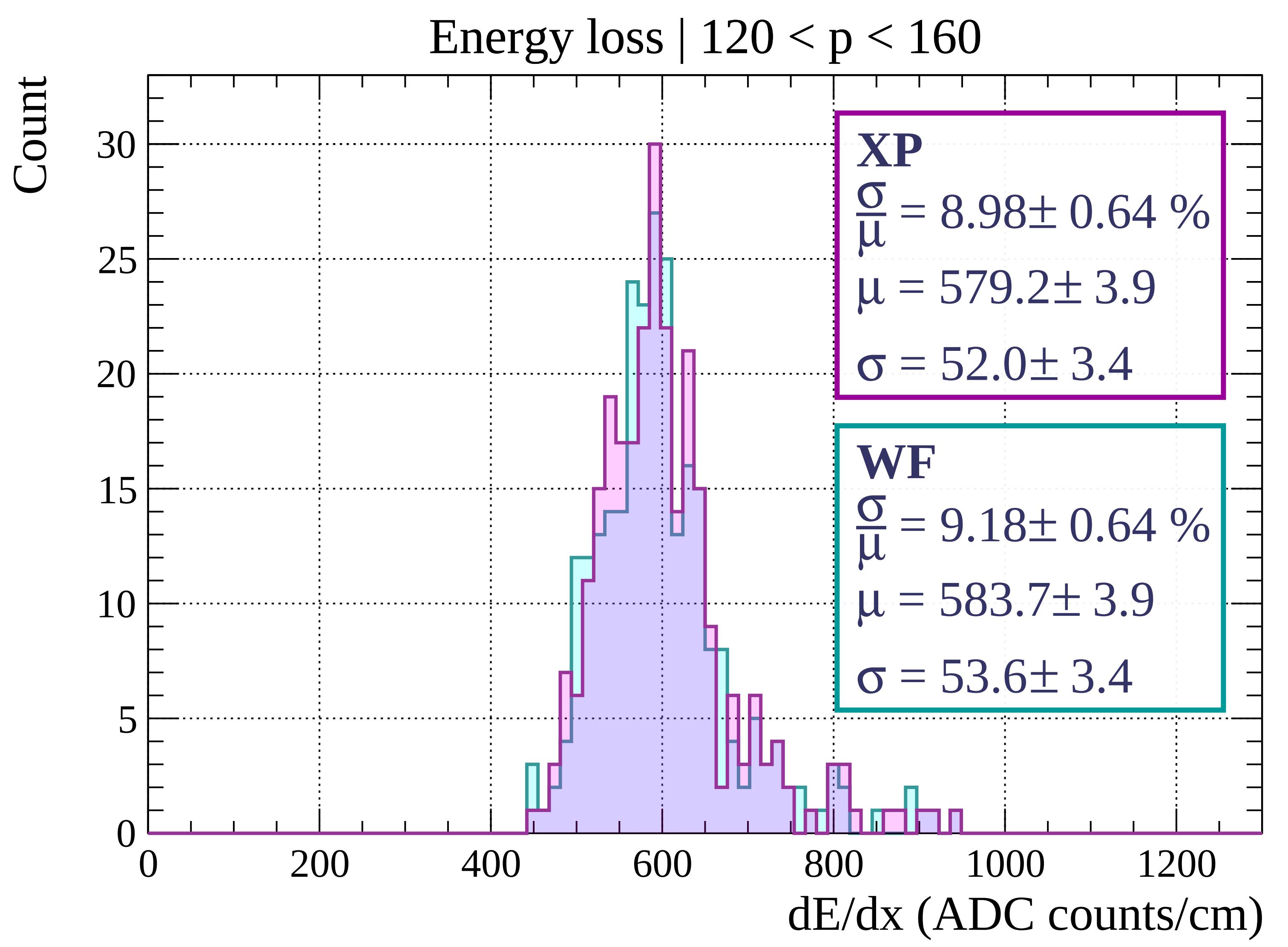


Energy loss | 0 < p < 40
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

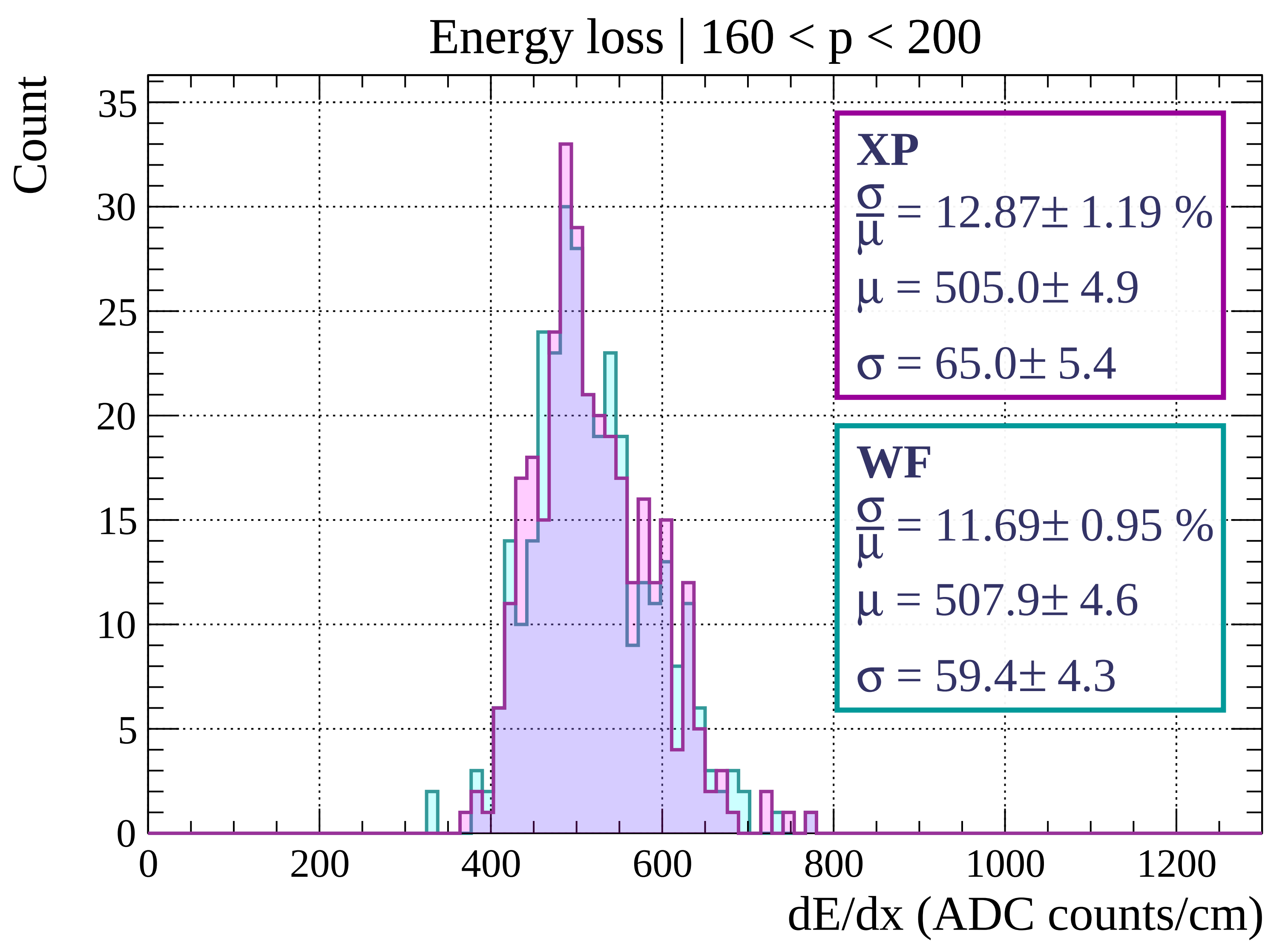


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 7.75± 0.73 %
μ = 540.0± 3.8
σ = 41.9± 3.7
WF
$\frac{\sigma}{\mu}$ = 7.81± 0.67 %
μ = 549.7± 4.1
σ = 42.9± 3.4

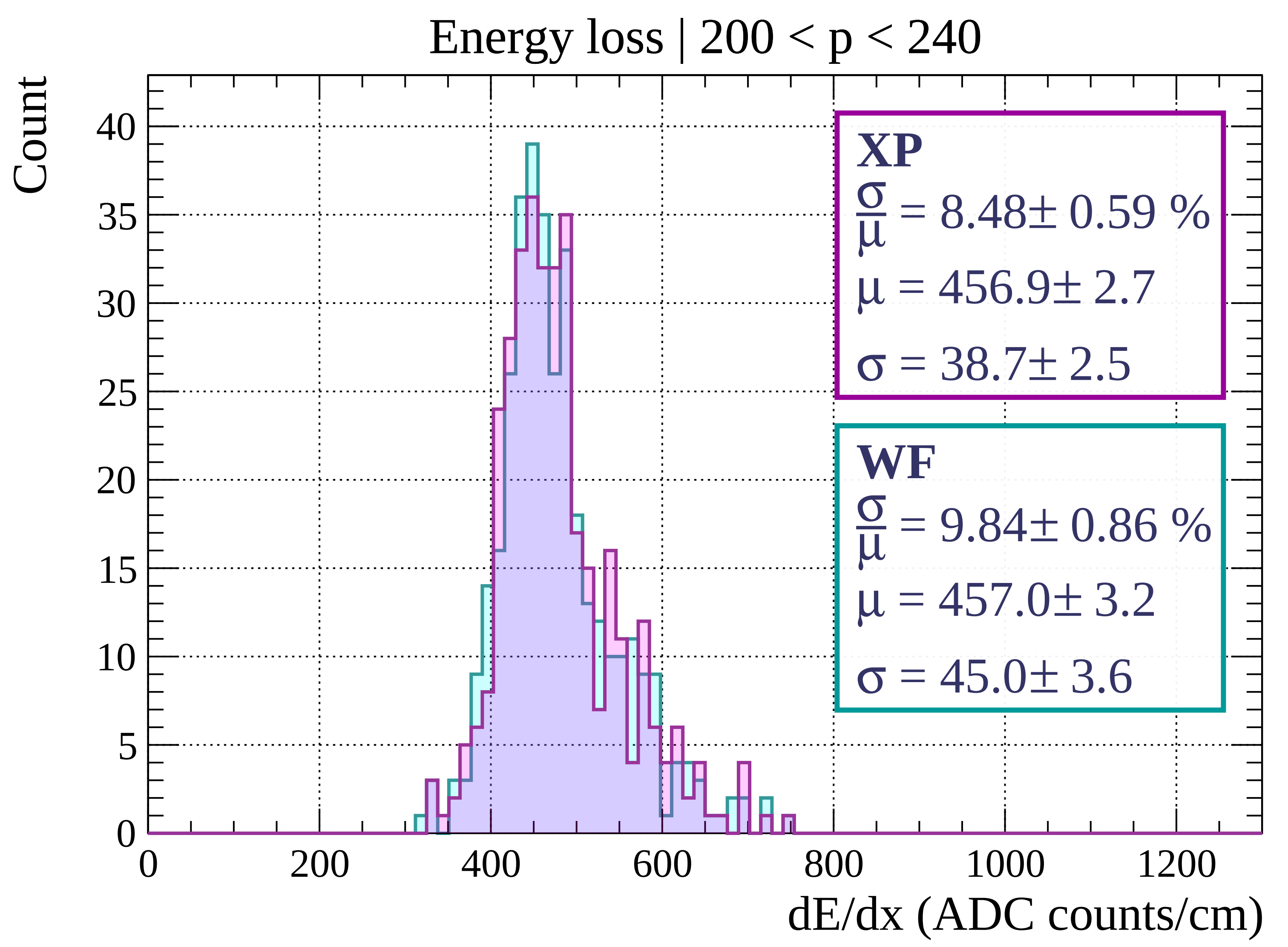

Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.18± 0.99 %
μ = 576.8± 5.1
σ = 41.4± 5.4
WF
$\frac{\sigma}{\mu}$ = 7.58± 1.12 %
μ = 574.7± 5.0
σ = 43.6± 6.1
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20

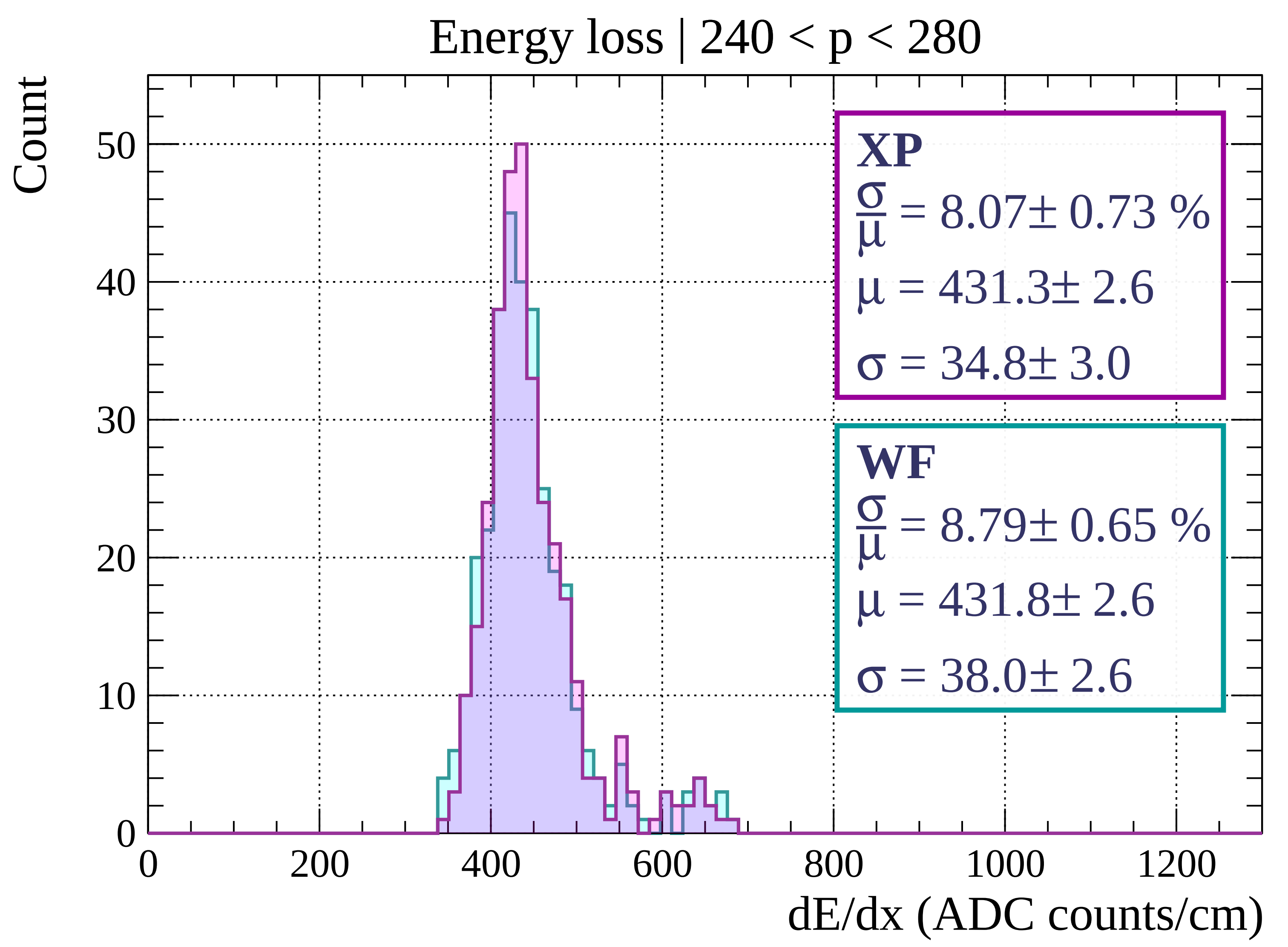


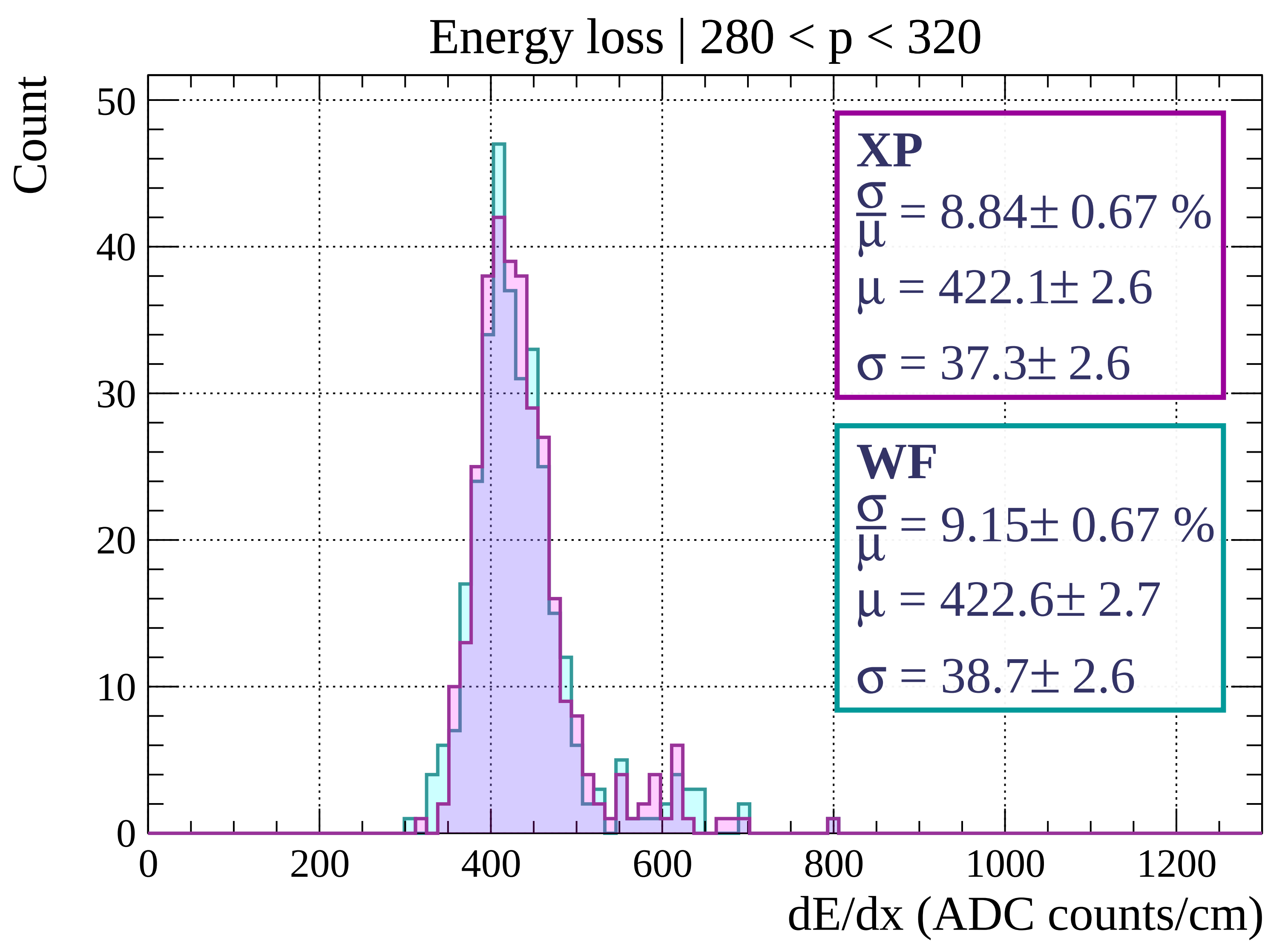
Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.98± 0.64 %
μ = 579.2± 3.9
σ = 52.0± 3.4
WF
$\frac{\sigma}{\mu}$ = 9.18± 0.64 %
μ = 583.7± 3.9
σ = 53.6± 3.4



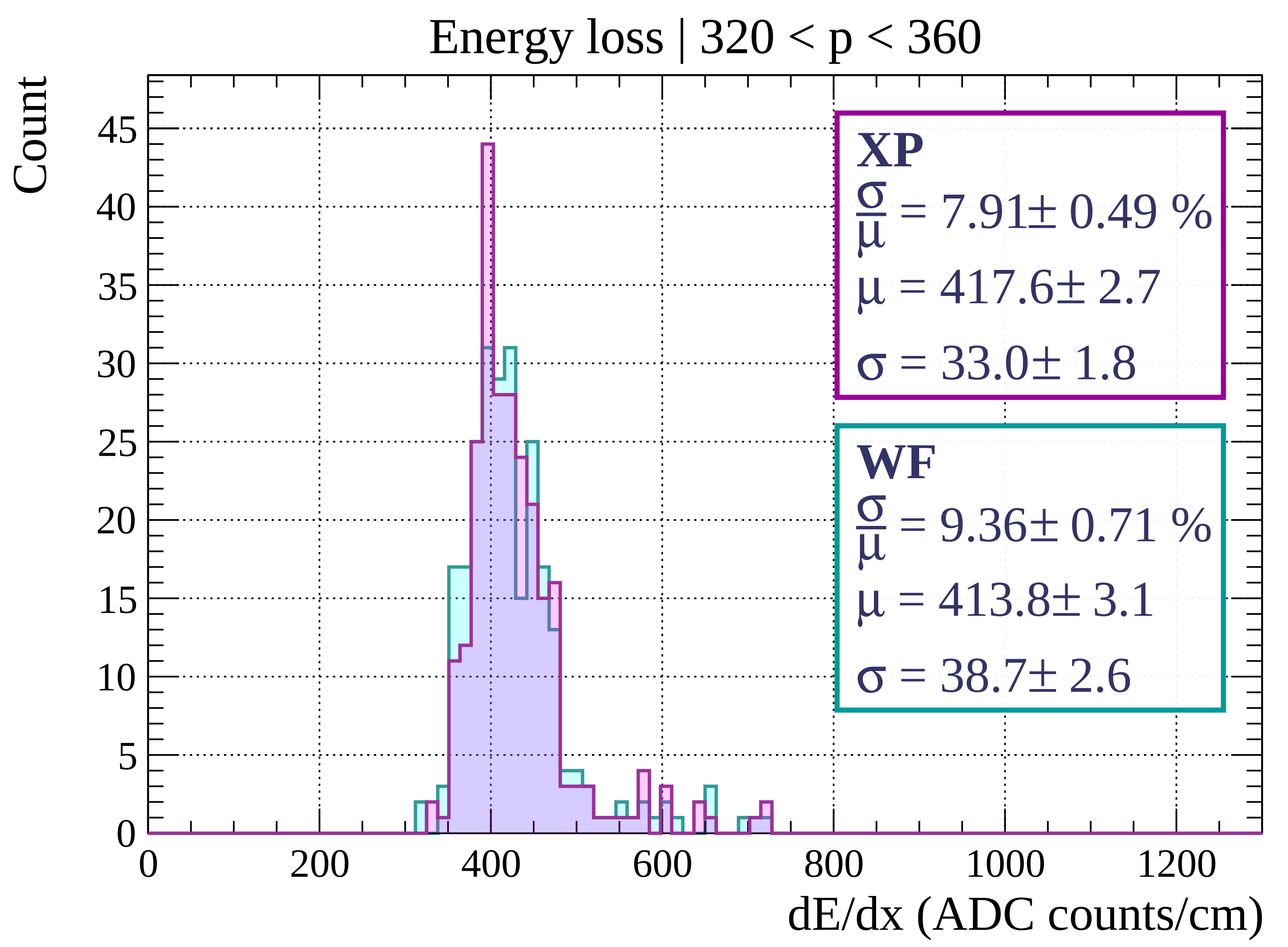
Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 12.87± 1.19 %
μ = 505.0± 4.9
σ = 65.0± 5.4
WF
σ/μ = 11.69± 0.95 %
μ = 507.9± 4.6
σ = 59.4± 4.3


Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
XP
σ/μ = 8.48± 0.59 %
μ = 456.9± 2.7
σ = 38.7± 2.5
WF
σ/μ = 9.84± 0.86 %
μ = 457.0± 3.2
σ = 45.0± 3.6

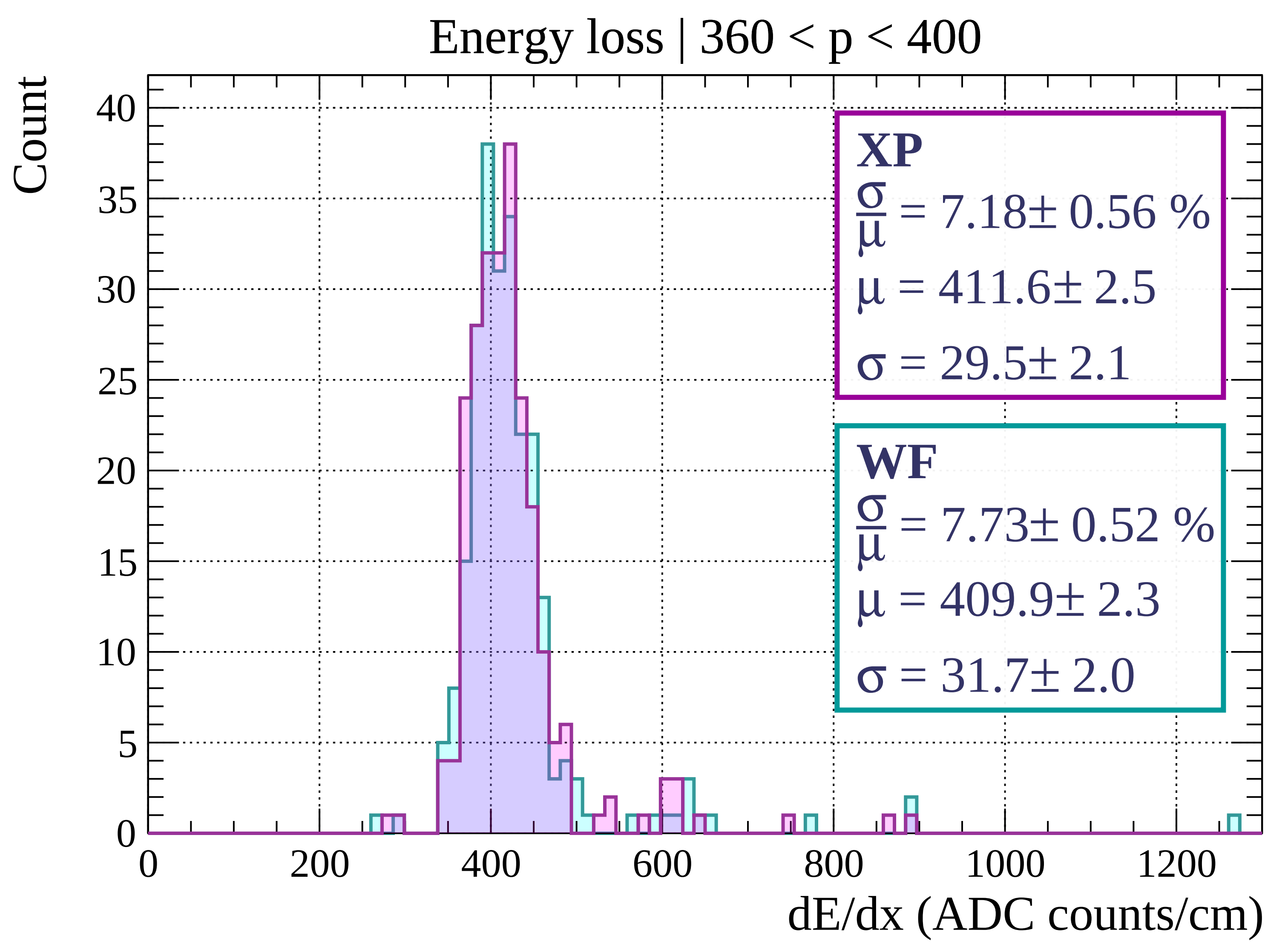

Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
$\frac{\sigma}{\mu}$ = 8.07± 0.73 %
μ = 431.3± 2.6
σ = 34.8± 3.0
WF
$\frac{\sigma}{\mu}$ = 8.79± 0.65 %
μ = 431.8± 2.6
σ = 38.0± 2.6



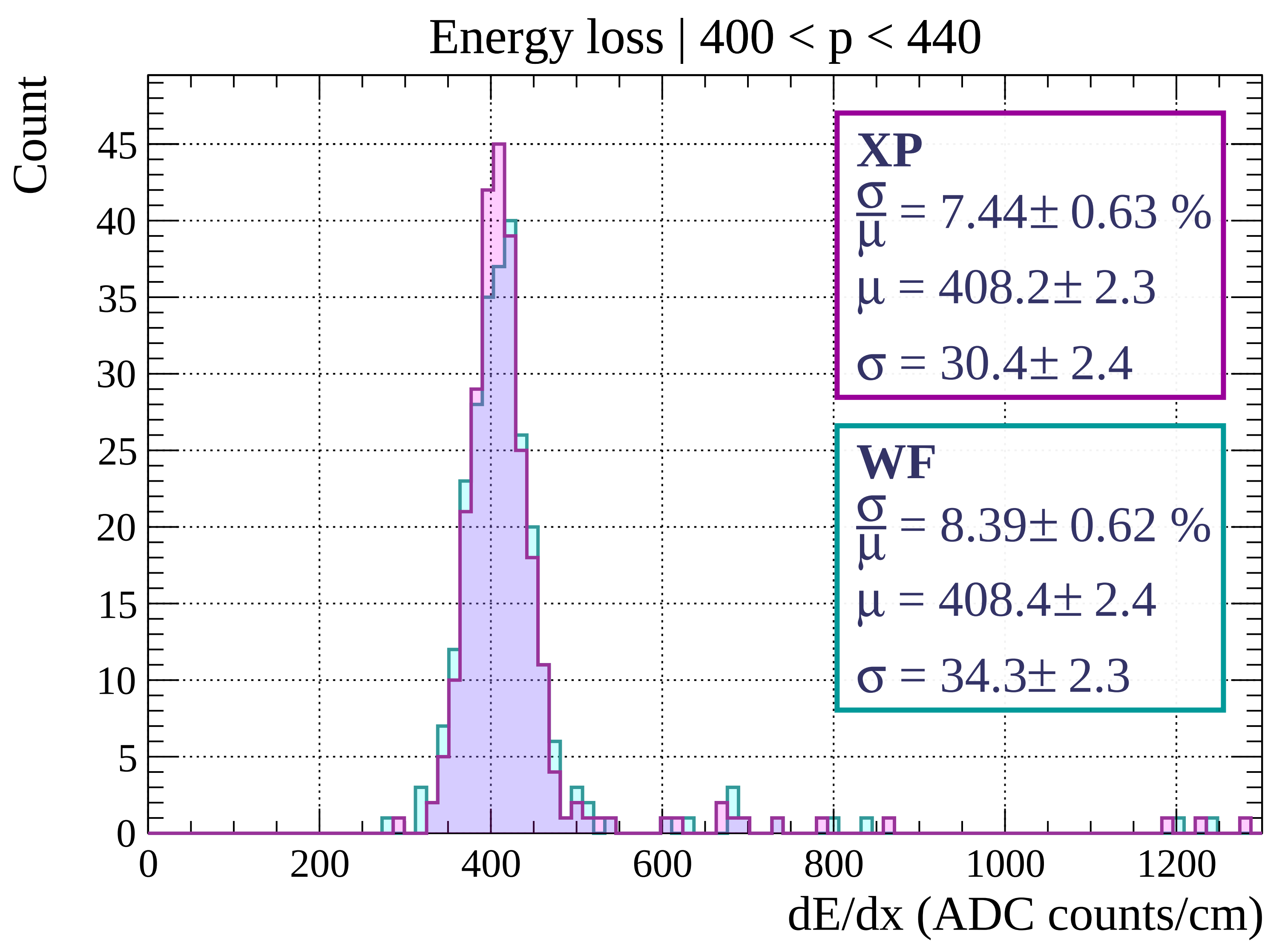
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.84± 0.67 %
μ = 422.1± 2.6
σ = 37.3± 2.6
WF
$\frac{\sigma}{\mu}$ = 9.15± 0.67 %
μ = 422.6± 2.7
σ = 38.7± 2.6



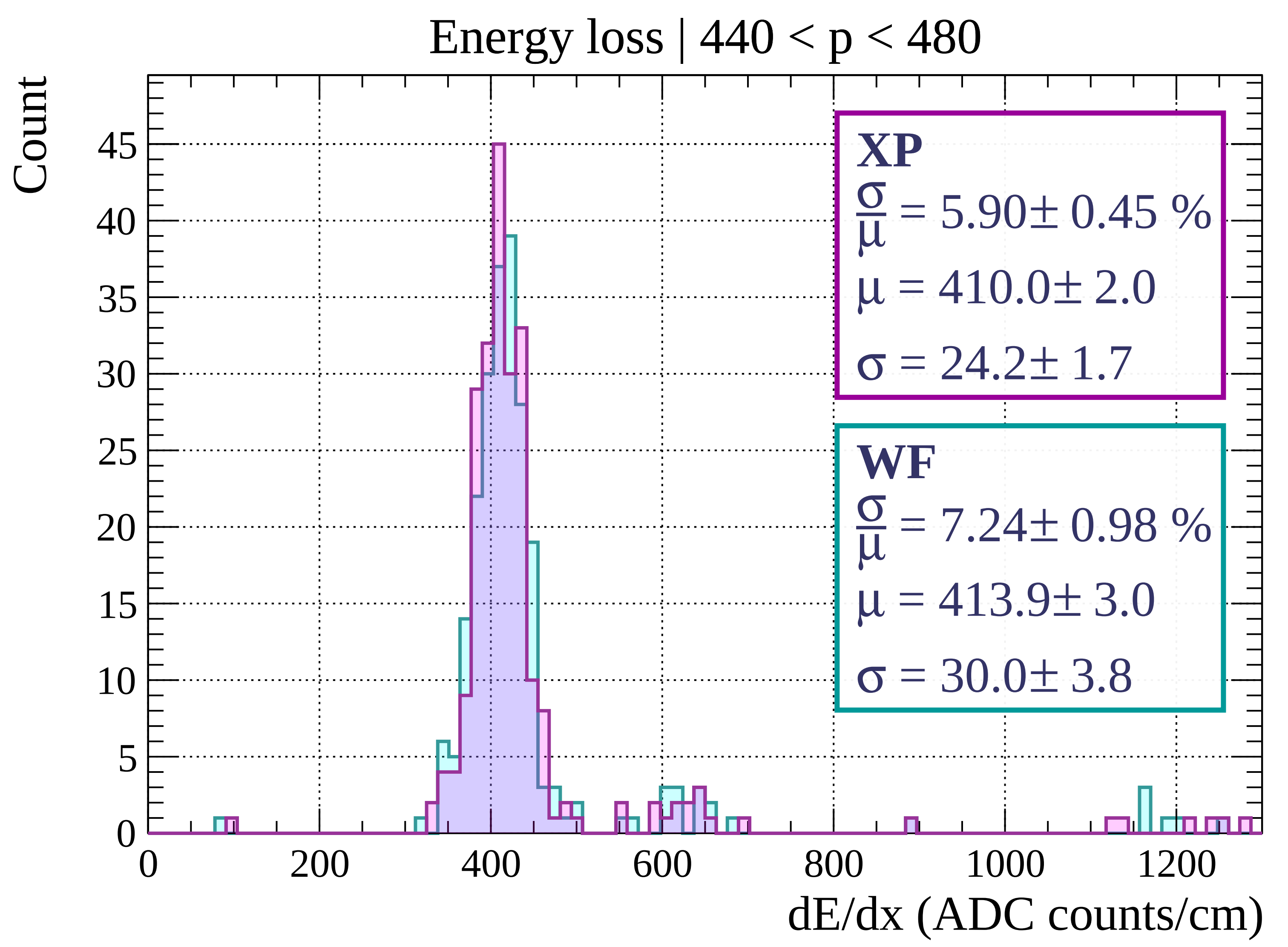
Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200
XP
σ/μ = 7.91± 0.49 %
μ = 417.6± 2.7
σ = 33.0± 1.8
WF
σ/μ = 9.36± 0.71 %
μ = 413.8± 3.1
σ = 38.7± 2.6



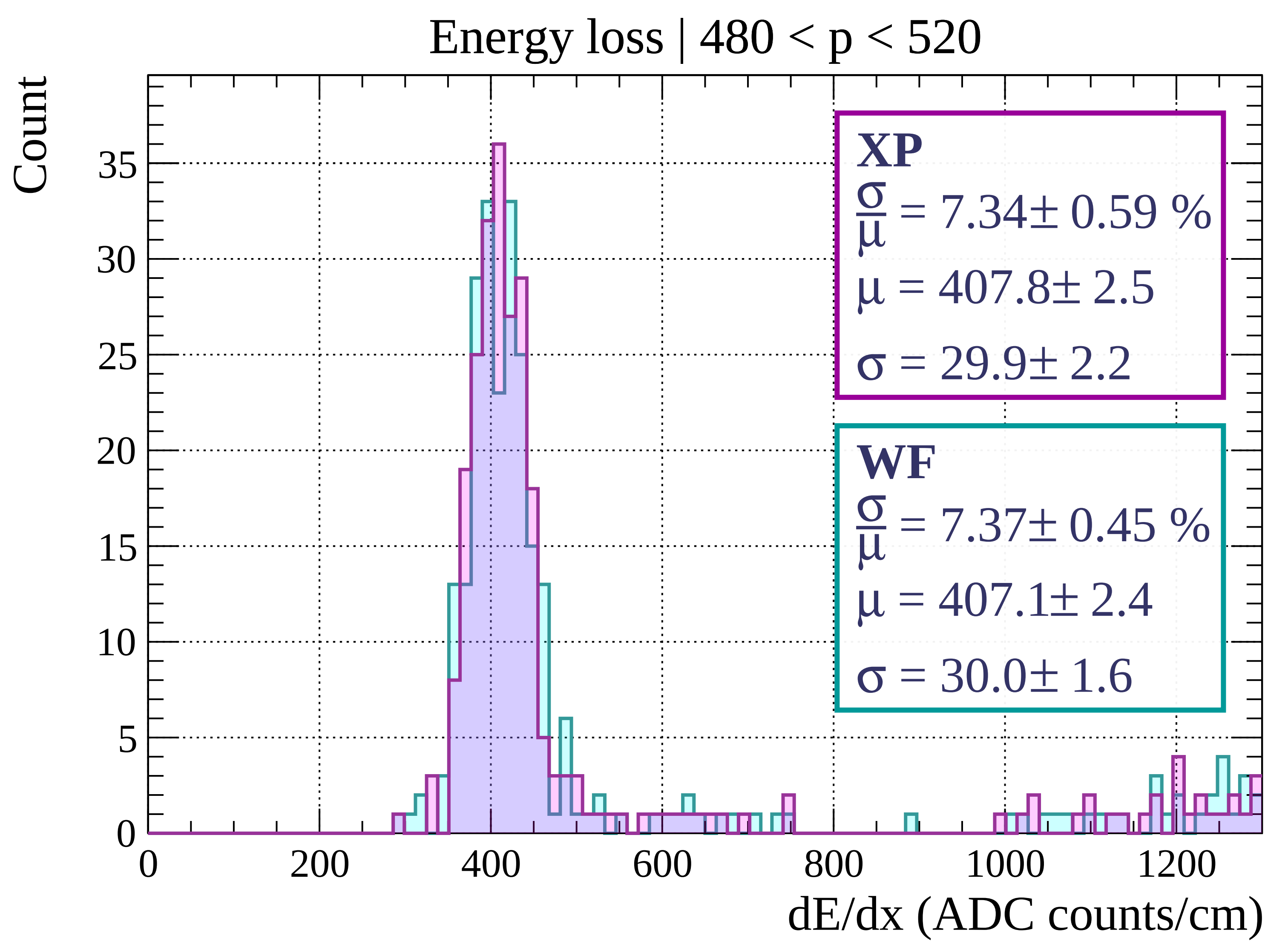
Energy loss | 360 < p < 400
Count
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.18± 0.56 %
μ = 411.6± 2.5
σ = 29.5± 2.1
WF
$\frac{\sigma}{\mu}$ = 7.73± 0.52 %
μ = 409.9± 2.3
σ = 31.7± 2.0

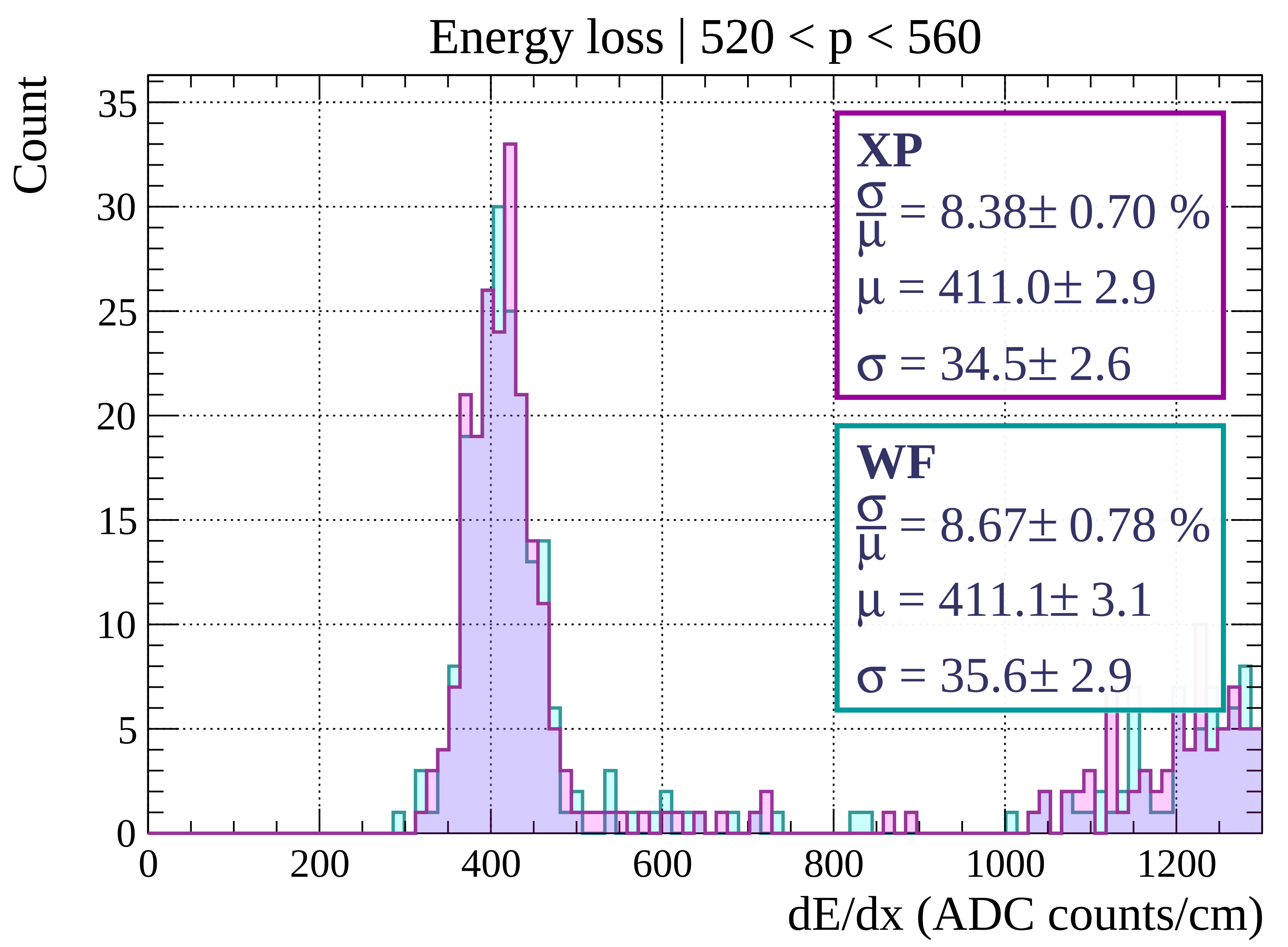


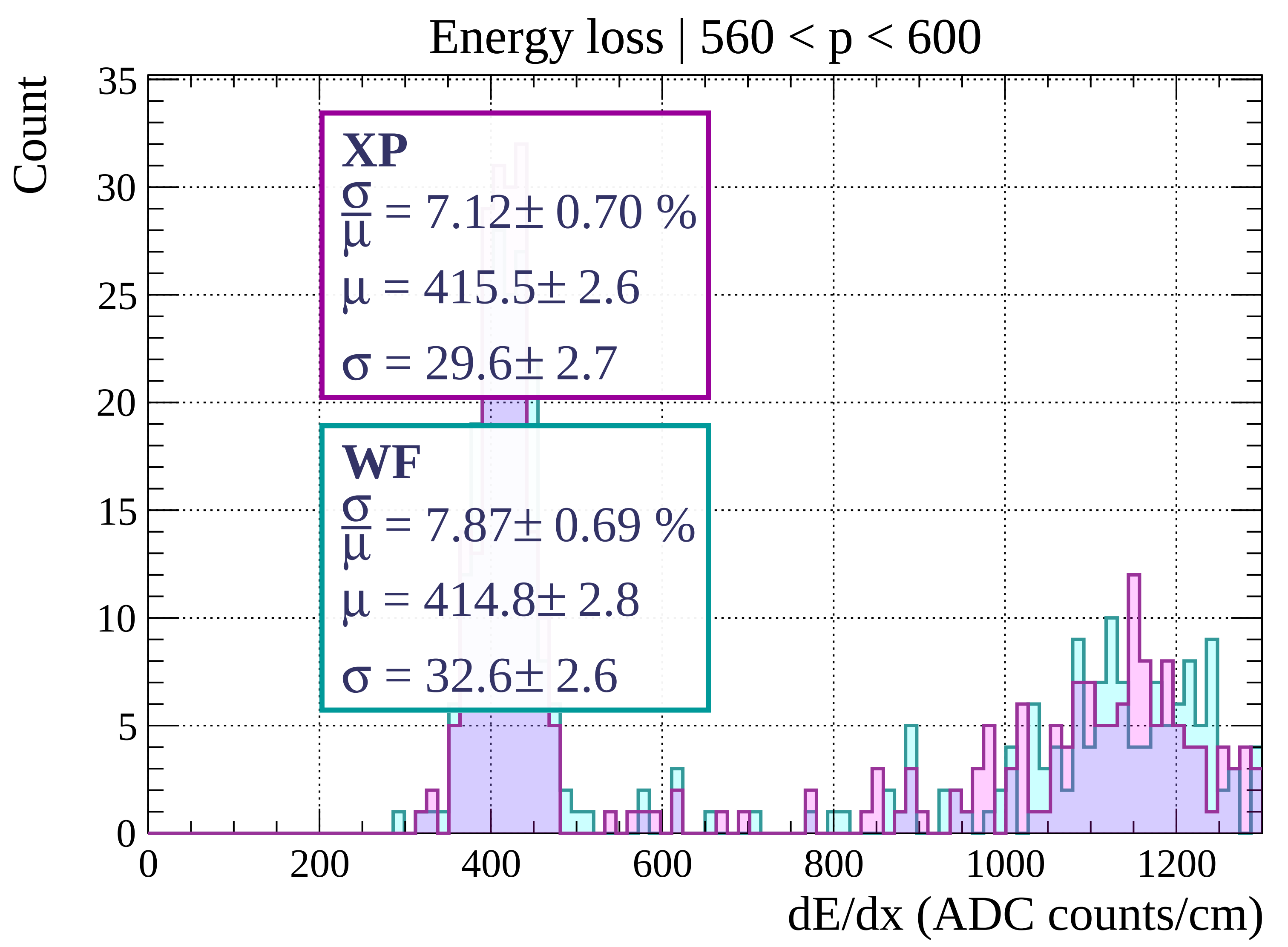
Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
45
XP
σ/μ = 7.44± 0.63 %
μ = 408.2± 2.3
σ = 30.4± 2.4
WF
σ/μ = 8.39± 0.62 %
μ = 408.4± 2.4
σ = 34.3± 2.3

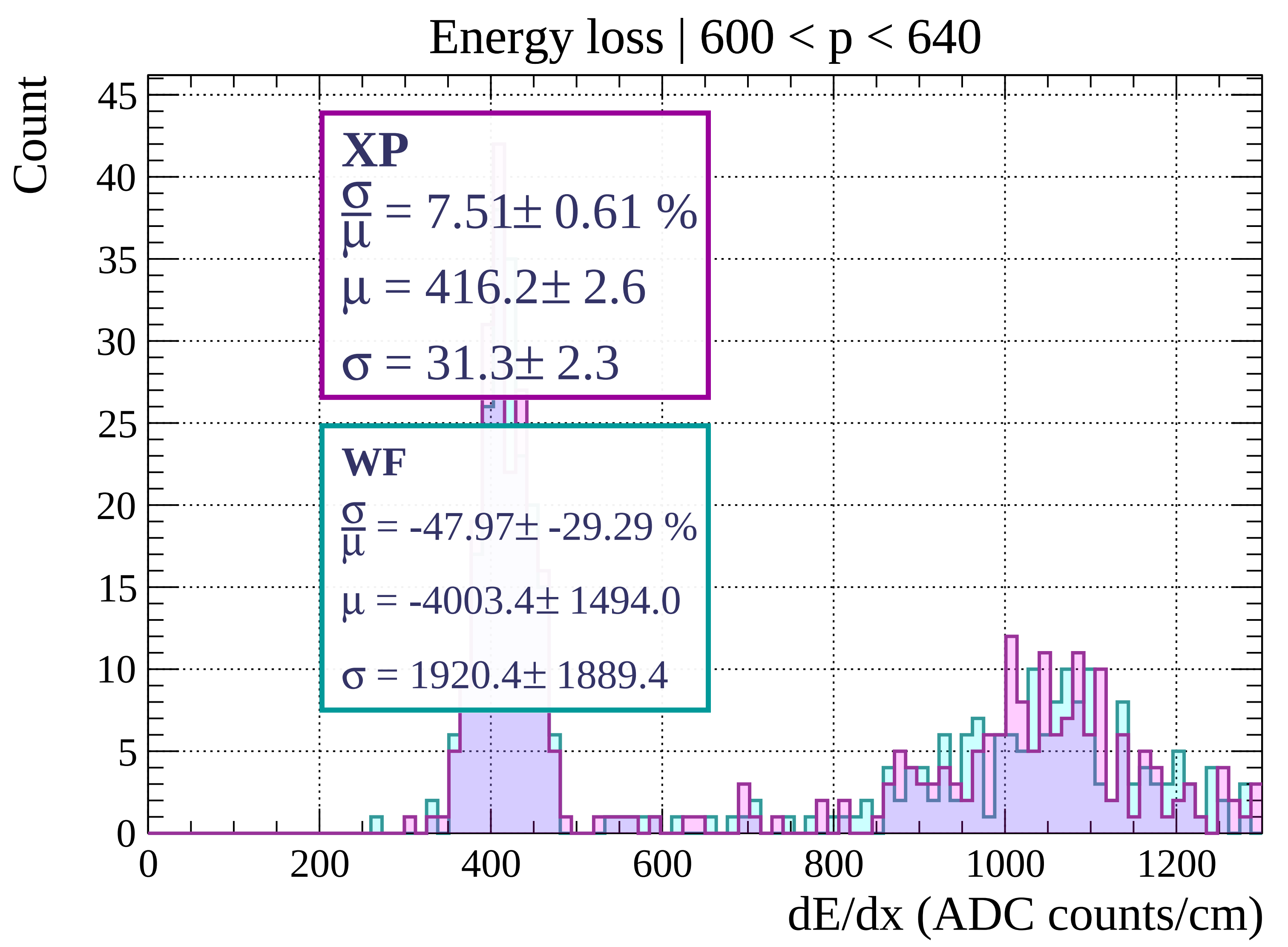


Energy loss | 440 < p < 480
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
40
45
XP
σ/μ = 5.90± 0.45 %
μ = 410.0± 2.0
σ = 24.2± 1.7
WF
σ/μ = 7.24± 0.98 %
μ = 413.9± 3.0
σ = 30.0± 3.8

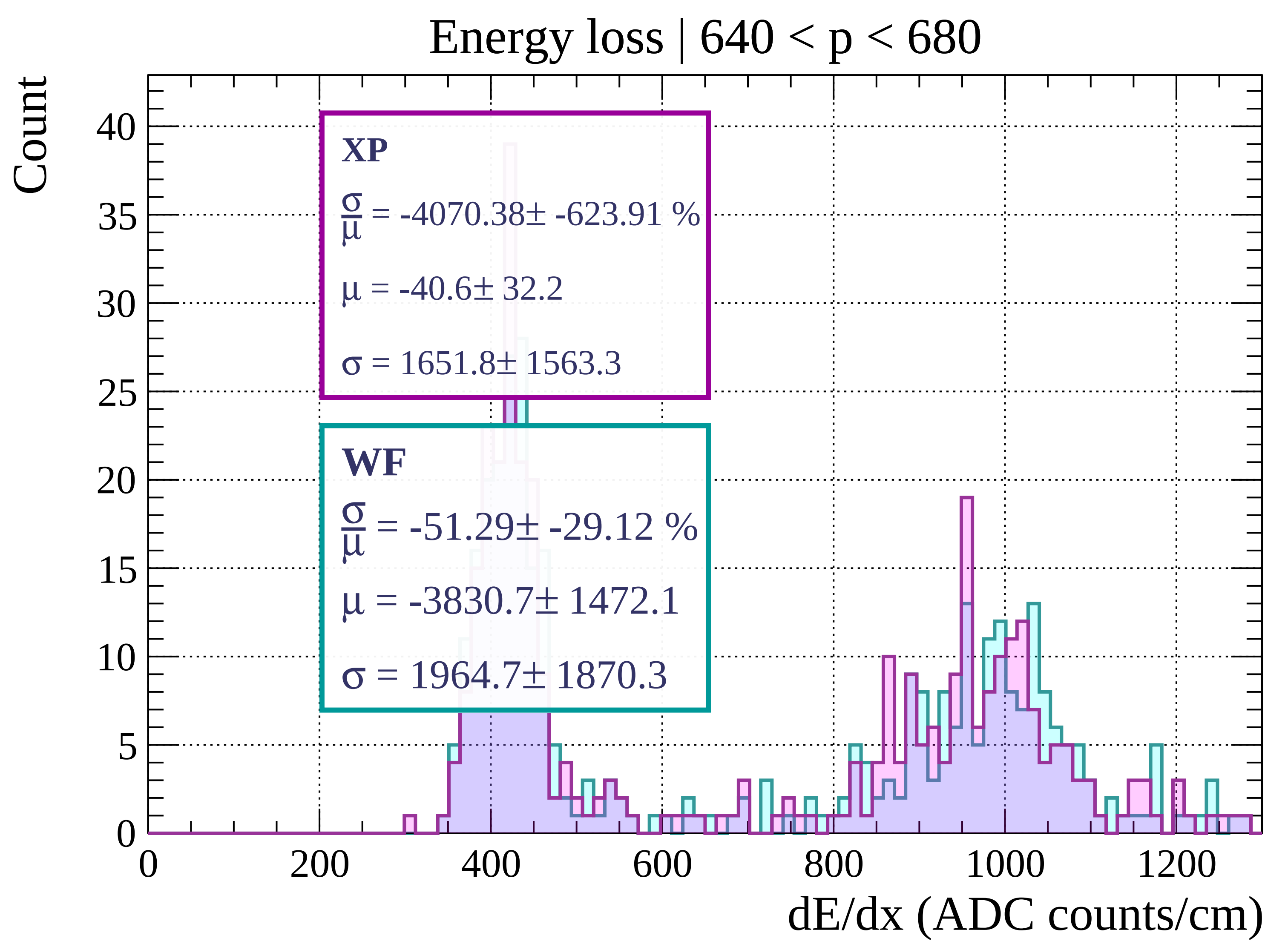

Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
$\frac{\sigma}{\mu}$ = 7.34± 0.59 %
μ = 407.8± 2.5
σ = 29.9± 2.2
WF
$\frac{\sigma}{\mu}$ = 7.37± 0.45 %
μ = 407.1± 2.4
σ = 30.0± 1.6

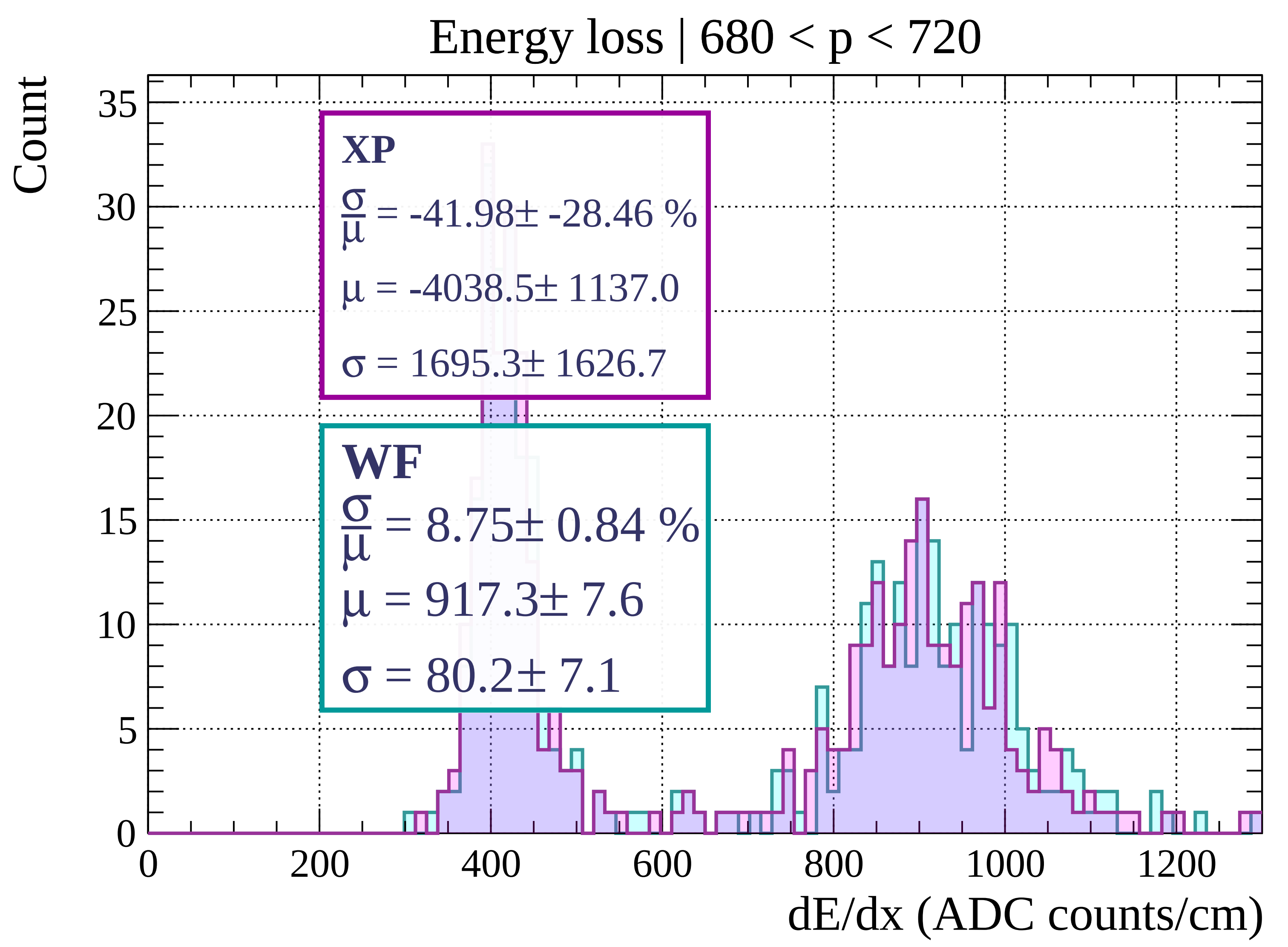

Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
$\frac{\sigma}{\mu}$ = 8.38± 0.70 %
μ = 411.0± 2.9
σ = 34.5± 2.6
WF
$\frac{\sigma}{\mu}$ = 8.67± 0.78 %
μ = 411.1± 3.1
σ = 35.6± 2.9

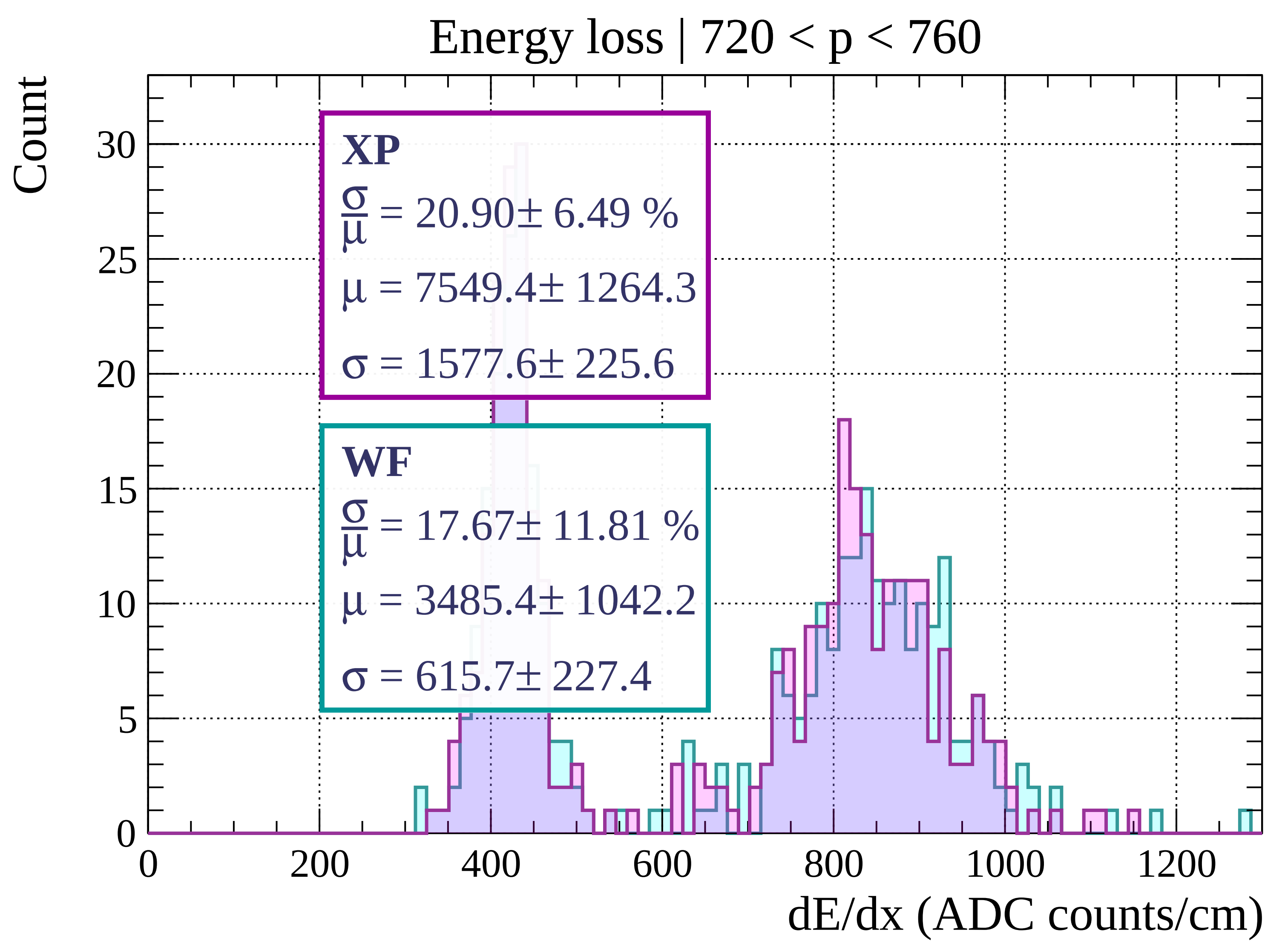

Energy loss | 560 < p < 600
Count
XP
σ/μ = 7.12± 0.70 %
μ = 415.5± 2.6
σ = 29.6± 2.7
WF
σ/μ = 7.87± 0.69 %
μ = 414.8± 2.8
σ = 32.6± 2.6
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



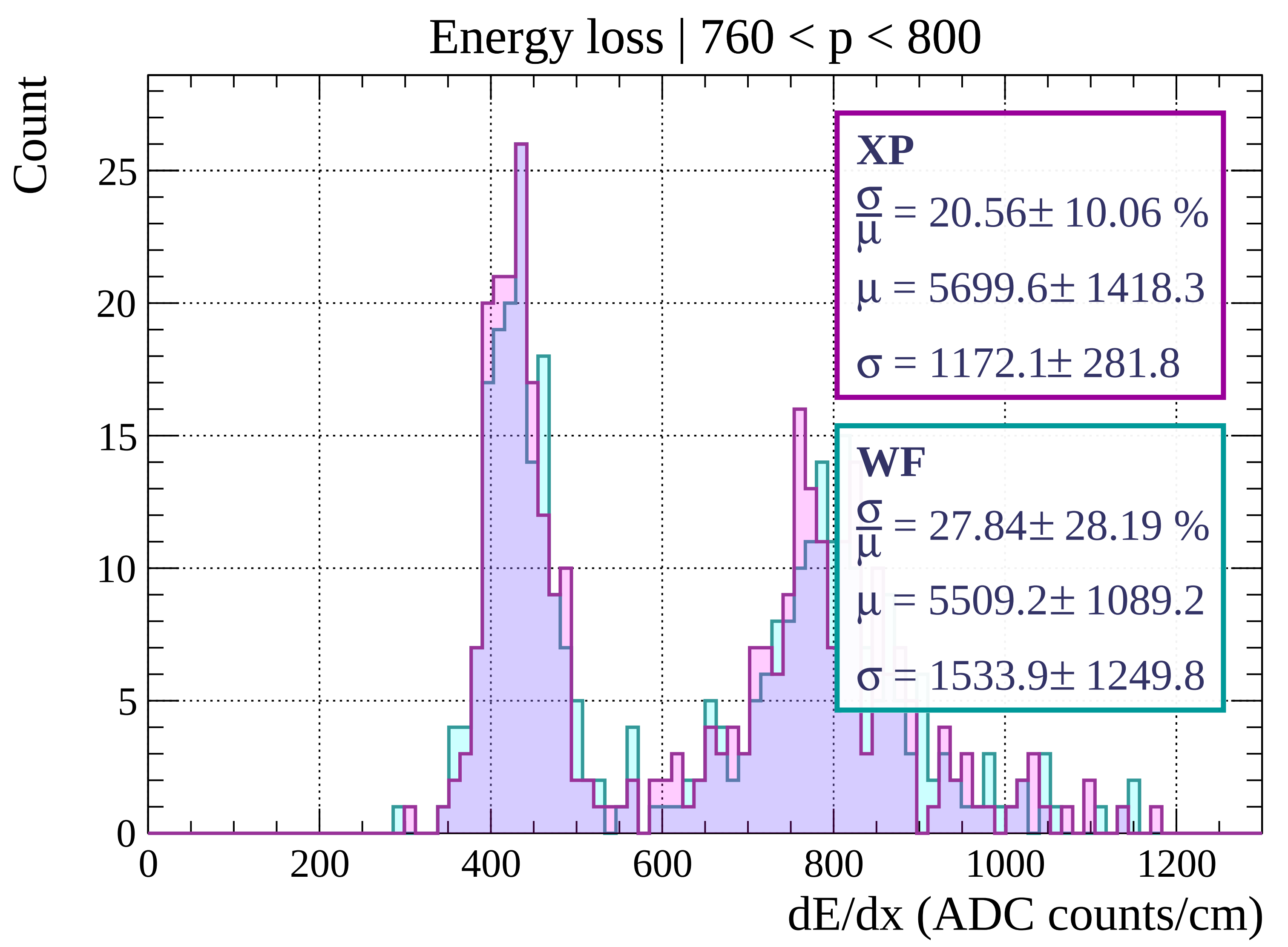
Energy loss | 600 < p < 640
Count
XP
σ/μ = 7.51± 0.61 %
μ = 416.2± 2.6
σ = 31.3± 2.3
WF
σ/μ = -47.97± -29.29 %
μ = -4003.4± 1494.0
σ = 1920.4± 1889.4
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45


Energy loss | 640 < p < 680
Count
XP
σ/μ = -4070.38± -623.91 %
μ = -40.6± 32.2
σ = 1651.8± 1563.3
WF
σ/μ = -51.29± -29.12 %
μ = -3830.7± 1472.1
σ = 1964.7± 1870.3
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200

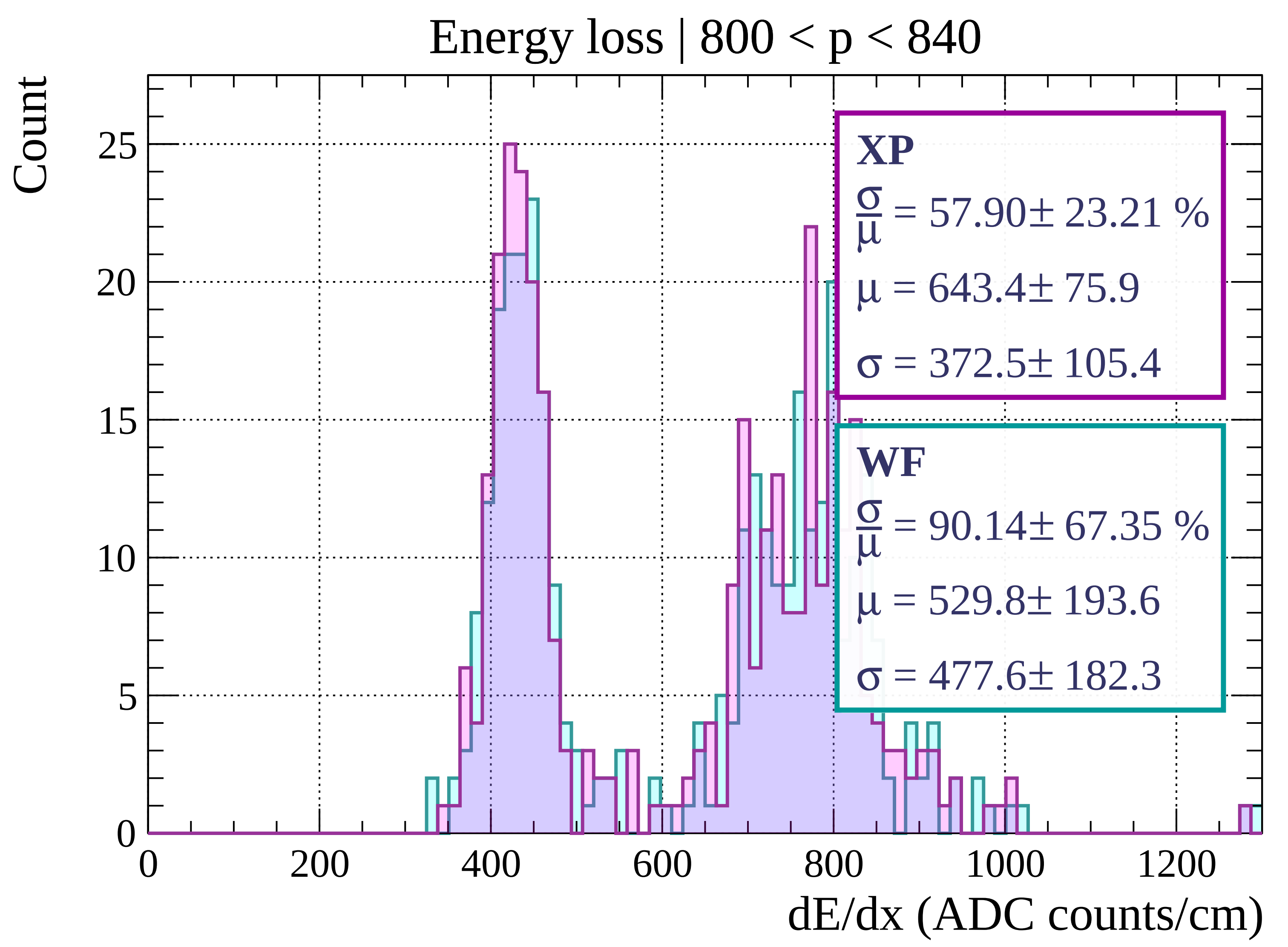

Energy loss | 680 < p < 720
Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -41.98± -28.46 %
μ = -4038.5± 1137.0
σ = 1695.3± 1626.7
WF
σ/μ = 8.75± 0.84 %
μ = 917.3± 7.6
σ = 80.2± 7.1



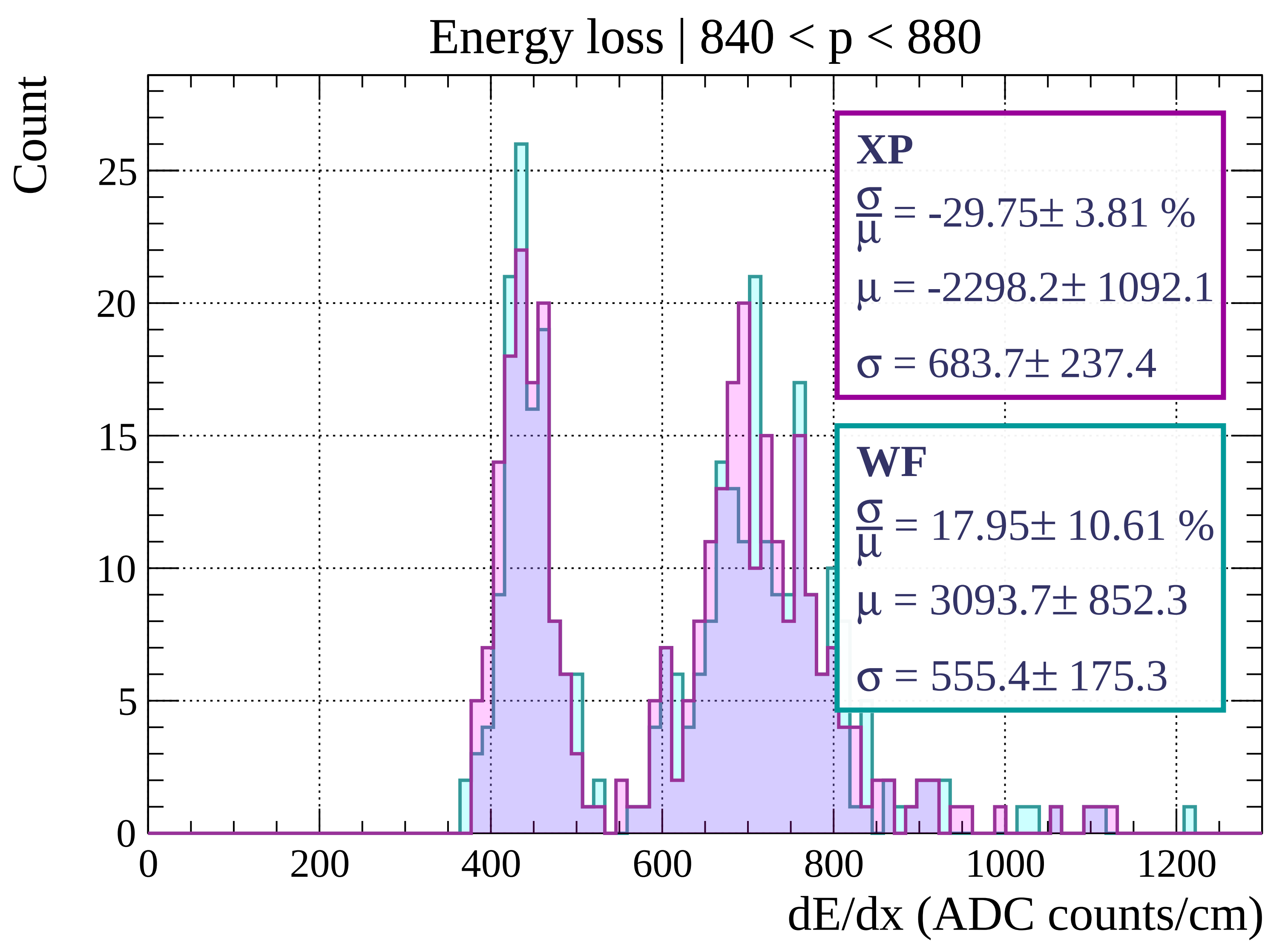
Energy loss | 720 < p < 760
Count
XP
$\frac{\sigma}{\mu}$ = 20.90± 6.49 %
μ = 7549.4± 1264.3
σ = 1577.6± 225.6
WF
$\frac{\sigma}{\mu}$ = 17.67± 11.81 %
μ = 3485.4± 1042.2
σ = 615.7± 227.4
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)

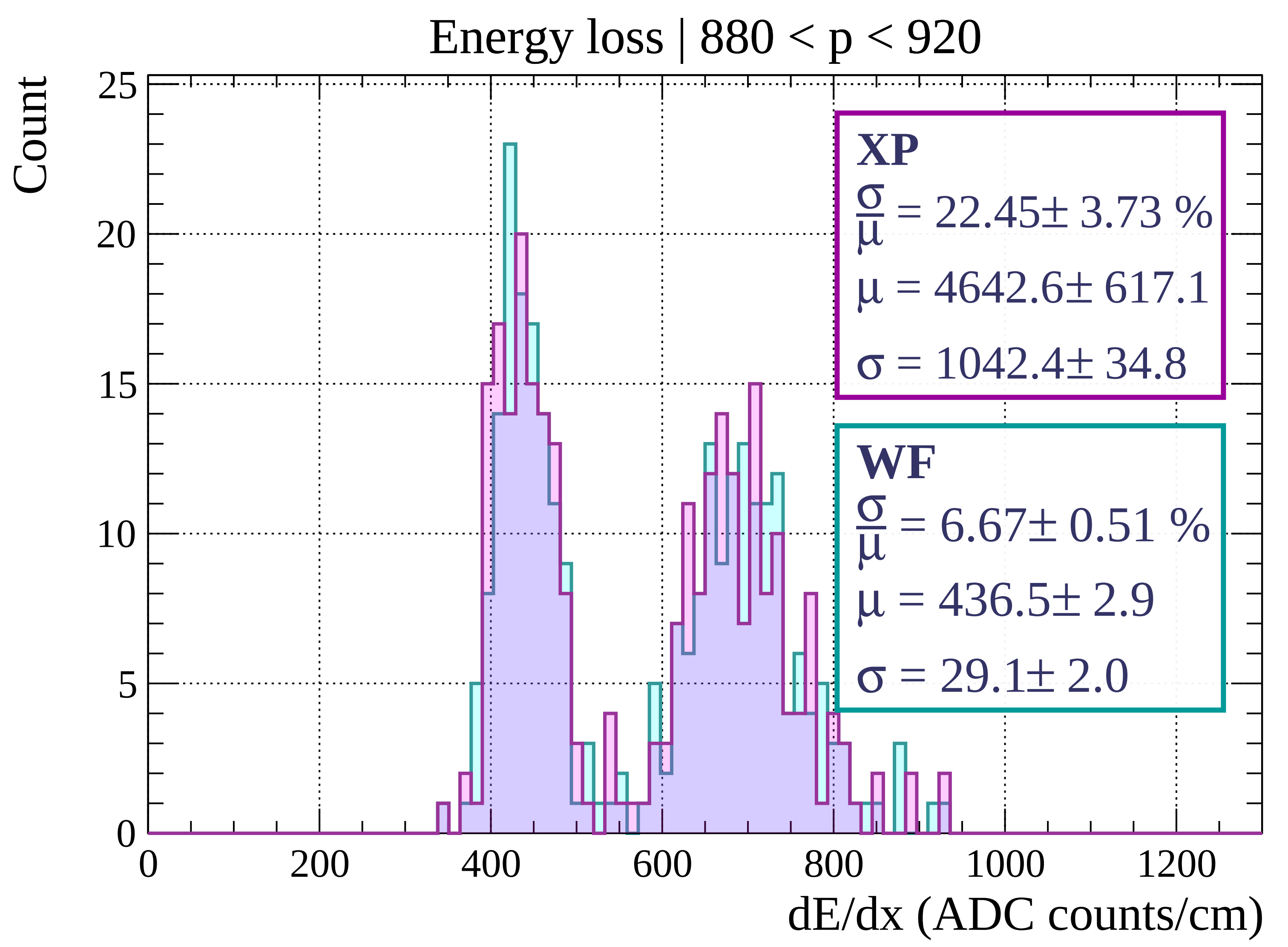


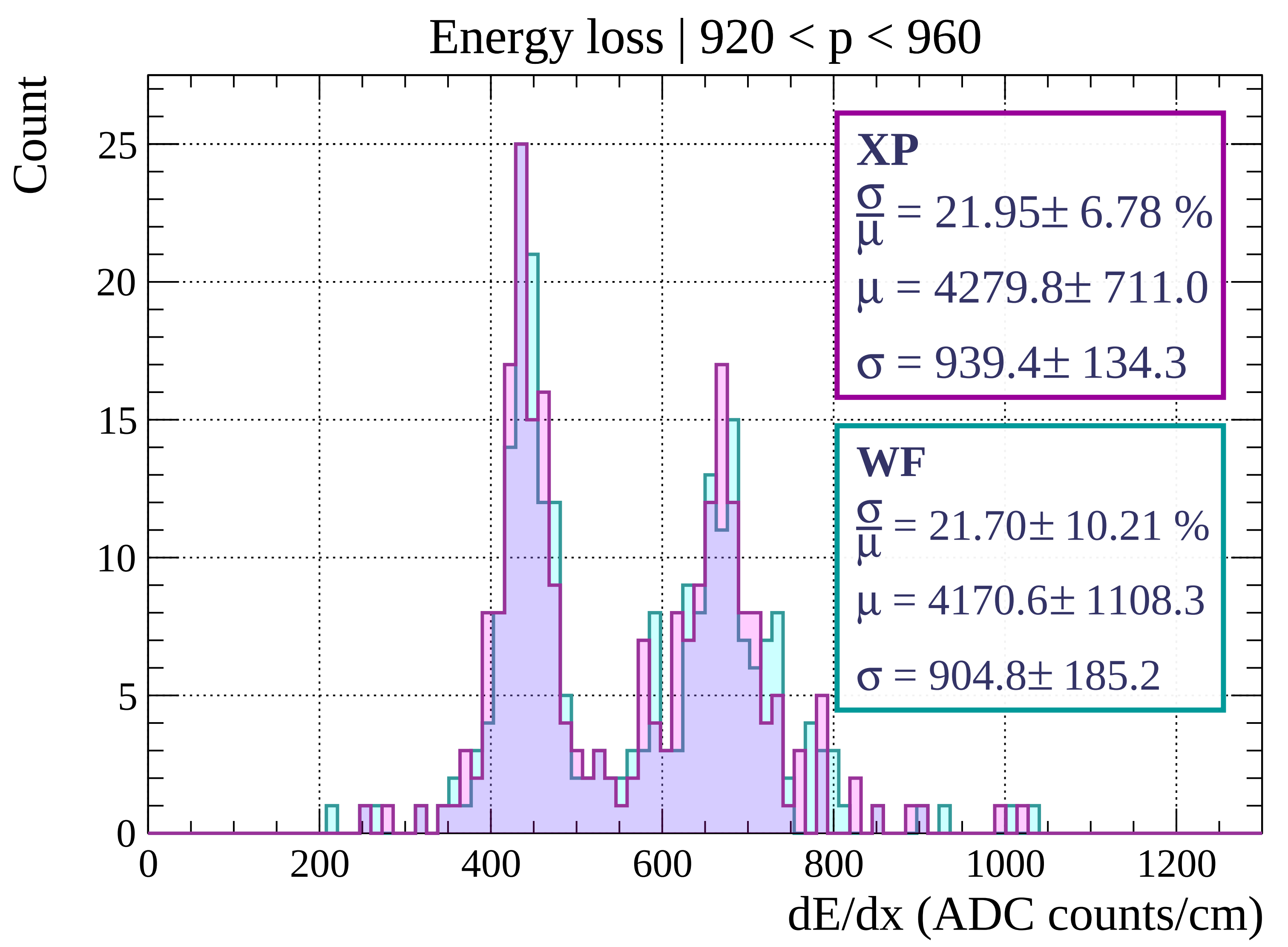
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 20.56± 10.06 %
μ = 5699.6± 1418.3
σ = 1172.1± 281.8
WF
σ/μ = 27.84± 28.19 %
μ = 5509.2± 1089.2
σ = 1533.9± 1249.8



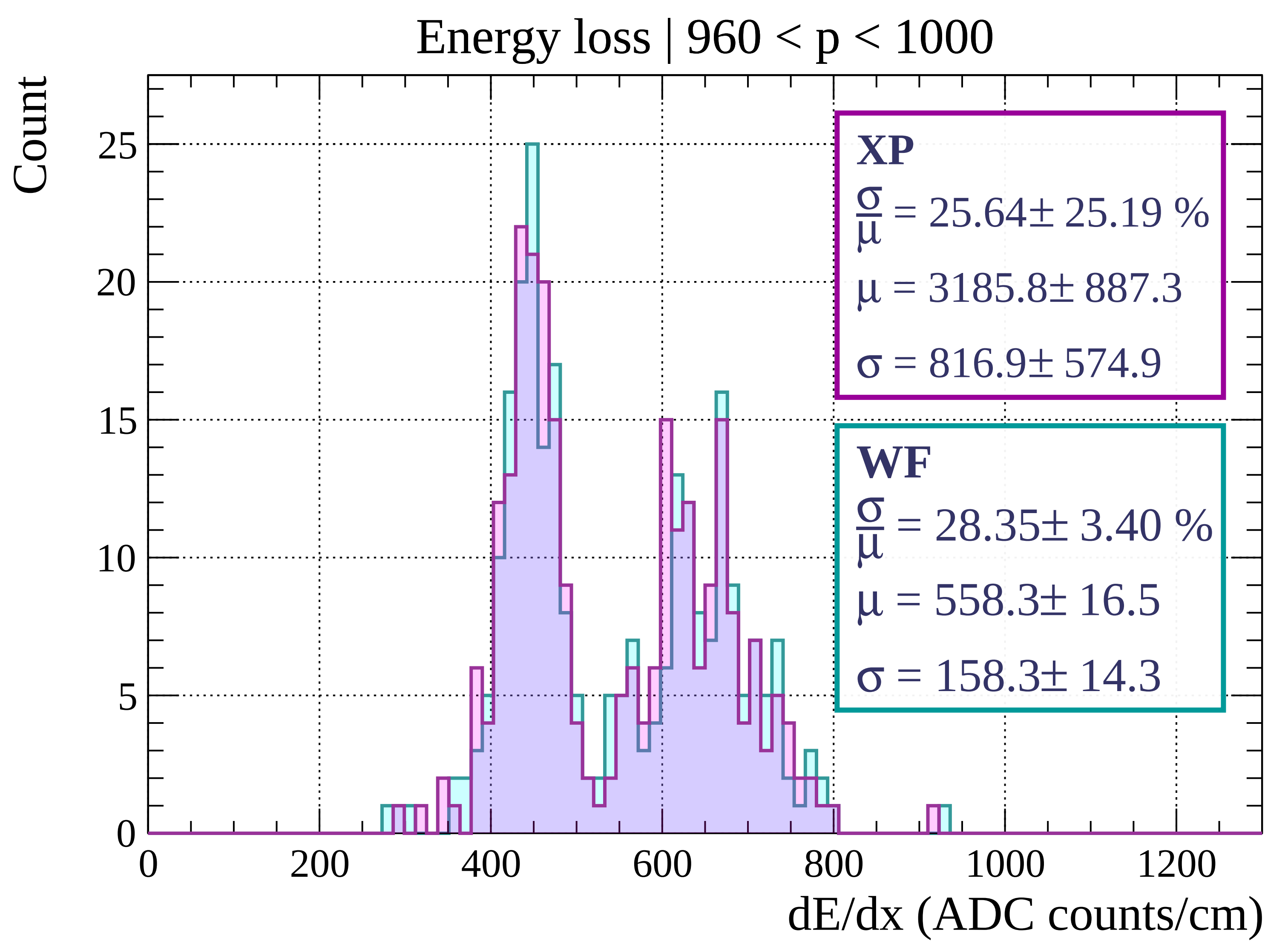
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 57.90± 23.21 %
μ = 643.4± 75.9
σ = 372.5± 105.4
WF
$\frac{\sigma}{\mu}$ = 90.14± 67.35 %
μ = 529.8± 193.6
σ = 477.6± 182.3

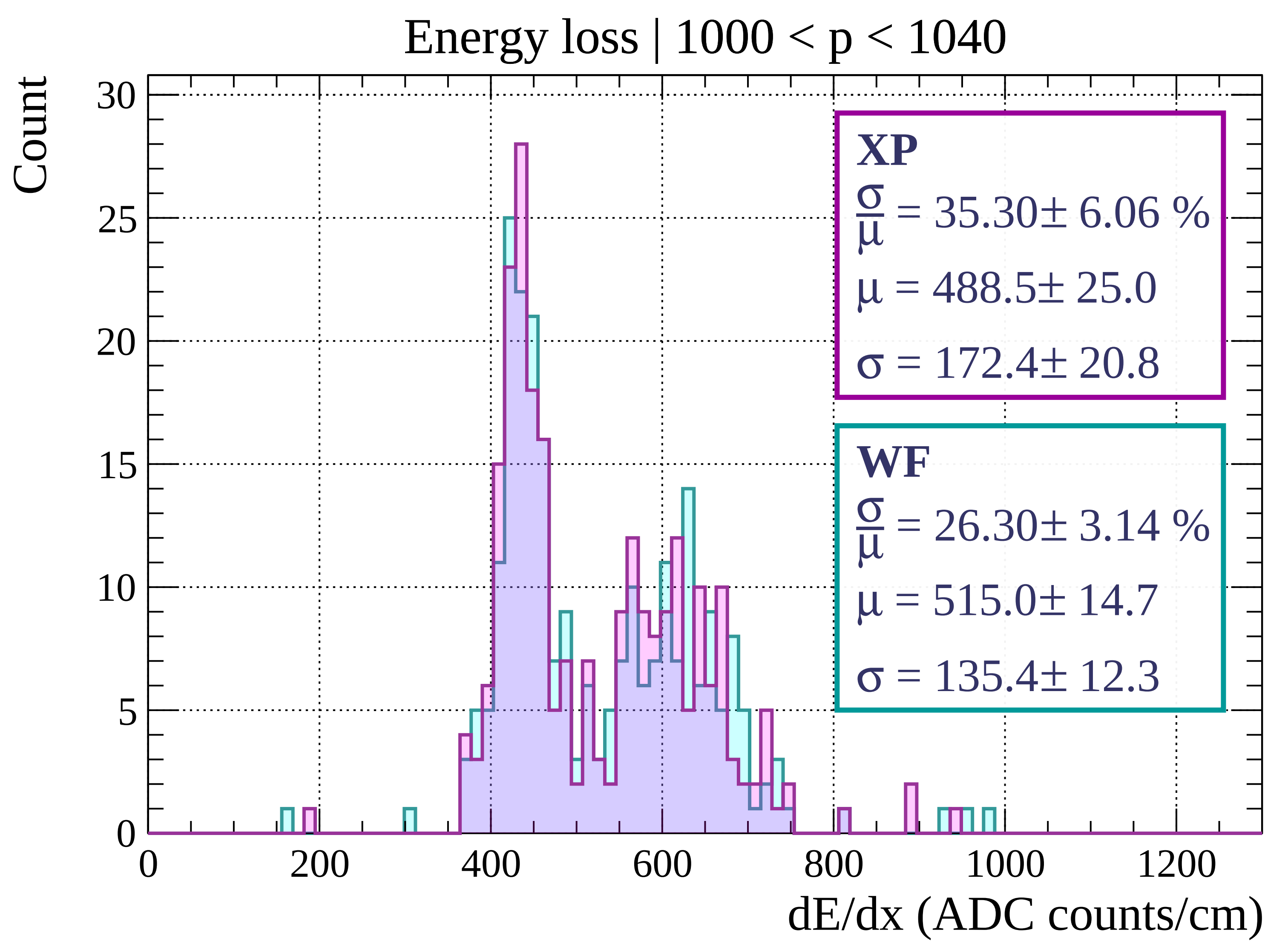

Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = -29.75± 3.81 %
μ = -2298.2± 1092.1
σ = 683.7± 237.4
WF
$\frac{\sigma}{\mu}$ = 17.95± 10.61 %
μ = 3093.7± 852.3
σ = 555.4± 175.3


Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 22.45± 3.73 %
μ = 4642.6± 617.1
σ = 1042.4± 34.8
WF
σ/μ = 6.67± 0.51 %
μ = 436.5± 2.9
σ = 29.1± 2.0

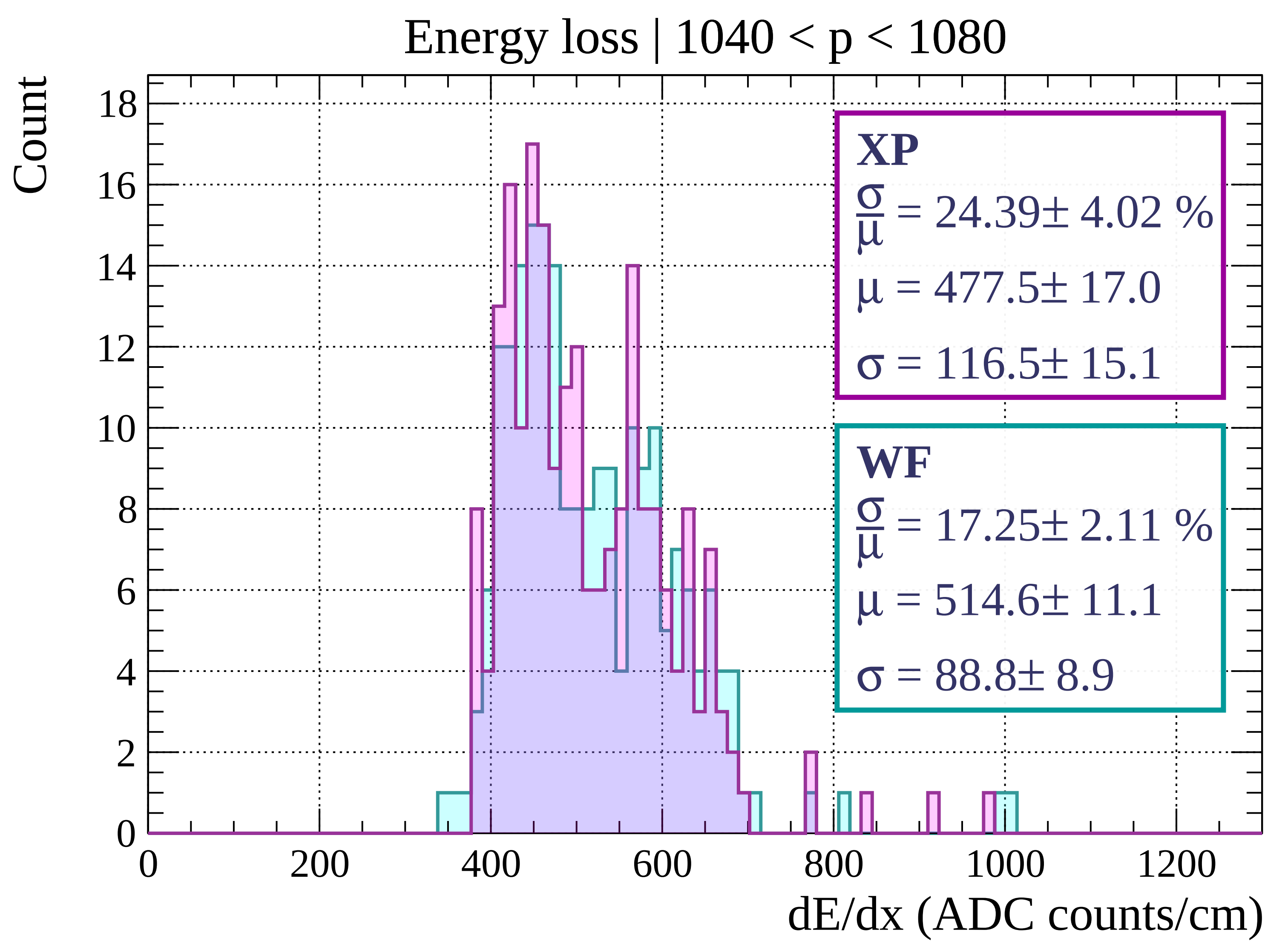


Energy loss | 920 < p < 960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
$\frac{\sigma}{\mu}$ = 21.95± 6.78 %
μ = 4279.8± 711.0
σ = 939.4± 134.3
WF
$\frac{\sigma}{\mu}$ = 21.70± 10.21 %
μ = 4170.6± 1108.3
σ = 904.8± 185.2

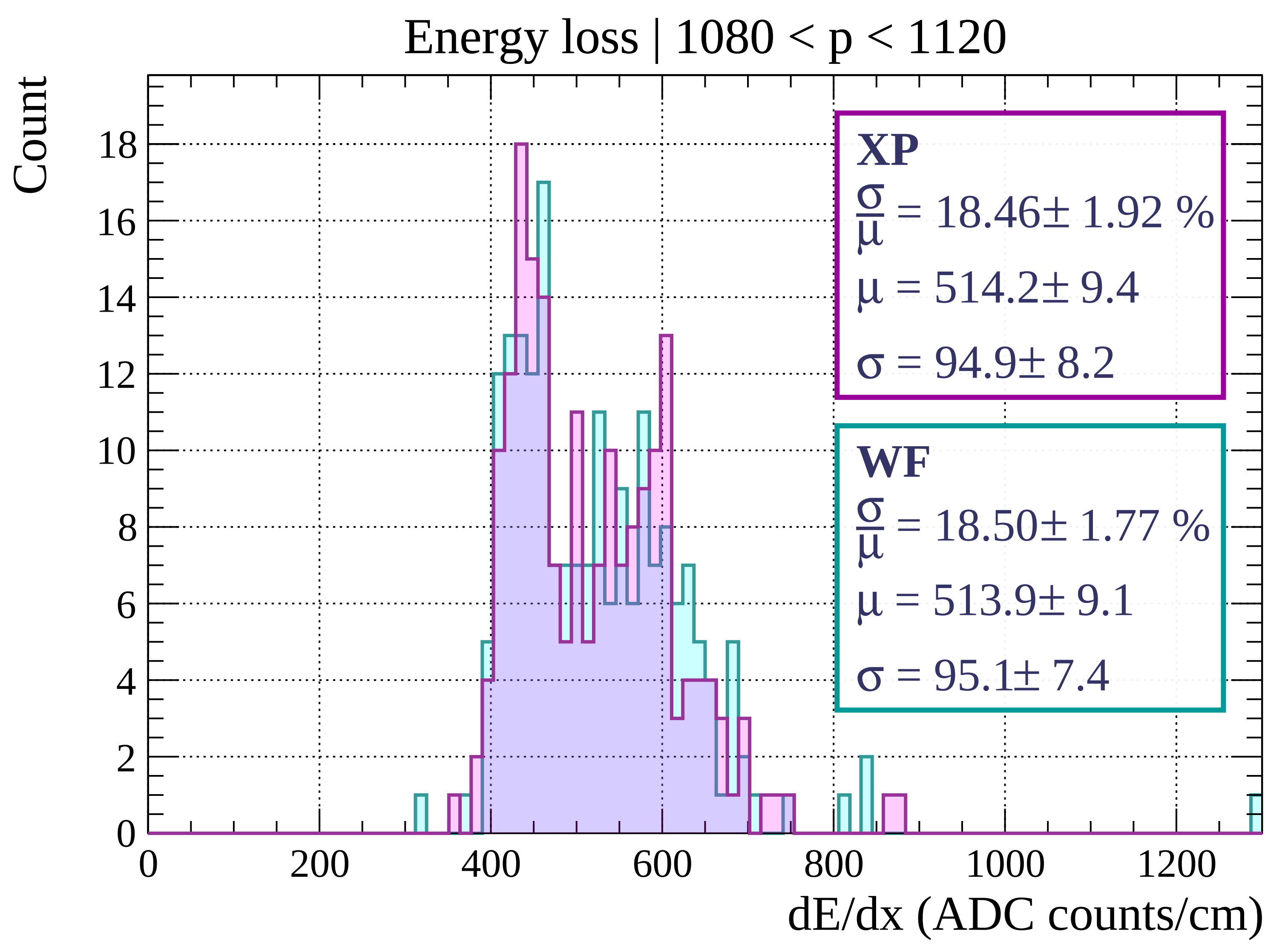

Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 25.64± 25.19 %
μ = 3185.8± 887.3
σ = 816.9± 574.9
WF
σ/μ = 28.35± 3.40 %
μ = 558.3± 16.5
σ = 158.3± 14.3



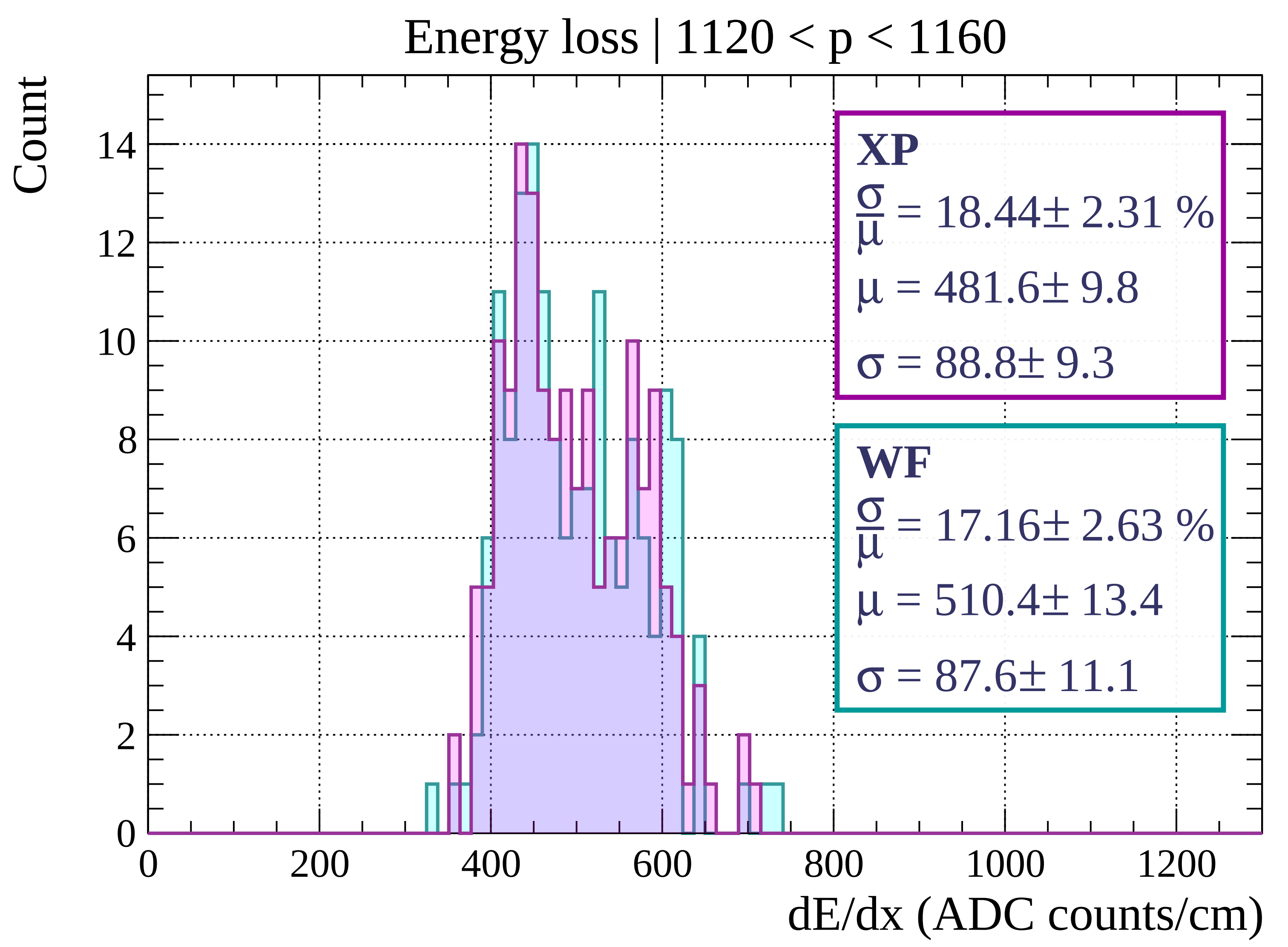
Energy loss | 1000 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
XP
σ/μ = 35.30± 6.06 %
μ = 488.5± 25.0
σ = 172.4± 20.8
WF
σ/μ = 26.30± 3.14 %
μ = 515.0± 14.7
σ = 135.4± 12.3



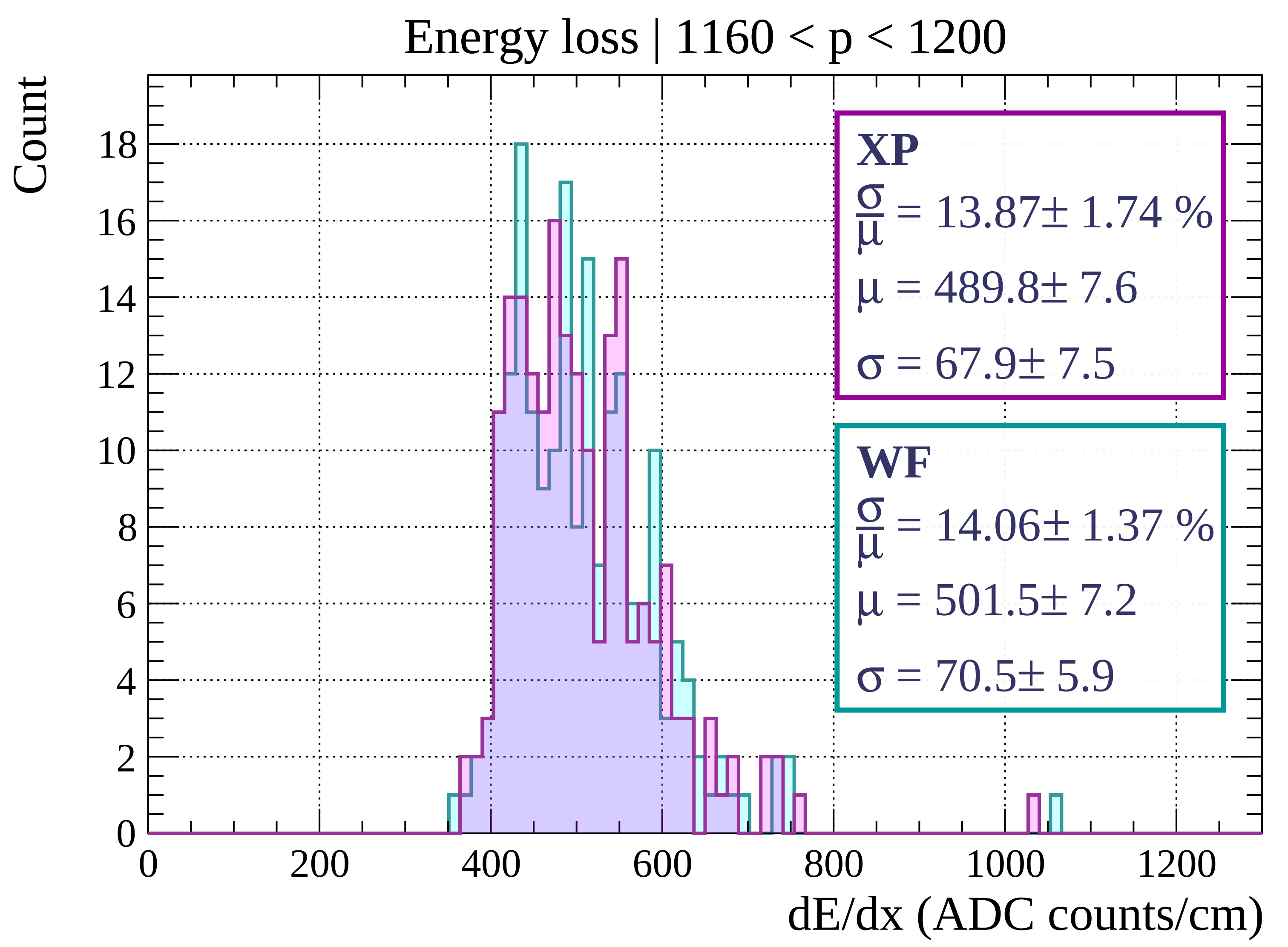
Energy loss | 1040 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 24.39± 4.02 %
μ = 477.5± 17.0
σ = 116.5± 15.1
WF
σ/μ = 17.25± 2.11 %
μ = 514.6± 11.1
σ = 88.8± 8.9

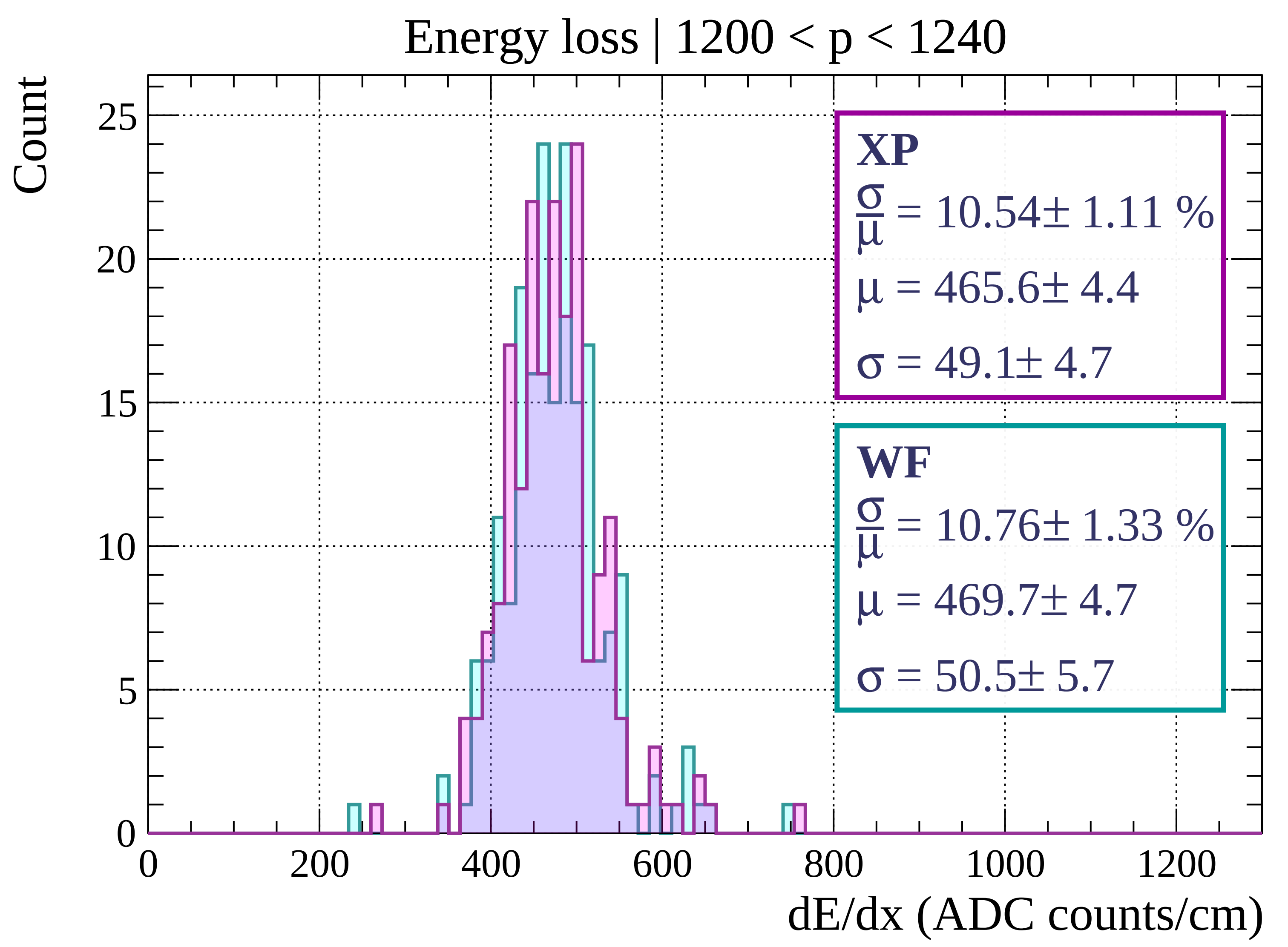


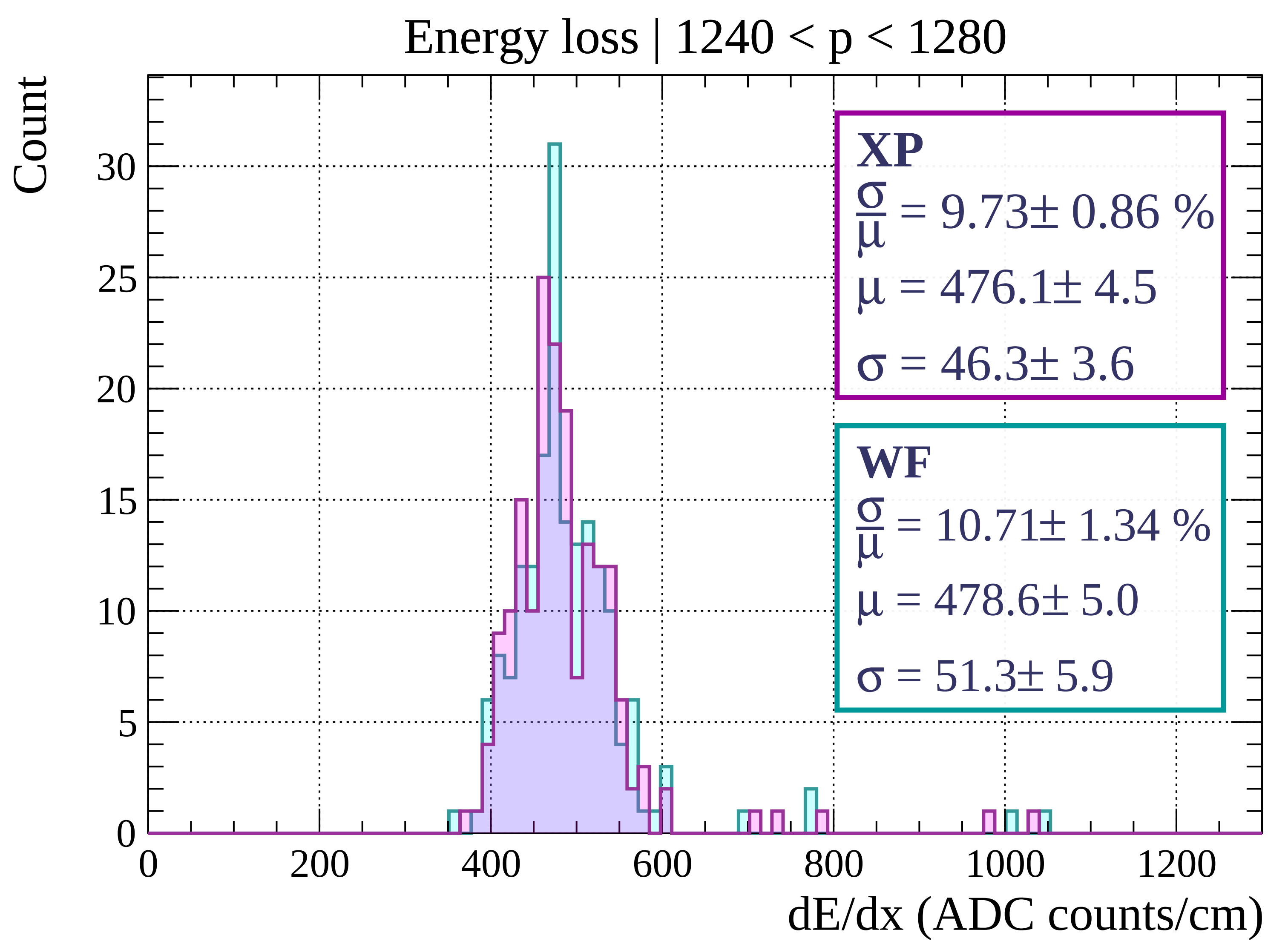
Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 18.46± 1.92 %
μ = 514.2± 9.4
σ = 94.9± 8.2
WF
σ/μ = 18.50± 1.77 %
μ = 513.9± 9.1
σ = 95.1± 7.4



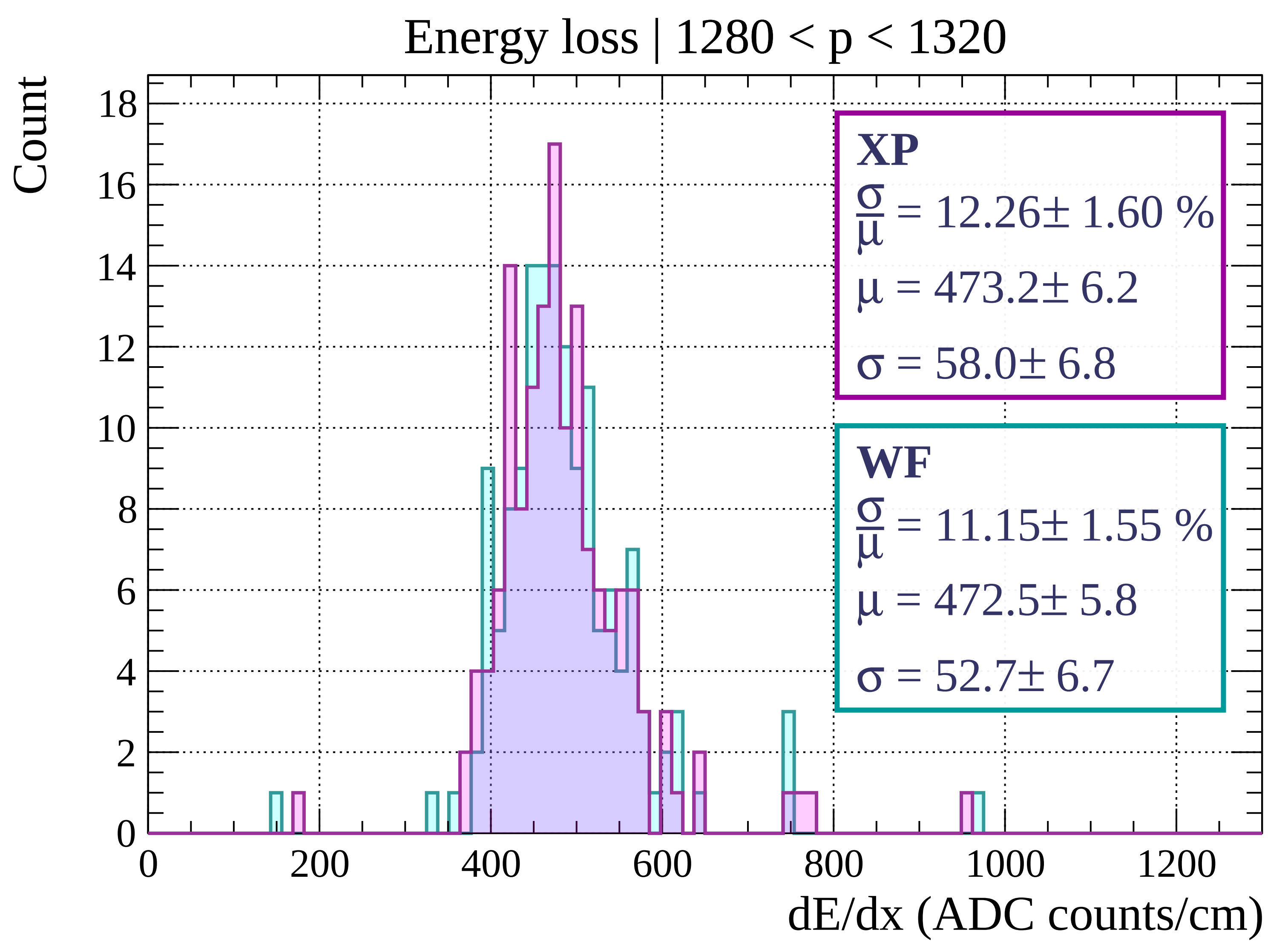
Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
XP
σ/μ = 18.44± 2.31 %
μ = 481.6± 9.8
σ = 88.8± 9.3
WF
σ/μ = 17.16± 2.63 %
μ = 510.4± 13.4
σ = 87.6± 11.1

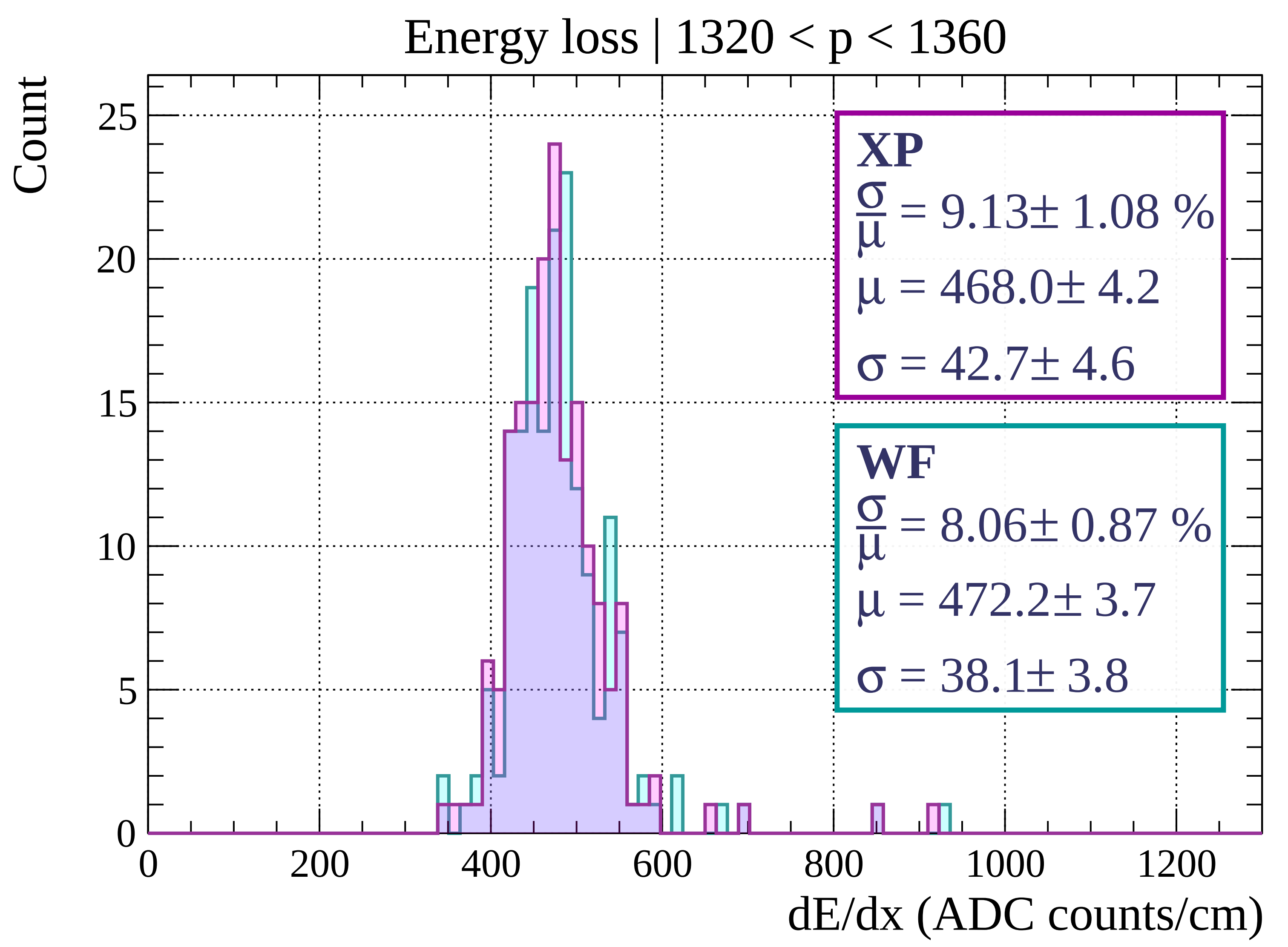

Energy loss | 1160 < p < 1200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 13.87± 1.74 %
μ = 489.8± 7.6
σ = 67.9± 7.5
WF
σ/μ = 14.06± 1.37 %
μ = 501.5± 7.2
σ = 70.5± 5.9
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200

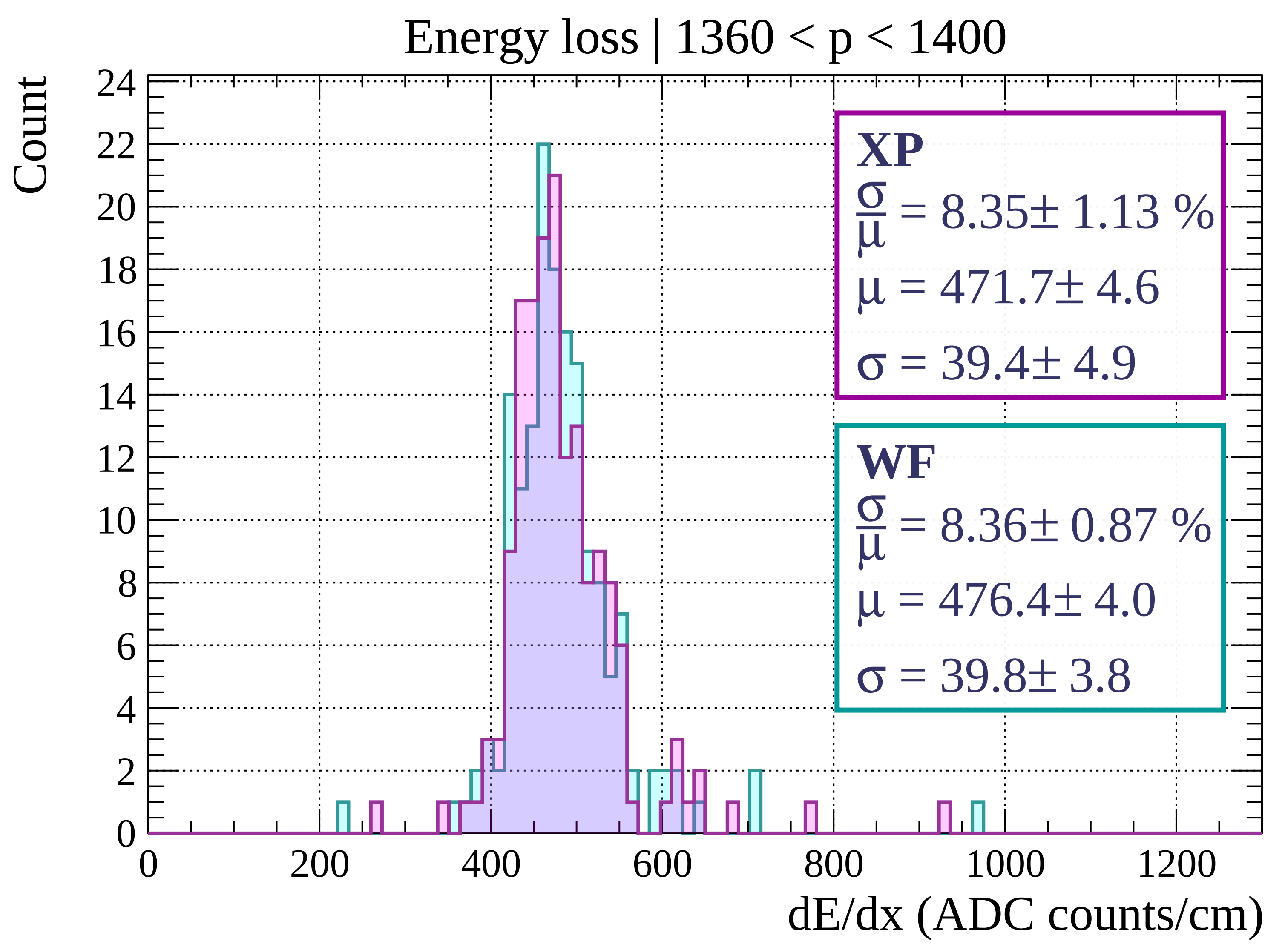

Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
XP
σ/μ = 10.54± 1.11 %
μ = 465.6± 4.4
σ = 49.1± 4.7
WF
σ/μ = 10.76± 1.33 %
μ = 469.7± 4.7
σ = 50.5± 5.7

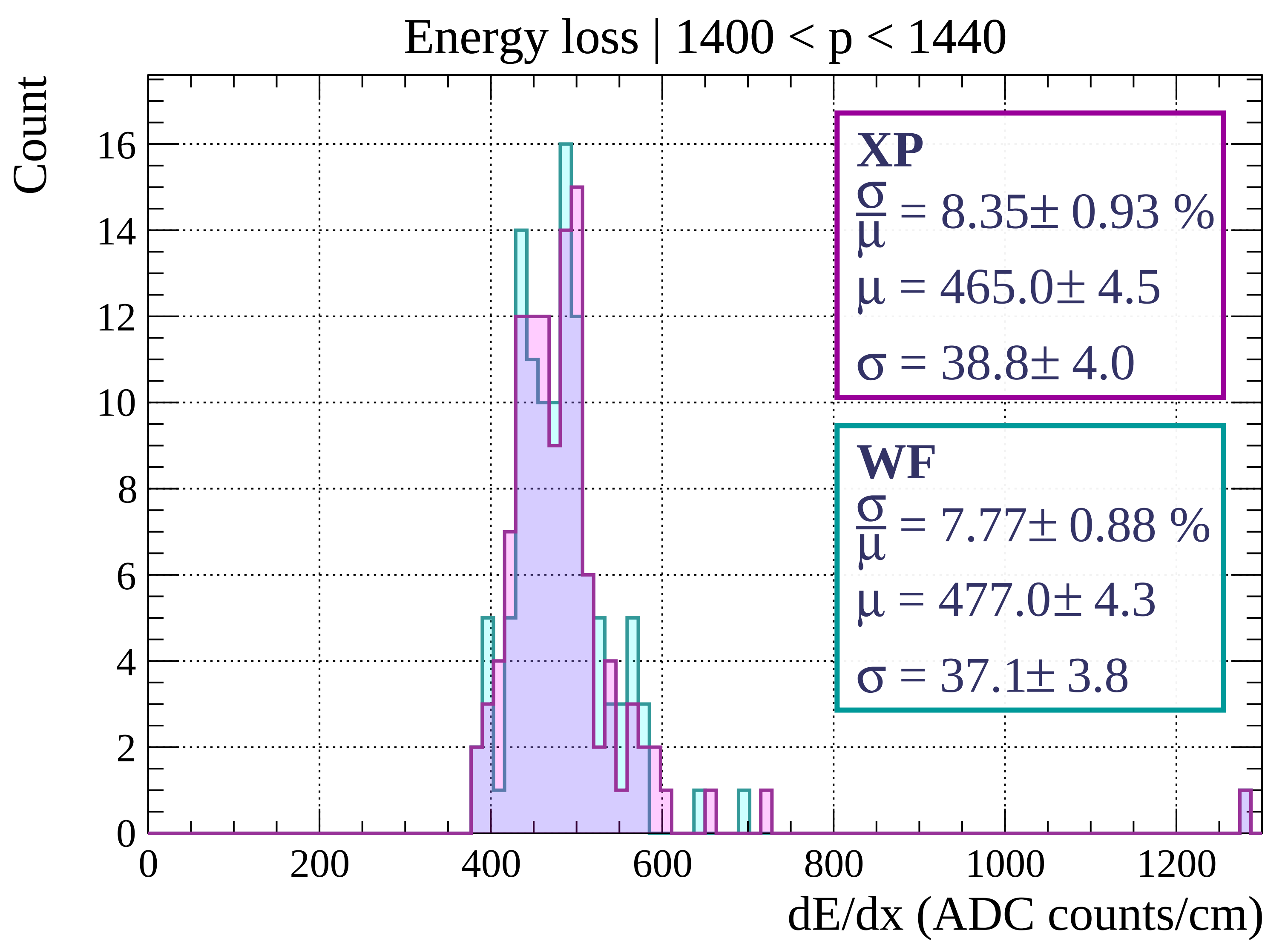


Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 9.73± 0.86 %
μ = 476.1± 4.5
σ = 46.3± 3.6
WF
σ/μ = 10.71± 1.34 %
μ = 478.6± 5.0
σ = 51.3± 5.9

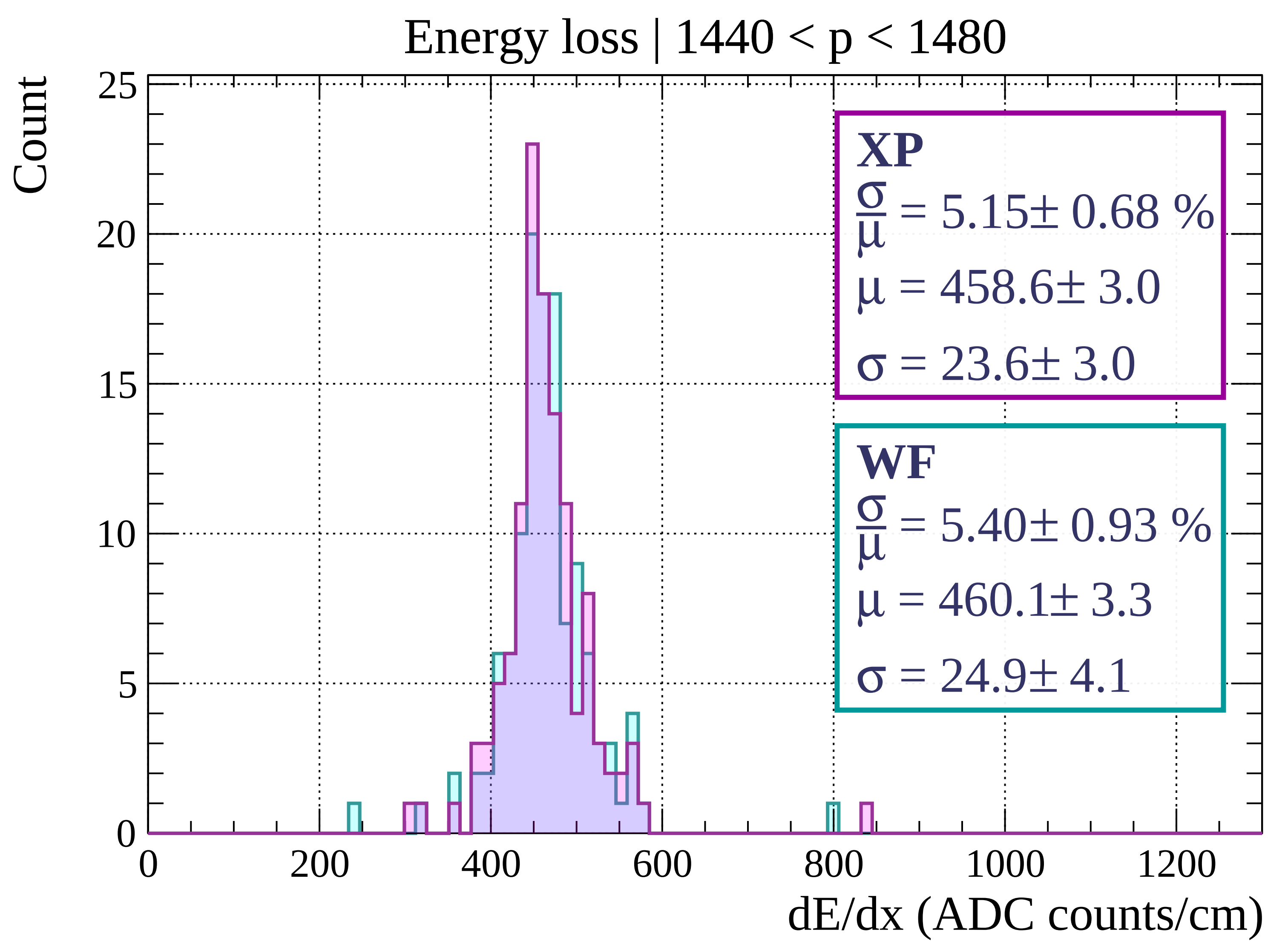

Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
XP
σ/μ = 12.26± 1.60 %
μ = 473.2± 6.2
σ = 58.0± 6.8
WF
σ/μ = 11.15± 1.55 %
μ = 472.5± 5.8
σ = 52.7± 6.7


Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
XP
σ/μ = 9.13± 1.08 %
μ = 468.0± 4.2
σ = 42.7± 4.6
WF
σ/μ = 8.06± 0.87 %
μ = 472.2± 3.7
σ = 38.1± 3.8

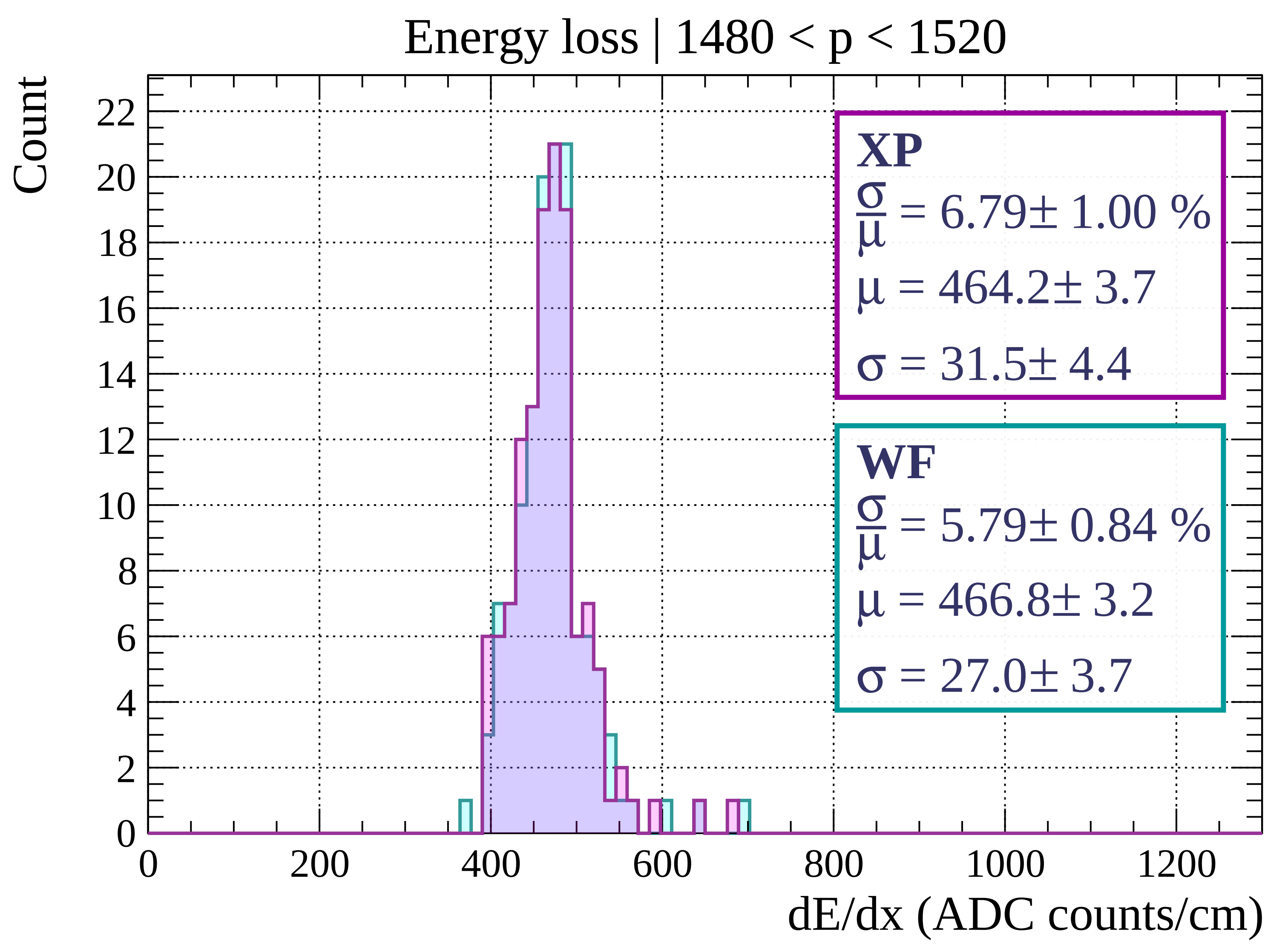

Energy loss | 1360 < p < 1400
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
24
XP
σ/μ = 8.35± 1.13 %
μ = 471.7± 4.6
σ = 39.4± 4.9
WF
σ/μ = 8.36± 0.87 %
μ = 476.4± 4.0
σ = 39.8± 3.8

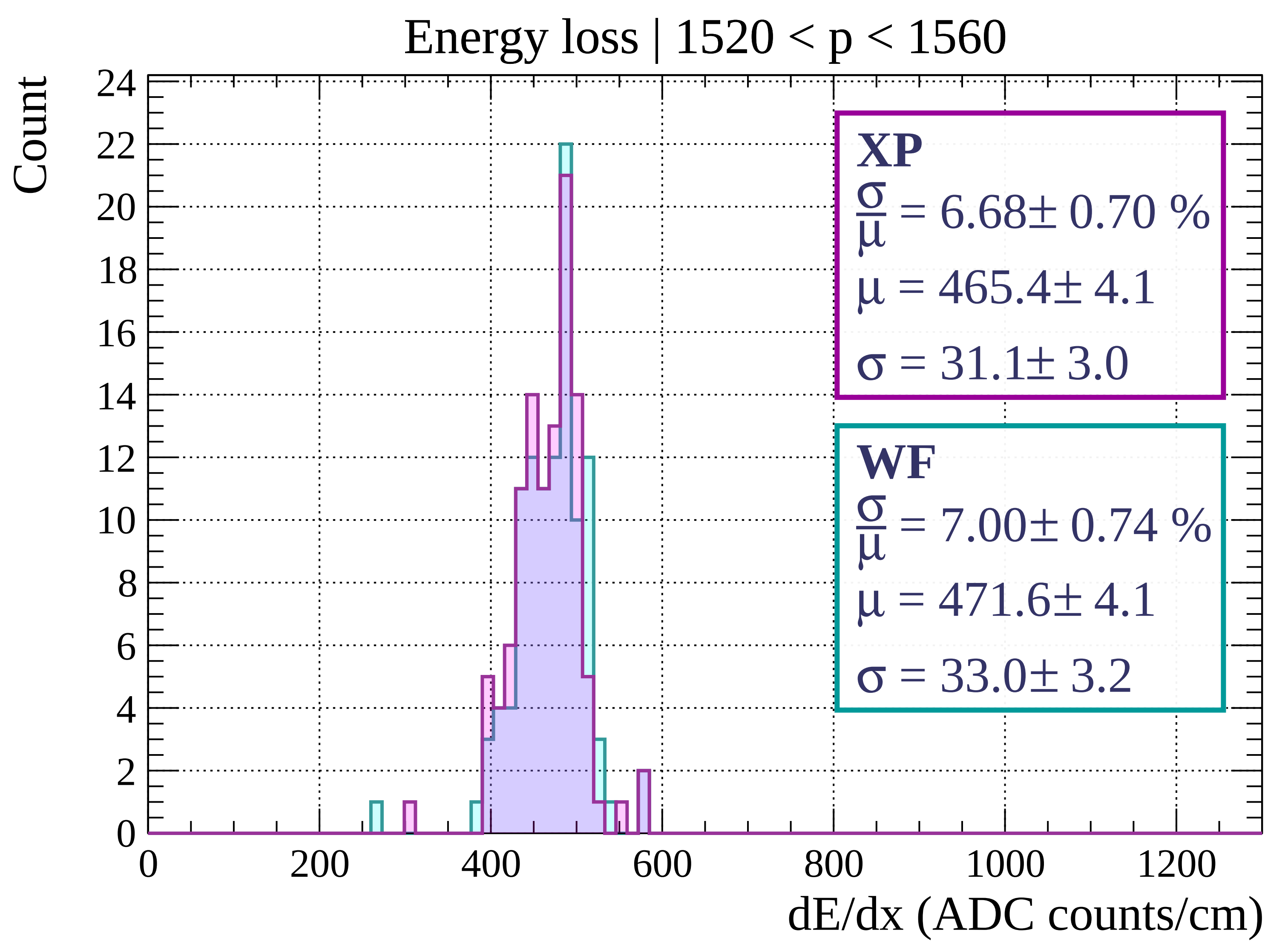


Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.35± 0.93 %
μ = 465.0± 4.5
σ = 38.8± 4.0
WF
$\frac{\sigma}{\mu}$ = 7.77± 0.88 %
μ = 477.0± 4.3
σ = 37.1± 3.8

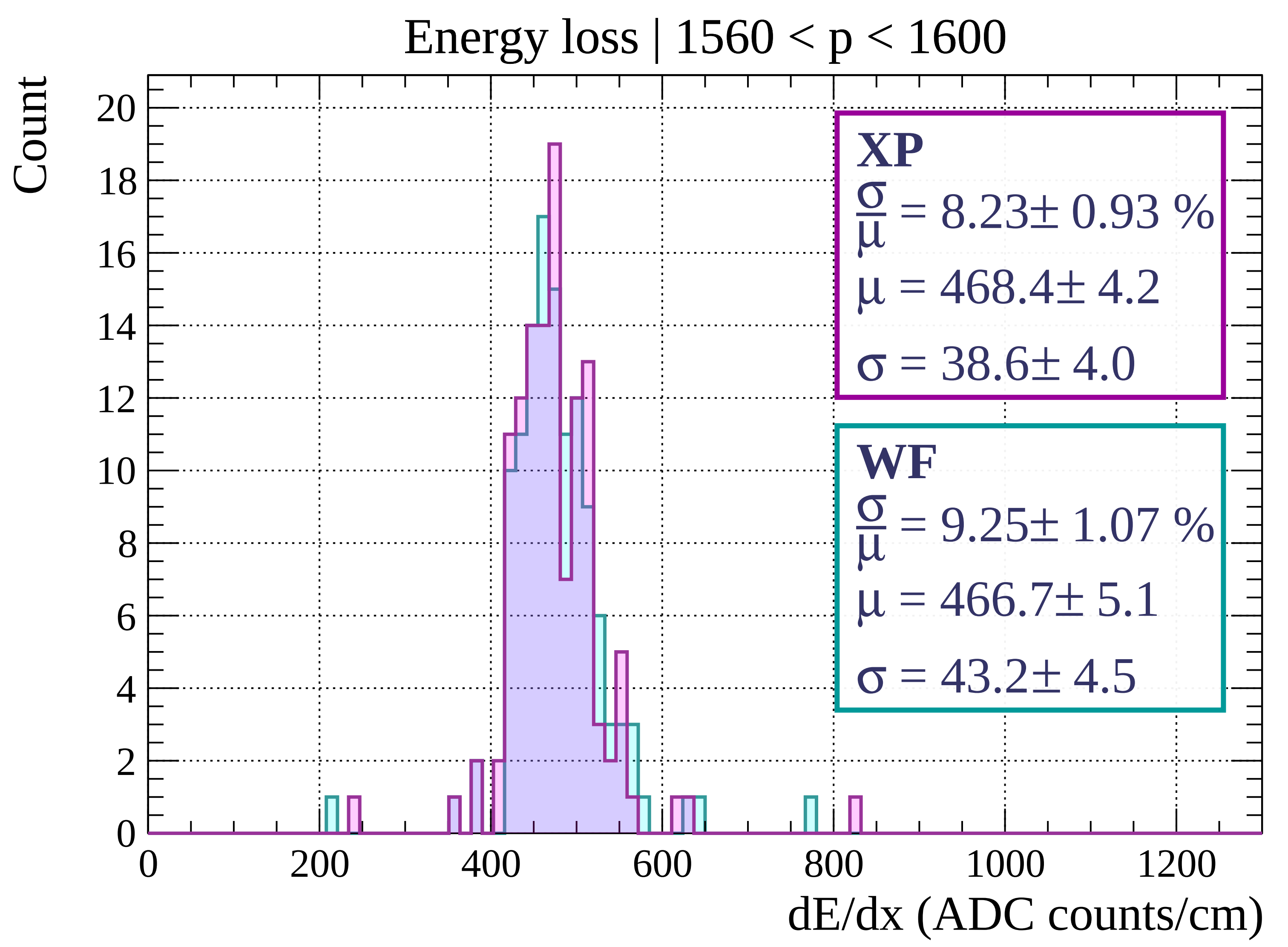


Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 5.15± 0.68 %
μ = 458.6± 3.0
σ = 23.6± 3.0
WF
σ/μ = 5.40± 0.93 %
μ = 460.1± 3.3
σ = 24.9± 4.1
0
200
400
600
800
1000
1200
5
10
15
20
25

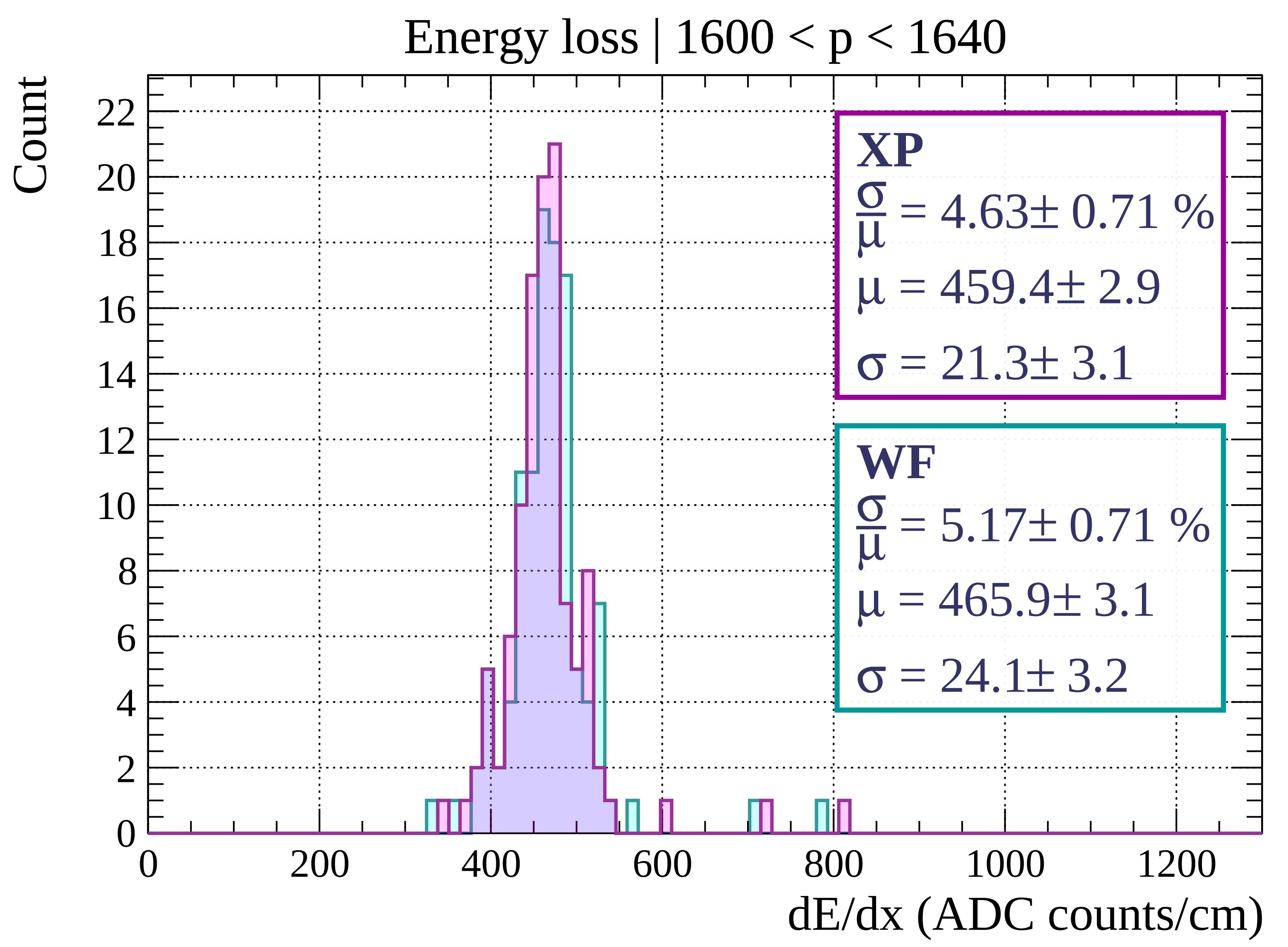

Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
22
XP
$\frac{\sigma}{\mu} = 6.79 \pm 1.00$ %
$\mu = 464.2 \pm 3.7$
$\sigma = 31.5 \pm 4.4$
WF
$\frac{\sigma}{\mu} = 5.79 \pm 0.84$ %
$\mu = 466.8 \pm 3.2$
$\sigma = 27.0 \pm 3.7$



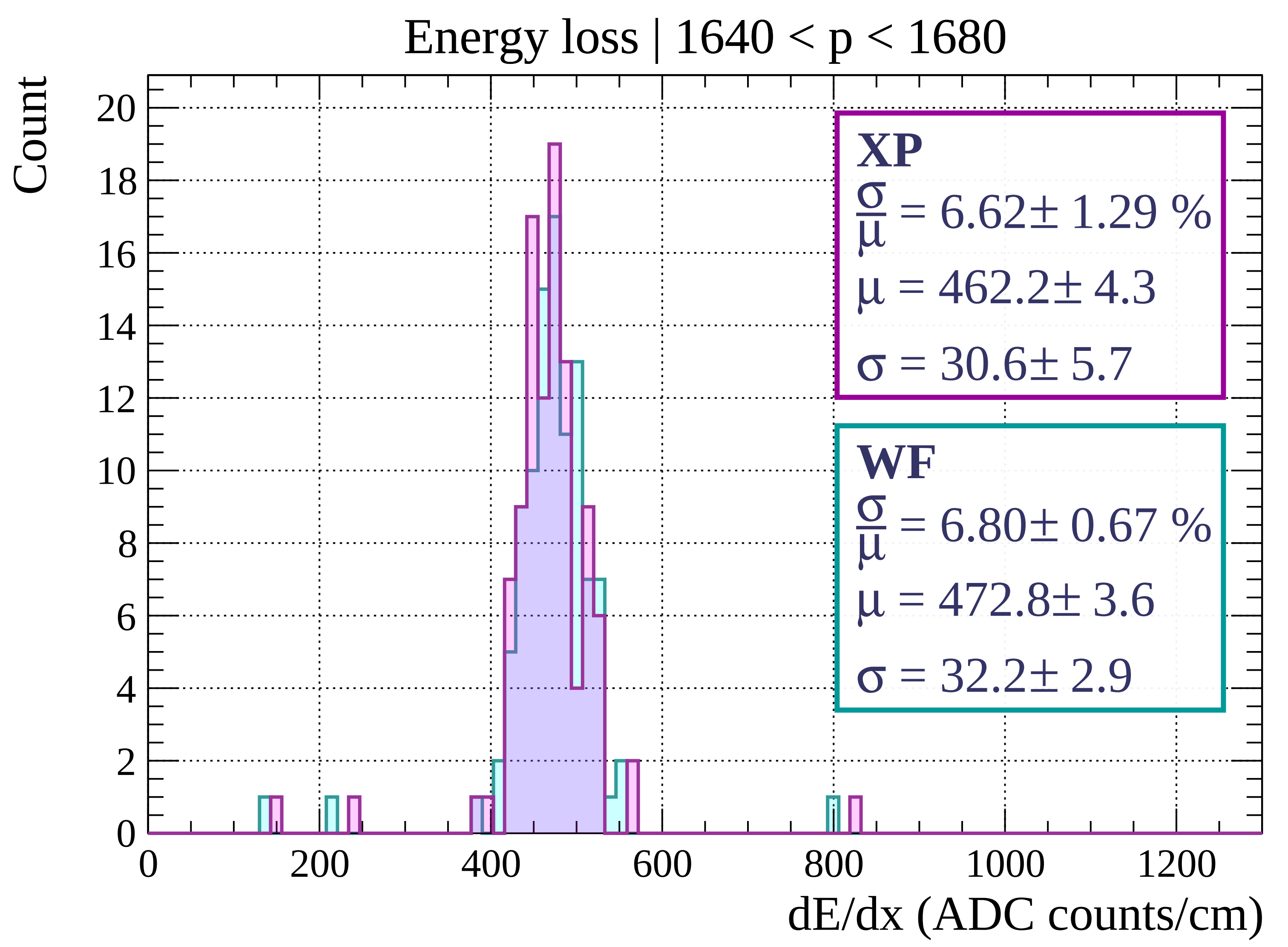
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
24
XP
$\frac{\sigma}{\mu}$ = 6.68± 0.70 %
μ = 465.4± 4.1
σ = 31.1± 3.0
WF
$\frac{\sigma}{\mu}$ = 7.00± 0.74 %
μ = 471.6± 4.1
σ = 33.0± 3.2

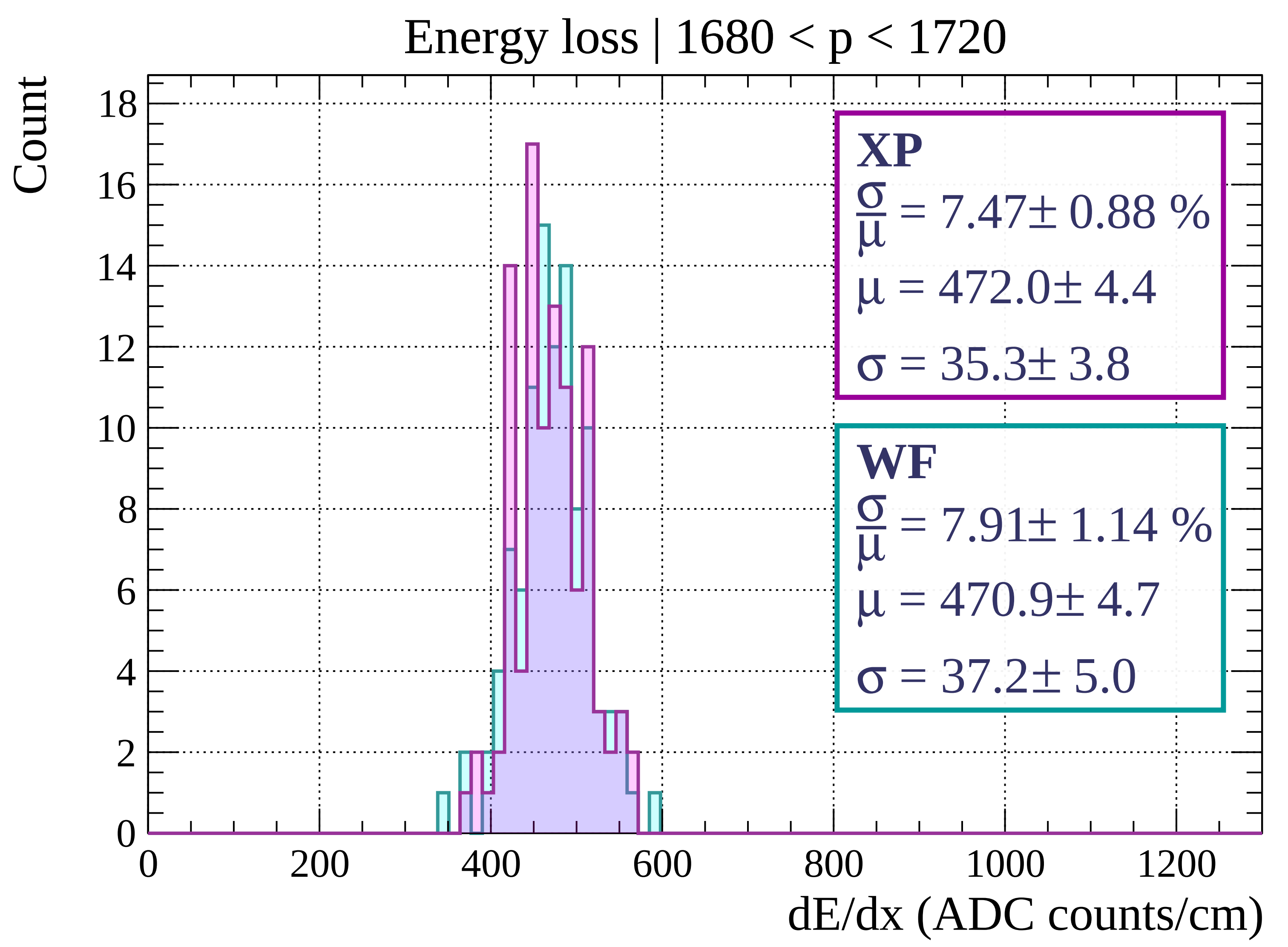


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
XP
$\frac{\sigma}{\mu}$ = 8.23± 0.93 %
μ = 468.4± 4.2
σ = 38.6± 4.0
WF
$\frac{\sigma}{\mu}$ = 9.25± 1.07 %
μ = 466.7± 5.1
σ = 43.2± 4.5

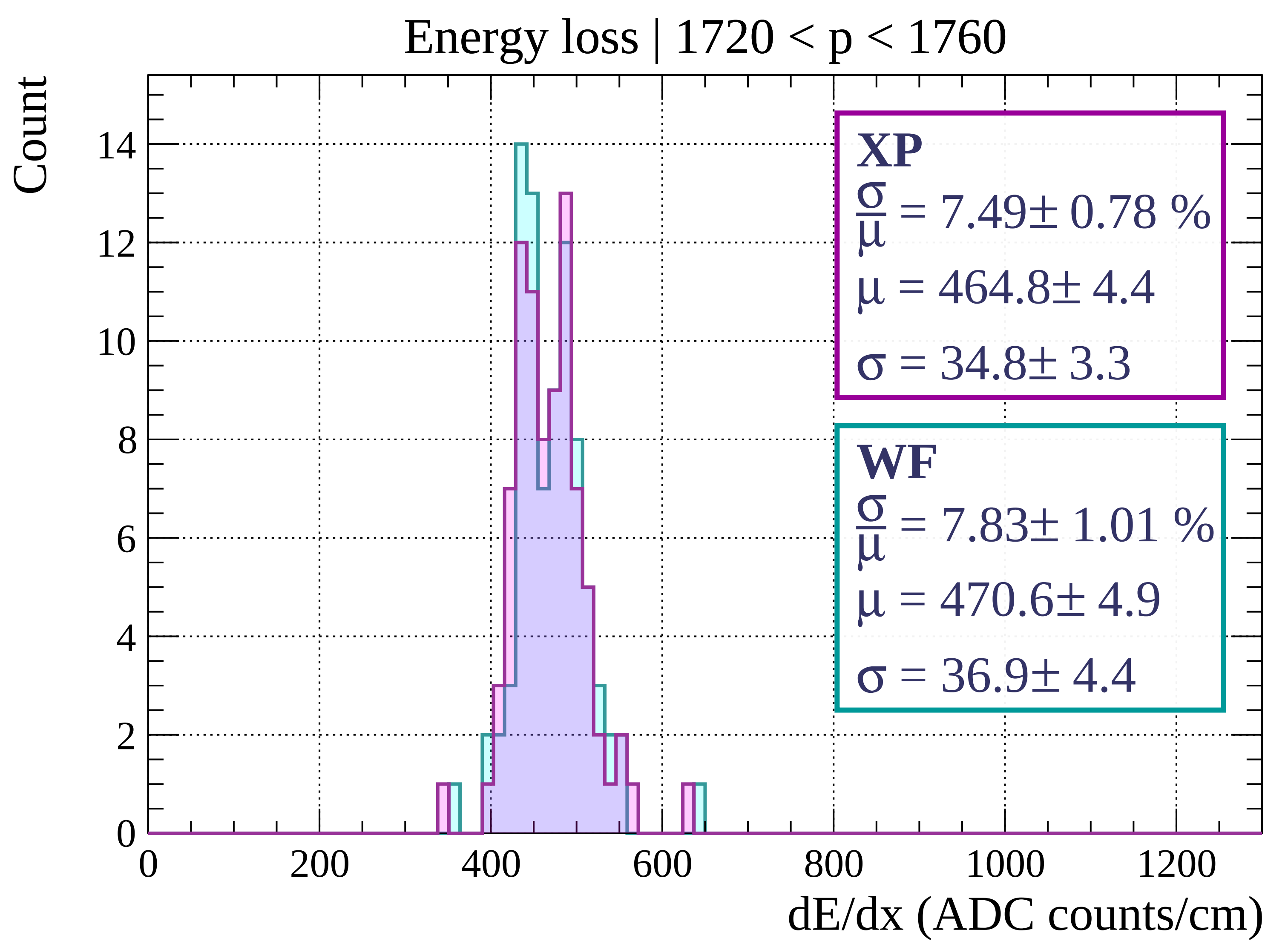


Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
XP
$\frac{\sigma}{\mu}$ = 4.63± 0.71 %
μ = 459.4± 2.9
σ = 21.3± 3.1
WF
$\frac{\sigma}{\mu}$ = 5.17± 0.71 %
μ = 465.9± 3.1
σ = 24.1± 3.2

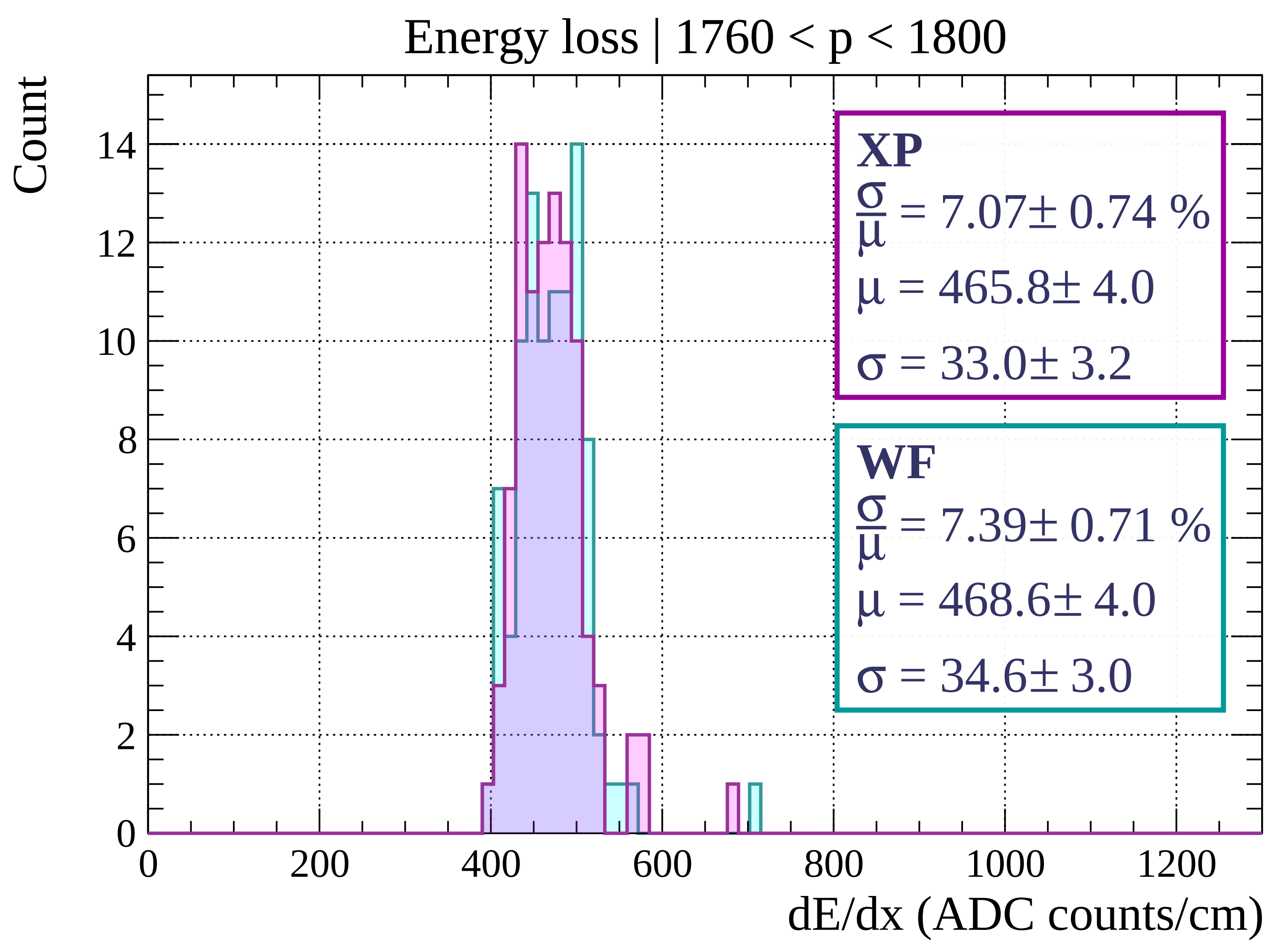

Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
XP
$\frac{\sigma}{\mu}$ = 6.62± 1.29 %
μ = 462.2± 4.3
σ = 30.6± 5.7
WF
$\frac{\sigma}{\mu}$ = 6.80± 0.67 %
μ = 472.8± 3.6
σ = 32.2± 2.9



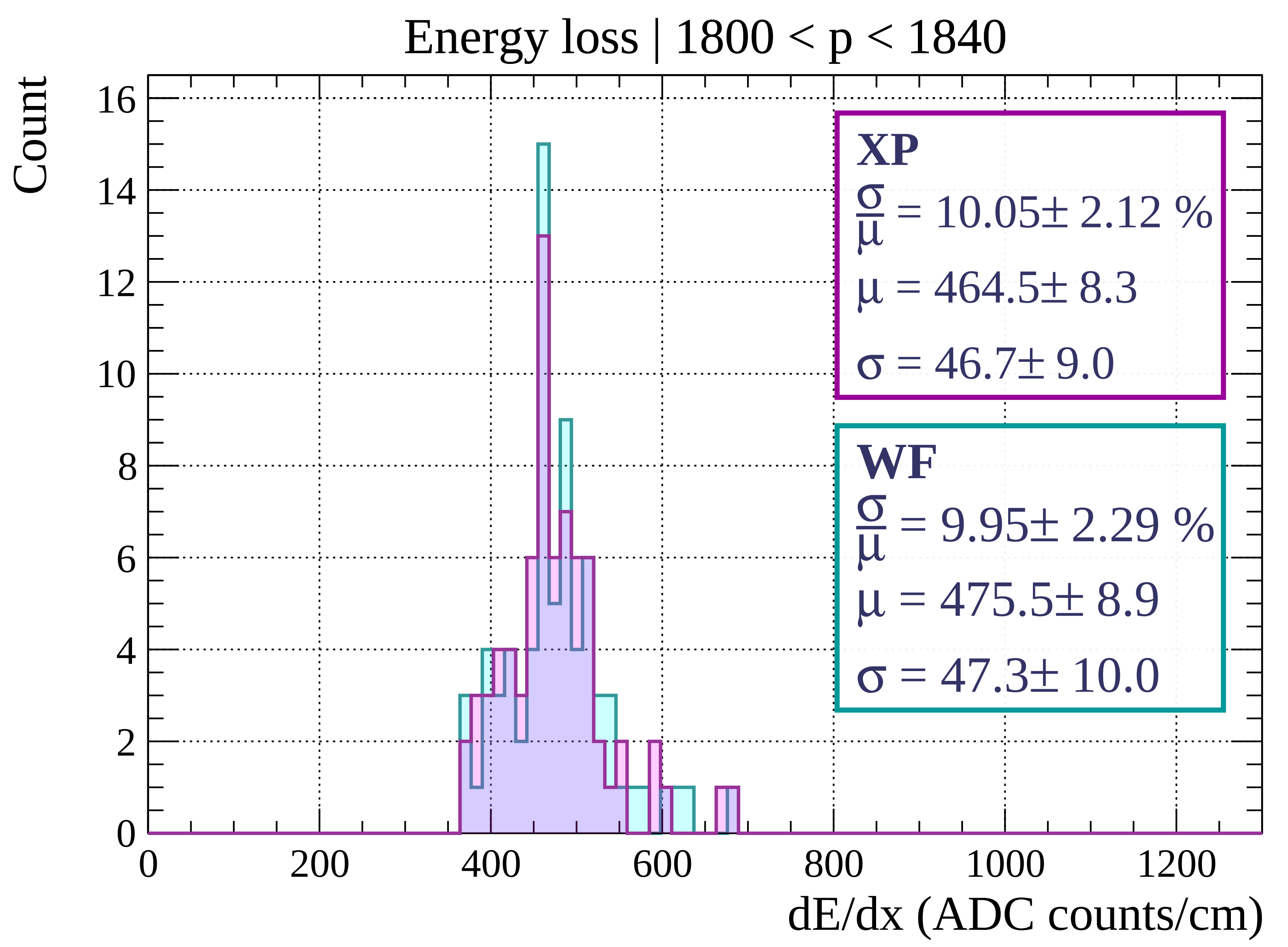
Energy loss | 1680 < p < 1720
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.47± 0.88 %
μ = 472.0± 4.4
σ = 35.3± 3.8
WF
$\frac{\sigma}{\mu}$ = 7.91± 1.14 %
μ = 470.9± 4.7
σ = 37.2± 5.0



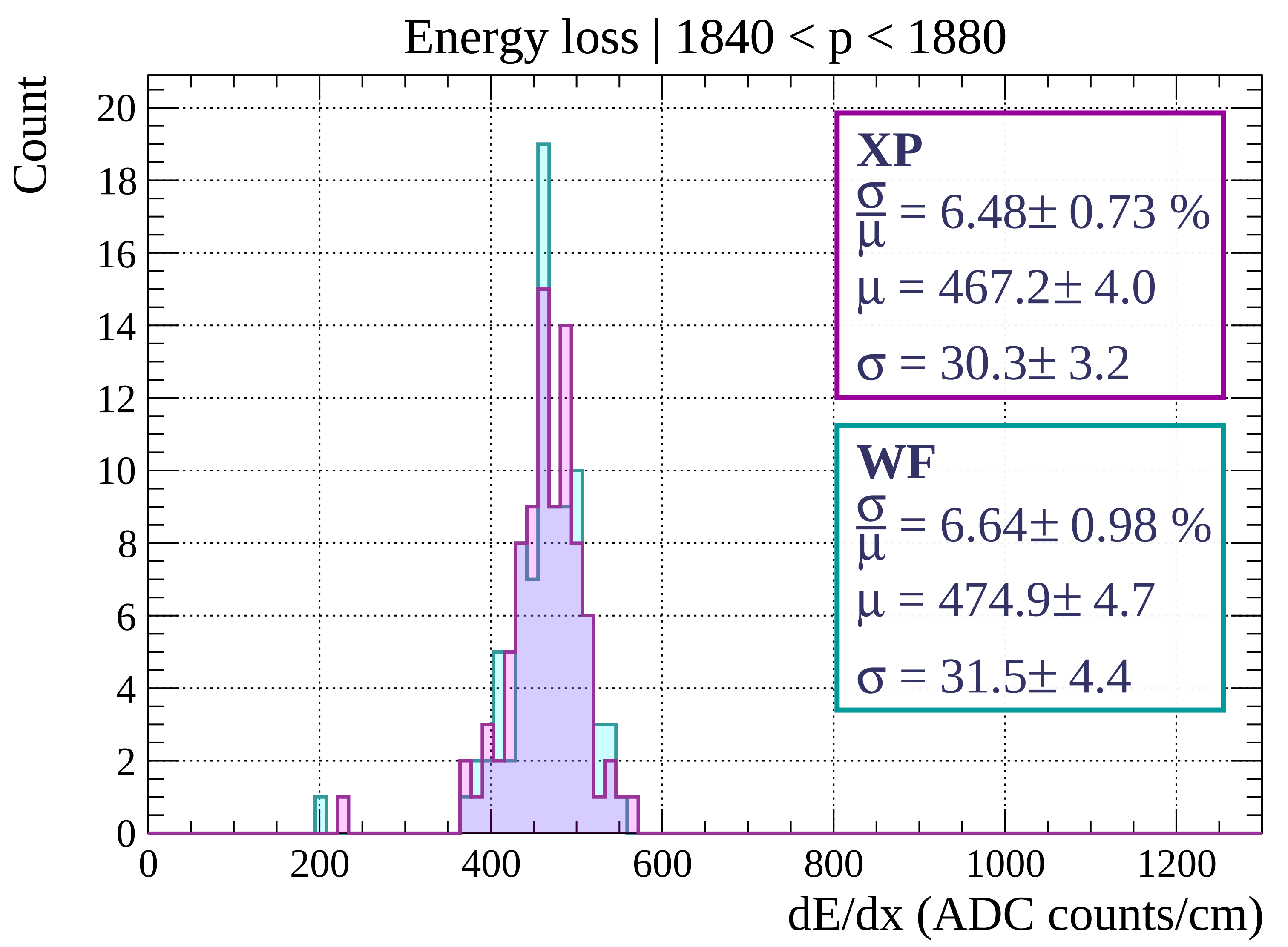
Energy loss | 1720 < p < 1760
Count
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.49± 0.78 %
μ = 464.8± 4.4
σ = 34.8± 3.3
WF
σ/μ = 7.83± 1.01 %
μ = 470.6± 4.9
σ = 36.9± 4.4



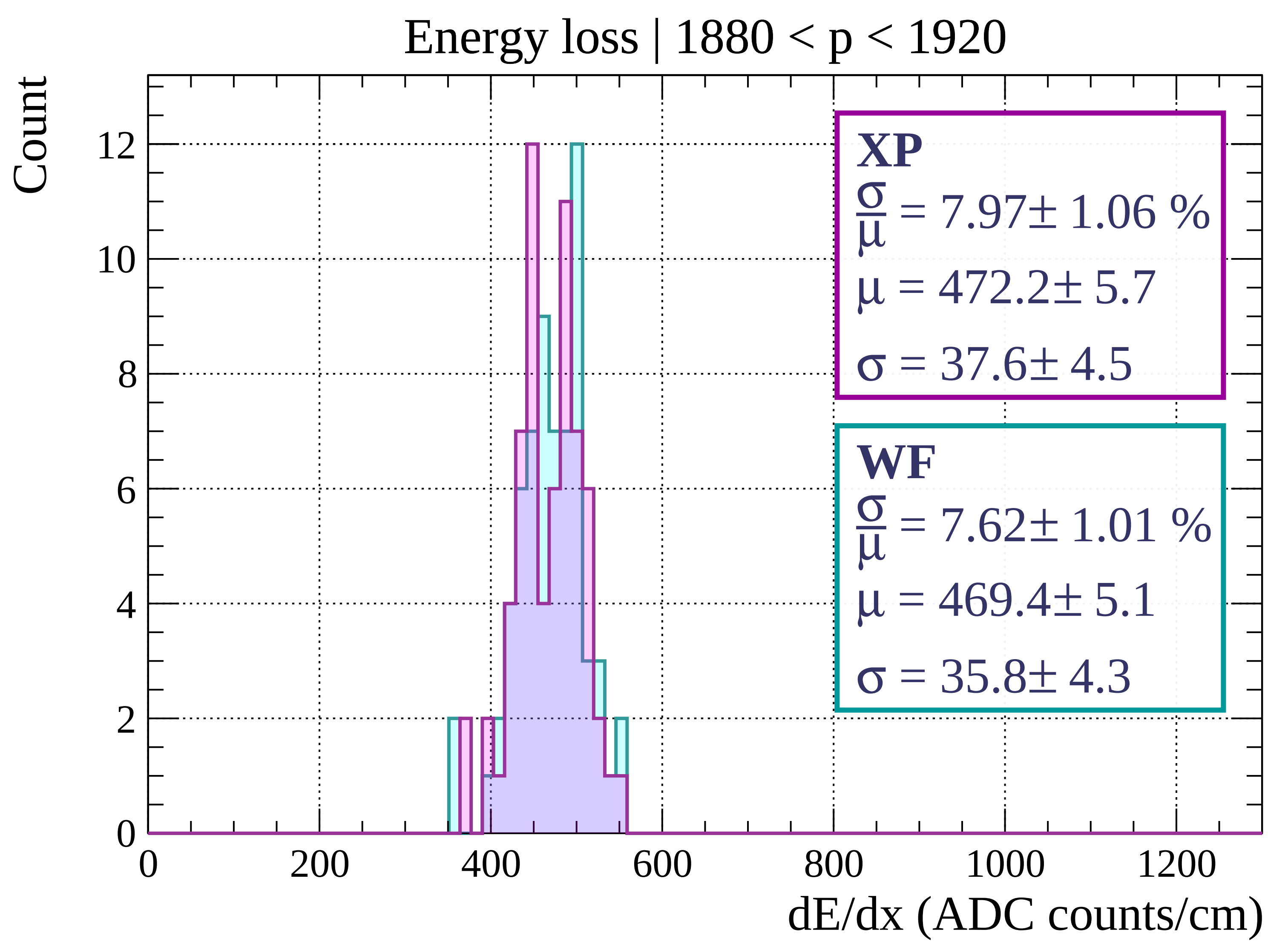
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
XP
$\frac{\sigma}{\mu}$ = 7.07± 0.74 %
μ = 465.8± 4.0
σ = 33.0± 3.2
WF
$\frac{\sigma}{\mu}$ = 7.39± 0.71 %
μ = 468.6± 4.0
σ = 34.6± 3.0

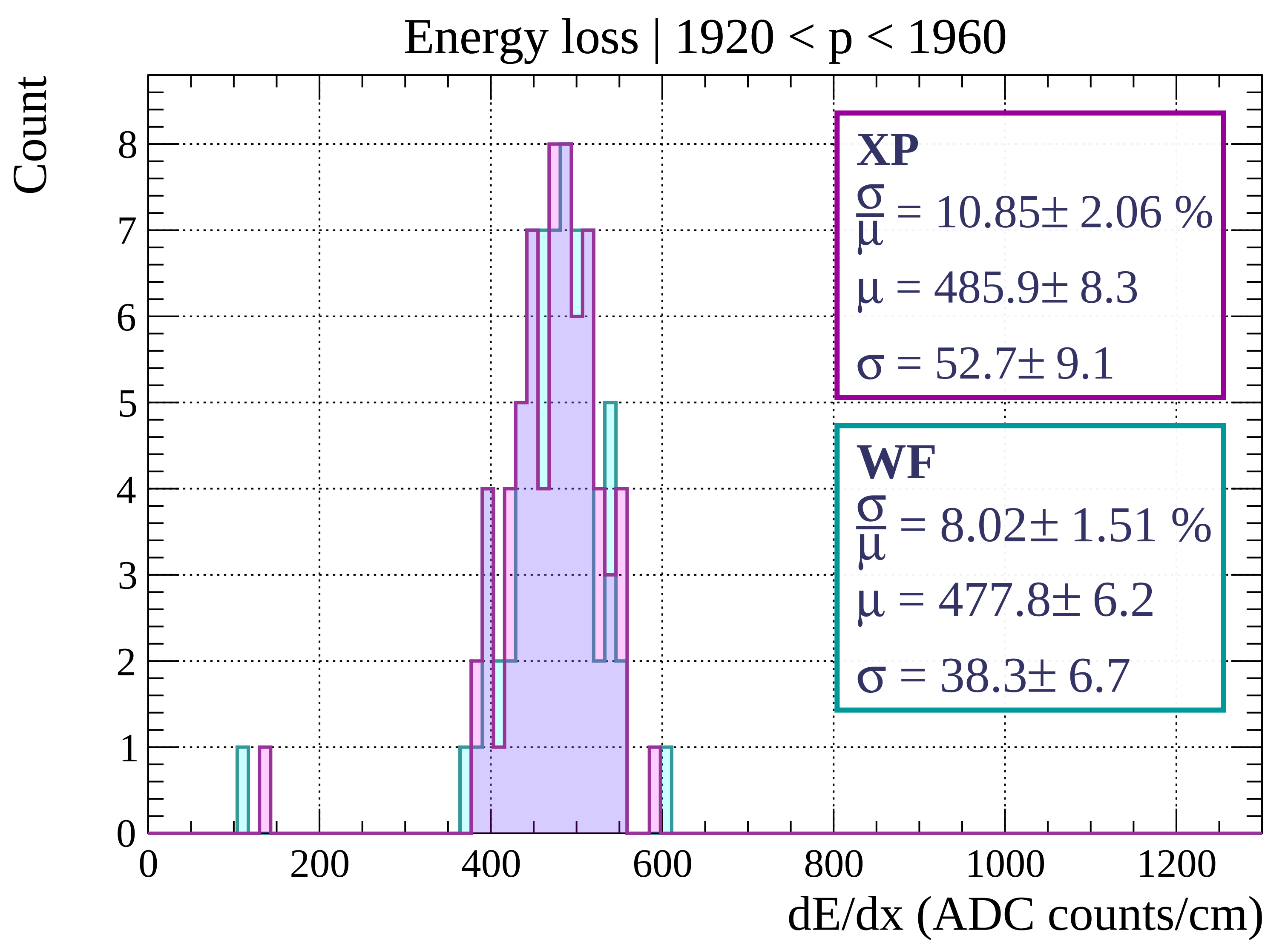


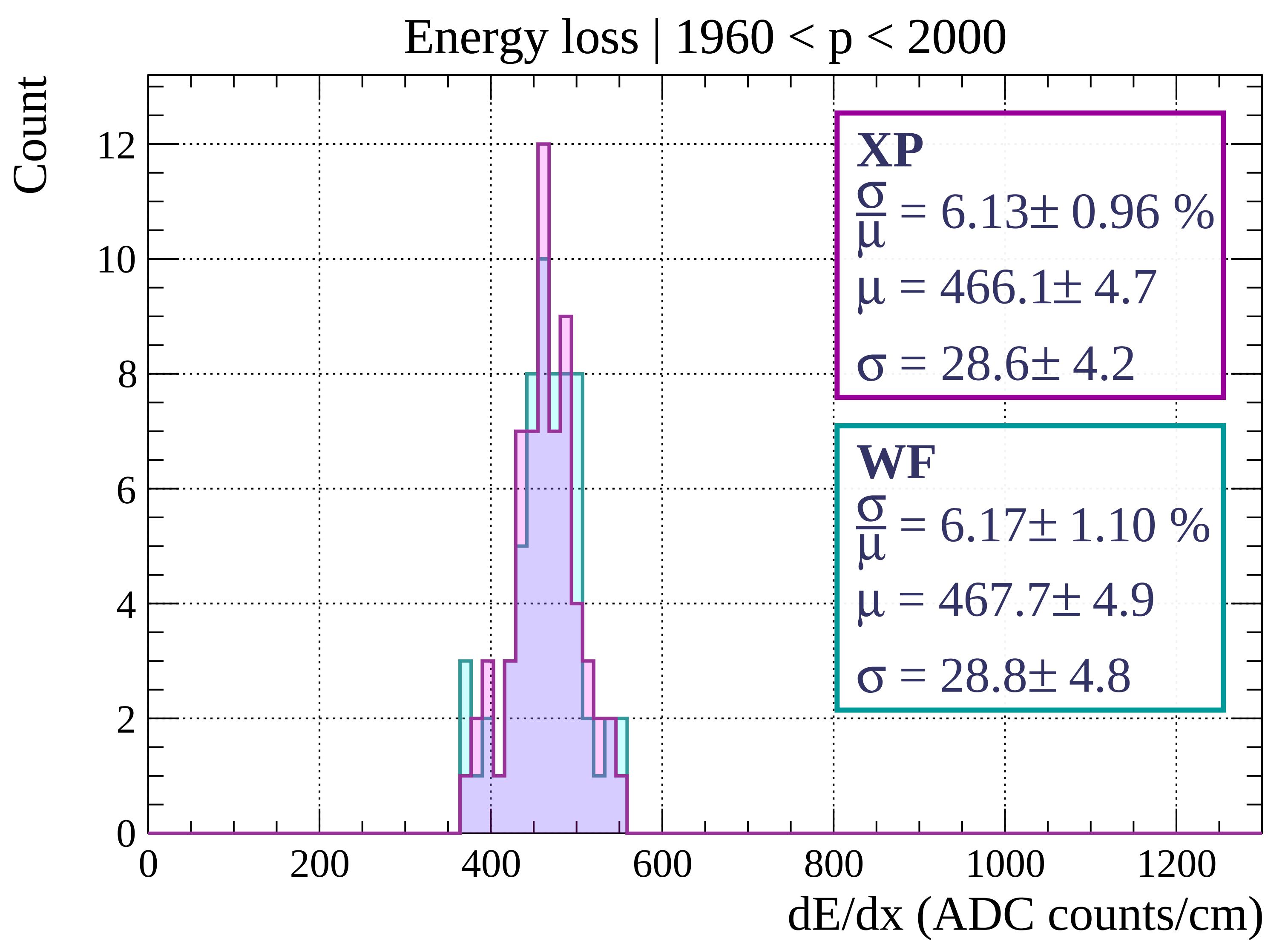
Energy loss | 1800 < p < 1840
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.05± 2.12 %
μ = 464.5± 8.3
σ = 46.7± 9.0
WF
$\frac{\sigma}{\mu}$ = 9.95± 2.29 %
μ = 475.5± 8.9
σ = 47.3± 10.0

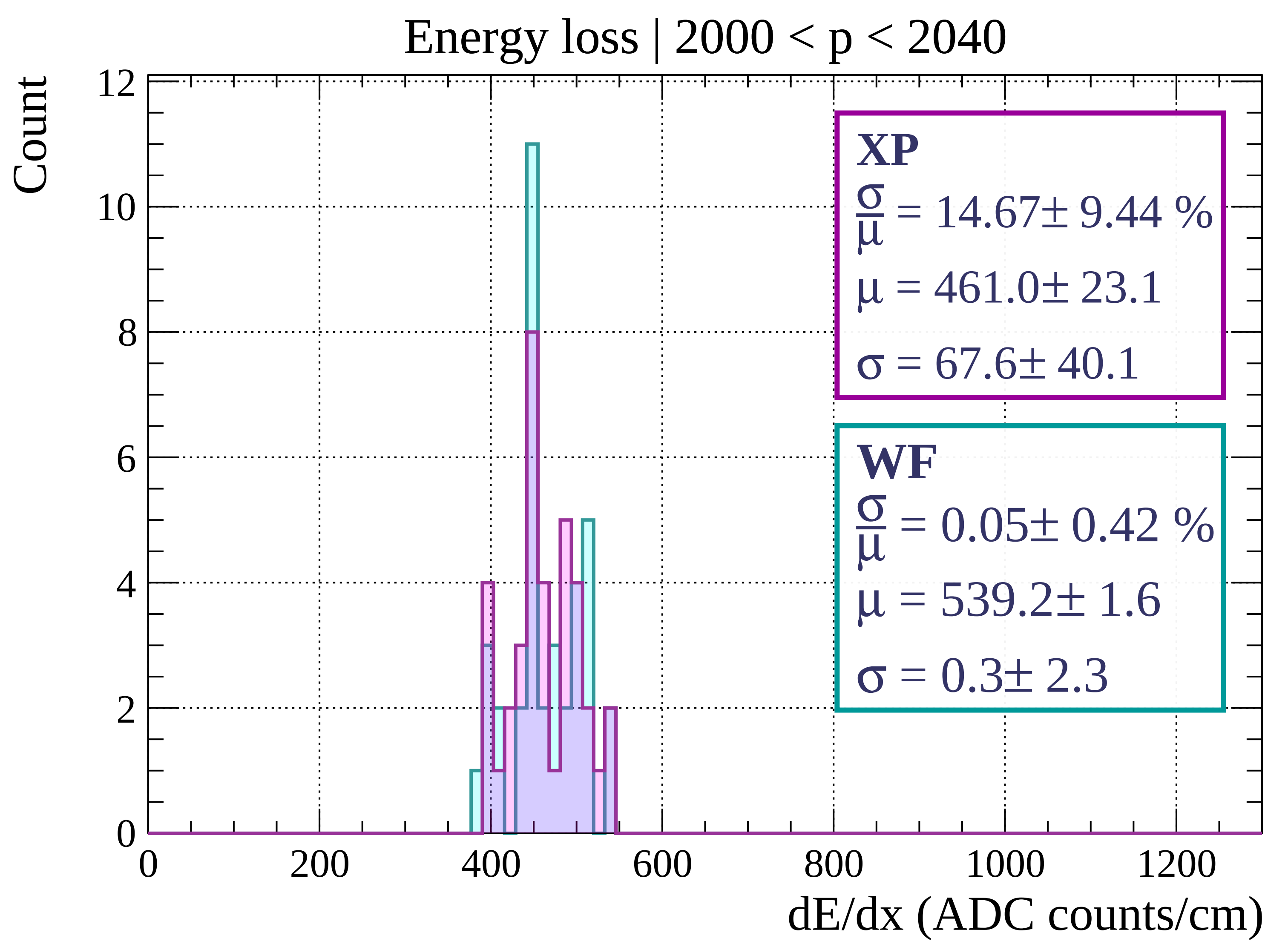


Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
2
4
6
8
10
12
14
16
18
20
XP
σ/μ = 6.48± 0.73 %
μ = 467.2± 4.0
σ = 30.3± 3.2
WF
σ/μ = 6.64± 0.98 %
μ = 474.9± 4.7
σ = 31.5± 4.4

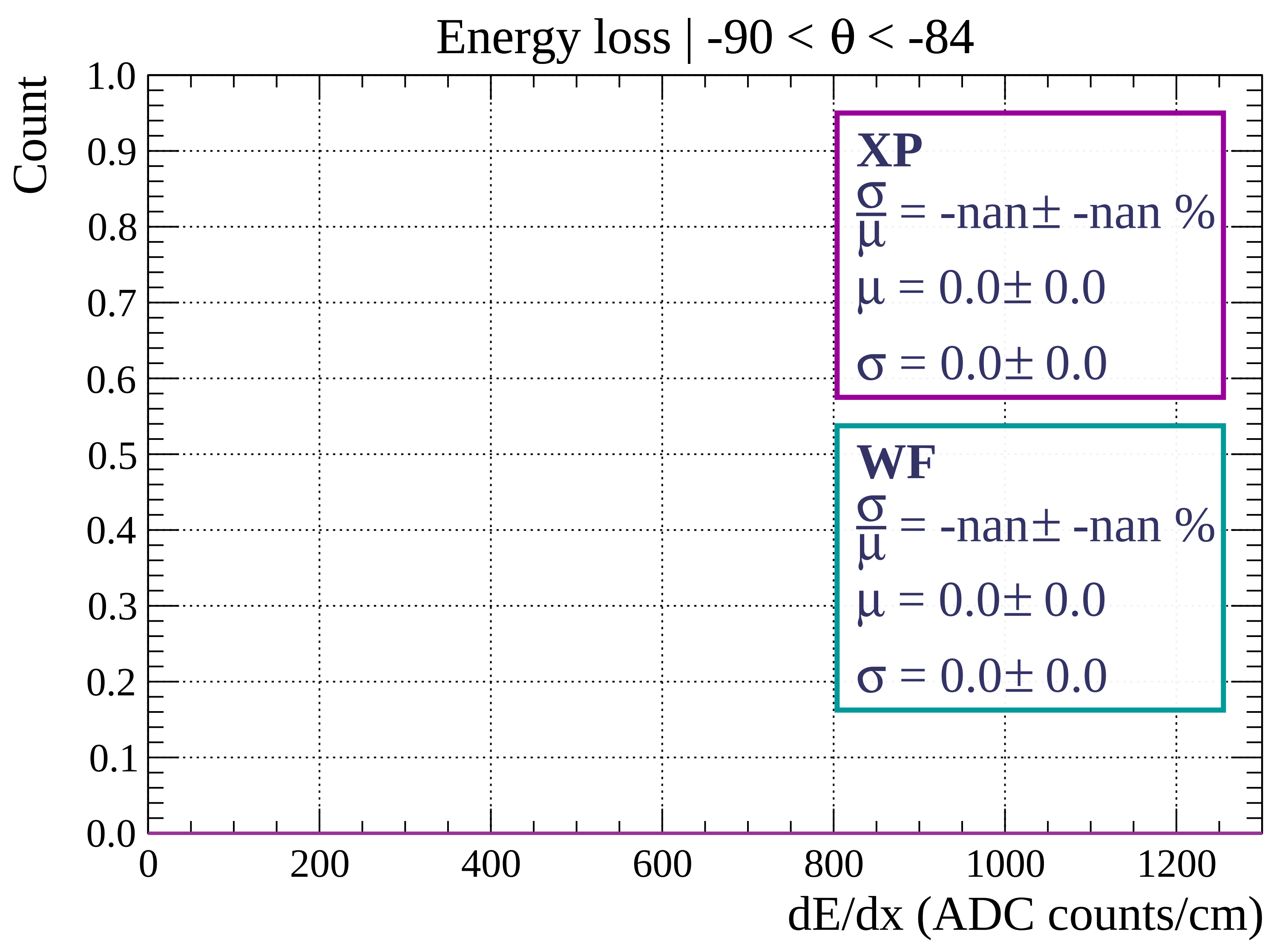

Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 7.97± 1.06 %
μ = 472.2± 5.7
σ = 37.6± 4.5
WF
σ/μ = 7.62± 1.01 %
μ = 469.4± 5.1
σ = 35.8± 4.3

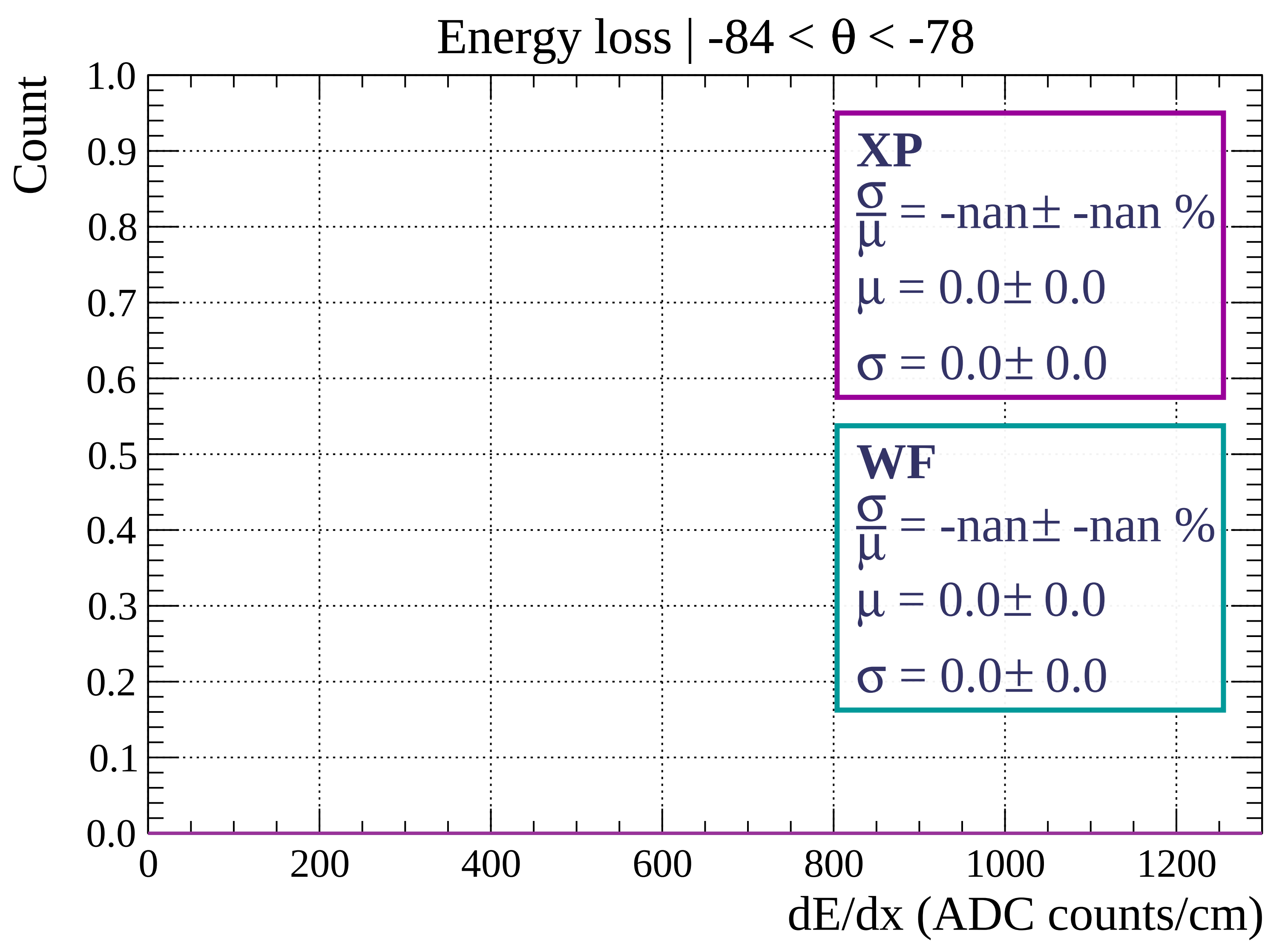

Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
σ/μ = 10.85± 2.06 %
μ = 485.9± 8.3
σ = 52.7± 9.1
WF
σ/μ = 8.02± 1.51 %
μ = 477.8± 6.2
σ = 38.3± 6.7


Energy loss | 1960 < p < 2000
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 6.13± 0.96 %
μ = 466.1± 4.7
σ = 28.6± 4.2
WF
σ/μ = 6.17± 1.10 %
μ = 467.7± 4.9
σ = 28.8± 4.8

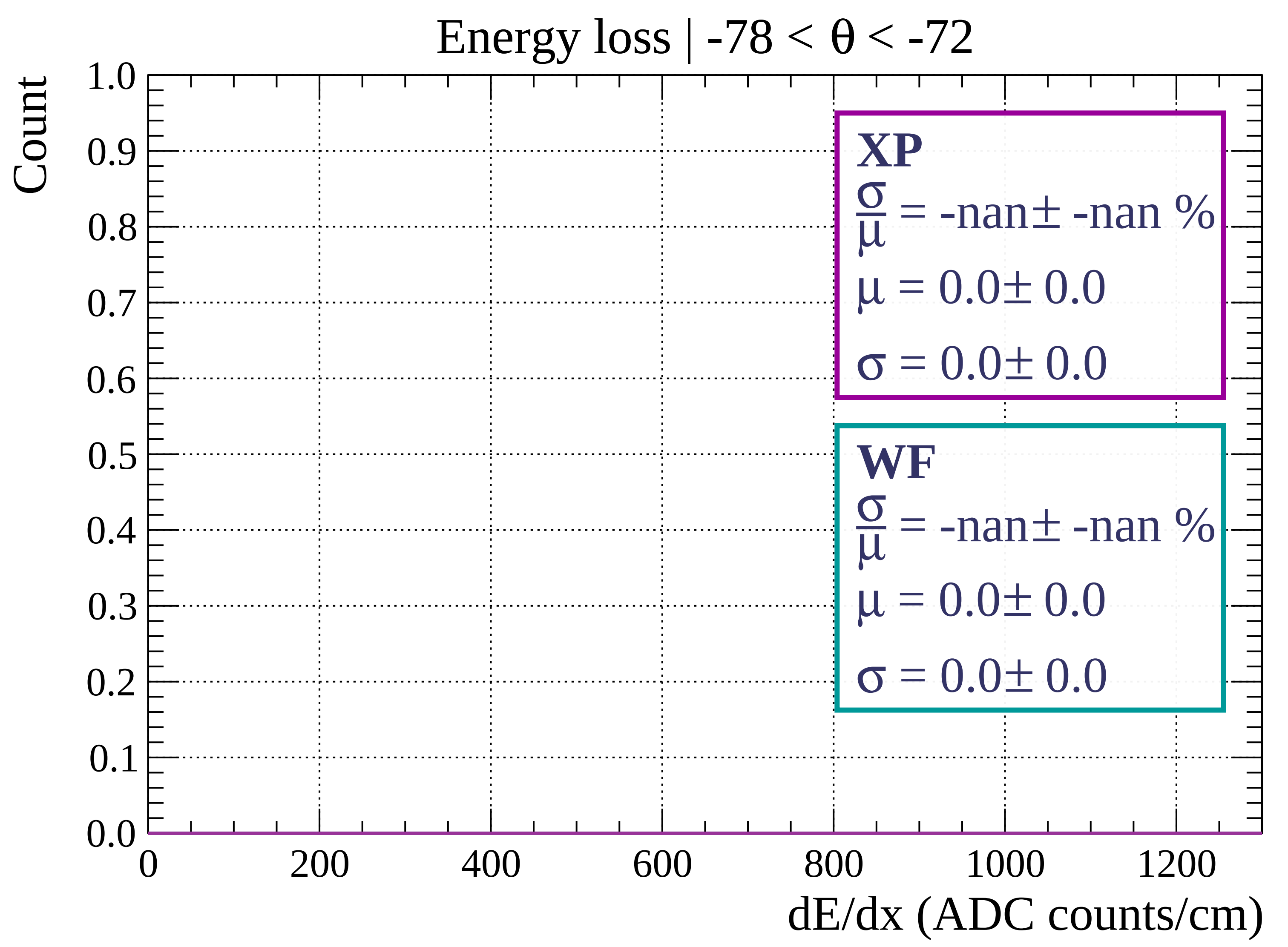


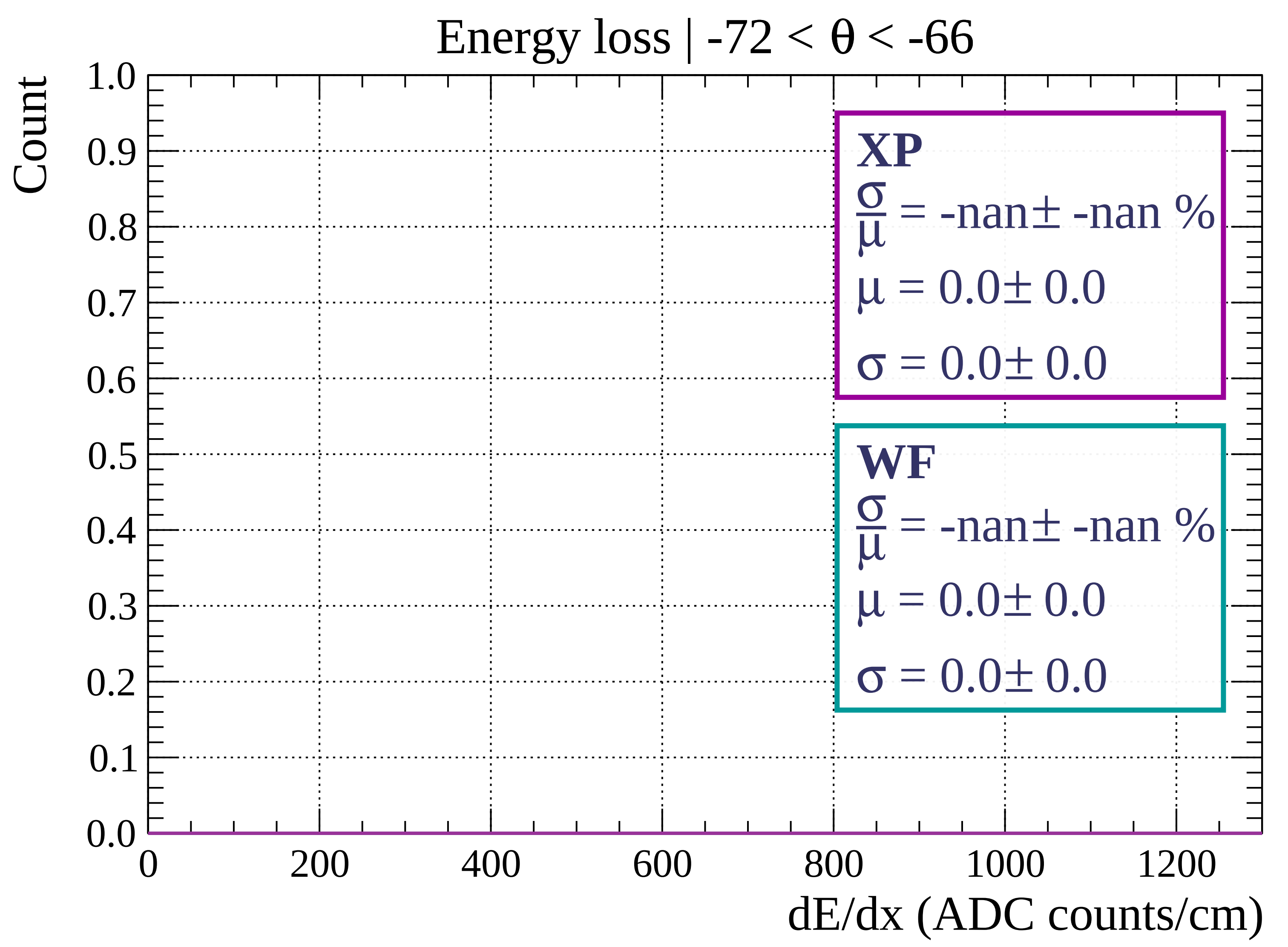
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
XP
$\frac{\sigma}{\mu}$ = 14.67± 9.44 %
μ = 461.0± 23.1
σ = 67.6± 40.1
WF
$\frac{\sigma}{\mu}$ = 0.05± 0.42 %
μ = 539.2± 1.6
σ = 0.3± 2.3


Energy loss | -90 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

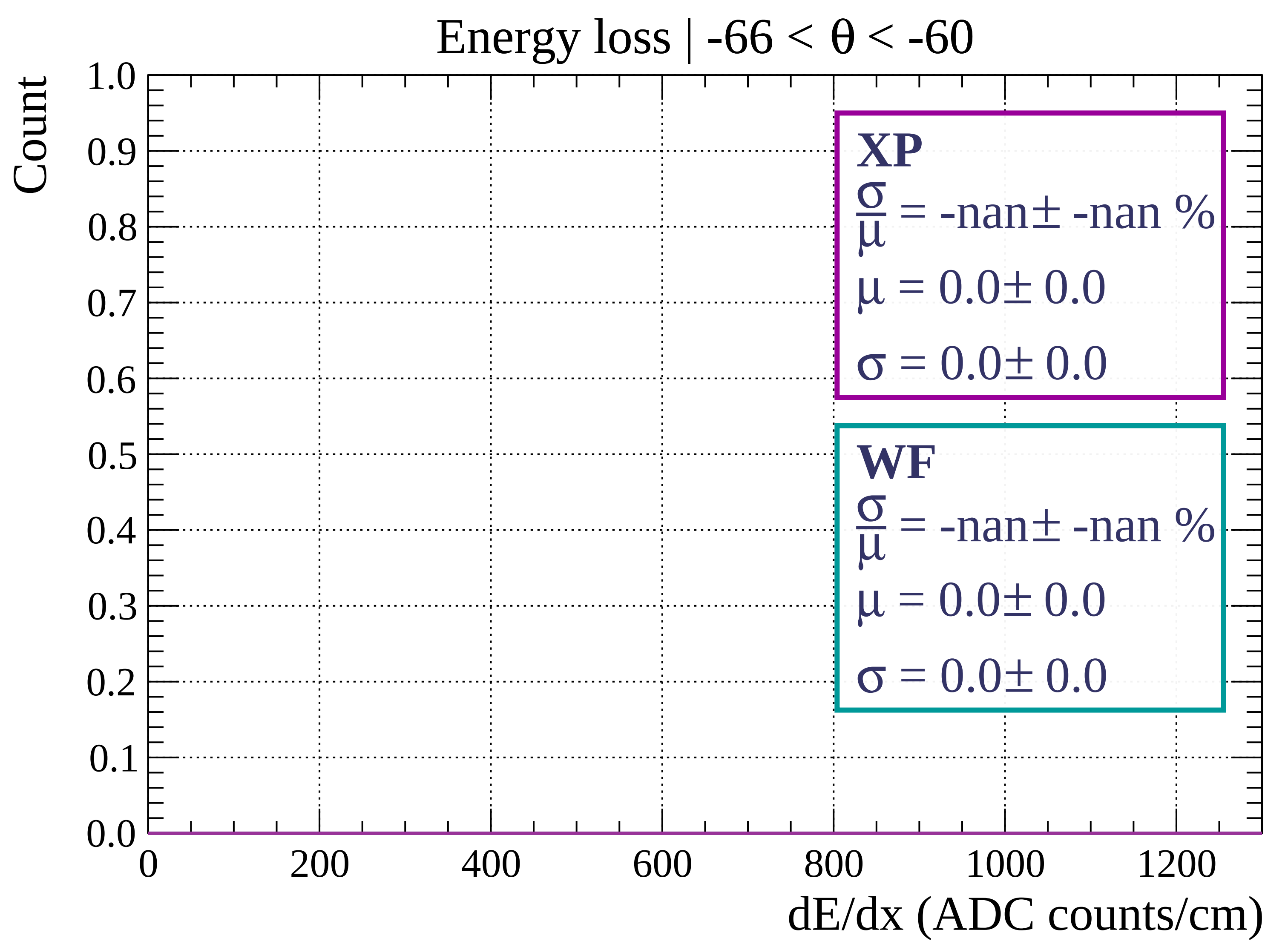


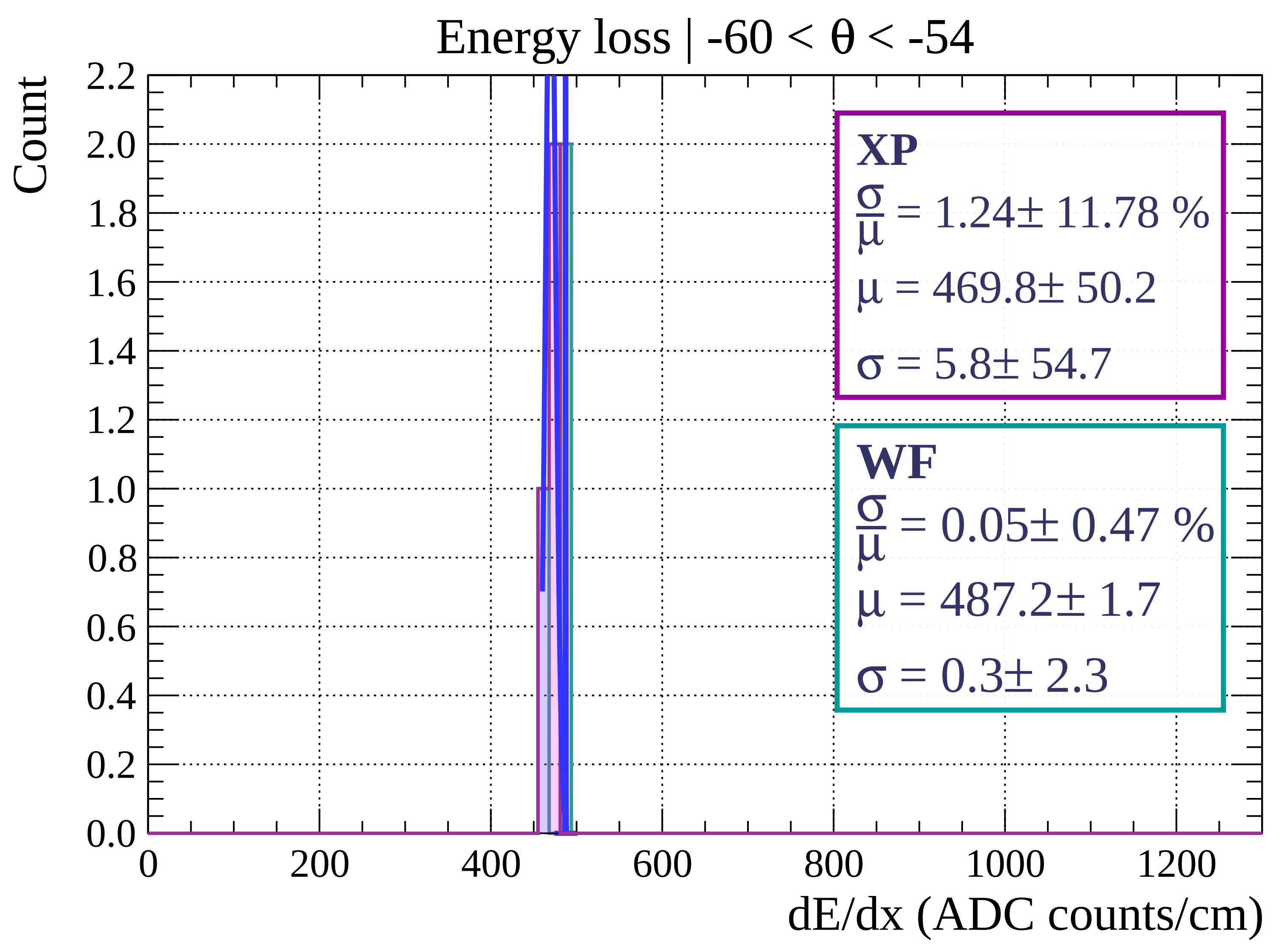
Energy loss | -84 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -78 < θ < -72
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

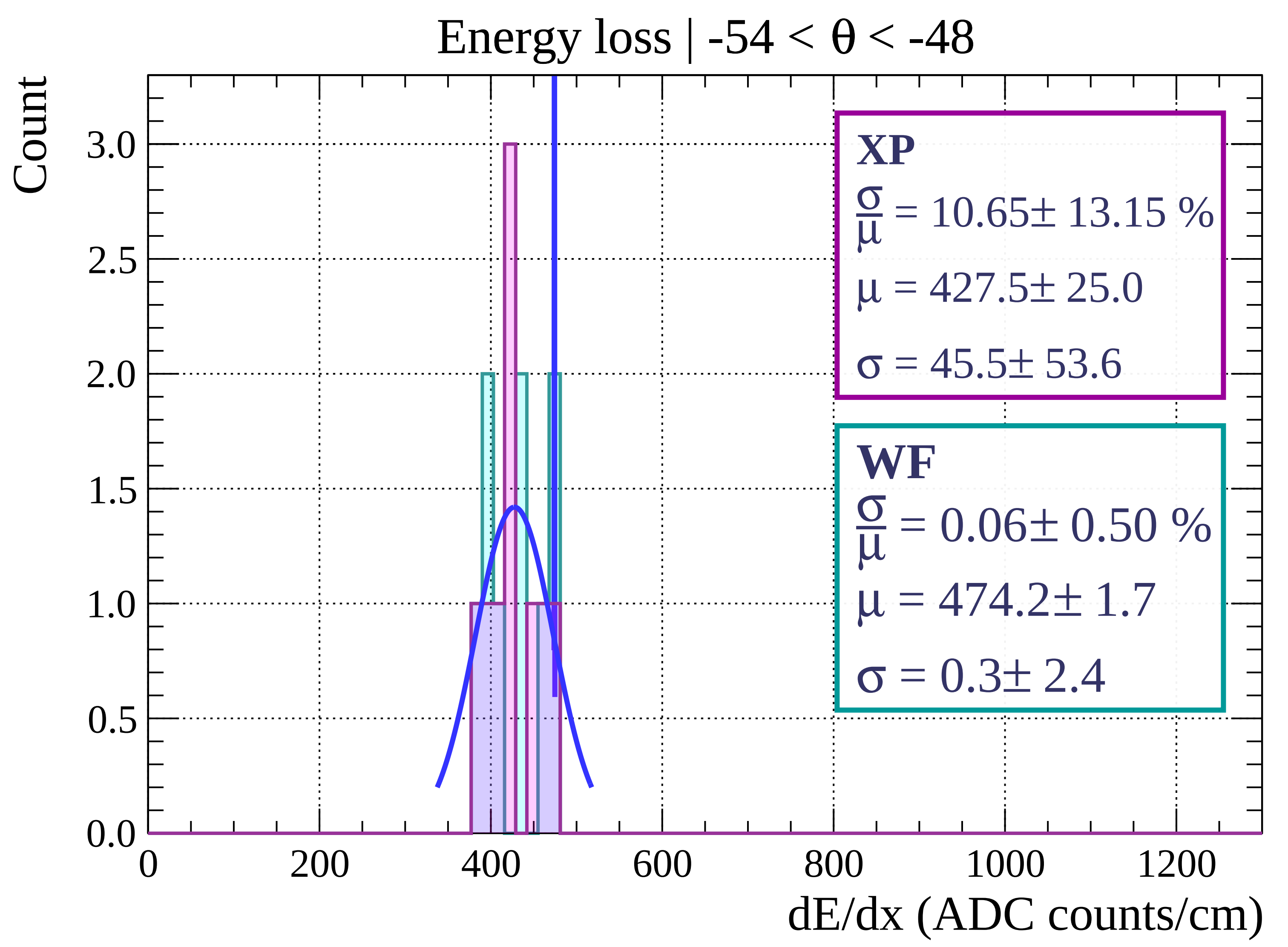


Energy loss | -72 < θ < -66
Count
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

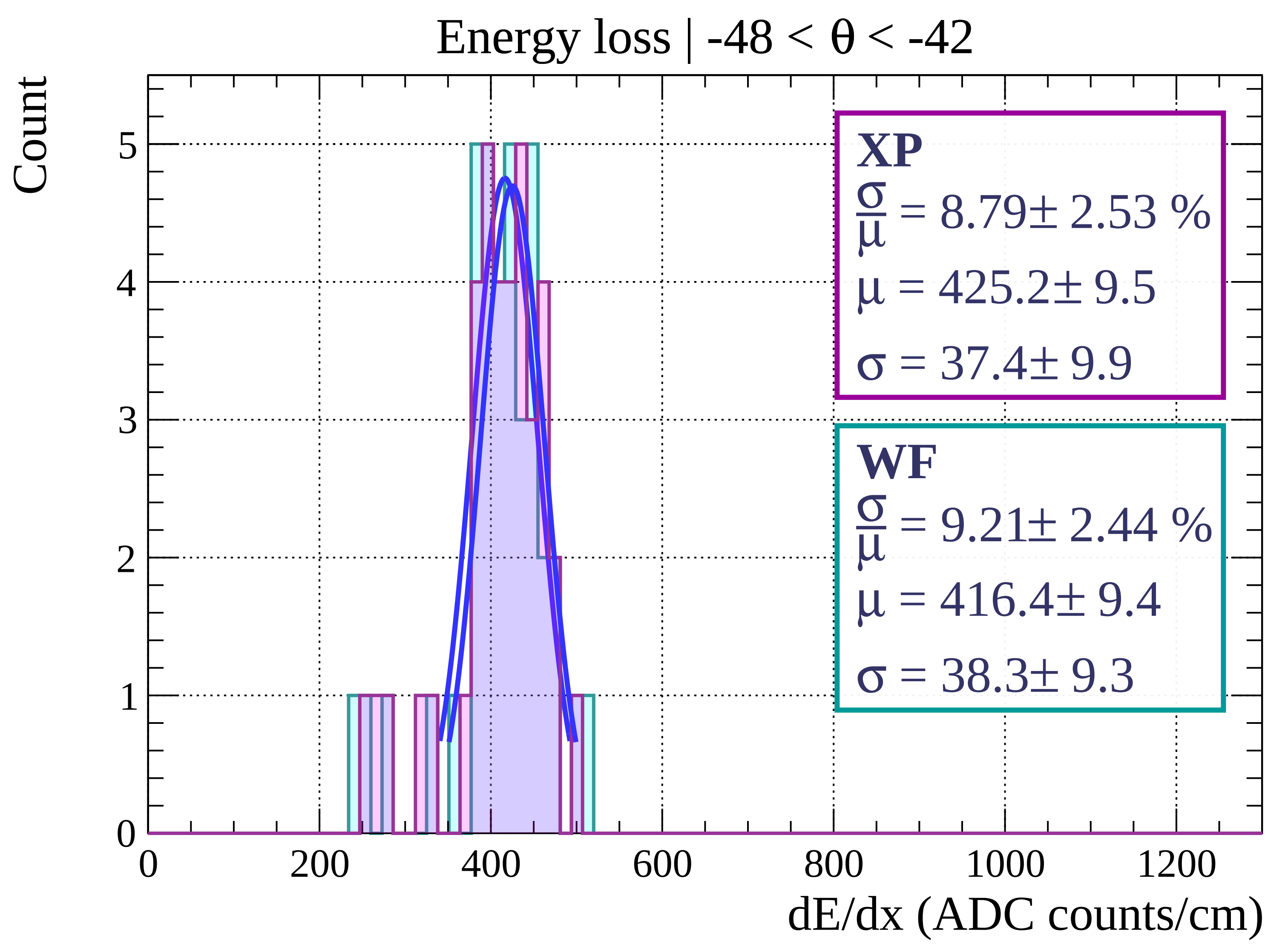

Energy loss | -66 < θ < -60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



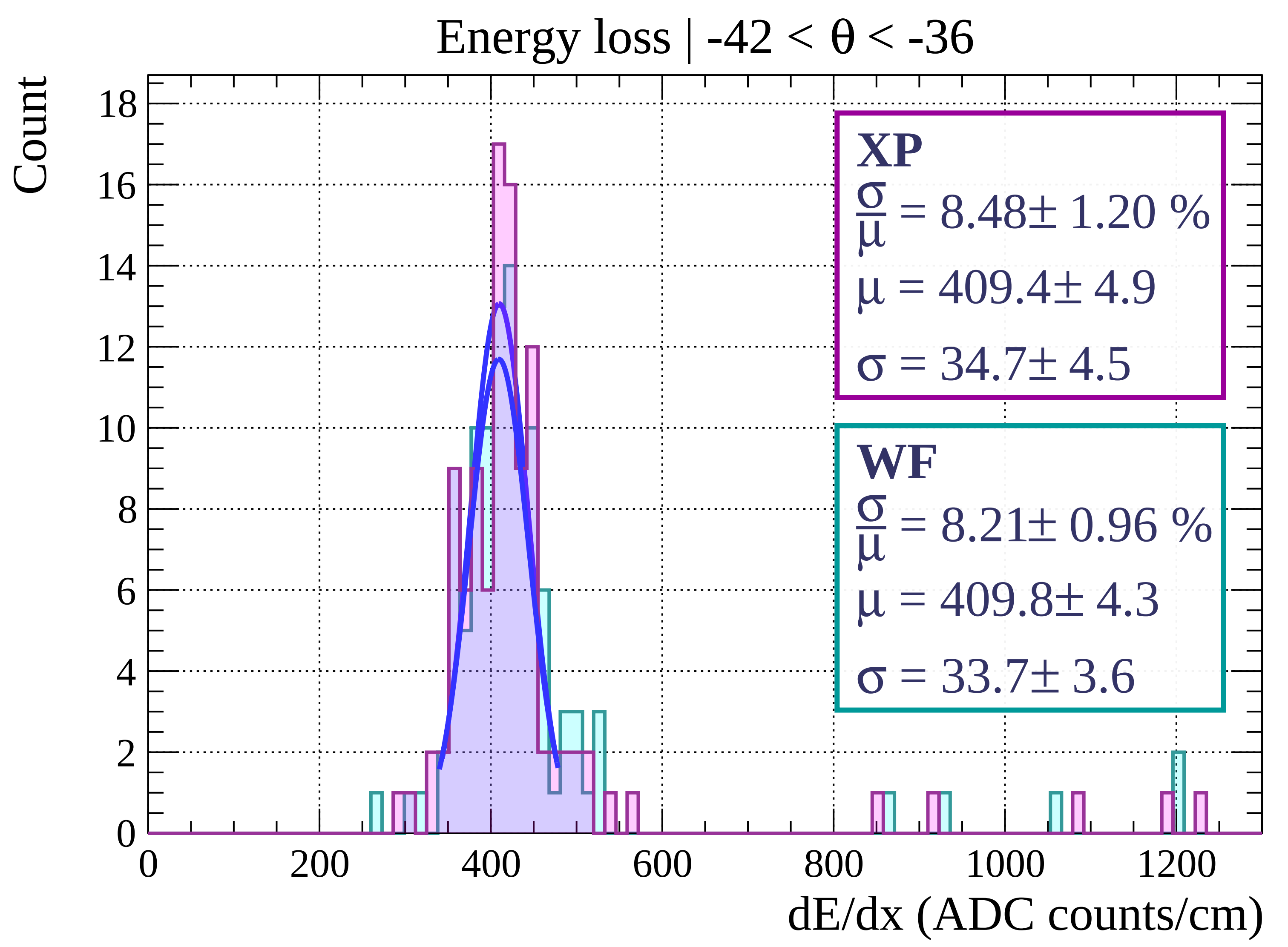
Energy loss | -60 < θ < -54
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = 1.24± 11.78 %
μ = 469.8± 50.2
σ = 5.8± 54.7
WF
σ/μ = 0.05± 0.47 %
μ = 487.2± 1.7
σ = 0.3± 2.3



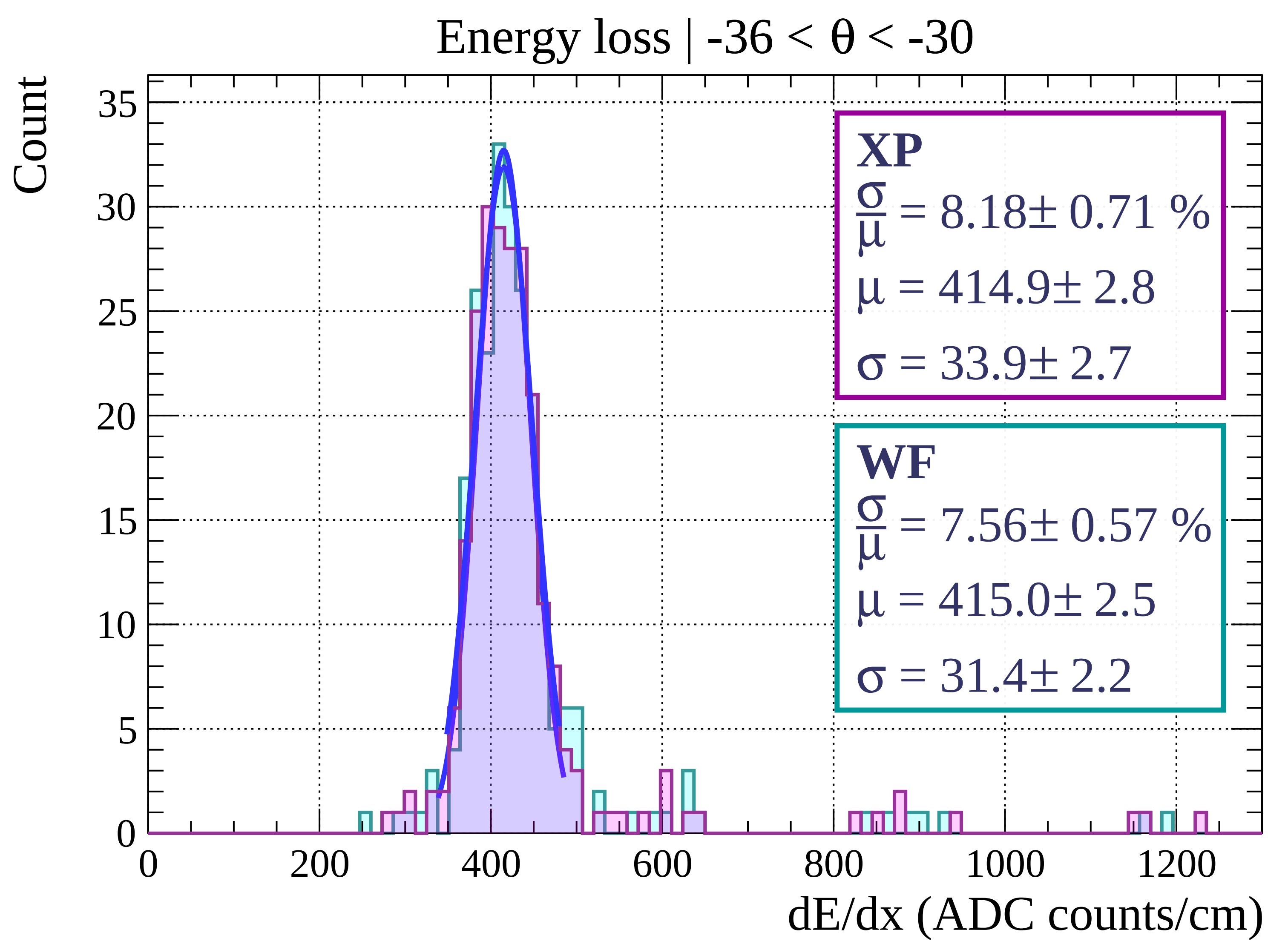
Energy loss | -54 < θ < -48
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.65± 13.15 %
μ = 427.5± 25.0
σ = 45.5± 53.6
WF
σ/μ = 0.06± 0.50 %
μ = 474.2± 1.7
σ = 0.3± 2.4



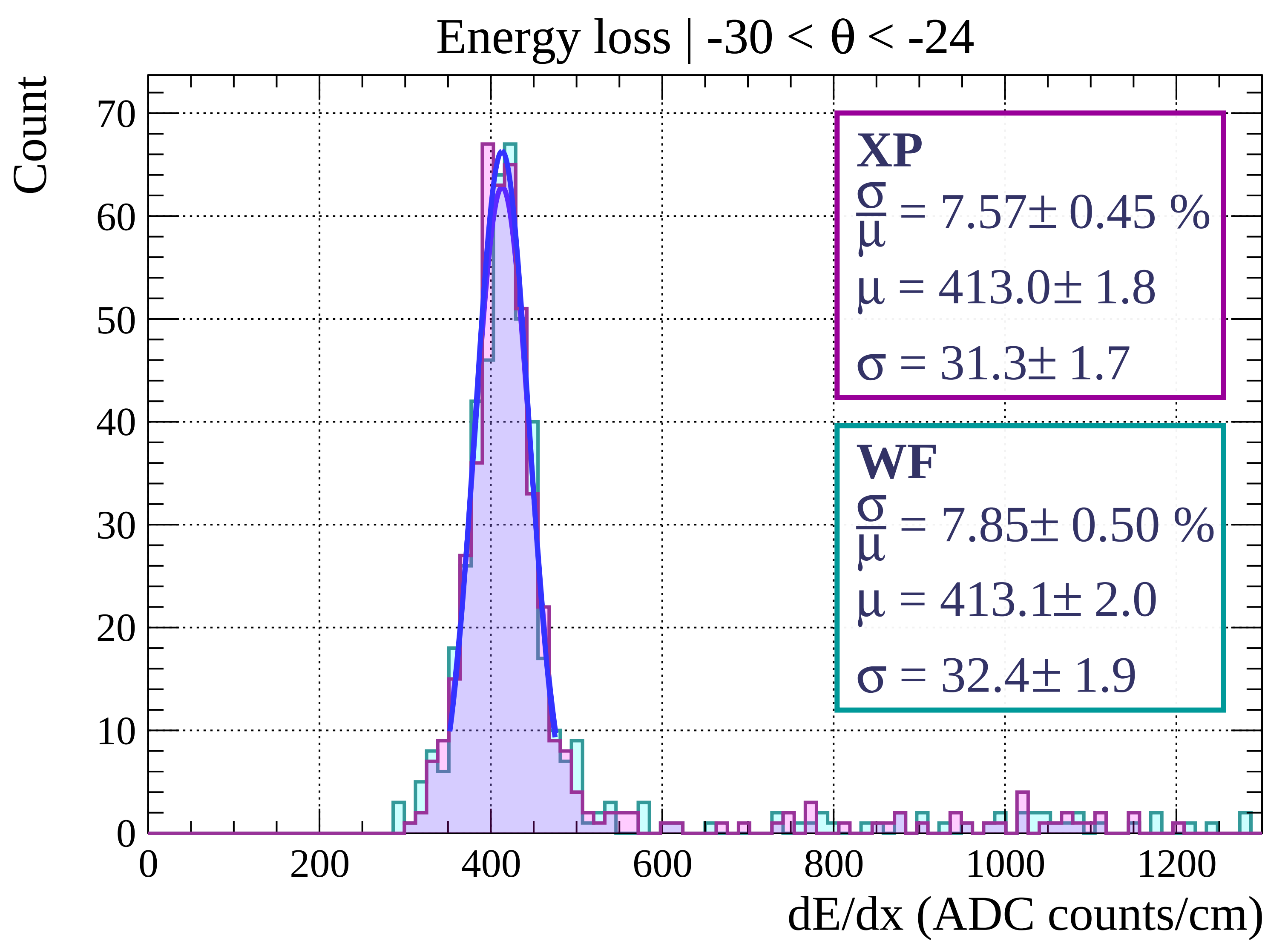
Energy loss | -48 < θ < -42
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 8.79± 2.53 %
μ = 425.2± 9.5
σ = 37.4± 9.9
WF
σ/μ = 9.21± 2.44 %
μ = 416.4± 9.4
σ = 38.3± 9.3

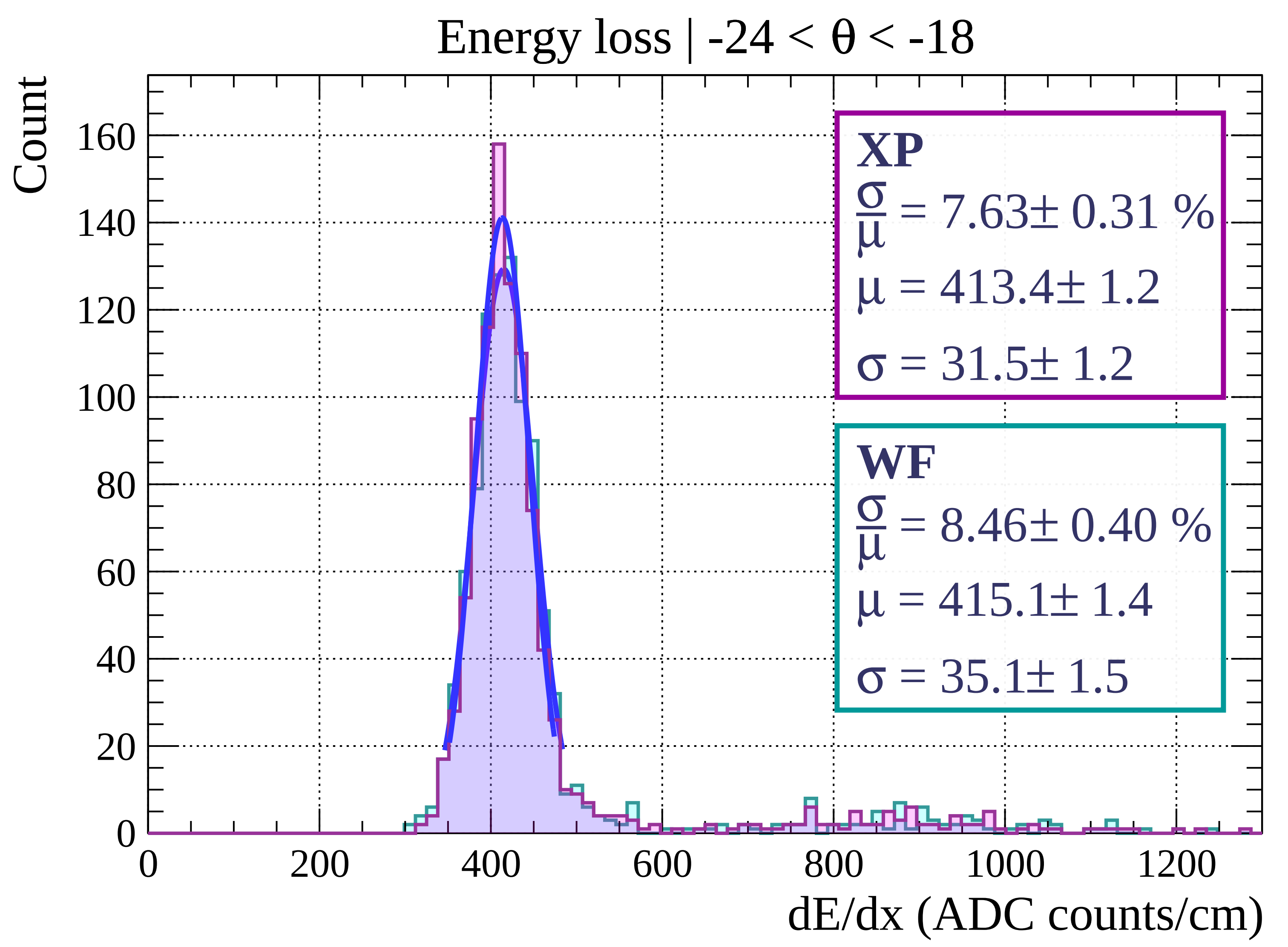


Energy loss | -42 < θ < -36
Count
18
16
14
12
10
8
6
4
2
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.48± 1.20 %
μ = 409.4± 4.9
σ = 34.7± 4.5
WF
$\frac{\sigma}{\mu}$ = 8.21± 0.96 %
μ = 409.8± 4.3
σ = 33.7± 3.6

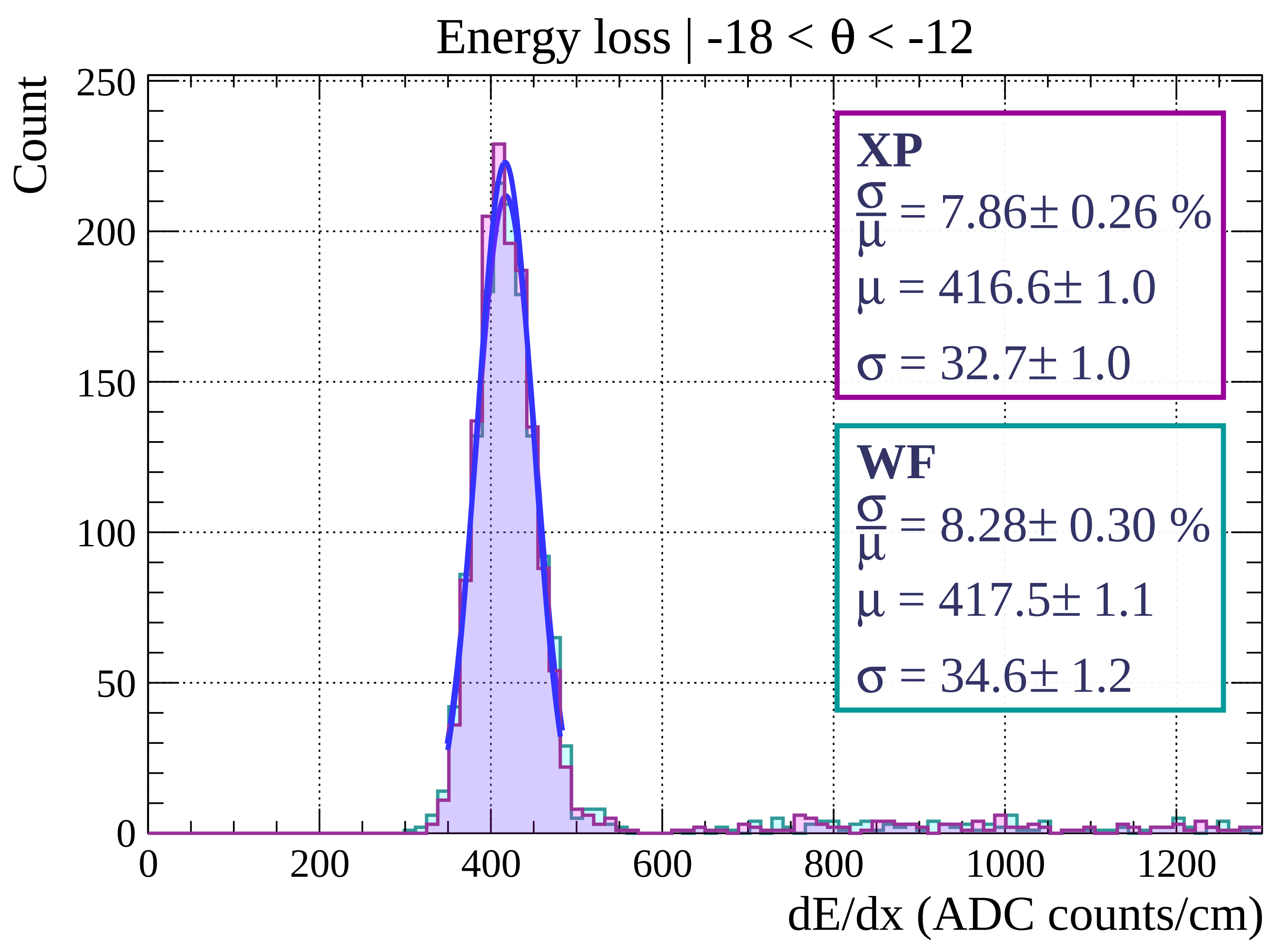


Energy loss | -36 < θ < -30
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
30
35
XP
σ/μ = 8.18± 0.71 %
μ = 414.9± 2.8
σ = 33.9± 2.7
WF
σ/μ = 7.56± 0.57 %
μ = 415.0± 2.5
σ = 31.4± 2.2

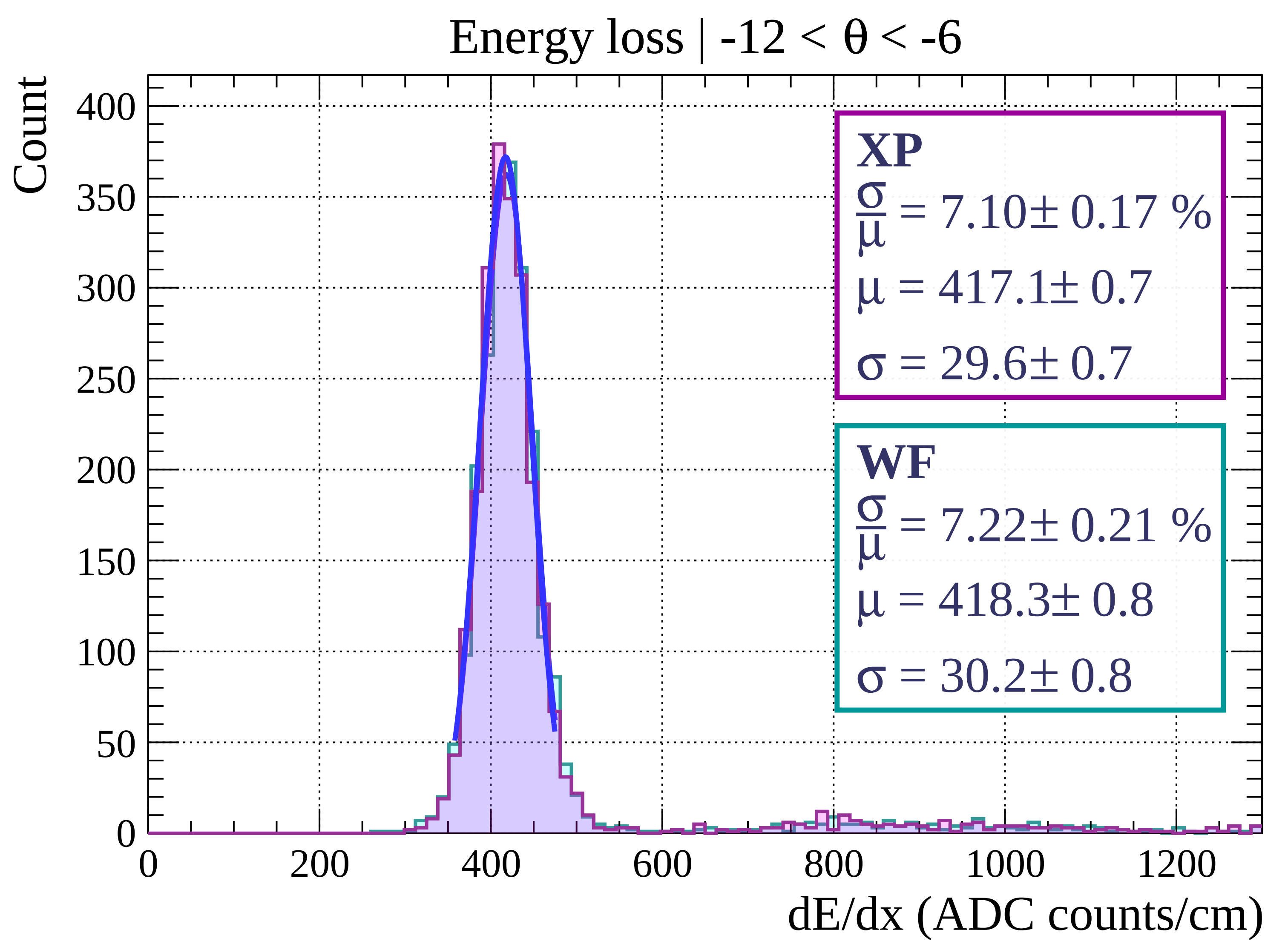

Energy loss | -30 < θ < -24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
XP
σ/μ = 7.57± 0.45 %
μ = 413.0± 1.8
σ = 31.3± 1.7
WF
σ/μ = 7.85± 0.50 %
μ = 413.1± 2.0
σ = 32.4± 1.9



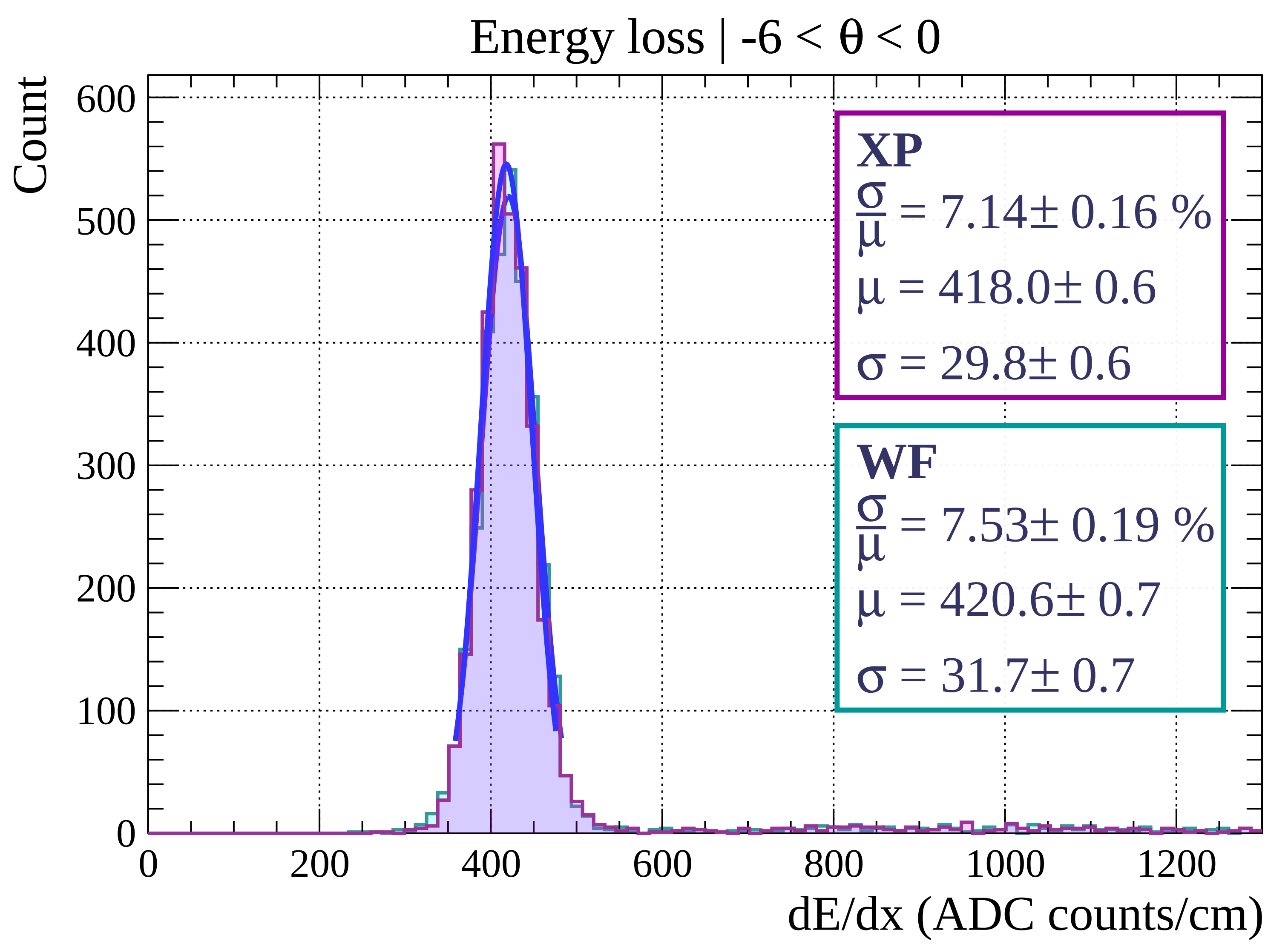
Energy loss | -24 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.63± 0.31 %
μ = 413.4± 1.2
σ = 31.5± 1.2
WF
$\frac{\sigma}{\mu}$ = 8.46± 0.40 %
μ = 415.1± 1.4
σ = 35.1± 1.5



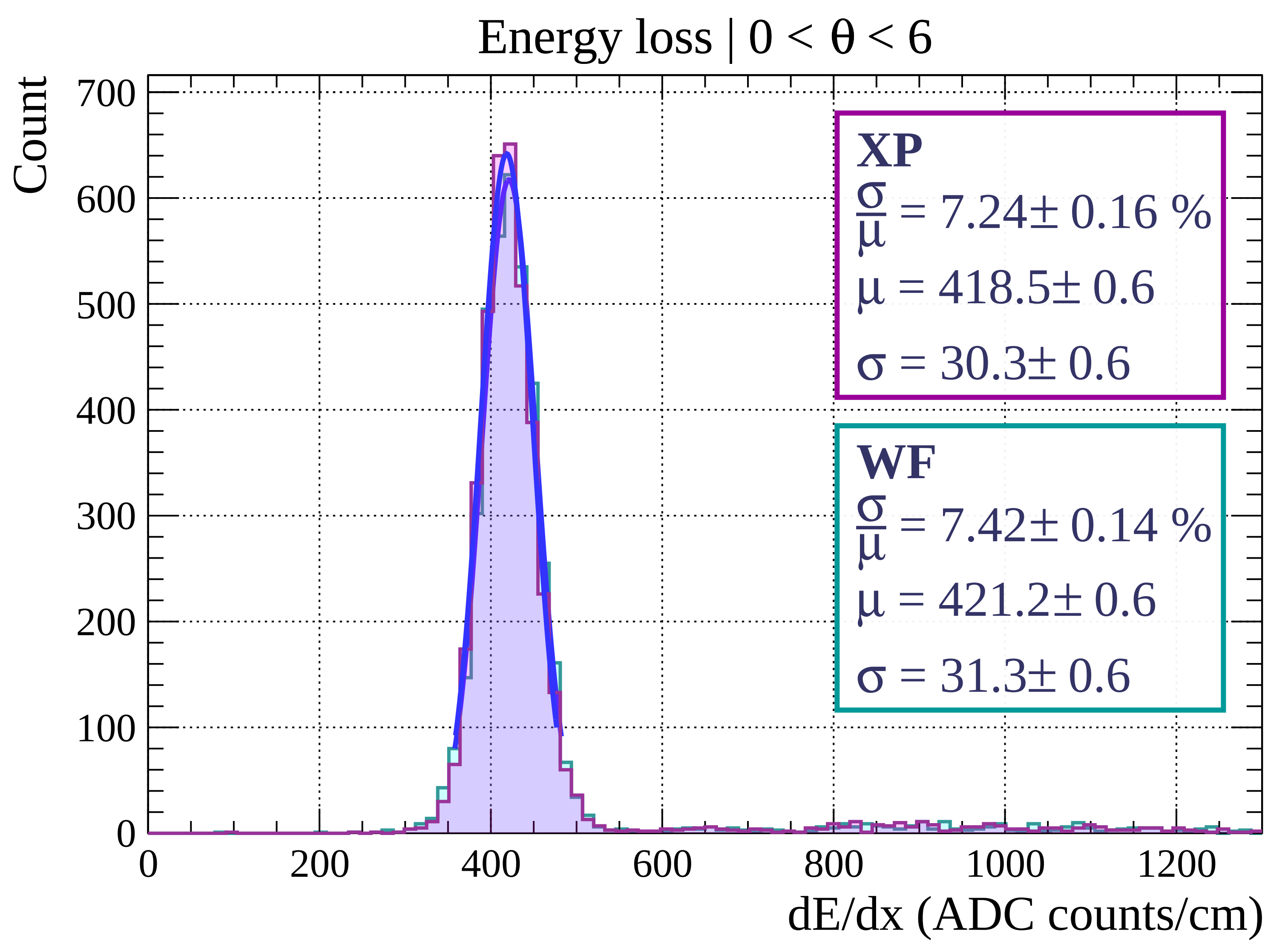
Energy loss | -18 < θ < -12
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 7.86± 0.26 %
μ = 416.6± 1.0
σ = 32.7± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.30 %
μ = 417.5± 1.1
σ = 34.6± 1.2

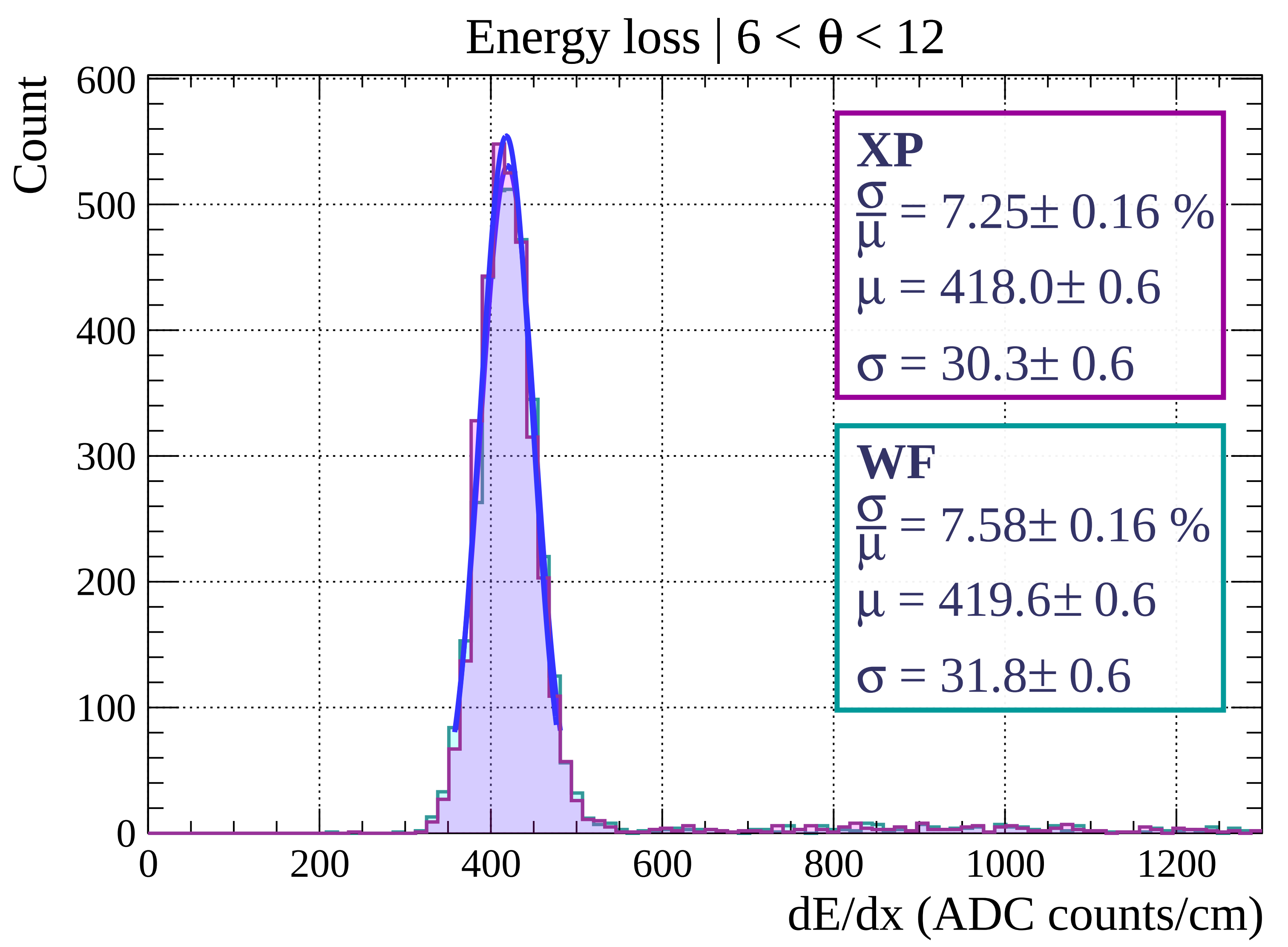


Energy loss | -12 < θ < -6
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.10± 0.17 %
μ = 417.1± 0.7
σ = 29.6± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.22± 0.21 %
μ = 418.3± 0.8
σ = 30.2± 0.8

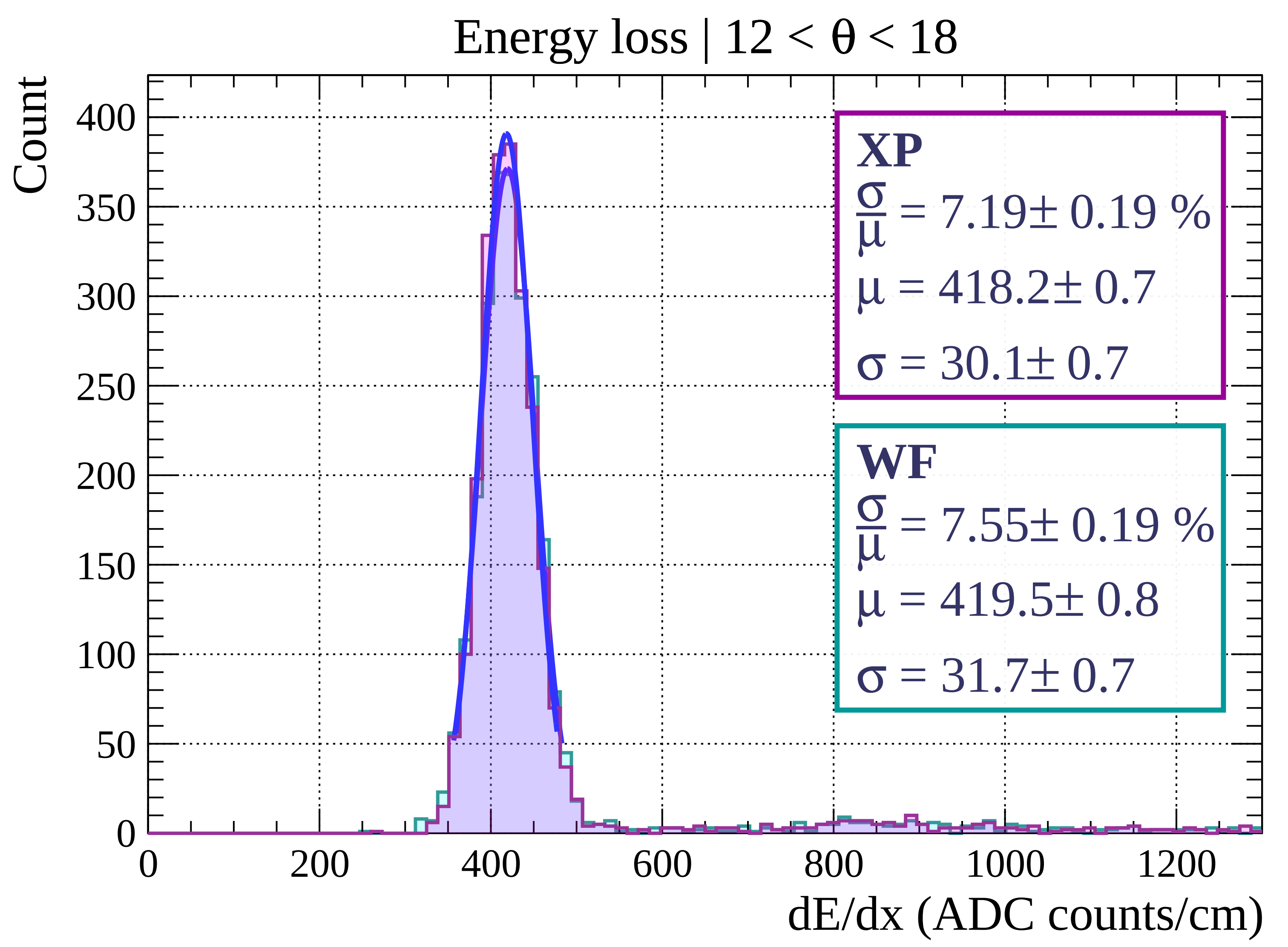


Energy loss | -6 < θ < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 7.14± 0.16 %
μ = 418.0± 0.6
σ = 29.8± 0.6
WF
σ/μ = 7.53± 0.19 %
μ = 420.6± 0.7
σ = 31.7± 0.7

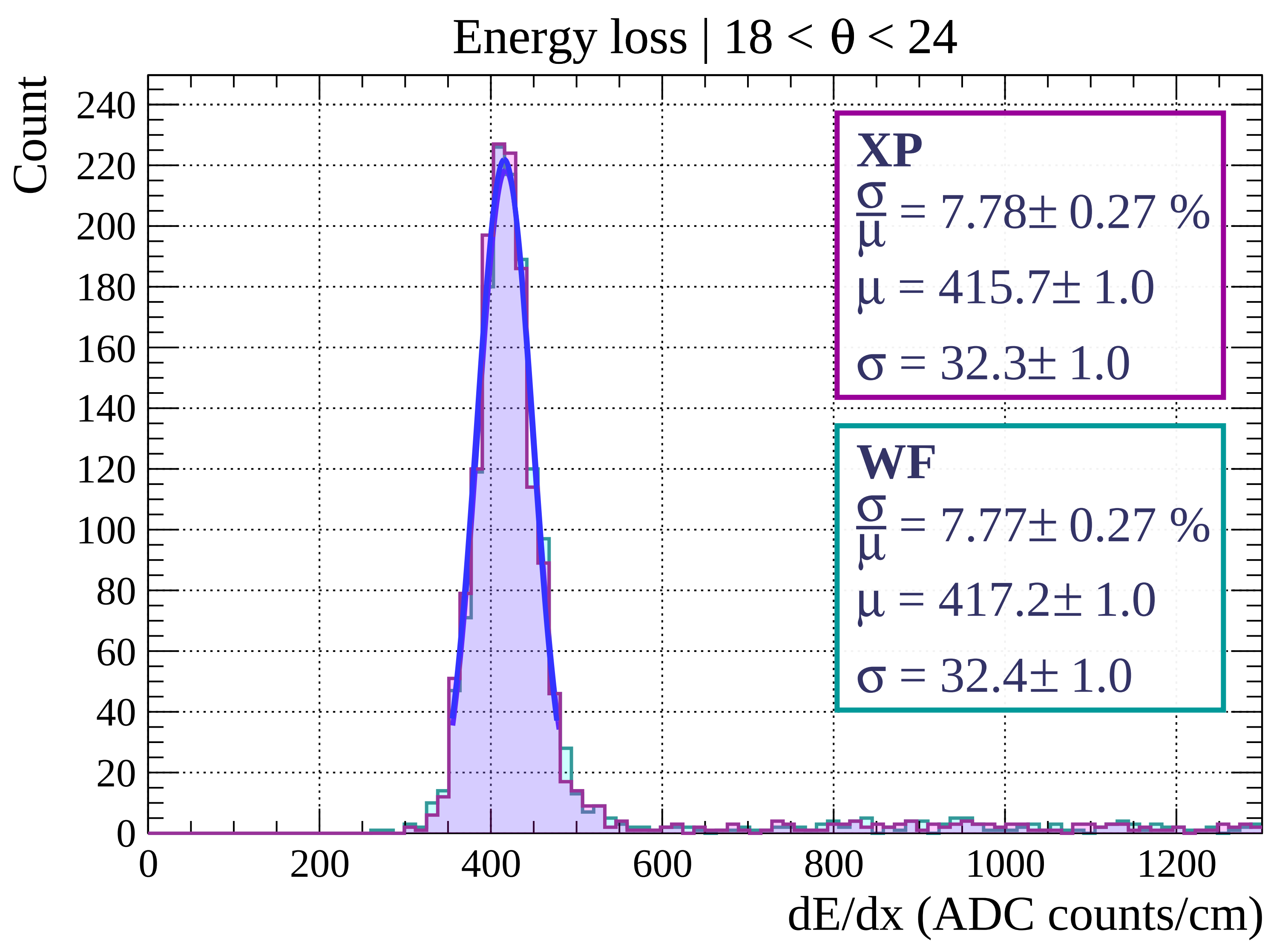

Energy loss | 0 < θ < 6
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 7.24± 0.16 %
μ = 418.5± 0.6
σ = 30.3± 0.6
WF
σ/μ = 7.42± 0.14 %
μ = 421.2± 0.6
σ = 31.3± 0.6



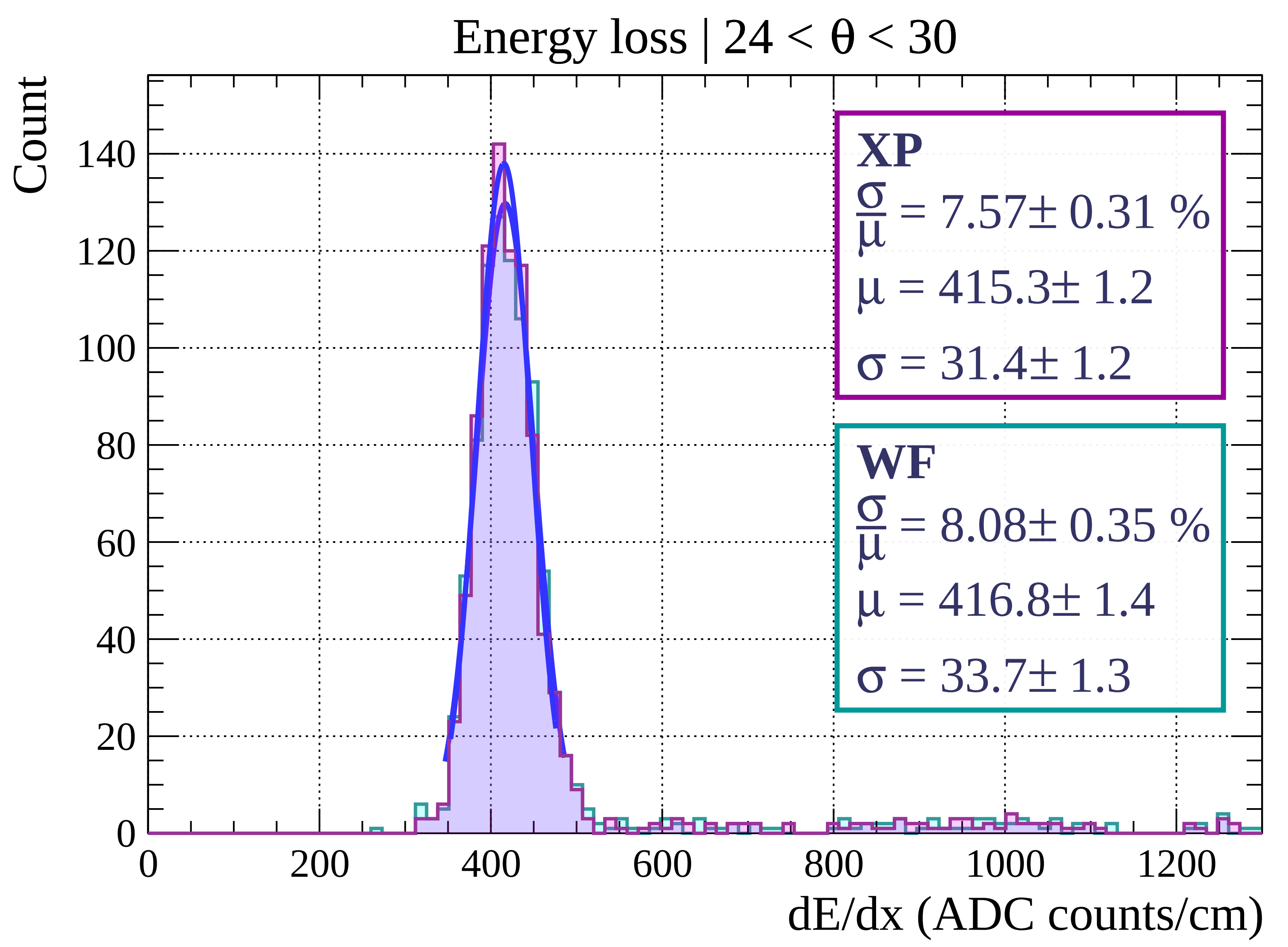
Energy loss | 6 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 7.25± 0.16 %
μ = 418.0± 0.6
σ = 30.3± 0.6
WF
$\frac{\sigma}{\mu}$ = 7.58± 0.16 %
μ = 419.6± 0.6
σ = 31.8± 0.6

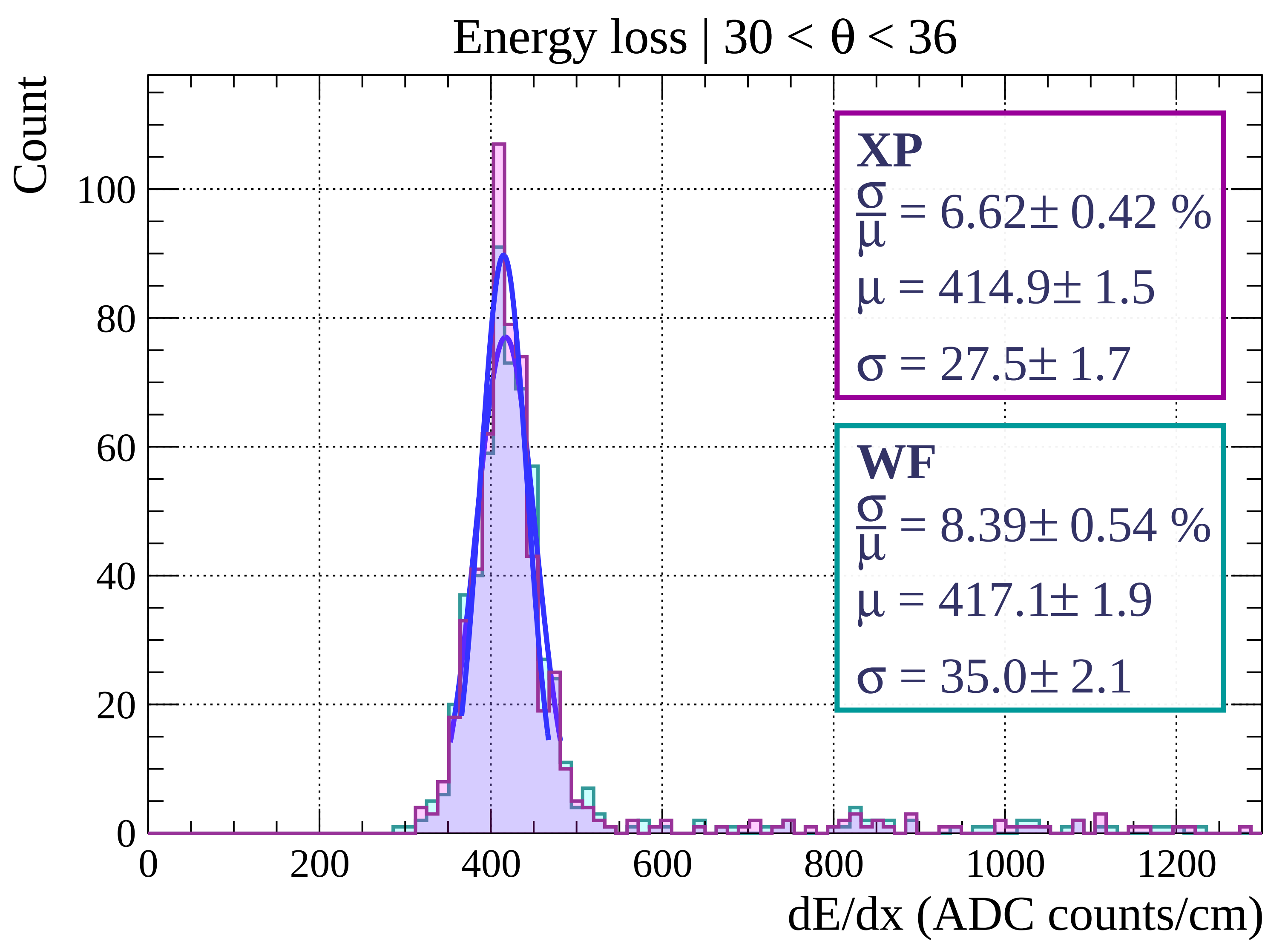


Energy loss | 12 < θ < 18
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 7.19± 0.19 %
μ = 418.2± 0.7
σ = 30.1± 0.7
WF
σ/μ = 7.55± 0.19 %
μ = 419.5± 0.8
σ = 31.7± 0.7

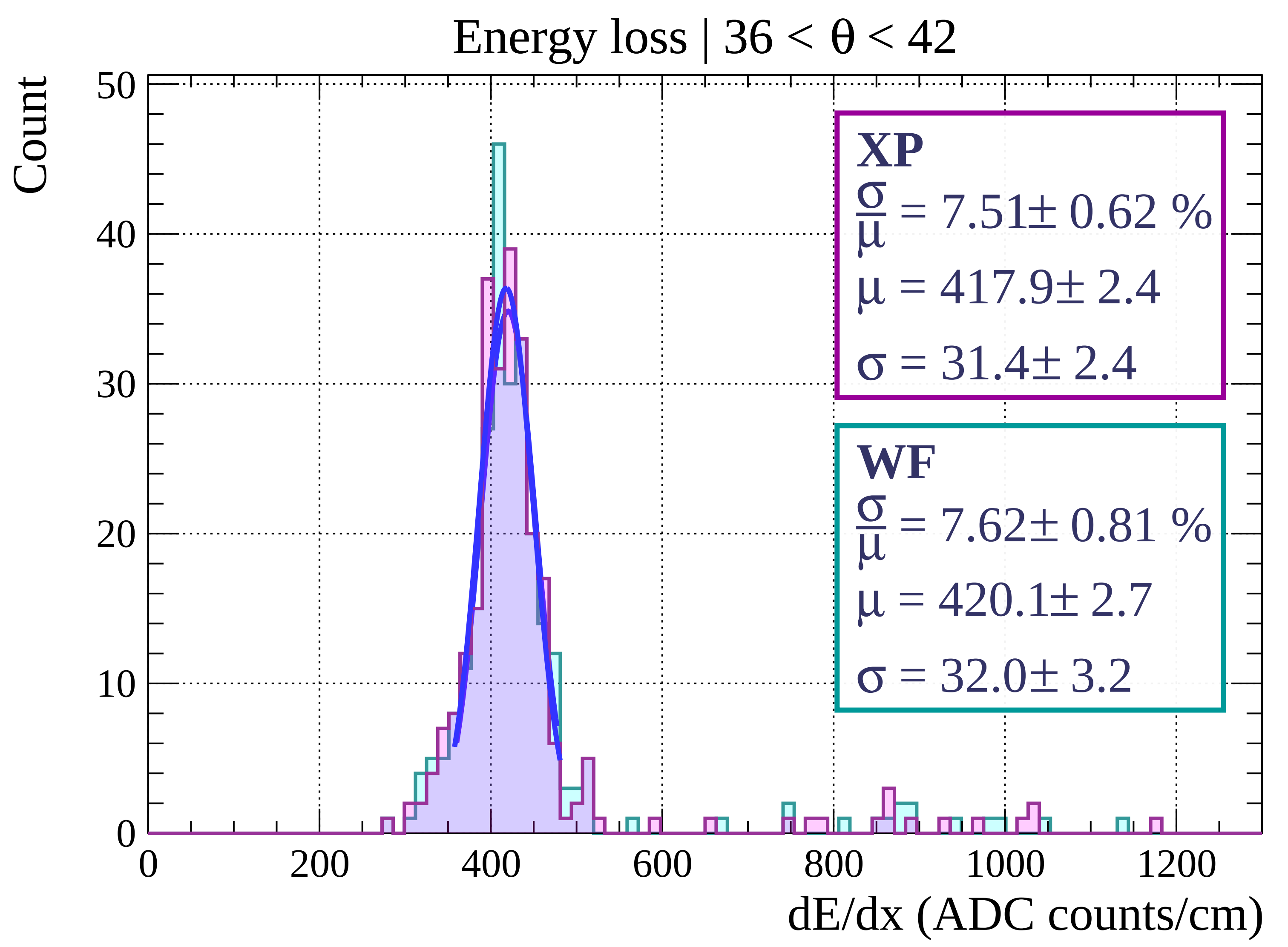


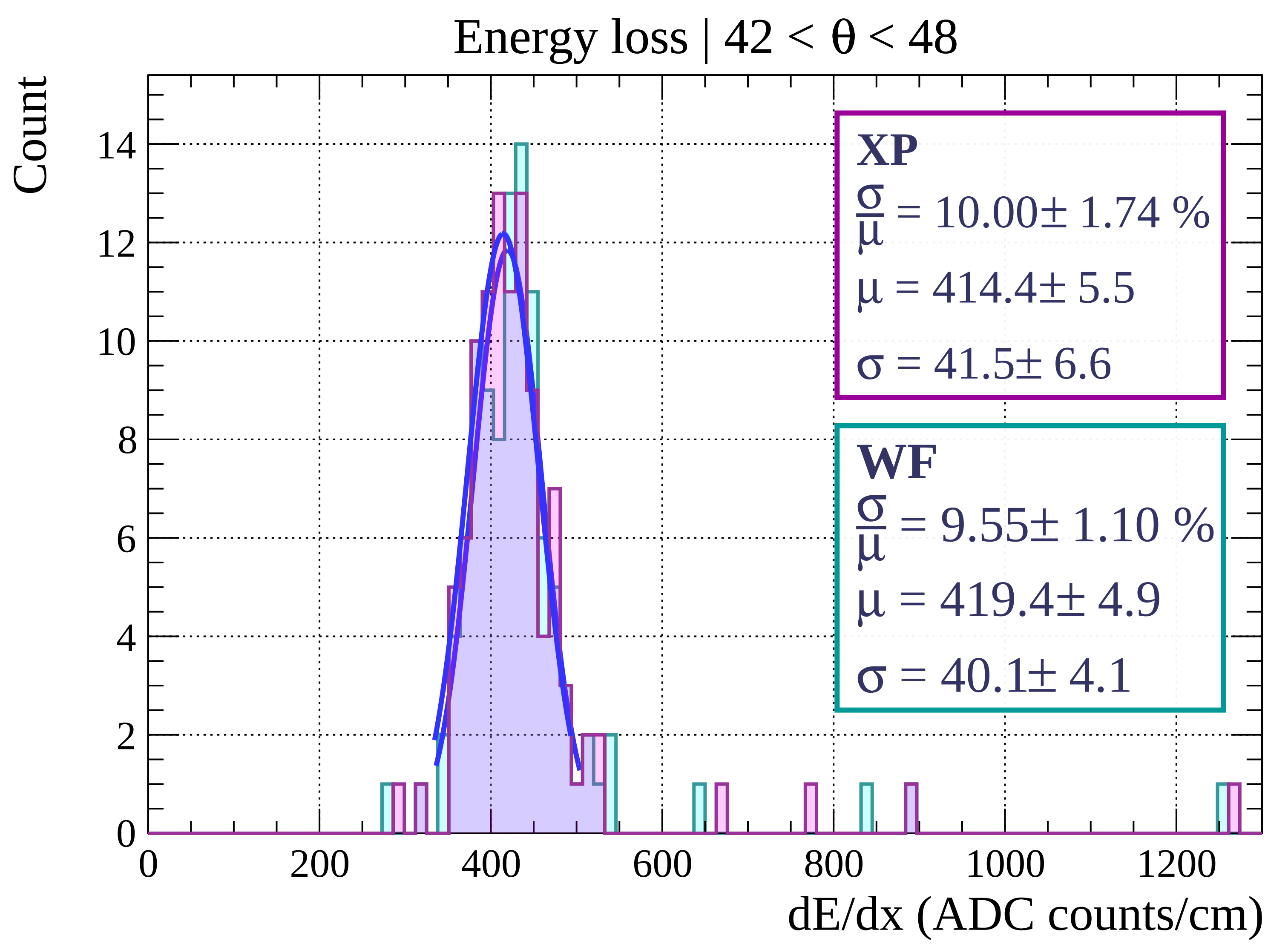
Energy loss | 18 < θ < 24
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 7.78± 0.27 %
μ = 415.7± 1.0
σ = 32.3± 1.0
WF
$\frac{\sigma}{\mu}$ = 7.77± 0.27 %
μ = 417.2± 1.0
σ = 32.4± 1.0


Energy loss | 24 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 7.57± 0.31 %
μ = 415.3± 1.2
σ = 31.4± 1.2
WF
σ/μ = 8.08± 0.35 %
μ = 416.8± 1.4
σ = 33.7± 1.3
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

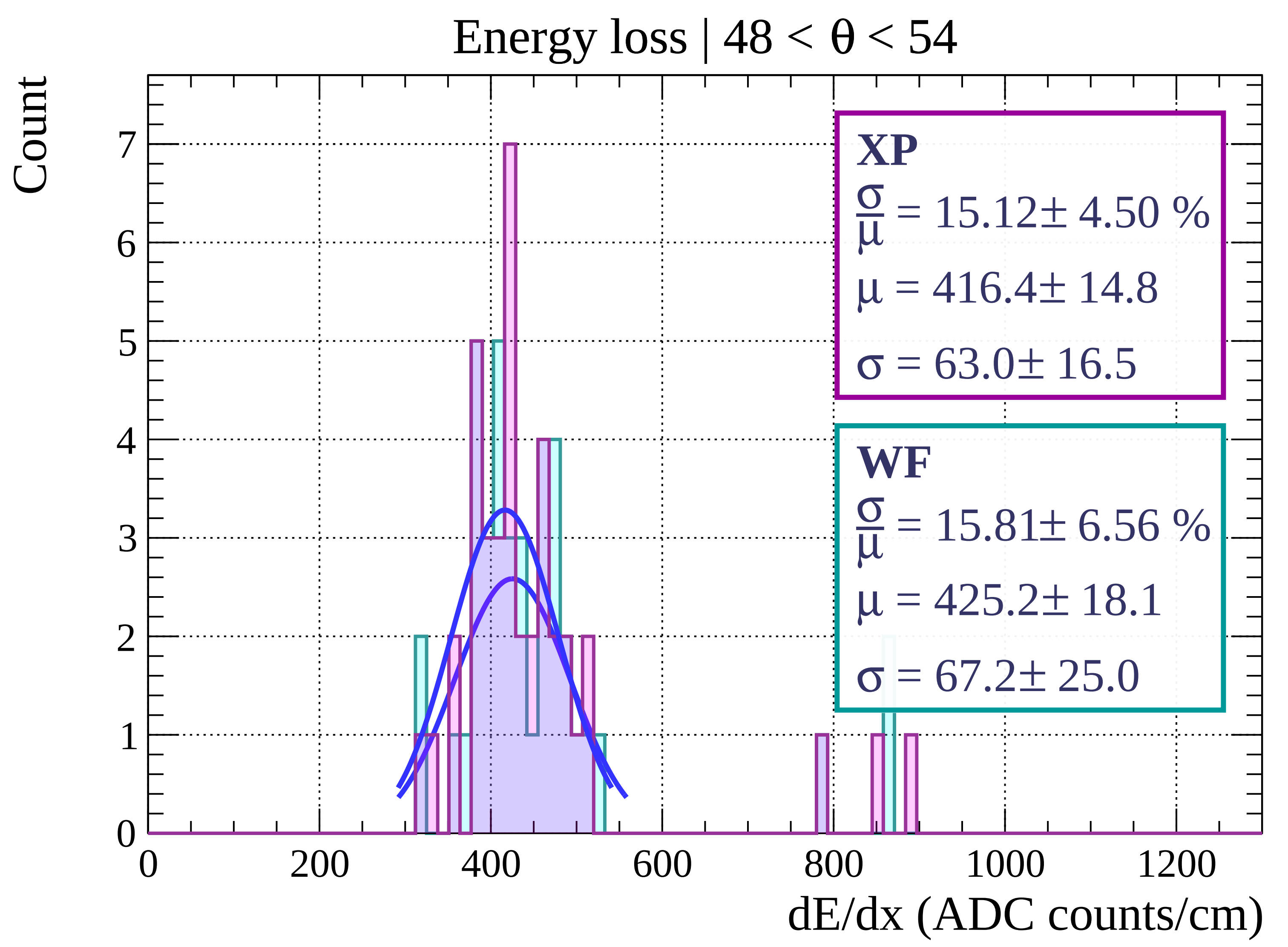


Energy loss | 30 < θ < 36
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.62± 0.42 %
μ = 414.9± 1.5
σ = 27.5± 1.7
WF
$\frac{\sigma}{\mu}$ = 8.39± 0.54 %
μ = 417.1± 1.9
σ = 35.0± 2.1

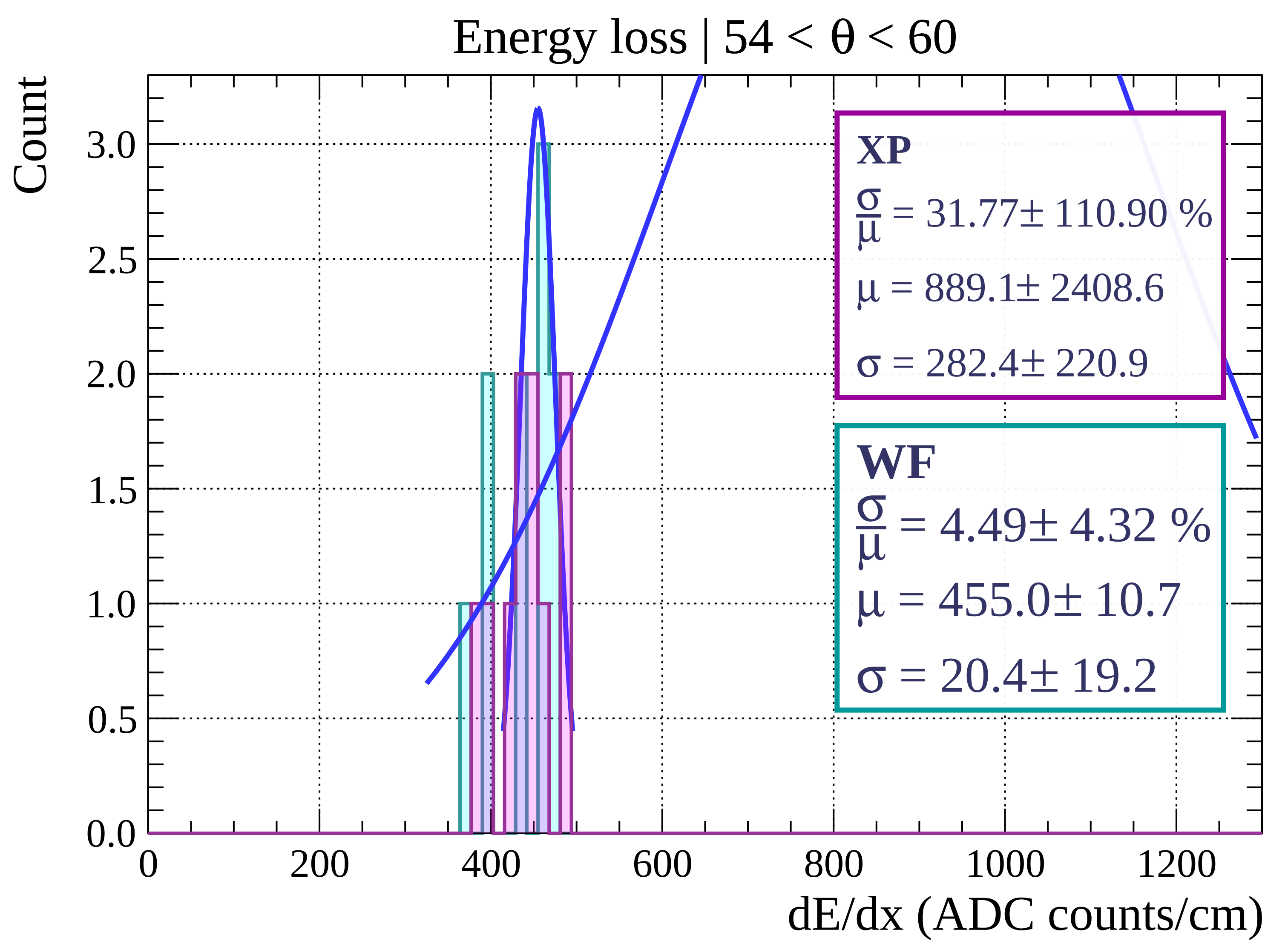

Energy loss | 36 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 7.51± 0.62 %
μ = 417.9± 2.4
σ = 31.4± 2.4
WF
σ/μ = 7.62± 0.81 %
μ = 420.1± 2.7
σ = 32.0± 3.2



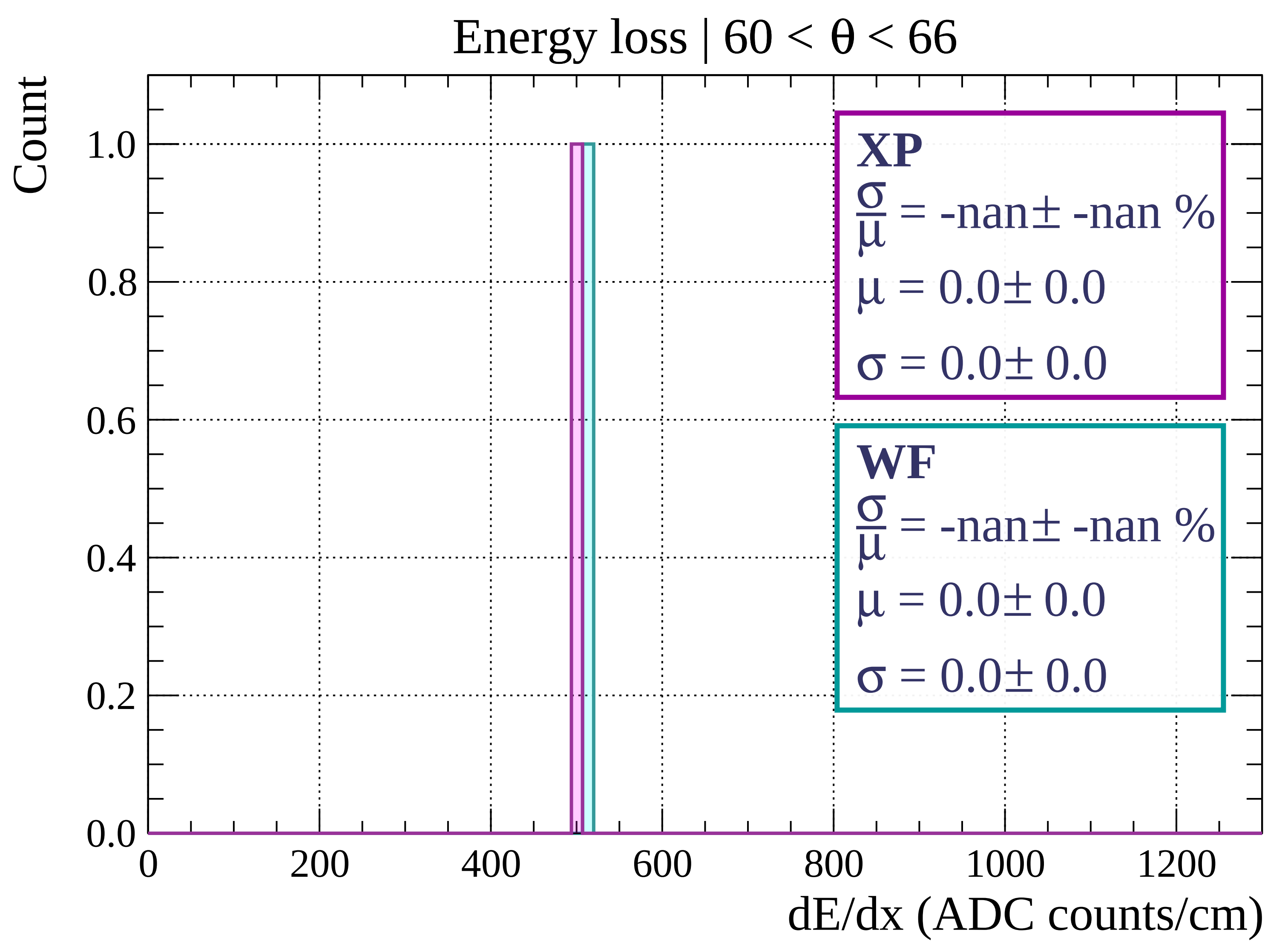
Energy loss | 42 < θ < 48
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.00± 1.74 %
μ = 414.4± 5.5
σ = 41.5± 6.6
WF
σ/μ = 9.55± 1.10 %
μ = 419.4± 4.9
σ = 40.1± 4.1
0
2
4
6
8
10
12
14
0
200
400
600
800
1000
1200



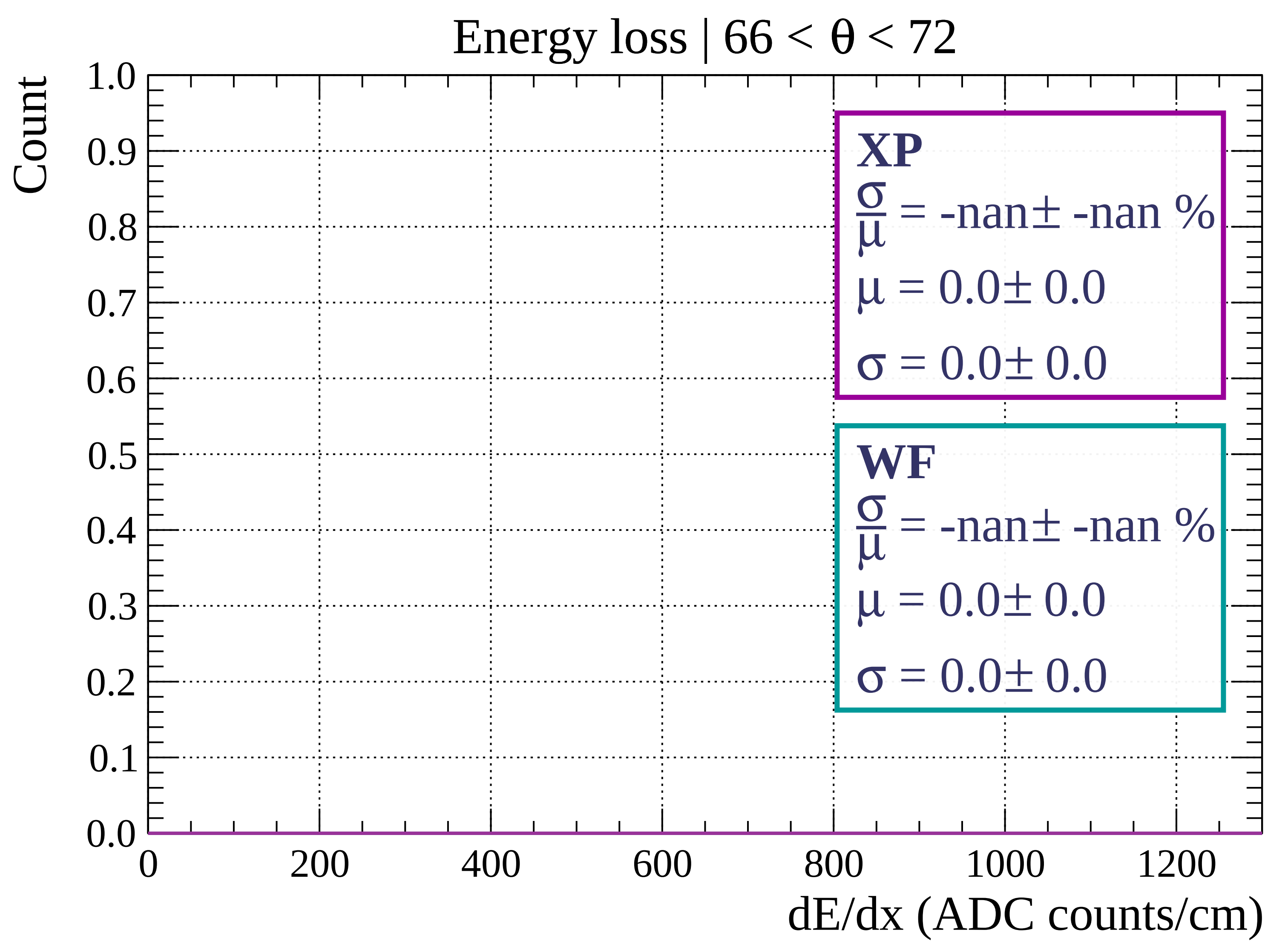
Energy loss | 48 < θ < 54
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
7
0
200
400
600
800
1000
1200
XP
σ/μ = 15.12± 4.50 %
μ = 416.4± 14.8
σ = 63.0± 16.5
WF
σ/μ = 15.81± 6.56 %
μ = 425.2± 18.1
σ = 67.2± 25.0

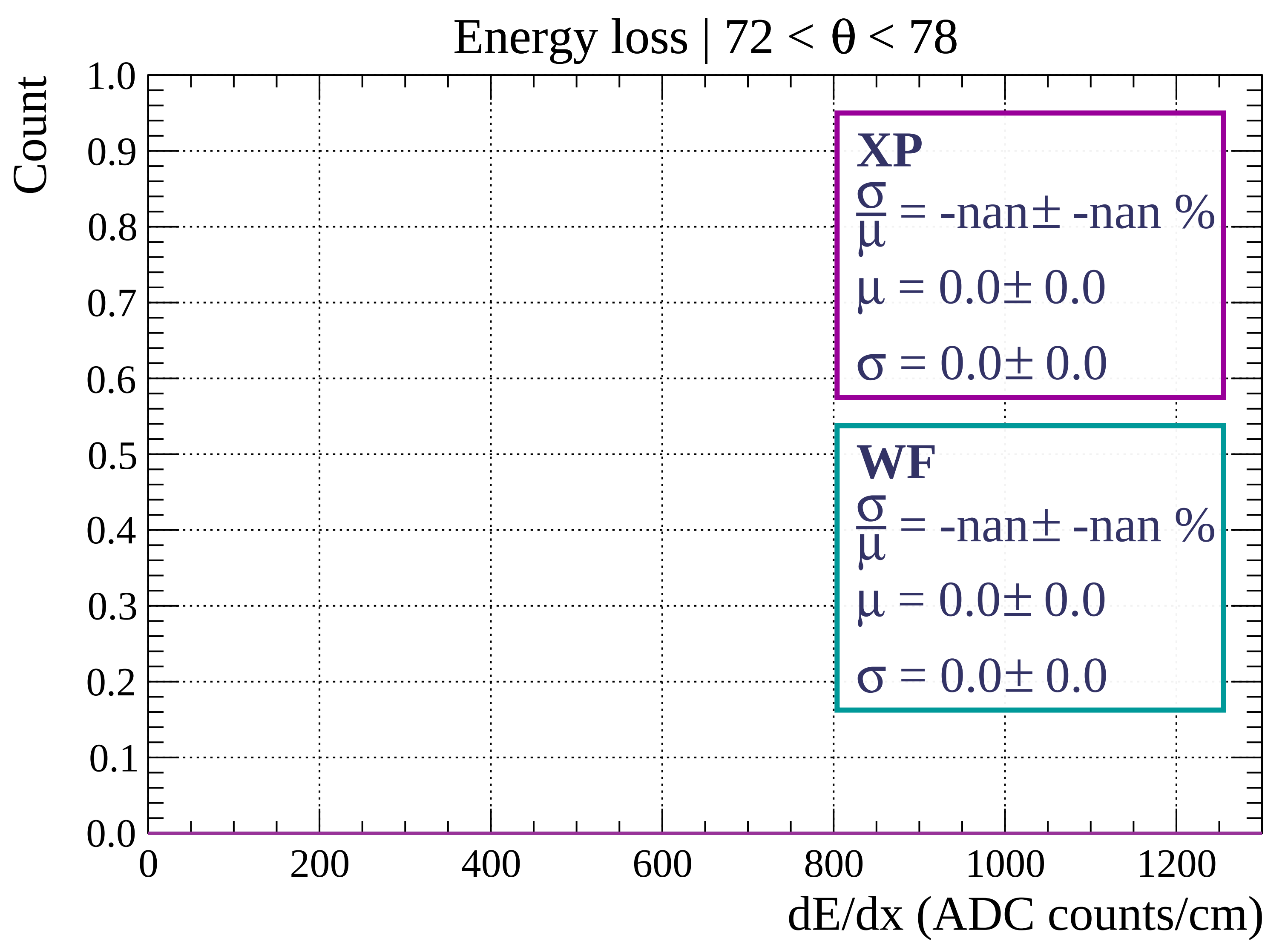


Energy loss | 54 < θ < 60
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 31.77± 110.90 %
μ = 889.1± 2408.6
σ = 282.4± 220.9
WF
σ/μ = 4.49± 4.32 %
μ = 455.0± 10.7
σ = 20.4± 19.2

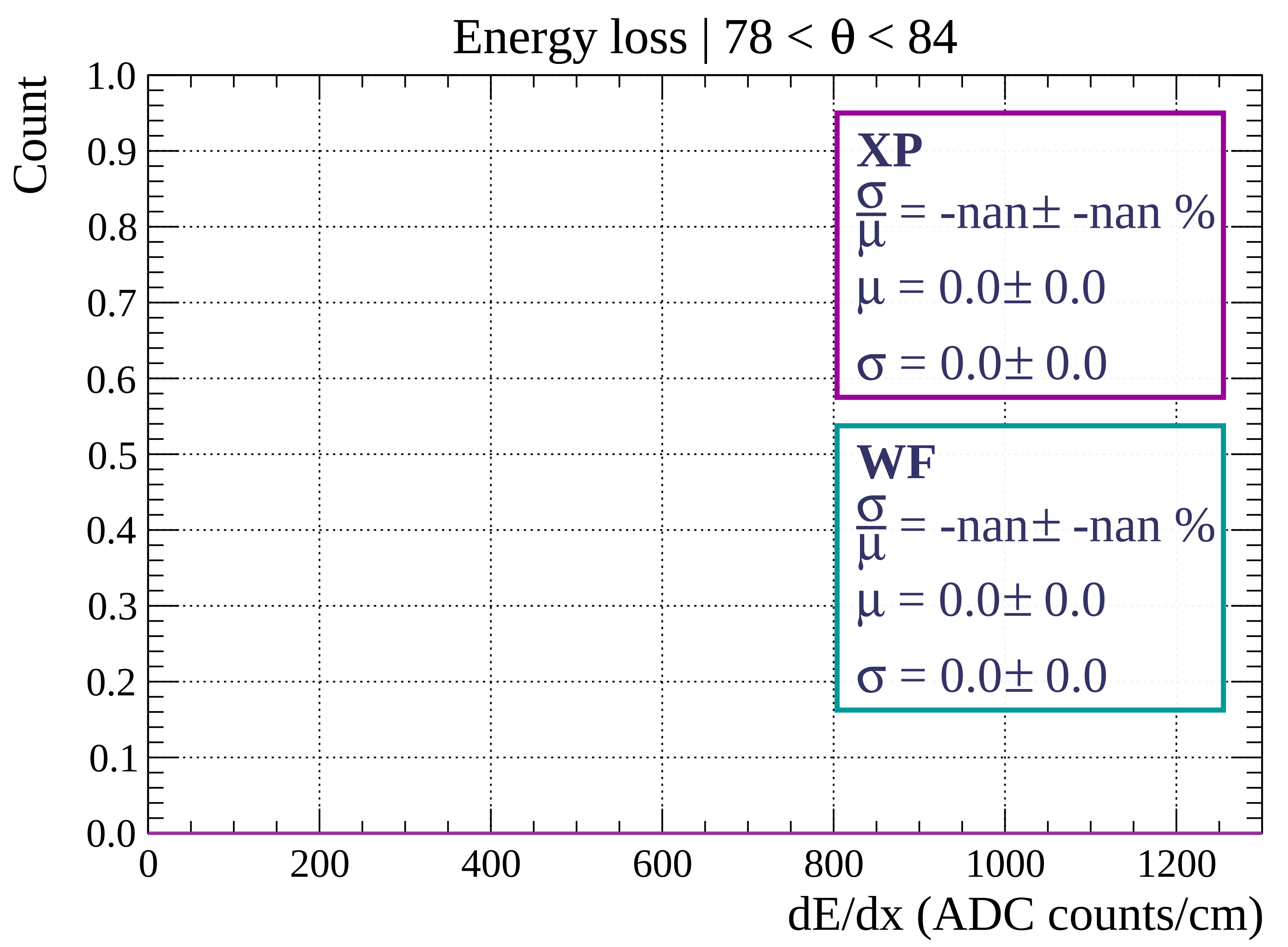


Energy loss | 60 < θ < 66
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

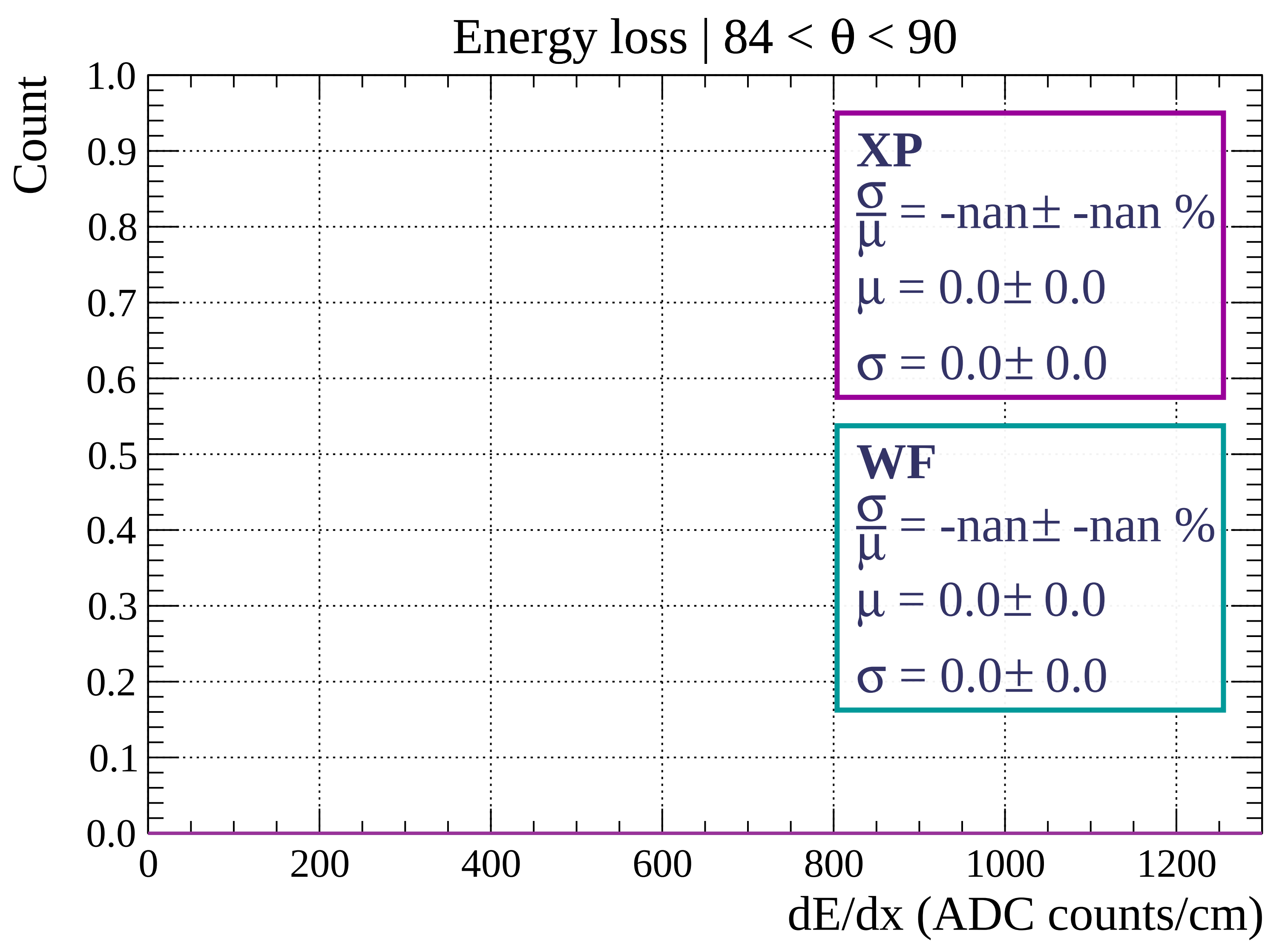

Energy loss | 66 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

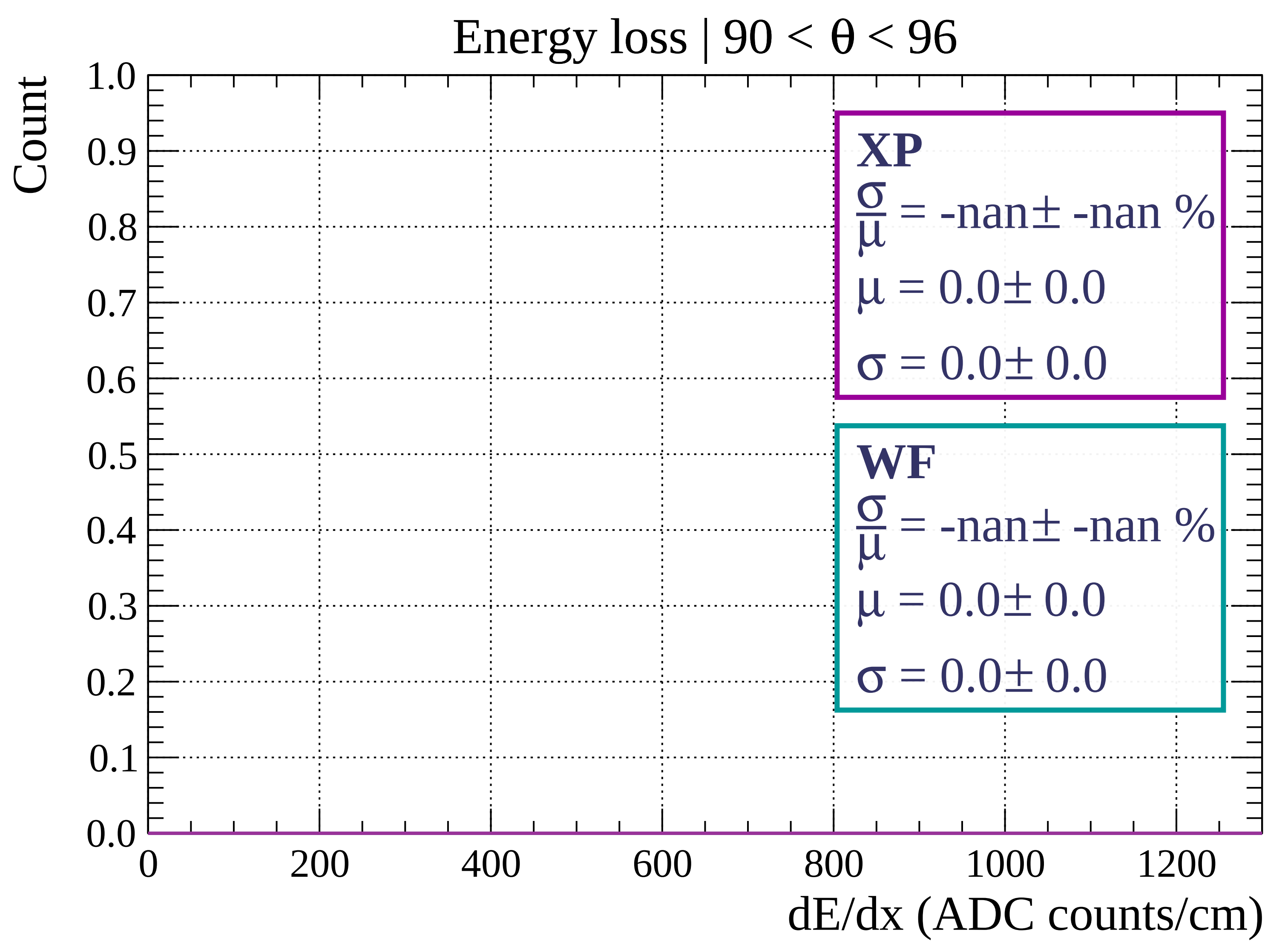


Energy loss | 72 < θ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

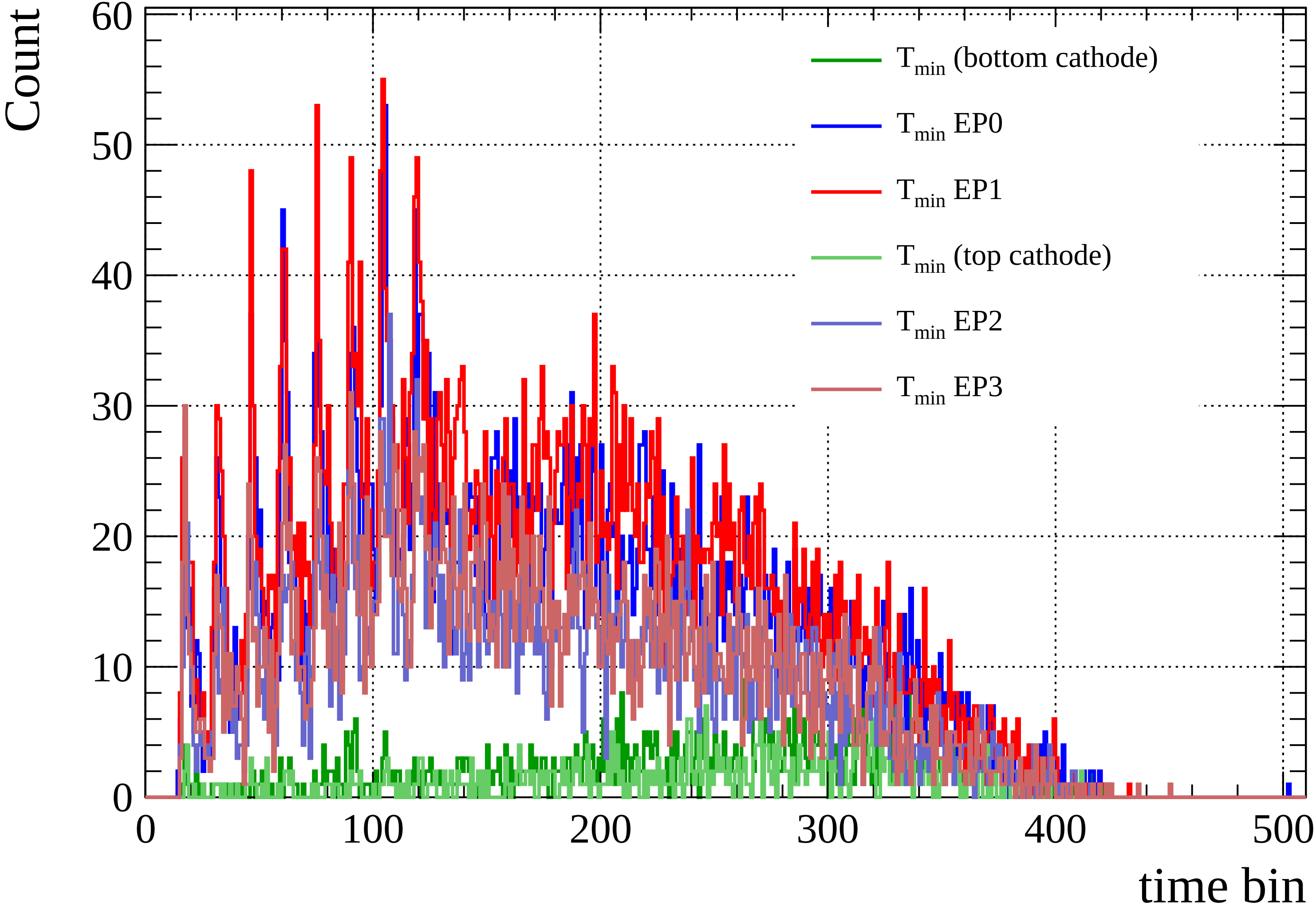


Energy loss | 78 < θ < 84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

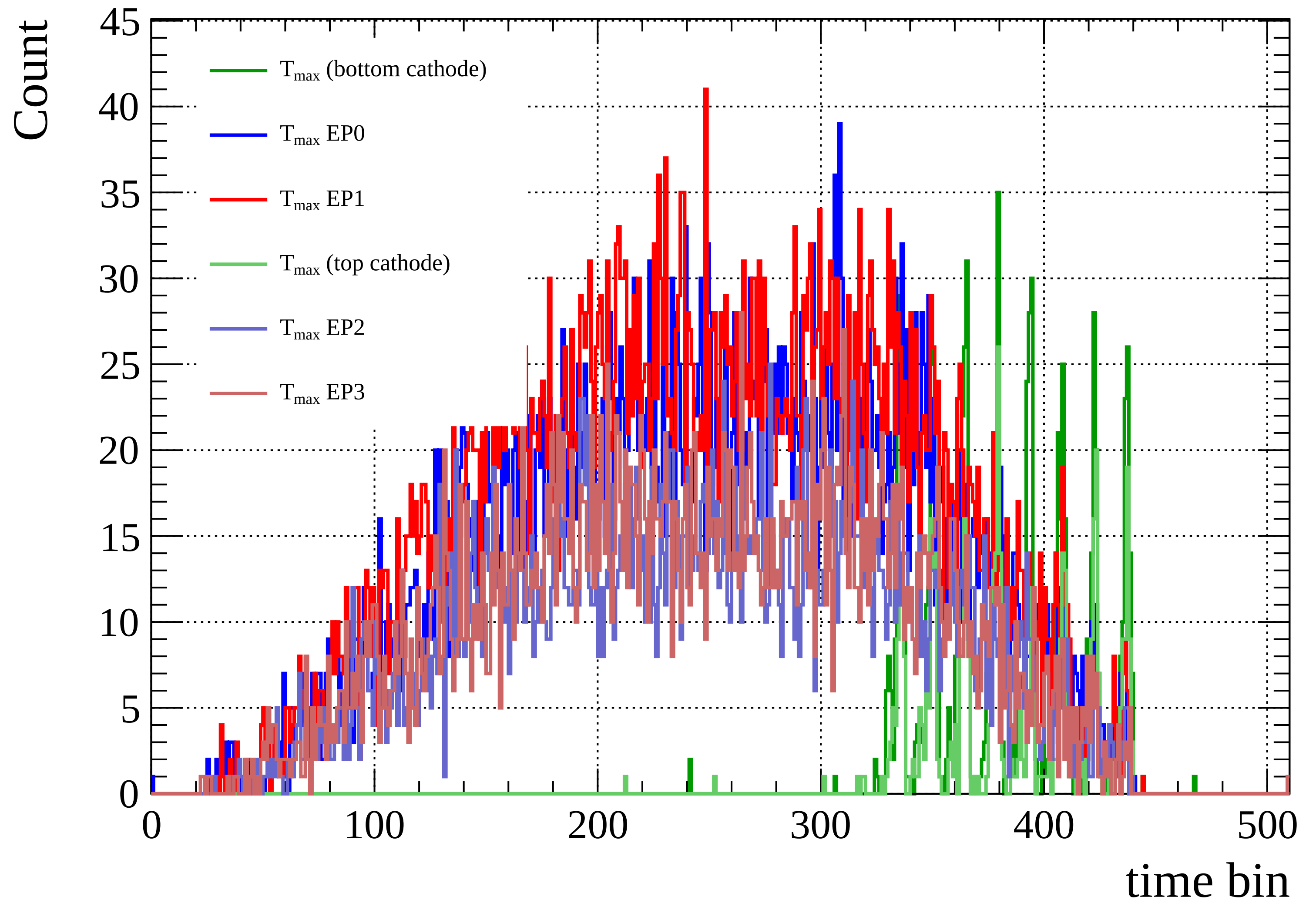

Energy loss | 84 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 90 < θ < 96
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
100
200
300
400
500
0
10
20
30
40
50
60
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3


Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
5
10
15
20
25
30
35
40
45
0
100
200
300
400
500